INTEGRATED SCIENCE

Pupil's Book 4

First Edition



Don't speak for quality, let quality speak for itself.

ESSAGE FROM THE AUTHOR

ır pupil, welcome to The Surekey Integrated ence Book Four, a resource carefully crafted to et the demands of the Competence-Based riculum (CBC). This book is designed to provide lear and engaging learning experience for every pupil of Primary r class.

h lesson in this book begins with well-organised **key words** to learners understand essential concepts before diving into main content. The **notes** are presented in a clear, precise and il-friendly manner to make learning of science enjoyable and ievable.

book includes **high-quality diagrams** that support lerstanding by illustrating key ideas and processes, helping rners to visualize and connect with real-world science. At the end ach lesson, carefully **structured exercises** ensure that learners e opportunities to assess their understanding and build critical king skills.

urther support mastery, the **meaning of key terms** is provided at end of each topic, enabling learners to grasp scientific guage and concepts with ease. This approach encourages ners develop knowledge, skills and values as demanded by the mpetence-Based Curriculum, preparing them for real-life blem-solving.

believe that The Surekey Integrated Science Book Four will serve valuable tool in building confident, capable and curious young ntists.

but not least, I regret for any typing error you will come across ading this book.

e: **In case of any questions, corrections, concerns and inquiries ut the book, please contact me using the telephone numbers -mail address on the back cover.**

hing You Happy learning!

ME HUMPHREY
AUTHOR

ACKNOWLEDGEMENT



I take this golden opportunity to cast a vote of thanks to the following people for their great contribution towards the compilation and production of this reference book.
Your guidance and consulations have played a fundermental role during the course of writing this academic material.

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"Your support in crafting this reference material in the world of science is truly commendable".

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CONTENT OUT OF CURRICULUM IN PRIMARY FOUR INTEGRATED SCIENCE

TOPIC	IRRELEVANT CONCEPTS
Plant life	⊙ Tropisms ⊙ Definition and experiment of osmosis ⊙ Testing for starch in a leaf.
Human body organs	⊙ Parts and functions of each body organ.
Sanitation	⊙ Types of toilets and latrines.
Weather changes around us	⊙ Clinical thermometer

Important points to note about The Surekey Integrated Science Book 4

1. The surekey Book 4 Integrated Science is a comprehensive study guide.
2. It is designed for primary four learners.
3. The book integrates different concepts of science.
4. It includes clear explanation of key science terms.
5. Contains engaging illustrations to simplify learning.
6. Includes practical activities to promote hands-on learning.
7. Features end-of-unit questions for practice.
8. Helps learners develop critical thinking skills.
9. Follows the national curriculum guidelines.
10. Supports independent learning and revision.

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1 THEME: THE WORLD OF LIVING THINGS
TOPIC 1: PLANT LIFE

Key Words:
• bear • bud • chlorophyll • cotyledon • moss
• dicotyledon

Plants are organisms that make their own food through photosynthesis and are typically rooted in the ground.

Characteristics of plants
(i) They make their own food. (ii) They contain chlorophyll.
(iii) They have many cells. (iv) They are rooted in the ground.

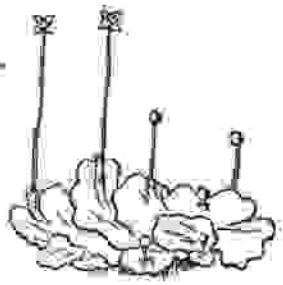
Groups of plants
* Flowering plants *Non flowering plants

Non-Flowering Plants
Non flowering plants are plants that do not bear flowers.
They reproduce by means of spores or seeds.
They make their own food because they contain chlorophyll.
Examples of non-flowering plants
mosses, liverworts, horsetails, hornworts, ferns, pine, fir and cedar.


pine


moss


fern


liverwort

FLOWERING PLANTS
Flowering plants are plants that bear flowers.
They reproduce by means of seeds.
They are the largest group of the plant kingdom.
Examples of flowering plants
• maize plants • bean plants • banana plants • cassava plants
• mango plants • blackjack plants • cabbage plants • eggplants
• tomato plants • pea plants



Maize plant

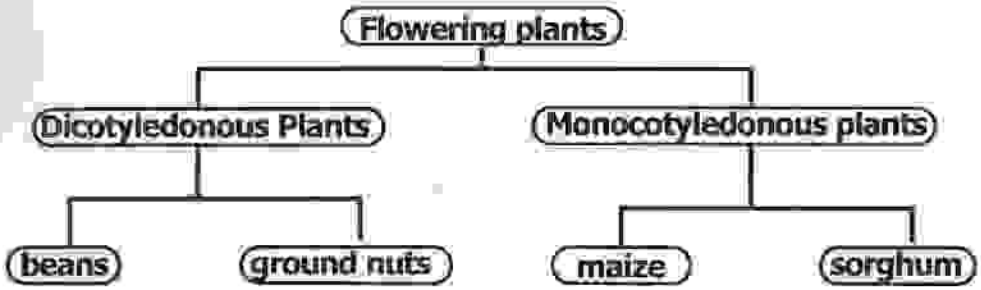


Bean plant



Banana plant

Table showing classification of flowering plants.



Groups of flowering plants

⊡ Dicotyledonous plants ⊡ Monocotyledonous plants

Dicotyledonous plants

These are flowering plants that bear seeds with two cotyledons

Characteristics of dicotyledonous plants

✓ They bear seeds with two cotyledons. ✓ They have network leaf venation.
✓ Their seeds store food in the cotyledon.
✓ Most dicot seeds undergo epigeal germination.
✓ They have taproot system.

Examples of dicot plants are

⊡ bean plants ⊡ ground nut plants ⊡ pea plants ⊡ mango plants
⊡ orange plants ⊡ cabbage plants ⊡ carrot plants ⊡ pawpaw plants.



pea plant



Ground nut plant



Bean plant



Soya bean plant

Monocotyledonous Plants

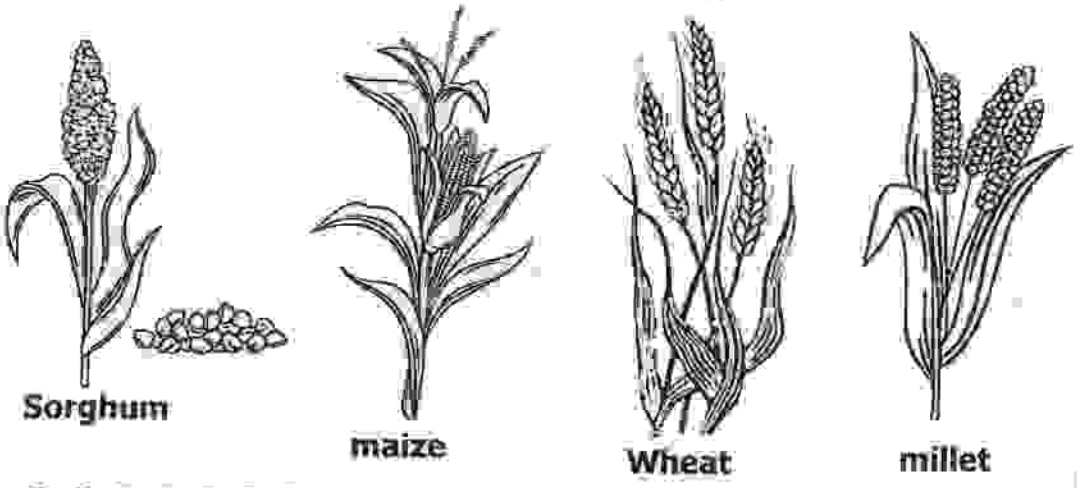
These are flowering plants that bear seeds with one cotyledon.

Characteristics of monocotyledonous plants

- They bear seeds with one cotyledon.
- They have parallel leaf venation.
- They store their food in the endosperm.
- Their seeds undergo hypogeal germination.
- They have fibrous root system.

Examples of monocotyledonous plants

maize plants, millet plants, sorghum plants, onion plants



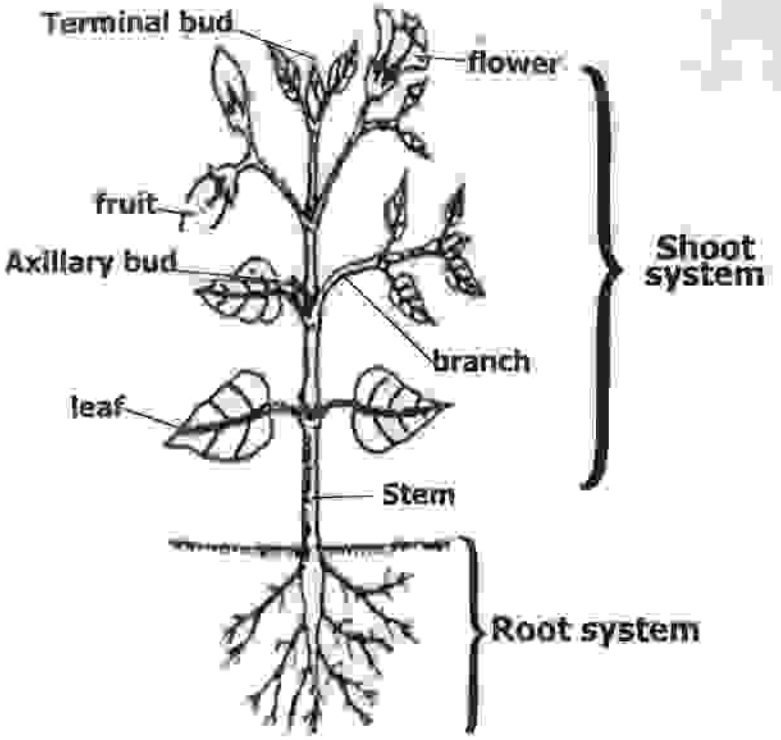
Evaluation activity 1.1

1. Name the process that enables plants to make their own food.
2. Mention **two** groups of plants.
3. State any **two** characteristics of dicotyledonous plants.
4. Give the reason why a carrot plant is grouped under flowering plant.
5. Name the group of flowering plants to which;
   i) Cowpeas belong
   ii) Millet belongs
6. State any **one** difference between a moss plant and a bean plant.
7. Mention any **one** way in which plants are important to people.
8. How can **one** identify a monocotyledonous plant from other plants?
9. State the way in which a tomato plant reproduces.
10. Which group of flowering plants undergo epigeal germination?

Key words:

- axillary
- terminal
- reproductive
- firm
- propagation
- bud

Parts of the flowering plant



Functions of the Parts of the Flowering Plant

Part	Function (s)
Terminal bud	It grows to increase the height of the plant.
Axillary bud	It grows into a side branch, leaf or flower
Stem	• It holds the leaves, flowers and fruits of a plant. • It transports water and mineral salts from roots to other parts of a plant. • It transports manufactured food from leaves to other parts of a plant.
Leaves	They make food for the plant.
Roots	They absorb water and mineral salts from the soil.
Flowers	• They are used for reproduction. • They develop into fruits and seeds.
Fruits	They store seeds.

Systems of the Flowering Plant

Shoot system.

It grows above the ground.

It is made up of stem, leaves, flowers and fruits.

Root system.

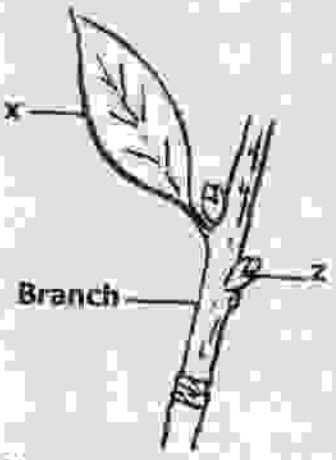
It grows under the ground/soil.

It is made up of; root cap, root hairs, lateral roots, tap root and fibrous roots

Evaluation activity 1.2

1. Name the part of a flowering plant which grows into a side branch.
2. State the main function of leaves to the plant.
3. Identify the system of flowering plants that enables the plant to absorb water from the soil.
4. Name the part of the flowering plant that stores seeds.
5. Give any **one** function of the flower to an orange plant.

The diagram below shows part of the flowering plant. Use it to answer questions **6** and **7**.



6. Name the part marked **X**.
7. Give the function of the part marked **Z** to a plant.
8. Name the system of a flowering plant that grows;
   (i) Above the ground (ii) Under the ground
9. The table below shows some of the parts of the flowering plant and the function of each part. Complete the table correctly.

Part	Function
.....	Manufactures food for the plant.
Roots	.....
.....	Develops into fruits and seeds.
Stem	.....

Key words:

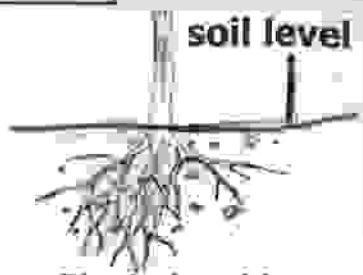
- lateral
- absorb
- radicle
- adventitious
- osmosis
- firmly

PLANT ROOTS

Roots are branching parts of a plant that grow downwards into the soil.
Roots develop from the radicle of the seed except adventitious roots.
Adventitious roots develop from other parts of the plant.
Roots have no leaves, nodes, internodes or buds.

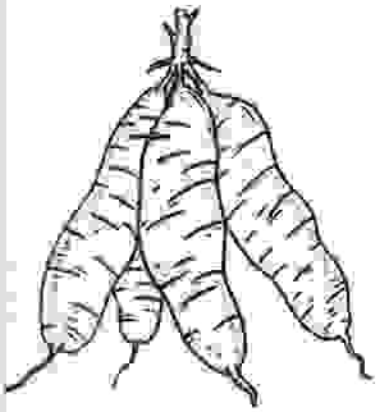
Functions of roots to a plant.

- Roots hold the plant firmly into the soil.
- Roots absorb water and mineral salts from the soil.
  This process is called **osmosis.**
- Root tubers store food for the plant.
  They include; cassava, turnips, sweet potatoes and carrots

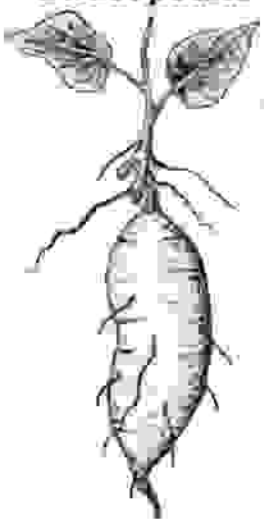


Plant absorbing water from the soil

Cassava



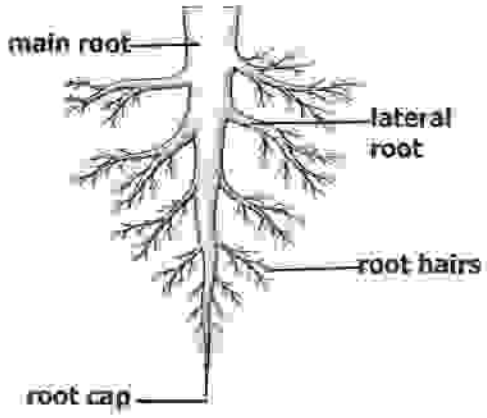
Sweet potato



Carrot



Parts of the root system



Functions of each part.

Root hairs. They absorb water and mineral salts.

Root cap. It protects the tip of the growing root.

Main root and lateral root. They hold the plant irmly into the soil.

Root Systems

Taproot system

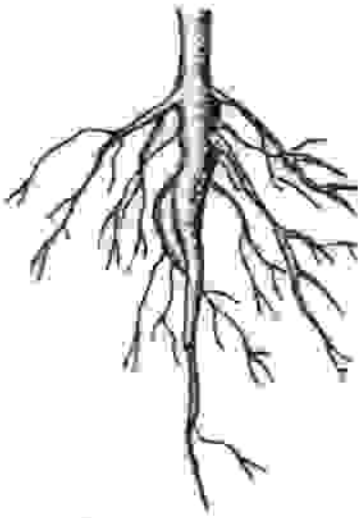
This is the type of root system with the main root from which lateral roots develop.

It is common in dicot plants like beans, peas, ground nuts, mangoes, cassava, carrots, oranges and ground nuts.

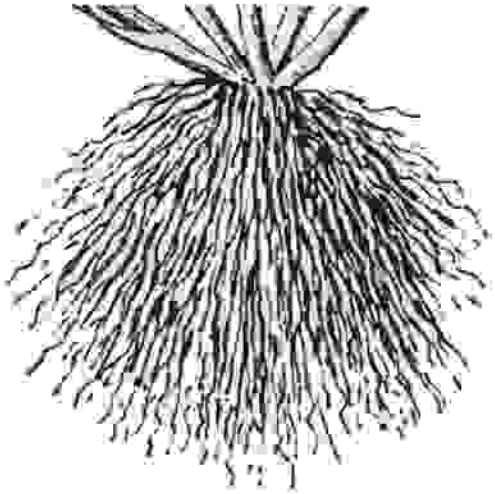
Fibrous root system

This is the type of root system in which all the roots develop from the same point and almost have the same size.

It is common in monocotyledonous plants like maize, millet, sorghum, onion and wheat.



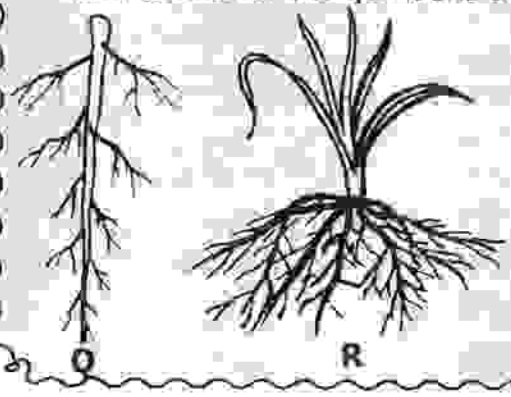
Taproot system



Fibrous root system

Evaluation activity 1.3

1. Name the part of a seed that develops into the roots.
2. State any **two** functions of roots to a plant.
3. State the function of the following parts of the taproot system to a plant.
   (i) Root hair (ii) Root cap
4. Identify the difference in root structure between a carrot plant and a maize plant.
5. State the meaning of the term adventitious root.
6. Name any **one** type of root system.
7. State the importance of radicle in the development of the plant.
8. The diagrams below show the types of root system. Study and use them to answer the questions that follow.



(a) Name the types of root system marked **Q** and **R**.
(b) Identify the group of flowering plants that develop the type of root system marked **Q**.
(c) Mention any one plant that grows the type of root system marked **R**.

Key words:

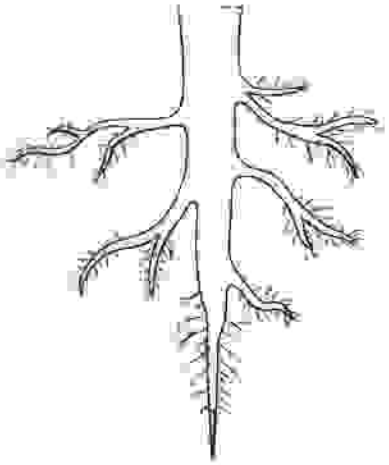
- buttress
- absorb
- clasp
- prop
- buttress
- adventitious

Types of Roots/Classification of Roots

Roots are classified basing on where they originate from in their growth.

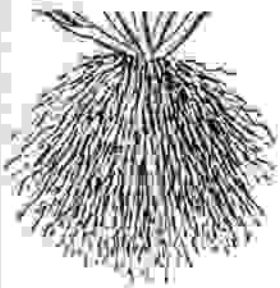
The types of roots are;

True roots. These are roots that originate from the radicle of a seed.
True roots develop in dicotyledonous plants.
An example of a true root is a **taproot.**
A tap root has the main root where lateral roots branch from bearing root hairs. Lateral roots of legumes bear both root hairs and root nodules.



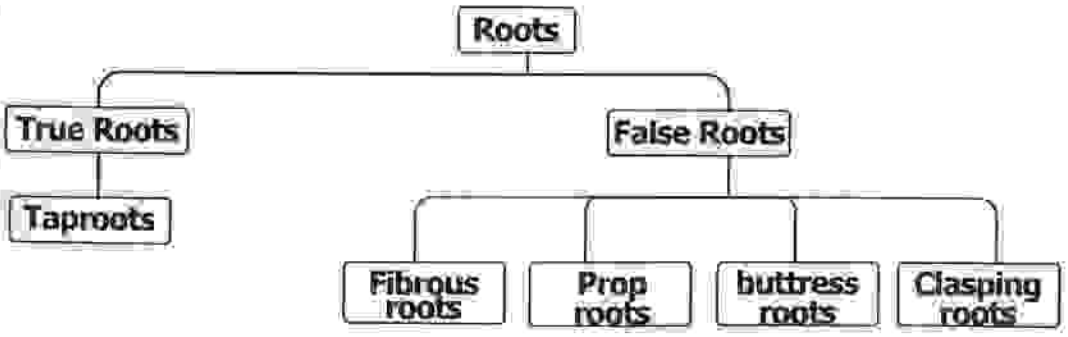
Examples of plants that grow taproots

-mango plants
-bean plants
-jackfruit plants
-pea plants
-soya bean plants
-avocado plants
-guava plants
-lemon plants
-ground nut plants



False roots/adventitious roots. These are roots that develop from other parts of the plant.
False roots grow in monocotyledonous plants.
Examples of false roots are; fibrous roots, prop roots, clasping roots, buttress roots...
Examples of plants that grow false roots
grass, millet, sorghum, maize, rice

Classification table for roots



False roots/adventitious roots

These are roots that develop from other parts of a plant.
These parts include; nodes, stems, branches, leaves and internode.

Common Examples of false/adventitious Roots

Aerial roots

These are roots that develop from the stem of a plant above the ground.
They absorb water directly from the atmosphere.



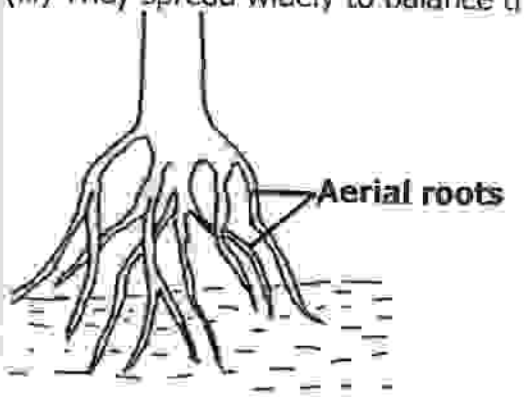
Aerial roots

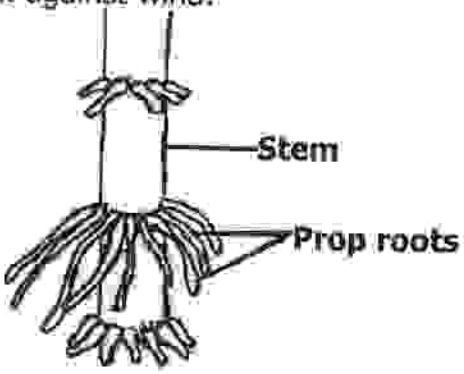
Prop roots

These are roots that develop at the base of the stem of some plants.
Prop roots provide extra support to the plant during flowering stage.
Plants that develop prop roots are; maize, sorghum, sugar cane and wheat.

Adaptations of prop roots to their function

(i) They are many to increase the stability of the plant.
(ii) They grow thick and strong for extra support.
(iii) They spread widely to balance the plant against wind.





Buttress roots

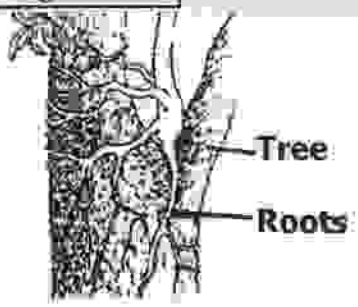
These are roots that grow at the base of bigger trees.
Buttress roots provide extra support to the plant.
They enlarge and form a triangular shape at the base of the plant.
They develop in plants like; jackfruit, Ficus trees, Mahogany, Mvule and silk cotton.

Clasping roots

Clasping roots are roots that develop on plants with weak stems that grow on top of other plants.
Clasping roots enable plants to climb other plants.
Plants that develop clasping roots are; morning glory, tomatoes and white yams



buttress roots



Clasping roots

Breathing roots

Breathing roots are common in plants which grow in swampy/waterlogged areas.

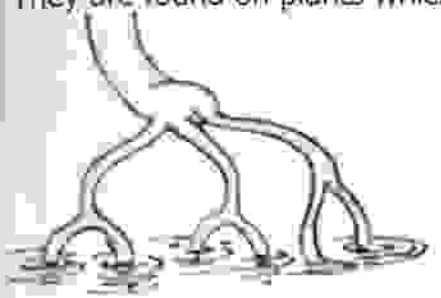
These roots grow upwards in air to enable the plant get enough oxygen for respiration.

Stilt roots

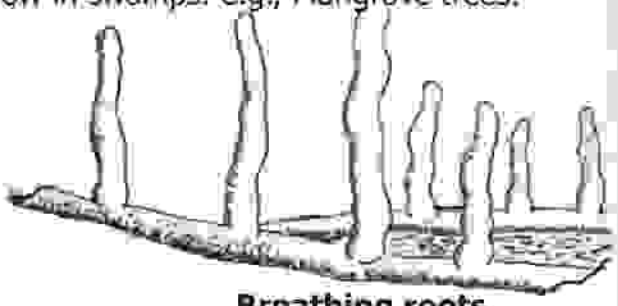
Stilt roots are roots that grow from lateral branches on different branches.

They provide extra support to the plant.

They are found on plants which grow in swamps. e.g., Mangrove trees.



Stilt roots



Breathing roots

Storage roots

These are swollen underground roots with stored food.

They are commonly known as root tubers.

Examples of storage roots are;

cassava, sweet potatoes, dahlia, carrots, turnips.

Carrot



Dahlia



Turnip



Point to note:

Carrots, turnips and beetroots are propagated by planting seeds while sweet potatoes and cassava are propagated by planting stem cuttings.

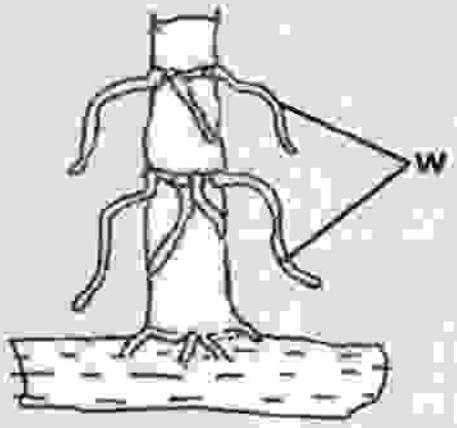
Ways in which roots are important to people.

- Some roots are eaten as food.
- Some roots are used as herbal medicine.

• Some roots are sold for money. • Some roots are used as wood fuel.

Evaluation activity 1.4

1. (a) At what stage does a maize plant develop prop roots?
   (b) Give any **two** ways in which prop roots are able to support the weight of a maize plant.
   (c) Apart from maize plants, name any **one** other plant that develops prop roots.
2. The diagrams below show adventitious roots. Use them to answer the questions that follow.





   (a) Name the adventitious roots marked **W** and **X**.
   (b) Give the function of the adventitious roots marked **W** to a plant.
   (c) Mention any **two** plants that develops the adventitious root marked **X**.
3. Name the adventitious roots that absorb water from the atmosphere.
4. Give any **one** way in which roots help to improve human health.
5. In the table below, part **A** shows some of the examples of adventitious roots and part **B** has their functions to the plant.

Part A: Adventitious roots	**Part B: Their functions to a plant**
Prop roots	Keep food for the plant.
Clasping roots	Absorb water from the atmosphere.
Aerial roots	Provide extra support to the plant.
Storage roots	Enable the weak plant to climb others.

Match the adventitious roots with their functions in the spaces provided below correctly

(i) Prop roots .....
(ii) Clasping roots .....
(iii) Aerial roots .....
(iv) Storage roots .....

Key words:

- rhizome
- twining
- garlic
- tendril
- foliage
- corm

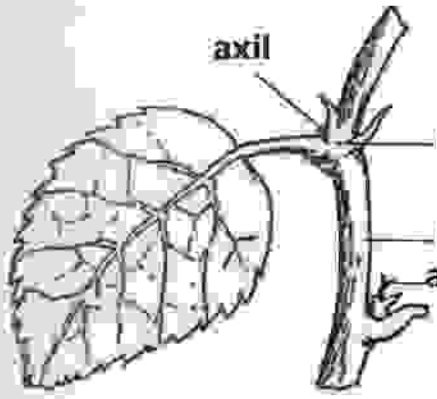
PLANT STEMS

A stem is the part of a plant that carries water and food between the roots and other parts of the plant.

Functions of the stems to the plant.

- ✓ Stems transport water and mineral salts from the roots to other parts of the plant.
- ✓ Stems transport food from leaves to other parts of a plant.
- ✓ Some stems store food for the plant.
- ✓ Stems hold leaves, flowers and fruits.

Parts of the stem



The **node** is the point on the stem where the leaves develop from.

The **internode** is the region between two nodes on the stem.

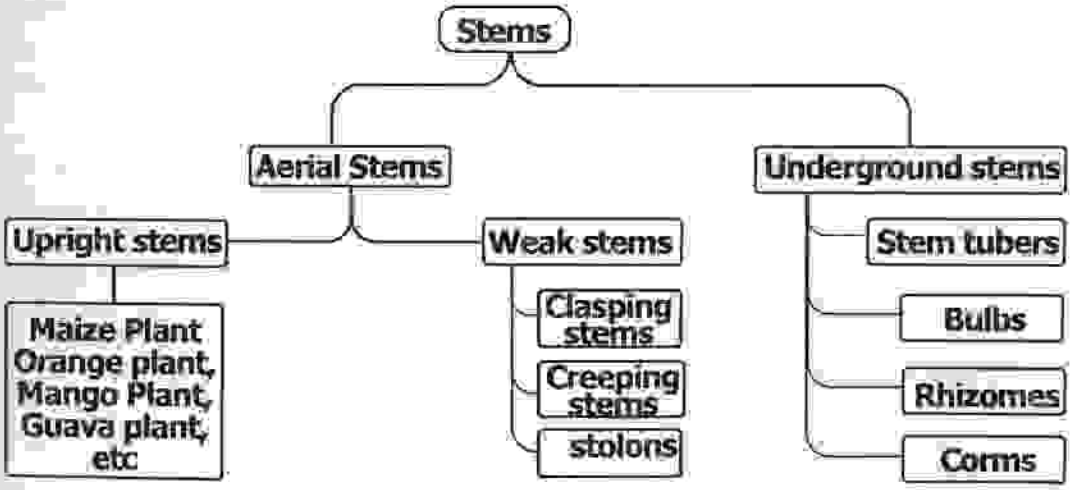
The **axil** is the angle between the leaf and the stem.

The **axillary** bud develops into a side branch.

Types of stems/classification of stems

Stems are classified basing on their position (location) of growth.

1. Aerial stems.
2. Underground stems.



1. Aerial stems

These are types of stems that grow above the ground.

Groups of aerial stems.

- upright stems
- weak stems

Upright stems

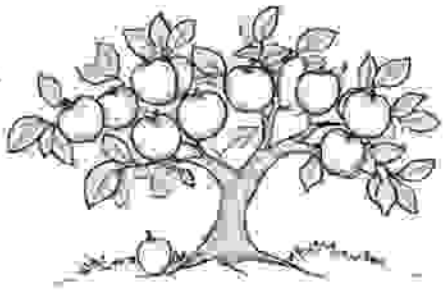
These are stems that grow vertically in space above the ground.
Plants which grow upright stems include; sugar canes, mangoes, oranges, avocados and apples.



Sugar cane



Mango plant



Apple plant

Weak stems/Climbing stems

Weak stems are stems that cannot support themselves upright and need support or spread along the ground.

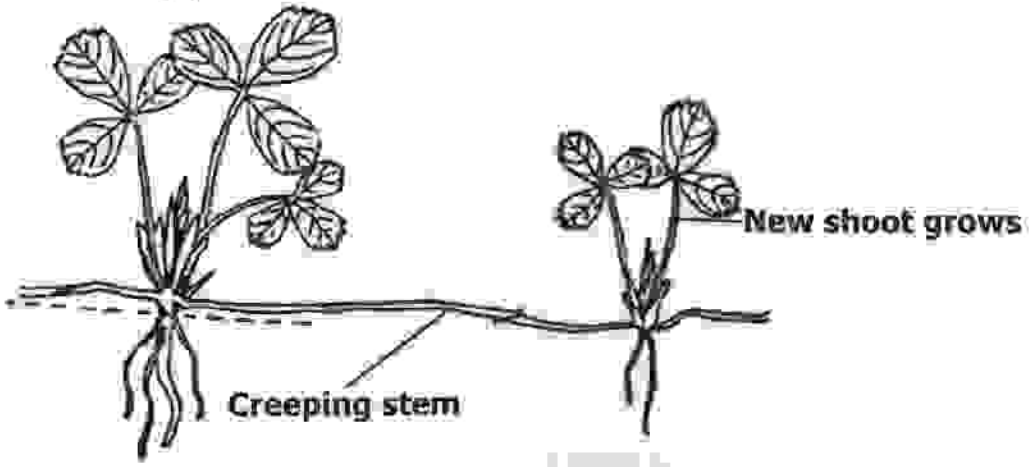
Examples of weak stems include; creeping stems, stolons and twiners

Creeping stems

These are stems that grow horizontally along the surface and give rise to new plants from axillary buds or terminal buds.
They are also called runners.
Creeping stems help the plant to spread and colonise.

Examples of plant with creeping stem Strawberry, watermelon and pumpkins.

A strawberry plant



Stolons

These are aerial side stems that give rise to a new daughter plant when the bud at its apex touches the soil e.g., strawberry and mint.

Points to note:

Plants with weak stems climb other plants to get extra support in order to get enough sunlight.

They climb other plants by clasping/twining, using tendrils or using hooks

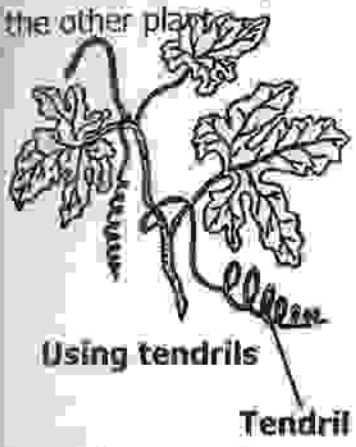
Plants that twine include; morning glory, tomatoes and white yams

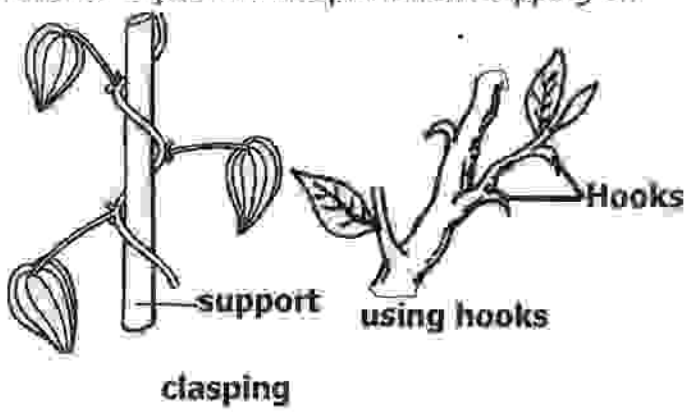
Plants that use tendrils include; passion fruits, beans, peas and cucumber

The tendril helps the plant to attach itself for support.

Plants that use hooks include; rose flowers and thorn apples.

The thorns/hooks point downwards to prevent the plant from slipping off the other plant.





2. Underground stems

These are food storage stems that grow under the ground.

Groups of underground stems

Stem tubers, bulbs, rhizomes and corms

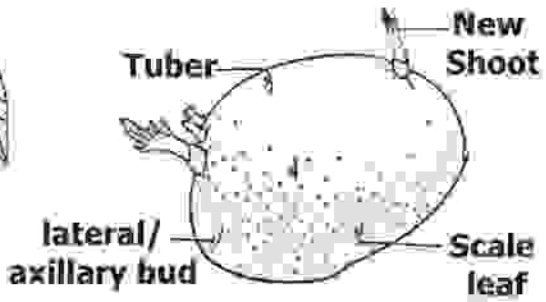
Stem tubers

These are swollen underground stems with stored food.

Examples of stem tubers are; Irish potatoes and cocoyams







Functions of each part

Tuber. It stores food for the plant.
It supplies food to the shoot.
Axillary bud. It develops into the new shoot.
Scale leaf. It protects the axillary bud from physical damage.

Bulbs

These are short underground stems with fleshy leaves that store food.
Examples of bulbs are onions, garlic and shallots.



Onion

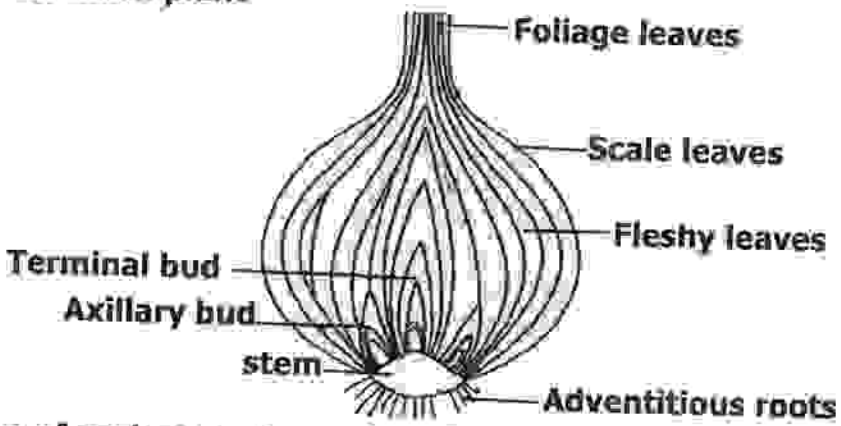


Garlic



Shallot

Parts of an onion plant



Functions of each part of an onion plant

Foliage leaves. They make food for the onion.
Terminal bud. It increases the growth height of an onion.
Fleshy leaves. They store food for the onion.
Scale leaves. They protect the fleshy leaves.
Adventitious roots. They hold the onion plant firmly into the soil.
Axillary bud. It develops into the new shoot.

<u>Points to note:</u> An onion can also be propagated by planting the bulb or planting seeds.

An onion has fibrous root system and adventitious type of roots.
It contains mineral salts (iodine), carbohydrates and vitamin C.
It undergoes hypogeal germination. It has parallel leaf venation and it is a monocot plant.

Rhizomes

These are horizontal underground stems with nodes from which roots and shoots grow.

Examples of rhizomes are; ginger, turmeric, zoyzia and cana lily.



Ginger

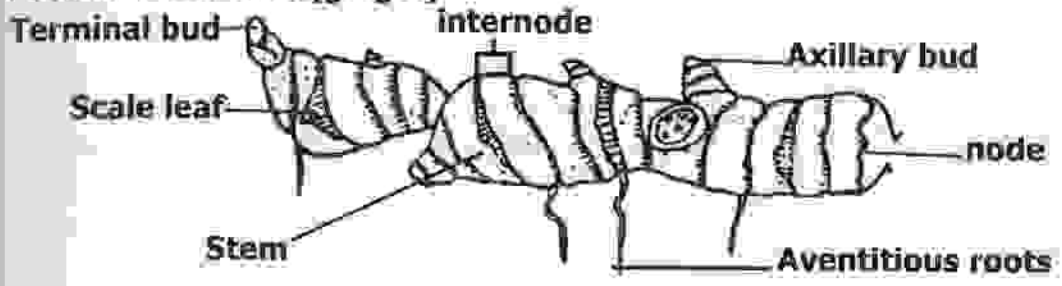


Turmeric



Cana lily

Parts of a rhizome (ginger)



Functions of each part

Terminal bud. It increases the growth height of the plant.

Axillary bud. It develops into the new shoot.

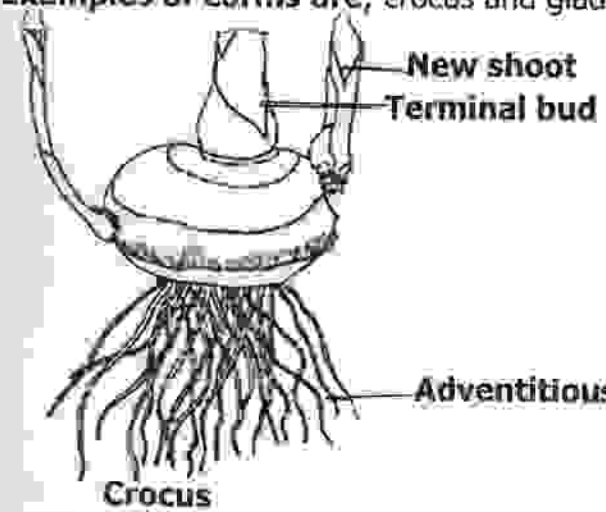
Stem. It supports the plant and transports water and nutrients.

Adventitious roots. They absorb water and mineral salts from the soil.

Scale leaf. It protects the bud.

Corms; These are short, vertical swollen underground stems that store food for the plant.

Examples of corms are; crocus and gladiolus.



Crocus



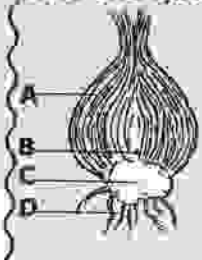
Gladiolus

Ways in which stems are useful to people.

(i) Stems are source of timber.
(ii) Stems are source of food.
(iii) Stems are source of herbal medicine.
(iv) Stems are sold for money.
(v) Stems are source of wood fuel.

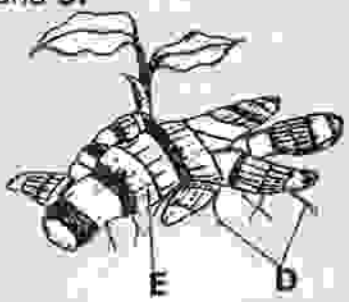
Evaluation activity 1.5

1. The diagram below shows an onion plant. Study and use it to answer the questions that follow.



(a) Name the parts of an onion marked **B** and **C**.
(b) Give the function of the part marked **A** to an onion.
(c) Identify any one substance absorbed by the part marked **D** from the soil.

The diagram below shows a ginger plant. Use it to answer questions **2** and **3.**

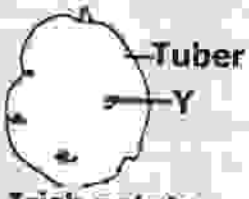


2. Name the part marked **E**.
3. Give any **one** function of the part marked **D** to a ginger plant.

The diagrams below show part of a cassava plant and an irish potato that are used in propagation. Study and use them to answer the questions that follow



Part of a cassava plant

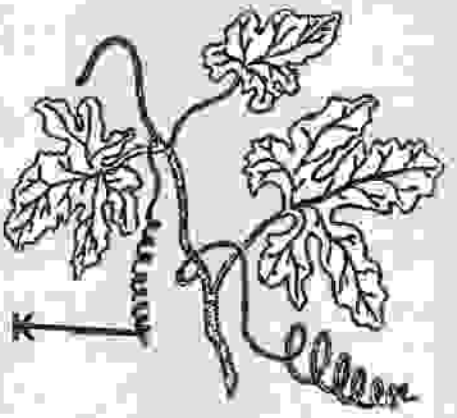


Irish potato

(a) Name the part of the cassava plant marked **X**.
(b) State the method of vegetative propagation where the part of cassava plant shown above is used.
(c) In which **one** is the function of part **X** of the cassava plant similar to that of part **Y** of the irish potato?
(d) Give the importance of the tuber in the propagation the irish potato.

Evaluation activity 1.5 Continued

The diagram below shows a weak plant. Use it to answer questions **6** and **7**.



6. Name any **one** plant that develops the part marked **K**.
7. State the function of part marked **K** to the plant.

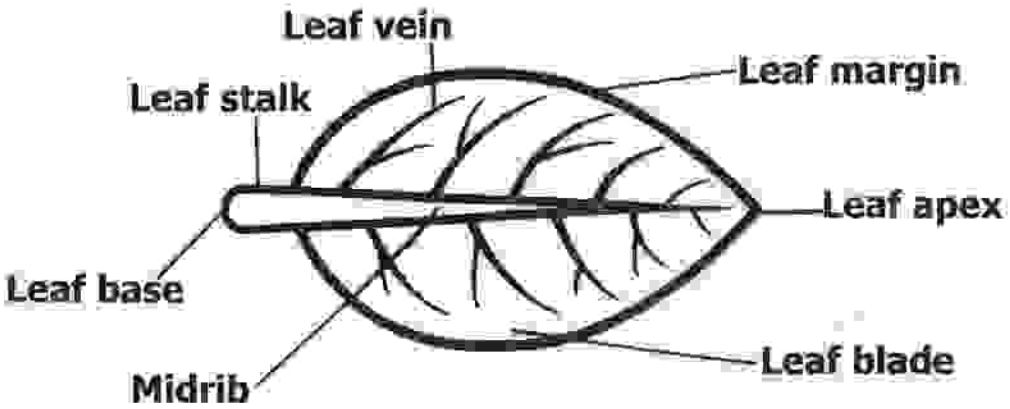
Keyword:

- parallel
- pinnate
- chlorophyll
- stomata
- trifoliate
- venation

PLANT LEAVES

A leaf is a structure of a plant, usually flat and green which grows from the stem.
Most leaves are green because they contain chlorophyll.
Leaves that have more than one colour are called **variegated leaves**.
Leaves are the main parts for photosynthesis.

Parts of the leaf



Functions of each part of the leaf

Leaf blade/lamina.

It is the flat surface of the leaf.

It is where photosynthesis takes place.
It contains chlorophyll which absorbs sunlight.
It has stomata for breathing.
Stomata are also used for transpiration.

Midrib/main vein.
It holds the leaf firmly.
It conducts water and mineral salts from the stem to the veins.

Leaf veins.
They transport water and mineral salts to all parts of the leaf.
They are hollow to allow distribution of water and mineral salts in the leaf.

Leaf stalk/petiole.
It holds the leaf to the branch or stem.

Leaf apex.
It is the tip of the leaf.
It protects the leaf e.g., aloe vera leaf and spear grass leaf.
Leaf margin. It gives the leaf protection.

Functions of leaves to a plant
- Leaves make food for the plant.
- Leaves have stomata for breathing.
- Leaves carry out transpiration.
- Some leaves store food for the plant. e.g., cabbages and onions

Types of leaves
Leaves are grouped according to the number of leaflets.

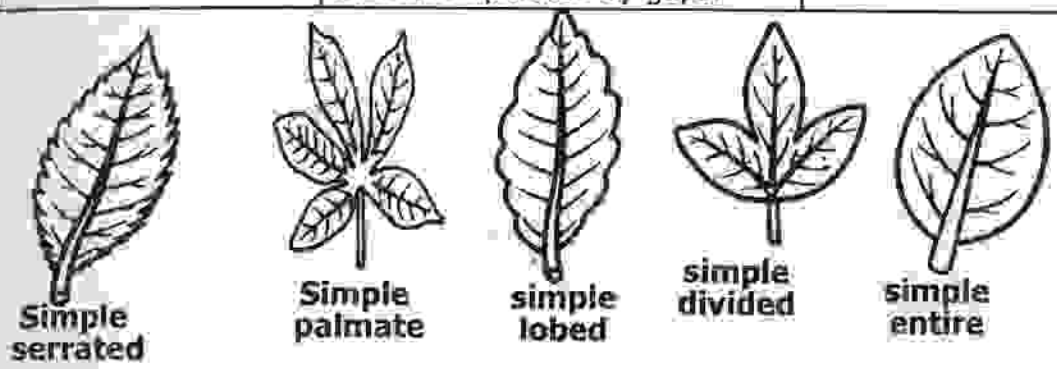
Simple leaves
These are leaves whose leaf blade/lamina is not divided.
They have one leaflet on the leaf stalk.

Characteristics of simple leaves
❖ The lamina is undivided ❖ They have single blades

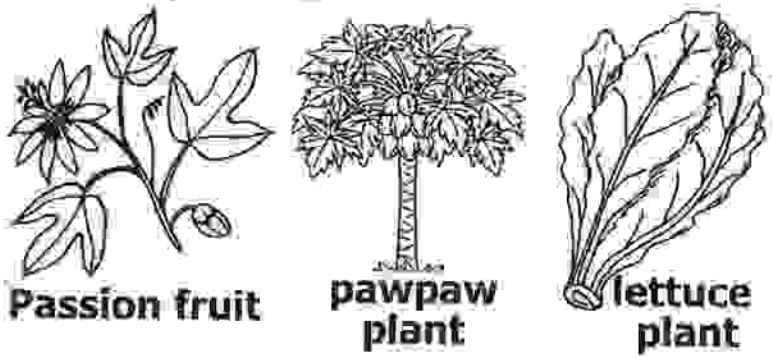
Examples of simple leaves

Simple leaf	Description	Examples of plants
Simple entire leaf	The margin is even and smooth around the entire leaf edge.	• mango • avocado • jackfruit • orange
Simple palmate	The leaf has several lobes or divisions that spread out from a single point resembling the palm. The leaf is undivided with a single blade.	▪ pawpaw ▪ sweet potato ▪ pumpkin

Simple leaf	Description	Examples of plants
Simple serrated	The leaf has saw-like edges on the margin.	• blackjack • tick berry
Simple divided entire	A leaf has a single blade that is either deeply lobed or dissected but not separated into multiple leaflets.	• passion fruit
Simple lobed	A leaf has distinct round or pointed projections called lobes that are separated by gaps.	• bryophyllum



Plants that have simple leaves



Compound leaves

These are leaves whose leaf blade is fully/completely divided into leaflets.
A leaflet is a small, leaf-like structure that is part of a compound leaf.
A leaflet consists of two parts:
Lamina. The flat, green part that performs photosynthesis.
Stalk. The part that attaches the leaflet to the central stem of the compound leaf.

Characteristics of compound leaves

-They have many leaflets on one stalk.
-They have rachis.
-They are divided into several leaflets.

Examples of compound leaves

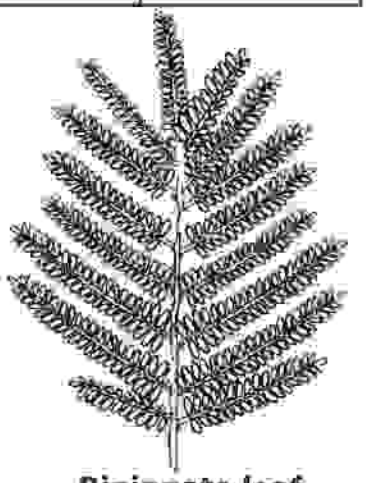
Compound leaf	Characteristics/description of the leaf	Examples of plants
Trifoliate leaf	This is a leaf with three divided leaflets.	• beans • peas • soya beans
Pinnate leaf	This is a leaf which has a main stalk and other rachis forming independent leaf blades.	• rose • neem plants
Bipinnate leaf	This is a leaf which has two pinnate altering on the main stalk.	• acacia • coffee • jacaranda
Digitate leaf	This is a leaf that has leaflets that spread out from a single point like fingers of a hand. They have a short rachis.	• silk cotton



Trifoliate



Pinnate leaf



Bipinnate leaf

Plants that have compound leaves



Acacia plant



Neem plant



Silk cotton

Leaf Venation

Leaf venation is the arrangement of veins in a leaf.

Types of leaf venation

Network venation.

This is where the veins cross each other in a leaf.

It is also called reticulate venation.

It is found in dicotyledonous plants.

Examples of plants with network veined leaves

Oranges, pumpkins, pawpaw, mangoes, beans

Parallel venation

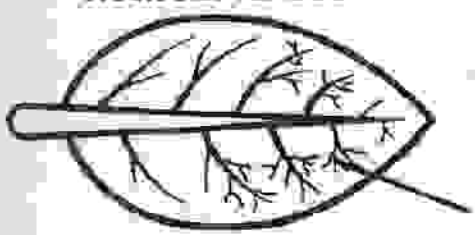
This is where veins run from the leaf stalk to the margins without crossing each other.

It is found in monocotyledonous plants.

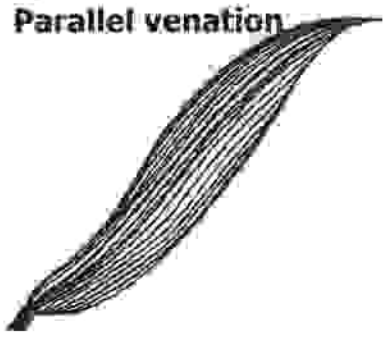
Examples of plants with parallel veined leaves

Maize, millet, sorghum, sugarcane, wheat, onion, banana and rice

Network venation



Parallel venation



Ways in which leaves are important to people.

- ✓ Leaves are eaten as food e.g., cabbage
- ✓ Spear grass is used for thatching houses.
- ✓ Leaves are source of herbal medicine e.g., aloe vera
- ✓ Dry leaves are used for mulching the garden.
- ✓ Leaves are sold for money e.g., tobacco leaves.
- ✓ Palm leaves are used to make crafts like mats and hats



Grass thatched house



mat



hut



Aloe vera



cabbage

Evaluation activity 1.6

1. (a) Name the part of the leaf that;
   (i) Holds the leaf to the stem.
   (ii) Transports water in the leaf.
   (b) Give any **two** functions of leaves to the plant.
2. The diagrams below show compound leaves in plants. Study and use them to answer the questions that follow.



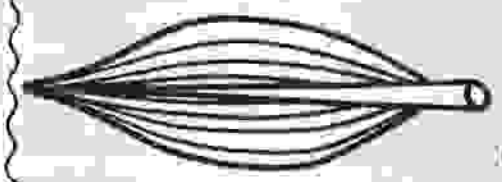


(a) Name the compound leaves marked **M** and **N**.
(b) Mention any **one** plant that has the compound leaf marked **M**.
(c) On the compound leaf marked **N**, label the leaf stalk.

3. The table below shows some of the parts of the leaf and the function of each each part. Complete the table correctly.

Part of the leaf	Function of the part
.....	Allows gaseous exchange in leaf.
Leaf stalk	.....
.....	Where photosynthesis takes place.
Midrib	.....

The diagram below shows the type of leaf venation. Use it to answer questions **4** and **5**.



4. Identify the type of leaf venation shown in the diagram above.
5. Mention any **one** plant that has the type of leaf venation above.
6. Apart from making food, mention any **two** other uses leaves to a plant.

Key words:

• starch • carbon dioxide • glucose • chlorophyll • photosynthesis • absorb

Processes that take place in plant leaves

Transpiration, photosynthesis, respiration and breathing.

Photosynthesis

Photosynthesis is the process by which plants make their own food.
Photosynthesis comes from two terms;
Photo. It means **light.**
Synthesis. It means to **make.**
During photosynthesis, glucose is formed. The excess glucose is stored in

plants as starch. Hence food made by plants is called **starch.**
Photosynthesis takes place in leaves, green stems and sepals of a flower.

Requirements for photosynthesis to take place

Raw materials for photosynthesis.

Water.

It combines with carbon dioxide to make food.
It is got from the soil.

Carbon dioxide

It combines with water to make food.
It is got from air.

Conditions necessary for photosynthesis

Presence of chlorophyll.

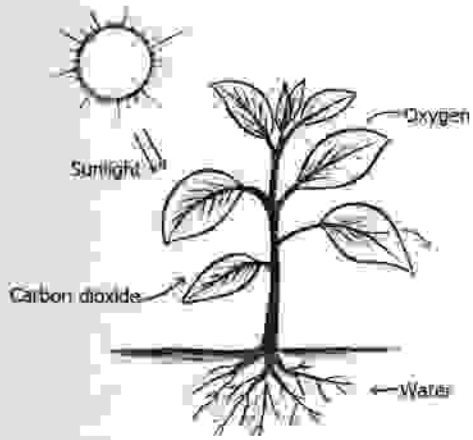
Chlorophyll absorbs sunlight.
Chlorophyll is the green pigment found in plants.
Chlorophyll determines the colour of the leaves.

Presence of sunlight.

It provides energy needed by plants to make food.

Note. Photosynthesis cannot take place at night.
Reason. There is no sunlight energy.

$$\text{Water + carbon dioxide} \xrightarrow[\text{chlorophyll}]{\text{sunlight}} \text{starch + oxygen}$$



Summary

Requirements

Raw materials
Carbon dioxide
Water

Conditions
Chlorophyll
Sunlight

Products
Starch-----main product
Oxygen ---- by product

Points to note:

Plants respire during the day, using oxygen to break down glucose for energy.
Plants use oxygen at night for respiration.

Factors that determine the rate of photosynthesis

- Amount of carbon dioxide in the atmosphere.
- Amount of chlorophyll in the leaves.
- Amount of water in the plant.
- Sunlight intensity.
- Temperature.

Adaptations of leaves to photosynthesis

- They have a broad shape to absorb enough sunlight.
- They have chlorophyll to absorb sunlight.
- They are many to increase the surface area for photosynthesis.
- They have many veins to supply water to all cells.

Importance of photosynthesis in the environment

To plants. It enables plants to get food.

To animals. It enables animals to get oxygen. It enables animals to get food.

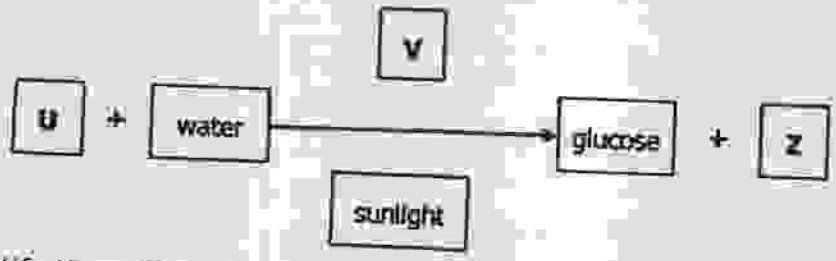
To the environment. It reduces the amount of carbon dioxide in the environment.

How plants purify/filter air in the atmosphere

Plants absorb carbon dioxide and release oxygen during photosynthesis.

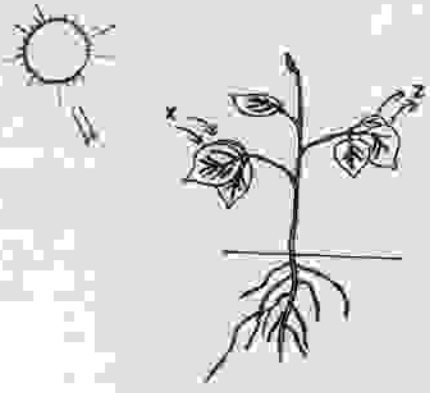
Evaluation activity 1.7

1. (a) What kind of food is made by plants during photosynthesis?
(b) State any **one** raw material used by plants during photosynthesis.
(c) How is chlorophyll useful during the process of photosynthesis?
(d) Give the reason why photosynthesis cannot take place at night.
2. The diagram below shows a plant process. Study and use it to answer the questions that follow.



(a) Identify the process in plants shown in the diagram above.
(b) Name the substances represented by letters **U** and **V.**
(c) How do animals benefit from the substance produced at **Z**?

3. The diagram below shows a process that takes place in plants. Study and use it to answer the questions that follow.



(a) Which plant process is shown in the diagram above?
(b) Name the gaseous substances marked **X** and **Z**.
(c) State any **one** condition necessary for the plant process above to take place.

Evaluation activity 1.7 Continued

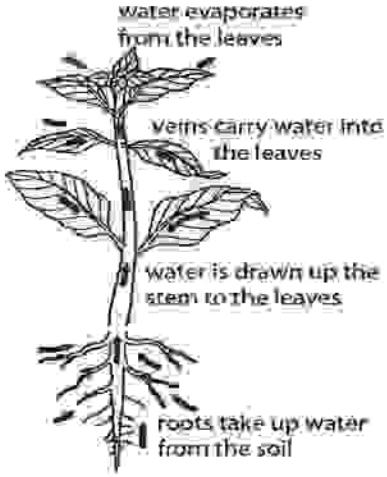
4. (a) Which component of soil is used by plants to make food?
   (b) Identify the part of air that increases around the plant during photosynthesis?
   (c) Name the chief product of photosynthesis.
   (d) State **one** way in which photosynthesis is useful to the atmosphere.
5. Name the process in plants that helps to purify air in the atmosphere.
6. How do plants help to purify air in the atmosphere?

Key words:

• excess • vapour • lose • humidity • shedding • intensity

Transpiration

Transpiration is the process by which plants lose water in form of vapour to the atmosphere. Transpiration takes place in the stomata, lenticels and cuticles.



An experiment to demonstrate transpiration in plants.

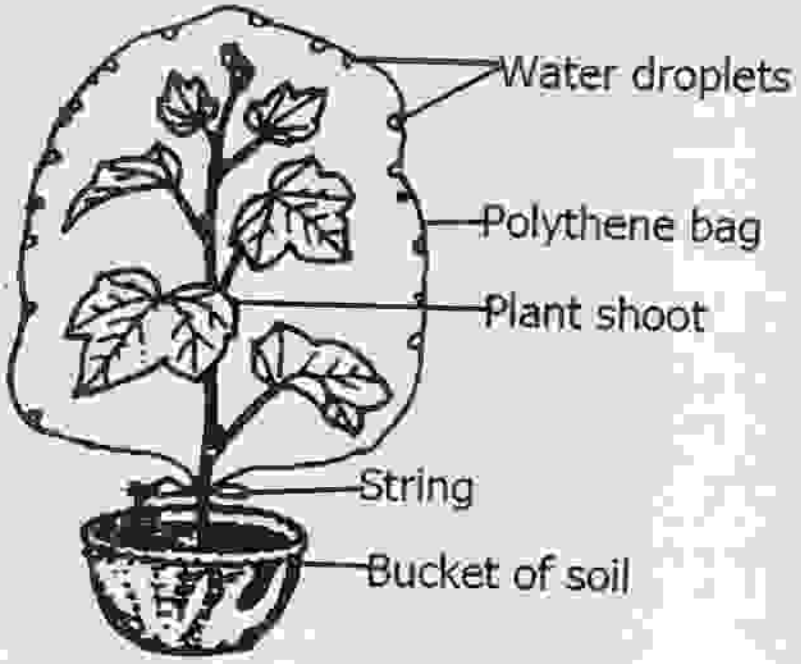
Materials needed: plant shoot, bucket of soil, string and transparent polythene bag

Steps taken:

(i) Place the plant shoot in the bucket of soil ensuring it is securely positioned.
(ii) Cover the plant shoot and soil with a polythene bag to create a controlled experiment.
(iii) Use the string to tie the polythene bag at the opening to trap moisture inside.
(iv) Observe the inside of the bag for moisture droplets, indicating transpiration.

An experiment to demonstrate transpiration in plants.

Set up:



Roles of the materials in the experiment

Polythene bag. It creates humid environment to collect moisture for the experiment.

It should be **transparent** to allow easy observation of water droplets on its sides.

Plant shoot. It is releases water vapour through its leaves to show transpiration.

String. It secures the polythene bag around the plant to prevent moisture from escaping.

Bucket of soil. It provides support and medium for the plant shoot to transpire.

Factors that affect the rate of transpiration in plants

Factor	Its effect
Temperature	High temperature increases the rate of transpiration and low temperature reduces the rate of transpiration.
Humidity	There is little or no transpiration in humid air.
Sunlight intensity	High amount of sunlight increases transpiration and low amount of sunlight reduces transpiration rate.

Factor	Its effect
Speed of wind	The rate of transpiration increases with the increase in the speed of wind and reduces with the decrease in the speed of wind.
Size of the leaves	A plant with larger leaves loses more water than the one with small leaves.
Number of leaves on a plant	A leaf with many stomata loses more water than a leaf with few stomata.

Factors that increase and decrease the rate of transpiration in plants

Increasing factors	Decreasing (reducing) factors
• High sunshine intensity • Low humidity • Strong wind • High temperature • Large surface area of leaves	• High humidity • Low speed of wind • Low temperature • Small surface area of the leaves • Low sunshine intensity

Importance of transpiration

To plants. It cools plants

It allows roots to absorb more water from the soil

To a crop farmer. It helps in rain formation which supports crop growth.

To the environment. It helps in rain formation which cools the environment.

Danger of transpiration.

• It leads to wilting of plants.

Wilting is a condition in which the plant loses more water than it gains from the soil.

• It causes drying up of the plant.

Ways in which plants reduce the rate of transpiration

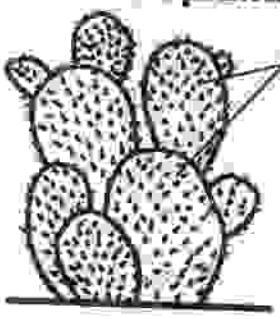
Way	Examples of plants
Shedding leaves during dry season. Such plants are called **deciduous plants**	Mvule trees, Ficus tree, Fig tree, Sycamore tree, oak tree
Modifying leaves into thorns/spines.	Cactus plant.
Secreting a layer of wax that covers stomata.	Pine, aloe vera, bananas and cacti
Having thick leaves	Sisal plant and cactus plant
Folding leaves during sunny weather	Maize plant and sorghum plant
Closing stomata during day time.	Cactus and pineapple

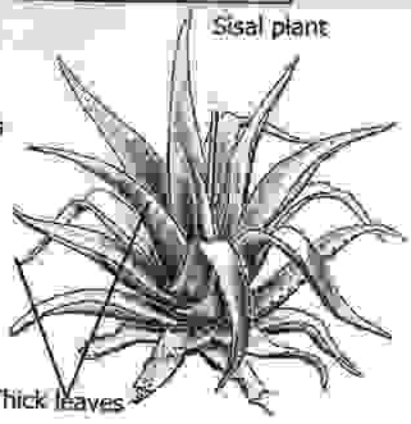
✍ ***Points to note.***

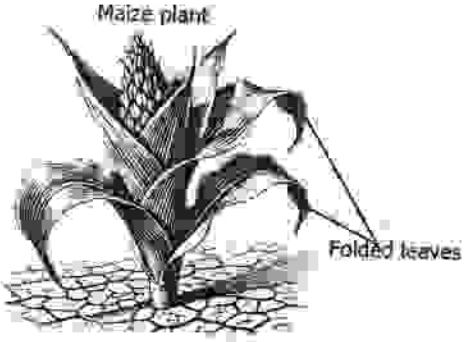
+ ***The cactus plant also has sunken stomata, wax cuticle and thick leaves which prevent it from losing water.***

+ ***Sweating in humans and panting in dogs are similar to transpiration. They are cooling processes.***

A cactus plant.









Evaluation activity 1.8

1. A primary four boy wanted to demonstrate a certain plant process. He wrapped a transparent polythene bag around a plant shoot and put it in sunshine for 12 hours. After 12 hours, droplets of water were seen on the sides of the polythene bag.
(a) Identify the plant process the boy demonstrated in the above experiment.
(b) Give the reason why the boy used a transparent polythene bag for his experiment.
(c) In the space provided below, draw a diagram to show the experiment demonstrated by the boy about the plant process. **(02 marks)**

The diagram below shows a condition in plants. Use it to answer questions **2** and **3**.



2. Name the condition in plants shown above.
3. State **one** cause of the condition in plants shown above.
4. Give **one** reason why a maize plant folds its leaves during sunny weather.
5. Which process in human beings is similar to transpiration in plants?
6. By what process do plants lose water to the atmosphere?

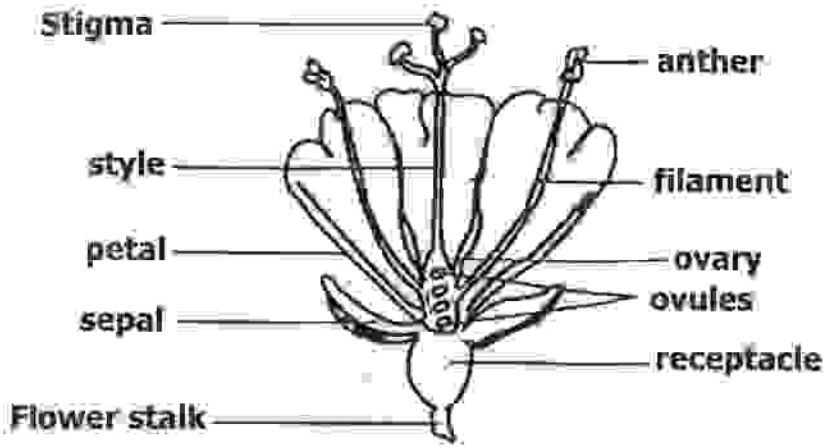
Key words

• gamete. • receptacle • reproduction • receive • pollen
• ovule

FLOWERS

A flower is the organ of sexual reproduction in flowering plants.
Flowers produce seeds and fruits.

Parts of the flower



Functions of each part of the flower

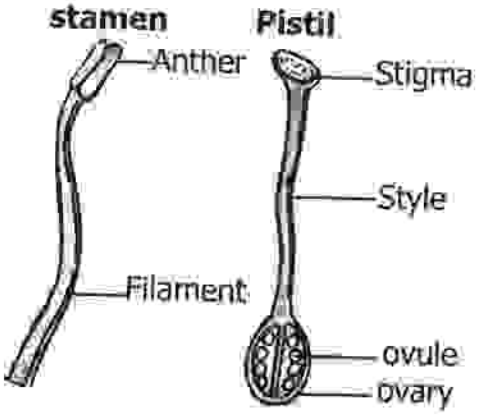
Part	Function
Anther	Produces and stores pollen grains. Pollen grains are male reproductive cells in a flower. Pollen is the yellow dust-like grains which contain the male sex cells of the plant.
Petal	It protects the inner parts of the flower. Petals of most flowers are brightly coloured to attract pollinators. Some petals produce scent. A group of petals is called **corolla.**
Sepal	It protects the flower before it opens/at the bud stage. A group of sepals is called **calyx**.
Stigma	It receives pollen grains during pollination. It is sticky to hold pollen grains firmly. When pollen grains land on the stigma, a pollen tube develops.
Filament	It holds the anthers in one position.
Style	It holds the stigma. It allows pollen grains to pass to the ovules.

Part	Function
Ovary	It produces ovules. The ovary develops into the fruit after fertilisation. Fertilisation in a flower takes place in the ovary.
Ovules	Ovules are female gametes. Ovules develop into seeds after fertilisation.
Flower stalk	It holds the flower on the stem or branch. It develops into the fruit stalk after fertilisation.
Receptacle	It develops into a fruit in some plants e.g., apples and pears.

Reproductive parts of the flower

Stamen.
It is the male part of the flower. It is made up of anther and filament.
A group of stamens is called **androecium**.

Pistil.
It is the female part of the flower. It is made up of ovary, ovules and stigma.
Pistil is also called **carpel**.
A group of pistils is called **gynaecium**

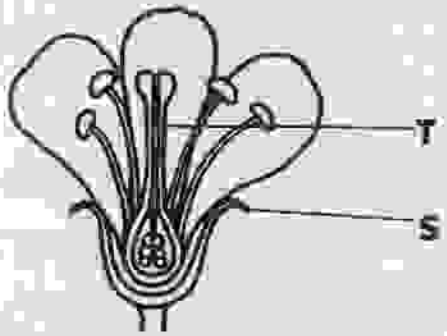


Evaluation activity 1.9

1. Name the part of the flower where fertilisation takes place.
2. Give any **one** reason why the petals of some flowers are brightly coloured.
3. State the function of flowers on a plant.
4. Identify the part of the flower that develops into;
(i) Fruit after fertilisation.....(ii) Seeds after fertilisation.....
5. State any **one** way in which the stigma of the flower is adapted to its function.
6. State the difference between the stamen and pistil of a flower.
7. Apart from the ovary, mention **one** other part of the flower that develops into the fruit after fertilisation.

Evaluation activity 1.9 Continued

The diagram below is of a flower. Use it to answer questions **8** and **9**.



8. Name the part labelled **T**.
9. Give the function of the part labelled **S** to a flower before it opens.
10. The table below shows part of the flower and their functions. Study and complete the table correctly.

Part	Function
Sepal	.....
.....	Attracts pollinators.
Stigma	.....
.....	Produces pollen grains.

The diagram below shows part of the plant. Use it to answer questions **11** and **12.**



12. Name the part marked **J.**
13. Give the function of the part marked **K** to the plant.
14. State what happens when pollen grains land on the stigma.
15. Name the male reproductive cells in a flower.

Key words:

• pollen • nectar • bright • anther • grains • species

Pollination in Flowering Plants

Pollination is the transfer of pollen grains from the anther to the stigma of a flower on a plant.

After pollination, pollen grains develop to form pollen tubes.

Pollination allows fertilisation to take place.

Types of pollination

The type of pollination is determined basing on the number of plants.

Self-pollination.

Self-pollination is the transfer of pollen grains from the anther to the stigma of the same flower or different flowers on the same plant.

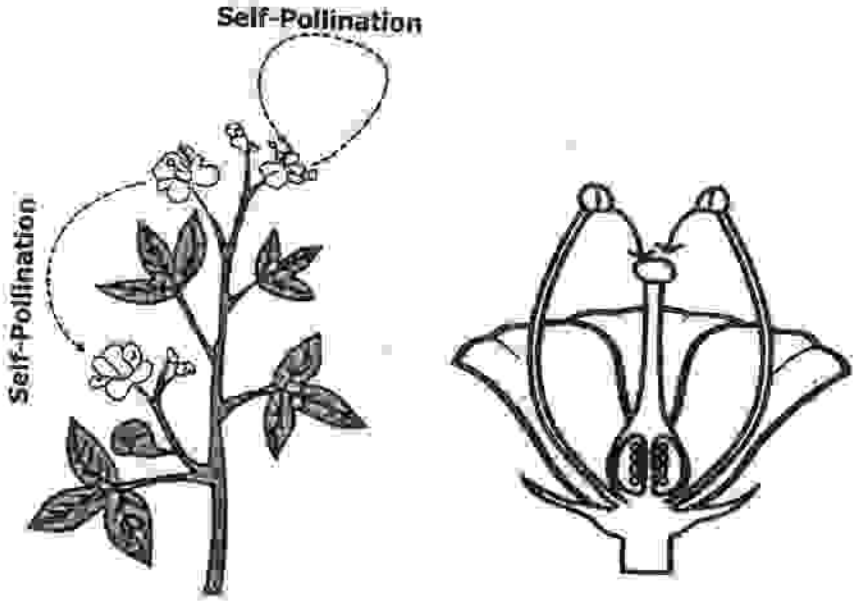
It does not depend on external factors e.g., wind or insects to carry pollen grains.

Plants that carry out self-pollination

tomatoes, potatoes, pumpkin, rice, soya beans and mangoes.

Ways in which plants are able to carry out self-pollination

- The flowers remain closed until self-pollination occurs.
- The anthers and the stigma mature at the same time.
- Flowers have both male and female parts.
- The flowers remain buried in the ground e.g., ground nuts



Characteristics of self-pollinated flowers

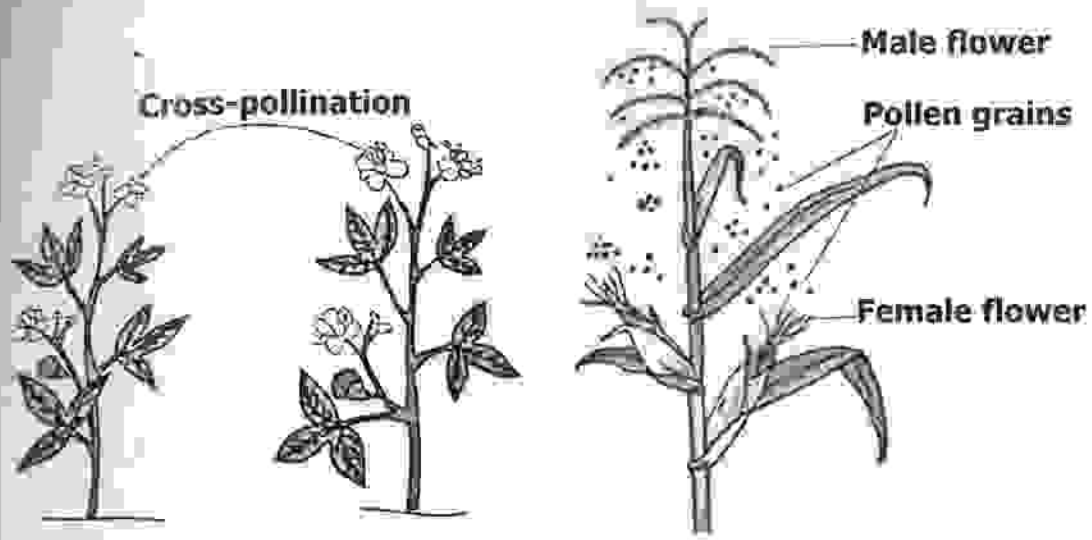
- Filaments are longer than the style.
- Flowers are small in size.
- Flowers produce few pollen grains.
- Pistil and stamen mature at the same time.

Cross pollination

Cross pollination is the transfer of pollen grains from the anther to the stigma of different flowers on different plants but of the same species/kind.
e.g., from flowers of one maize plant to flowers of another maize plant.

Plants that undergo cross pollination

maize, papaw and passion fruits



On the maize plant above, the male flower produces pollen grains while the female flower receives pollen grains.

Characteristics of cross-pollinated flowers

- The filaments are shorter than the style.
- Pistil and stamen mature at different times. e.g., maize
- The male and female flowers occur on separate plants e.g., pawpaw.
- Pollen grains cannot germinate on the stigma of the same flower e.g., passion fruits.

Agents of pollination

These are things which carry pollen grains from the anthers to the stigma.

Things that attract pollinators to flowers

brightly coloured petals, strong scent, nectar, pollen.

Insects. They include; butterfly, moths and bees

A moth pollinates flowers at night. It has a strong sense of smell to detect scent from flowers.

Flowers are able to be pollinated by moths at night.

Reason. . Flowers produce scent which attracts moths.

A worker bee has a hairy body and pollen basket which carry pollen grains from one flower to another.

Note: Bees visit flowers to get food. This food can be nectar or pollen.

Birds. They include; sun bird and humming bird.

Adaptation. They have a long slender curved beak for sucking nectar from flowers.

Pollen grains stick onto their legs and bodies and move it to other flowers.

Wind. Wind blows pollen grains from the anther to the stigma.

Water. Pollen grains float on water surface until they find the stigma of a an aquatic plant.

Bats. They have a hairy body on which pollen grains stick. These are fruit eating bats





Characteristics of Insect Pollinated flowers and Wind Pollinated Flowers

Insect pollinated flowers	Wind pollinated flowers
• They have brightly coloured petals. **Reason.** To attract pollinators. • They produce scent and nectar. **Reason.** To attract pollinators. • They produce few pollen grains. • They have sticky stigmas. **Reason.** To hold pollen grains firmly • They are large and easily seen. **Reason.** To be easily identified by pollinators.	• They produce a lot of pollen grains. **Reason.** To increase chances of pollination. • They produce light pollen grains. **Reason.** To be easily blown by wind. • They have hairy stigmas. **Reason.** To increase chances of pollination. • They have weak filaments. • Filaments are longer than the styles.

Fertilisation in a flower

Fertilisation is the union of male and female gametes to form a **zygote.** Gametes are reproductive cells. Male gametes of a flower are called **pollen grains.** Female gametes of a flower are called **ovules.**

Fertilisation in flowers takes place in the **ovary**.

The process of fertilisation in a flower

(i) The stigma receives pollen grains from the anthers.

(ii) Pollen grains develop pollen tube.

(iii) A pollen tube carries pollen grains to the ovary where there are ovules.

(iv) In the ovary, pollen grains unite with ovules hence fertilisation occurs.

Results after fertilisation in a flower

• The ovary develops into the fruit.

• The ovules develop into seeds.

- The petals, sepals and stigmas wither, dry and fall off.

Ways in which flowers are important to people.

(i) Flowers are used for decoration.
(ii) Some flowers are eaten as food e.g., cauliflower.
(iii) Flowers are used to make perfumes and dye.
(iv) Flowers are sold for money.
(v) Flowers are used as presents/gifts.
(vi) Flowers are used to make insecticides e.g., pyrethrum flowers



Wearing perfume from flowers



Brides holding a flower



Flowers for agift



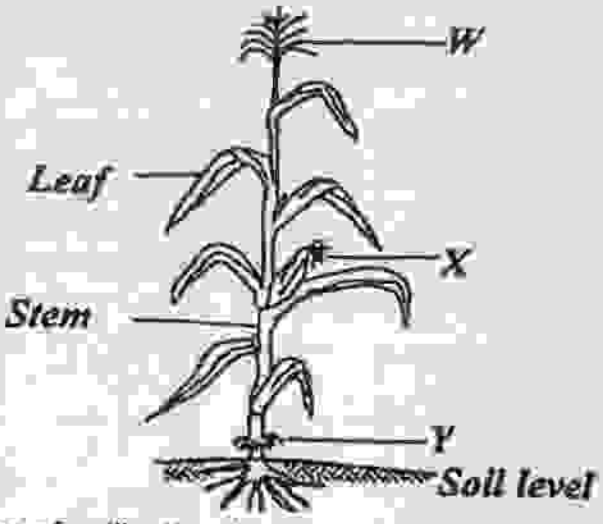
Decoration using flowers



Selling flowers

Evaluation activity 1.10

1. Apart from having bright colours, state any **one** other thing that attracts pollinators to flowers.
2. Give the reason why a moth is able to pollinate flowers at night.
3. (a) Name **two** types of pollination in flowering plants.
   (b) State any **two** differences between insect pollinated flowers and wind pollinated flowers.
4. (a) Apart from bees, name any **one** other agent of pollination.
   (b) Give any **one** thing that attracts bees to visit flowers.
   (c) Explain how a bee pollinates flowers **(02 marks)**
5. The diagram below shows a flowering plant. Study and use it to answer the questions that follow.



   (a) Identify the type of pollination that the flowering plant above undergoes.
   (b) Give the reason why the flowering plant above produces a lot of pollen grains.
   (c) State the difference in function between the part marked **W** and the part marked **X**.
6. Name the edible flower shown in the diagram below.



Key words:

• cotyledon • dicotyledonous • plumule • micropyle • fertilise • ovule

SEEDS

A seed is a fertilised ovule.

A seed develops into the new plant.

Types of seeds.

Seeds are classified according to the number of cotyledons.

Monocotyledonous seeds

These are seeds which have one cotyledon. The word mono means one.

Characteristics of monocotyledonous seeds

- They undergo hypogeal germination.
- They have one cotyledon.
- They store food in the endosperm.
- They are grain seeds.

Examples of monocotyledonous seeds

Maize, millet, sorghum, rice, wheat and barley



Maize grains

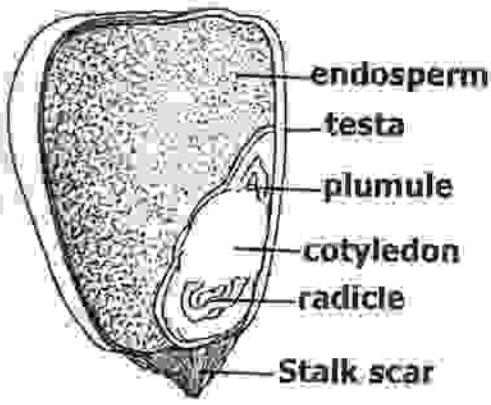


Sorghum grains



Rice grains

Parts of a monocot seed (maize grain)



Functions of each part of the maize grain.

Endosperm. It stores food for the embryo.

Cotyledon. It absorbs food from the endosperm and supplies it to the embryo.

Testa/seed coat. It protects the inner parts of a seed.

Plumule. It grows into the shoot system.

Radicle. It grows into the root system.

Stalk scar. It attaches the grain to the cob.

Both a fruit and a maize grain have two scars as a common feature.

Dicotyledonous seeds

These are seeds which have two cotyledons. The word **Di** means **two.**

Characteristics of dicotyledonous seeds.

- Most dicotyledonous seeds undergo epigeal germination.
- They have two cotyledons.
- They store food in the cotyledon.

Examples of dicotyledonous seeds

beans, peas, soya beans, ground nuts, mango seeds, oranges seeds, bambara nuts



Beans

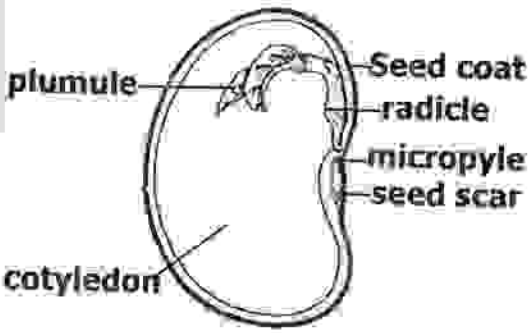


Cowpeas



Ground nuts

Parts of a dicot seed (bean seed)



Functions of each part of the bean seed.

Cotyledon. It stores food for the embryo.
It supplies food to the embryo during germination.

Testa/seed coat. It protects the inner parts of a seed.

Plumule. It grows into the shoot system.

Radicle. It grows into the root system.

Micropyle. It lets water and air into the seed during germination.

Hilum. It attaches the seed to the pod.

Points to note.

Both the cotyledon in a bean seed and the endosperm in a maize grain store food for the embryo.

The cotyledon in a maize grain supplies food to the embryo while the cotyledon in a bean seed stores food for the embryo.

The plumule and the radicle make up an embryo. An embryo is a young plant in a seed coat.

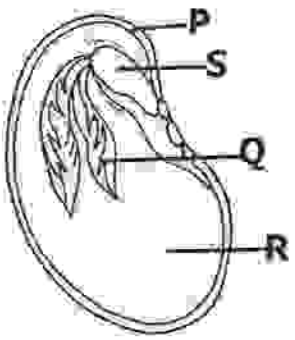
Differences between a seed and a fruit.

A seed has one scar while a fruit has two scars.

A seed is a fertilised ovule while a fruit is a fertilised ovary

Evaluation activity 1.11

1. Which part of a maize grain has a similar function as the cotyledon of a bean seed?
2. State **one** feature common to both a maize grain and a fruit.
3. Give any **two** ways in which millet seeds are different from pea seeds.
4. Which part of the maize grain absorbs and supplies food to the embryo?
5. Apart from storing food for the embryo, state **one** other function of the cotyledon in a ground nut seed.
6. The diagram below shows a bean seed. Study and use it to answer the questions that follow.



(a) Name the parts marked **P** and **Q**.

(b) Identify the plant system which part **S** develops after germination.

(c) State the function of part labelled **R** in a bean seed.

Key words:

• epigeal • warmth • viability • dissolves • respiration • cotyledon

SEED GERMINATION

Germination is the development of a seed into a seedling.

A seedling is a young plant after germination.

Stages of germination.

(i) The seed takes in air and water.

(ii) The seed swells and the testa breaks.

(iii) The radicle emerges first to form root system.

(iv) The plumule sprouts to form shoot system.

Reasons why the radicle comes out first during germination

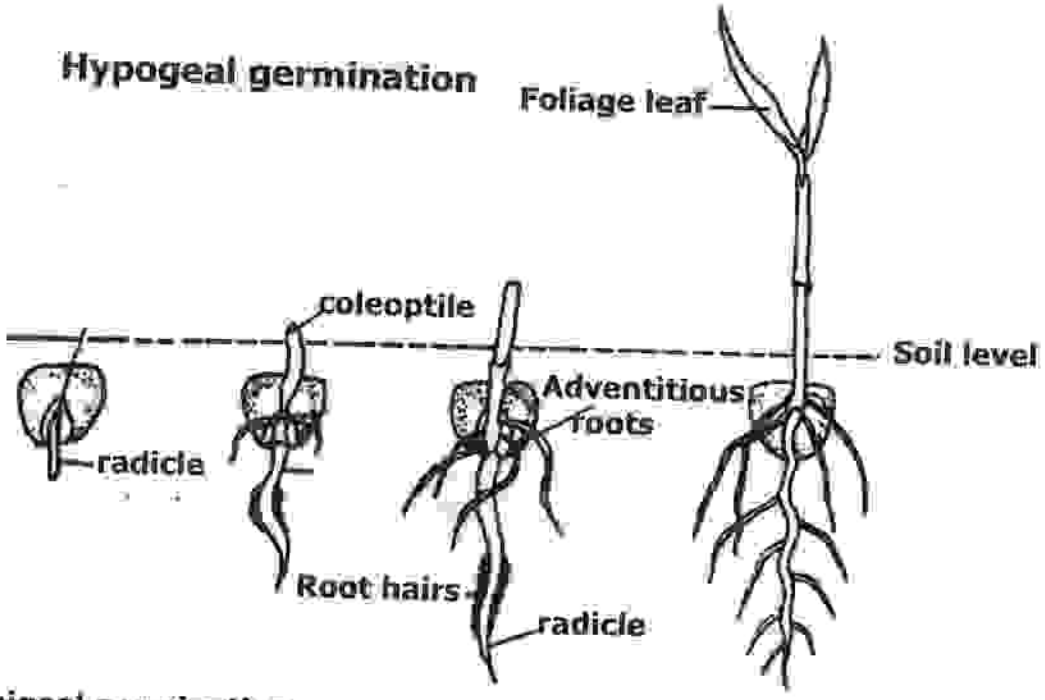
- To absorb water and mineral salts for the seedling.
- To anchor the seedling into the ground.

Types of germination.

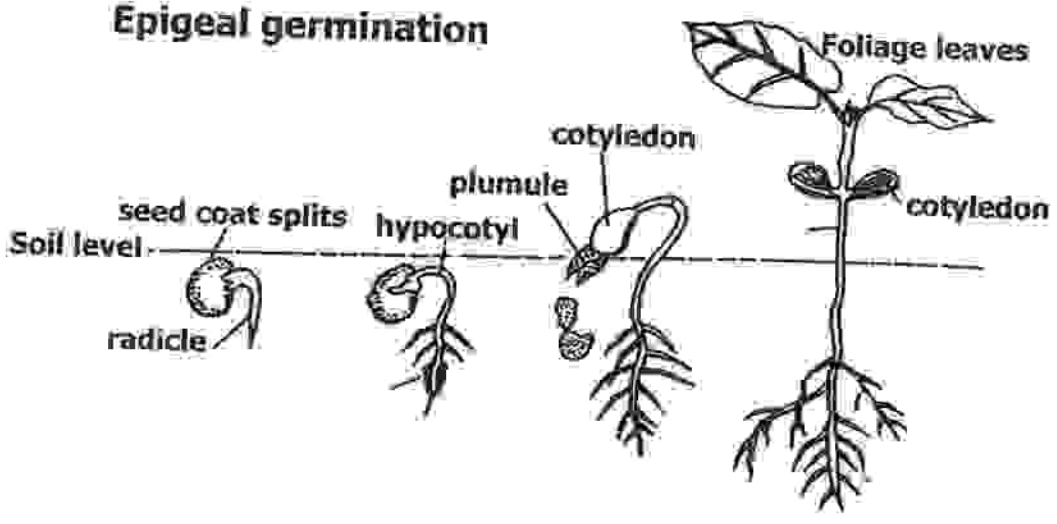
Hypogeal germination

This is the type of germination where the cotyledon remains under the ground.

"Hypo" means below or under.
Monocotyledonous seeds undergo hypogeal germination.
They include; maize, millet, sorghum, rice and wheat.
Note: Some dicotyledonous seeds undergo hypogeal germination.
They include; pea seeds and ground nut seeds.



Epigeal germination
This is the type of germination where the cotyledon comes above the ground.
"Epi" means above or on
Dicotyledonous seeds undergo epigeal germination.
They include; beans, soya beans, ground nuts and coffee.



Important points to note:

- The cotyledons supply food to the embryo. The cotyledon is also called seed leaf.
- They turn green after coming out of the soil to make food for the seedling.
- Foliage leaves are the first leaves to appear on a seedling after germination.
- They make food for the seedling.
- At the last stage as shown in the diagrams above, cotyledons are useless.

Reason. The seedling has developed foliage leaves to make food.

- The coleoptile protects the tip of the plumule.
- The hypocotyl pushes plumule out of the soil.

Importance of germination

- It prevents plant extinction.
- It enables animals to get food.
- It increases plant population in the environment.

Conditions necessary for germination

Presence of oxygen. It is used for respiration by an embryo.

Presence of water. It dissolves food for the embryo. It softens the seed coat.

Presence of warmth. It gives the right temperature for seeds to germinate.

Experiment to demonstrate the conditions necessary for seed germination.

Materials needed: bean seeds, 4 empty tins, oil, cotton wool, water, ice cubes

Steps taken:

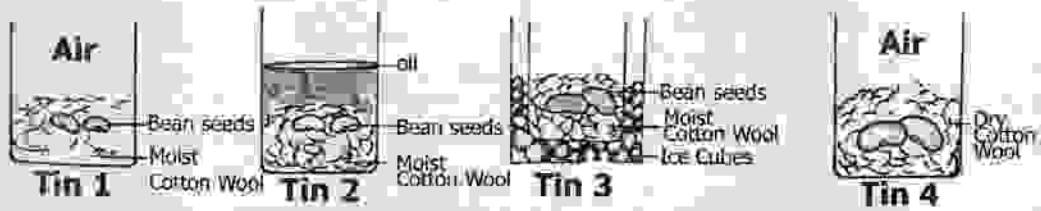
(i) Place cotton wool in each of the four tins and moisten it with water, except for one tin that remains dry.

(ii) In Tin 1, place seeds with moist cotton wool and keep it in a warm place.

(iii) In Tin 2, place seeds with moist cotton wool but cover them with oil to block oxygen.

(iv) In Tin 3, place seeds with moist cotton wool and put it in a cold place with ice cubes, and in Tin 4, place seeds with dry cotton wool in a warm place.

Set Up:



Observations and explanation:

Seeds in **Tin 1** germinate.

Reason. Water, oxygen and warmth are all present.

Seeds in **Tin 2** do not germinate.

Reason. Oil blocks oxygen, which is used for respiration of the embryo.

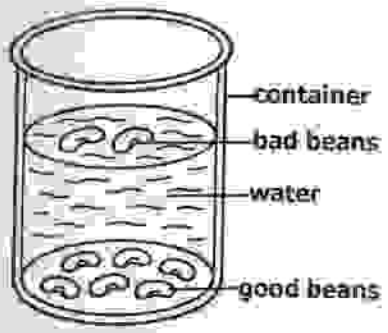
An experiment to test for seed viability.

Materials needed: 5 bad and 5 good bean seeds, transparent glass and clean water

Steps taken:

(i) Take a transparent glass and fill it halfway with clean water.
(ii) Drop five good beans and five bad beans into the glass of water.
(iii) Observe which beans float and which ones sink after 5-10 minutes.
(iv) Conclude that sinking beans are viable while floating beans are non-viable.

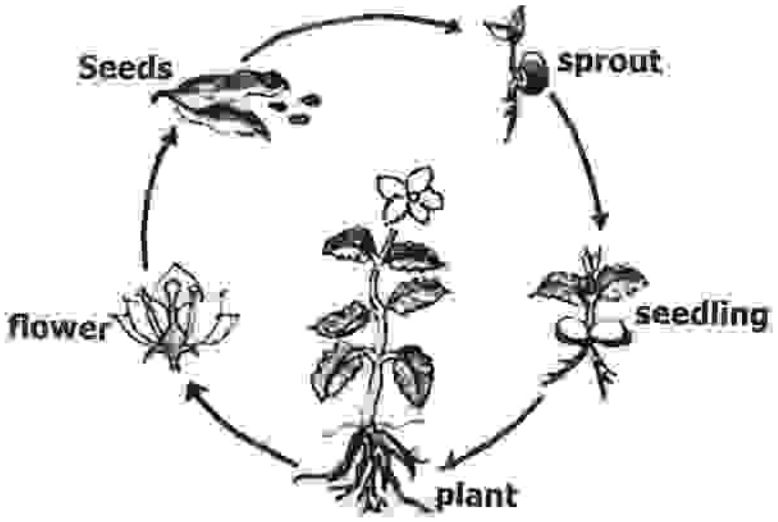
Set up:



Roles of the materials in the experiment

Good beans. Represent seeds with viable embryos for germination.
Bad beans. Represent non-viable seeds without functional embryos.
Transparent glass. Holds the water and allows observation.
Water. Helps distinguish between viable and non-viable seeds based on density.

The life cycle of a plant



A sprout is a young growth from a seed, plant or tree. It refers to the first stage of development when a plant begins to emerge.
Sprouts can grow from seeds, tubers or roots.

Seed dormancy.

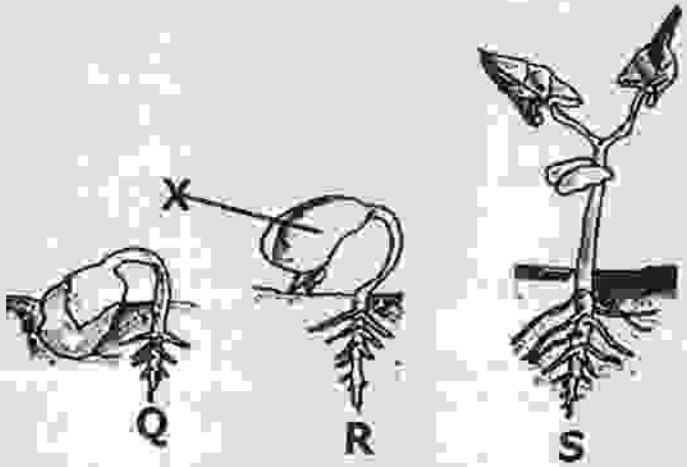
Seed dormancy is the condition when seeds fail to germinate due to lack of favourable conditions.

Factors that can make seeds fail to germinate

- Damaged seeds.
- Immature seeds.
- Broken seeds.
- Rotten seeds.
- Poorly dried seeds.
- Lack of water, oxygen and warmth.
- Overstayed seeds.

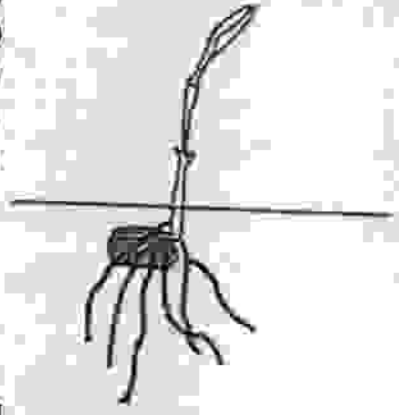
Evaluation activity 1.12

1. Name the part of the seed that comes out first during germination.
2. State any **one** way in which water is important during seed germination.
3. Apart from the lack of necessary conditions for germination, give any **two** other factors that can make a seed fail to germinate.
4. State the reason why the cotyledons that come out of the ground after germination appear green
5. In the diagram below, **Q**, **R** and **S** show the stages of germination of a seed. Study and use it to answer the questions that follow.



(a) Name the part marked **X**.
(b) How is part marked **X** useful to the seedling at stage **R**?
(c) Why is the part marked **X** useless at stage **S**?
(d) What type of germination is shown by the seed?

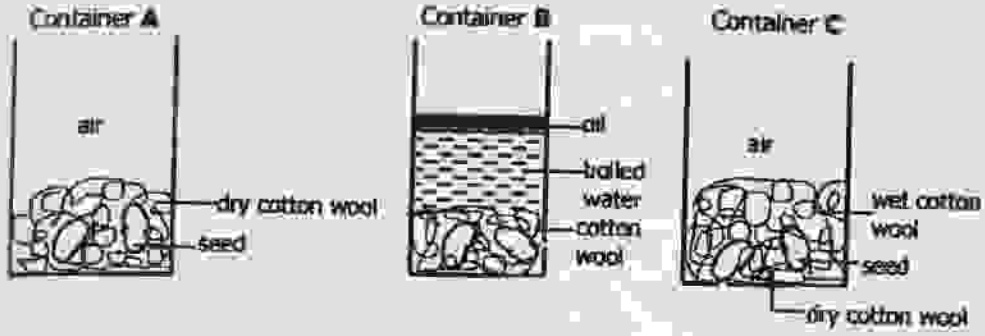
6. The diagram below shows a germinating seed. Use it to answer the questions that follow.



(a) Identify the type of germination shown above.
(b) Where does the seed above store the food useful during germination?
(c) Give any **two** examples of plants with the same type of germination.

Evaluation activity 1.12 Continued

7. The diagrams below show an experiment about germination. Study and use them to answer the questions that follow.



(a) In which container will the seeds germinate after some days?
(b) Why would the seeds in the container you have identified germinate?
(c) Give **one** effect of oil on the seed in the container marked **B**.
(d) State any **one** role of the condition which is absent in container **A** during germination.

8. The seeds were placed in four test tubes. The findings were recorded in the table below. Study it carefully and use it to answer the following questions.

Test tubes	**A**	**B**	**C**	**D**
Conditions				
Water	×	✓	✓	✓
Oxygen	✓	×	✓	✓
Warmth	✓	✓	✓	×

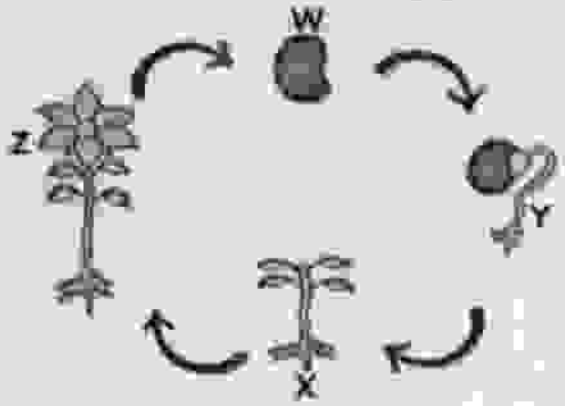
✓ Condition present × Condition absent

(a) Identify the **two** conditions which were present in test tube B.
(b) Give **one** reason why the seeds in test tube **D** did not germinate.
(c) State what happened to seeds in test tube **C** as recorded in the table.

9. (a) Give the role of each of the following conditions during germination.
(ii) Oxygen
(iii) Water
(b) Apart from oxygen and water, mention **one** other condition necessary for a seed to germinate.
(c) State the difference in germination between a maize grain and a bean seed.

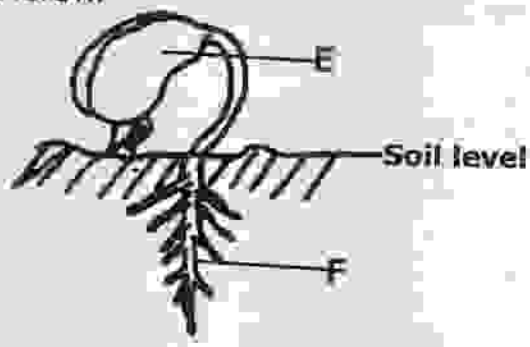
Evaluation activity 1.12 Continued

10. The diagram below shows the stages in growth of a plant. study and use it to answer the questions that follow.



(a) Name the stages of the plant marked **W** and **Y**.
(b) Identify the process that leads to the formation of stage marked **X**.
(c) State any **one** role of the stage marked **Z** in the growth of the plant.

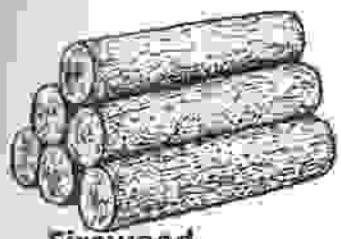
The diagram below shows a germinating seed. Study and use it to answer the questions that follow.



(a) Name the part labelled **F**.
(b) State the role of part marked **E** during germination.
(c) What type of germination is shown in the diagram above?
(d) To which group of flowering plants does the germinating seed above belong?

Ways in which plants are important to people.

- ✓ Plants are source of food.
- ✓ Plants are source of herbal medicine.
- ✓ Plants provide electric poles.
- ✓ Some plants are sold for money.
- ✓ Plants are used to make crafts.
- ✓ Plants are used for practical studies.
- ✓ Plants are source of manure.
- ✓ Plants provide shade to people.
- ✓ Plants purify air in the atmosphere.
- ✓ Plants are used to control soil erosion.
- ✓ Plants like Ficus trees are used to make bark cloth.
- ✓ Plants provide wood fuel like firewood and charcoal.
- ✓ Plants provide timber for making furniture and roofing houses.



Firewood



Wooden table



Electric poles



Playing a xylophone



Weaving a basket



Cooking food got from plants

Ways in which plants help to support human life.

- Plants are source of food.
- Plants provide oxygen during photosynthesis.
- Plants are source of herbal medicine.
- Plants purify air.

Ways in which plants can be harmful to people.

- Some plants are poisonous.
- Some plants cause irritation on the body.
- Some plants have thorns which pierce people.
- Plant harbour dangerous animals.
- Plants are hiding places for mosquitoes and tsetse flies.
- Plants fall and cause accidents to people.

Ways of conserving plants in the environment.

- Planting more trees.
- Using alternative sources of fuel.
- Use of energy saving stoves.
- Proper harvesting of trees.
- Gazetting forests.
- Controlling bush burning.
- Setting and reinforcing laws against deforestation.

Evaluation activity 1.13

1. (a) Give any **two** ways in which plants support human life.
   (b) Mention any **two** ways in which humans can conserve plant life.

2. (a) Write down any **two** forms of wood fuel got from plants
   (b) Give any **two** ways in which plants are harmful to people.

3. (a) State any **one** way in which plants are able to purify air.
(b) Apart from purifying air, mention any **two** other ways in which plants are useful to people.
(c) Name any **one** gas purified by plants in the atmosphere.

4. (a) Name any **two** parts of the plant that provide food to humans.
(b) Give **one** reason why people are encouraged to use energy saving stoves when cooking food.
(c) In which way are Ficus trees useful to people?
5. (a) Write down any **two** examples of medicinal plants.
(b) Identify the fuel from plants commonly used by people in villages.
(c) Name the process in plants that helps to purify air in the atmosphere.

MEANING OF KEY TERMS IN PLANT LIFE

Seed. A fertilised ovule.
Fruit. A fertilised ovary.
Pistil. The female part of a flower.
Starch. Food made by plants.
Tendril. Thread-like structure used by some plants to climb others.
Stamen. The male part of a flower.
Flower. A reproductive part of a flowering plant.
Stem tubers. Swollen underground stems with stored food.
Chlorophyll. The green pigment found in plants.
Dicotyledons. Flowering plants whose seeds have two cotyledons.
Monocotyledons. Flowering plants whose seeds have one cotyledon.
Photosynthesis. The process by which plants make their own food.
Transpiration. The loss of water in form of vapour in plants to the atmosphere.
Pollination. The transfer of pollen from the anther to the stigma of a flower.
Fertilisation. The fusion of male and female gametes to form a zygote.
Flowering plants. Plants that bear flowers.

2 THEME: SCIENCE IN HUMAN ACTIVITIES AND OCCUPATIONS

TOPIC 2: GROWING CROPS

Key words:

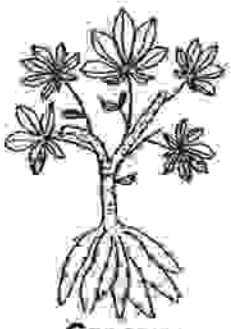
• perennial • consumption • annual • coffee • sale • season

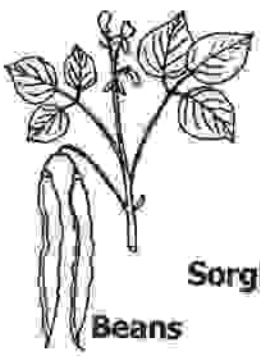
Crops are plants grown in the garden for consumption or sale.
Crops grown for home consumption are called **food crops.**
They include; maize, banana, beans, sweet potatoes, cassava and tomatoes.
Crops grown purposely for sale are called **cash crops.**
Examples are; tobacco, cotton, tea, vanilla and coffee.

Common crops grown in our community

Maize, sorghum, peas, ground nuts, bananas, beans, cassava, coffee, tea and cotton.


Maize


Cassava

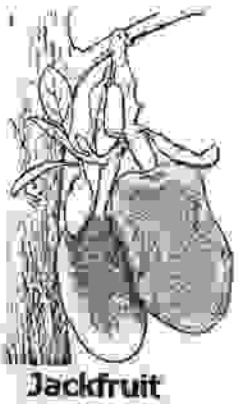

Beans


Sorghum


Banana


Coffee


Tea


Jackfruit

Groups of Crops.

Crops are grouped basing on their growth and maturity period/lifecycle time.

- Annual crops
- Perennial crops

Annual crops

Annual crops are crops that grow and mature in one season.
They are also called seasonal crops.

Characteristics of annual crops.

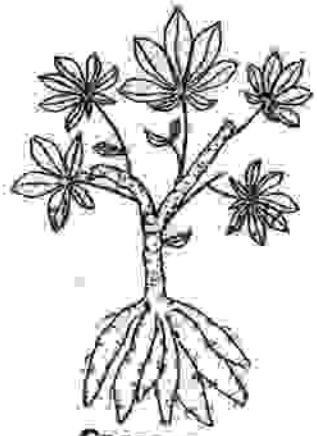
- They take less than a year to mature.
- Their life cycle lasts for a season.
- Most annual crops are harvested once.

Examples of annual crops

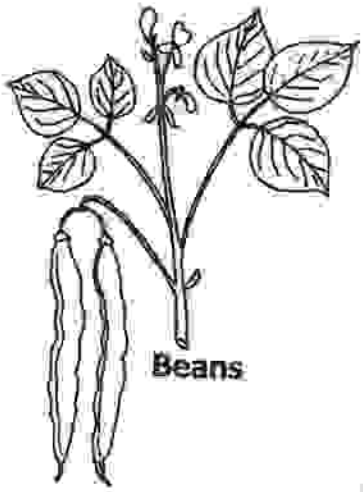
- tomatoes
- maize
- sorghum
- cassava
- beans
- cotton
- peas
- millet



Tomato



Cassava



Beans

Perennial crops

Perennial crops are crops that grow and mature in more than one season.

Characteristics of perennial crops.

- They take more than one year to mature.
- They are harvested year after year or season after season.
- They provide multiple harvests.
- Their life cycle lasts for many years

Examples of perennial crops

- coffee
- tea
- cocoa
- bananas
- mangoes
- jackfruits



Banana



Coffee



Tea

Evaluation activity 2.1

1. (a) Name the group of crops that are harvested year after year.
(b) Give any **two** examples of crops that belong to the group you have named in (a) above.
(c) State any **one** in which the above crops are harvested.
2. Why are tomatoes grouped as annual crops?
3. Name any **one** group of crops.
4. Name any **one** example of a perennial food crop grown by people.
5. State any **one** characteristics of annual crops.

The diagram below shows a crop commonly grown by people. Use it to answer questions **6** and **7**.



6. Name the crop shown in the diagram above.
7. Identify the group of crops to which the crop above belongs.

Key words:

• cereal • legume • annual • lettuce • nodules • cauliflower

Classification of annual crops

Cereal crops

Cereal crops are crops that produce grains.

Characteristics of cereal crops

- ✓ They are annual plants.
- ✓ They are rich in carbohydrates.
- ✓ They produce large quantities of seeds.
- ✓ They have fibrous root system
- ✓ They have parallel veined leaves.

Examples of cereal crops

Millet, maize, sorghum, wheat, rice, oats and barley



sorghum



maize



wheat



millet

Leguminous crops (legumes)

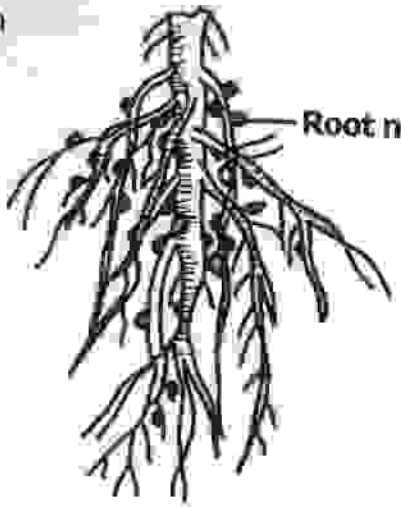
Legumes are crops that have nodules on their roots.

Root nodules store nitrogen fixing bacteria.

Nitrogen fixing bacteria fix nitrogen into the soil.

This helps to improve soil fertility.

Root hairs absorb water and mineral salts from the soil.



Examples of legumes

beans, peas, ground nuts, soya beans, French beans and castor

Points to note:

Legumes help to add nitrogen into the soil.

Legumes can be identified by checking for root nodules on their roots.


Pea plant


Groundnut plant


Bean plant


Soya bean plant

How legumes differ from other dicotyledonous crops.

- Legumes have root nodules unlike other dicotyledons.
- Legumes store their seeds in pods while other dicots store their seeds in the fruits.

Fruit crops

These are crops that provide fruits.

Examples are; apple, oranges, mangoes, pineapples, pawpaw, avocados and lemons.


Apples


Oranges

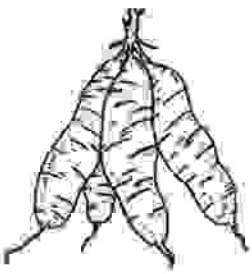

Pineapple

Root crops

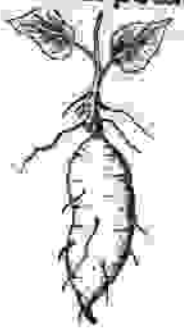
These are crops that have edible roots.

Examples are; cassava, sweet potatoes and carrots

Cassava



Sweet potato



Carrot



Vegetable crops

Vegetables are plants or parts of a plant eaten as food.

Types of vegetables

Leafy vegetables. They include; cabbage, spinach, lettuce and kales



cabbage



Pigweed



lettuce

Fruit vegetables. They include; tomatoes, eggplants and cucumber



Tomato



Eggplant



Cucumber

Root vegetables e.g., carrots and turnips



Carrot



Turnip

Flower vegetables e.g., cauliflower



Cauliflower

Evaluation activity 2.2

1. Name any **two** types of annual crops.
2. How are legumes different from other groups of annual crops?
3. State any **one** characteristic of cereal crops.
4. Mention any **one** vegetable crop used for spicing sauce during frying.
5. Identify the group of annual crops that adds nitrogen into the soil.

The diagram below shows roots of a bean crop. Study and use it to answer questions **6** and **7**.

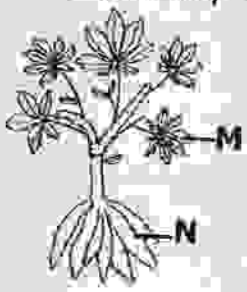


6. Name the structures marked **K**.
7. State the role of the organisms that live in the part marked **K**.
8. The list below shows crops commonly grown in our community. Study and use it to answer the questions that follow.

Spinach, maize, cabbage, carrot

(a) Give **two** crops from the list that can be grouped together.
(b) State any **one** reason why most people grow maize on the list above.
(c) Identify the crop from the list above that has edible swollen underground part.

9. The diagram below shows a food crop commonly grown by people in our community. Study and use it to answer the questions that follow.



(a) Name the parts marked **M** and **N**.
(b) State the function of the part marked **N**.
(c) Give any **one** other crop that has part marked **N**.

Key words:

• secateurs • stony • pruning • knapsack • manure • lubricate

Garden Tools, Equipment and Materials

Garden tools	Garden equipment	Garden materials
▪ hoe ▪ rake ▪ trowel	▪ wheelbarrow ▪ watering can ▪ knapsack sprayer	❖ string ❖ tape measure ❖ pegs

Garden tools	Garden equipment	Garden materials
• spade • pick axe • axe	• pail • pot • jerry can	

Hoe.

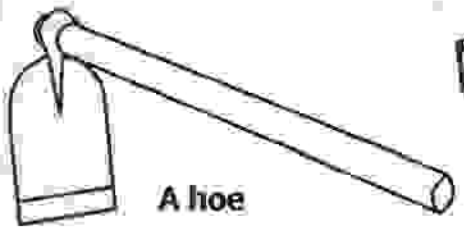
- It is used for digging.
- It is used for ploughing land.
- It is used for earthing up.
- It is used for weeding.
- It is used for sowing.
- It is used for harvesting tuber crops.

Adaptation. It has a wide, flat metal blade attached to a long handle, making it easy to dig, break soil and remove weeds.

Rake.

- It is used for collecting cut grass.
- It is used for levelling the soil.
- It is used for harrowing.
- It is used for collecting small samples of soil.

Adaptation. It has a strong metal with teeth-like structures that help in gathering weeds and levelling soil.



A hoe



A rake

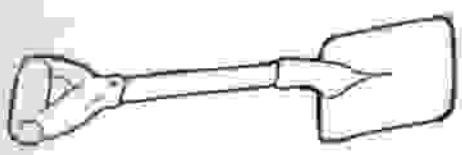
Spade

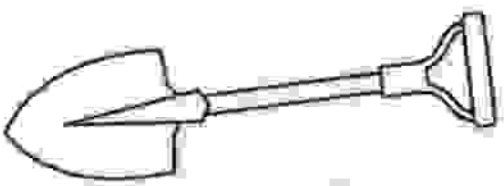
- It is used for moving loose soil.
- It is used for loading manure on a wheelbarrow.
- It is used for levelling the soil.
- It is used for mixing manure.

Adaptation. It has a flat strong metal blade and a long handle, making it easy to loosen soil and load manure.

Shovel

- It is used for scooping and moving loose soil.
- It is used for making irrigation channels.
- It is used for loading soil and gravel.





shovel

Wheelbarrow.

- It is used for transporting manure,
- It is used for transporting harvested crops.

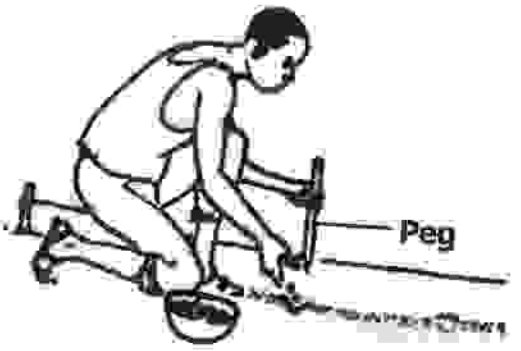
• It is used for transporting soil and other garden tools.

Peg.

It is a material used to do some activities in the garden.

• It is used to mark planting spots.
• It is used to align rows.
• It is used to support crops





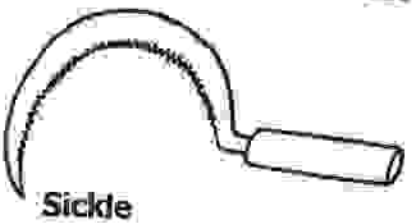
Sickle.

• It is used for harvesting millet, sorghum, rice and simsim.
• It used for cutting grass.

Adaptation. It is curved with sharp serrated edges for cutting cereals easily.

Garden trowel. It is used for transplanting seedlings.

Adaptation. It has a small, curved metal blade that helps in scooping seedlings from the soil without damaging the roots.

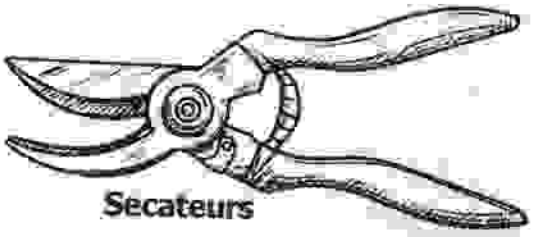




Hand fork. It is used for light weeding.

Secateurs. It is used for pruning crops.

Adaptation. It has sharp strong blades that help in cutting small branches and stems easily.





Panga

- It is used for cutting small trees.
- It is used for harvesting crops like sugarcanes and bananas.

Adaptation. It has a long sharp metal blade for cutting through thick plants and small branches.

Garden line (string)

- It helps to make straight lines when row planting.
- It is used for spacing crops.



Panga



Gardenline/string

Pick axe.

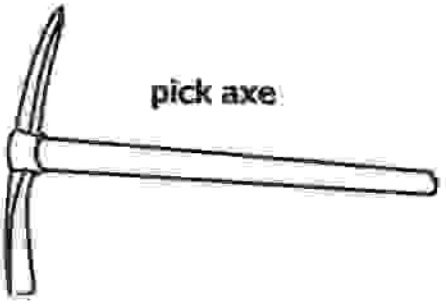
- It is used for breaking stones in the garden into small pieces.
- It is used for digging manure pit.

Mattock.

- It is used for digging stony grounds.
- It is used for removing stones from the garden.



Mattock



pick axe

Garden fork.

- It is used for turning/mixing manure.
- It is used for loading manure.

Forked hoe.

- It is used for digging stony grounds/hard soil.
- It is used for removing cut roots from the soil.

Adaptation. It has strong pointed metal prongs that help in digging stony grounds.



Garden fork



Forked hoe

Axe

- It is used for felling big trees.
- It is used for splitting wood.

Adaptation. It has a heavy, sharp metal head that helps in cutting thick wood and trees.

Dibber. It used for making holes in the soil for planting seeds or seedlings



Axe



Dibber

Knife. It is used for harvesting cereals like millet, sorghum and wheat.
Slasher. It is used for slashing bushes or weeds.
Adaptation. It has a long, sharp blade that helps in clearing bushes and cutting grass quickly.
Watering can. It is used for watering crops. It provides gentle, controlled flow of water, preventing soil erosion and damaging young crops.
Knapsack sprayer/backpack sprayer.
- It is used for spraying pesticides on crops.
- It is used for spraying liquid fertilisers.
- It is used for spraying herbicides to kill weeds.

Adaptation. It has a tank for holding chemicals, a pump for exerting pressure and a nozzle for spraying chemicals easily.



Knapsack sprayer



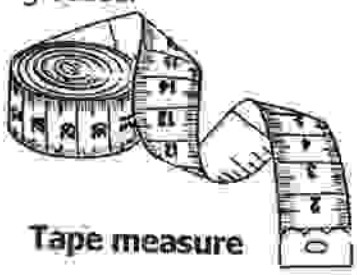
Watering can

A pail and jerrycan.
- They are used for carrying water.
- They are used for storing water.

Tape measure.
- It is used for spacing crops in the garden.
- It is used to make straight lines for planting seeds.



Pail and jerrycan



Tape measure

Billhook.
- It is used for pruning crops like banana.

Pruning saw. It is used for cutting thicker branches from the plant.

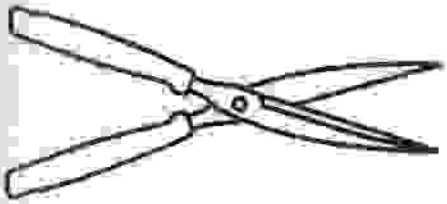




Shears. They are used for cutting twigs and leaves from the plant.

Pot.

- It can be used to raise seedlings.
- It can be used to carry seedlings to the garden.
- It can be used to hold water used for watering crops.



Shears



Pot

Diagrams of people Using some of the garden tools













Ways of caring for simple garden tools.

- Washing garden tools.
- Repairing damaged garden tools.
- Painting metallic garden tools.
- Use the garden tools for their right purpose.
- Lubricating moving parts of some garden tools.
- Keep metallic garden tools in cool dry places.
- Sharpening cutting garden tools.
- Regular serving of garden tools.

Evaluation activity 2.3

1. Name the correct garden tool for transplanting seedlings.
2. Write down any **one** activity done in the garden using a garden fork.
3. State the importance of a tape measure in crop growing.
4. Give **one** reason why farmers usually remove soil on the hoe after digging.

The diagram below shows a garden equipment. Use it to answer questions **5** and **6**.



5. Identify the garden equipment shown in the diagram above.
6. Give any **one** way in which the garden equipment above is useful to a crop farmer.
7. Give any **two** ways of caring for simple garden tools.
8. State the reason why farmers should paint their metallic garden tools.

The diagram below shows a garden tool. Use it to answer questions **9** and **10.**



9. Name the garden tool above.
10. Mention any **one** way in which the above garden tool is important to a crop farmer.
12. The table below shows some of the garden tools and the use of each tool. Complete the table correctly.

Garden tool	Use
....	For harvesting rice.
Dibber	....
.....	For mixing manure.
Forked hoe	....

13. Name the crop growing activity in the garden that requires the use of;

(i) Hand fork

(ii) Secateurs

Crop Growing Practices

- land clearing
- planting
- thinning
- transplanting
- spraying
- storing
- ploughing/digging
- gap filling
- mulching
- pruning
- pricking out
- watering
- seed selection
- weeding
- manuring
- staking
- harvesting
- record keeping

Land preparation

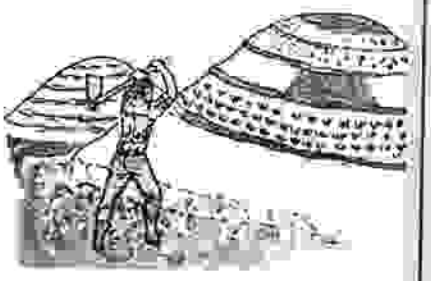
This is the first stage in crop growing.
It is done to remove bushes, trees and grass from the land.
It is usually done during dry season.

Reasons why land preparation is done during dry season.

(i) To enable weeds dry before planting.
(ii) To improve soil aeration.
(iii) To prevent weeds from regrowing after digging.
(iv) To prevent soil from sticking onto the hoes or oxen-plough.

Methods used to prepare land.





Slashing. It is done to remove weeds.

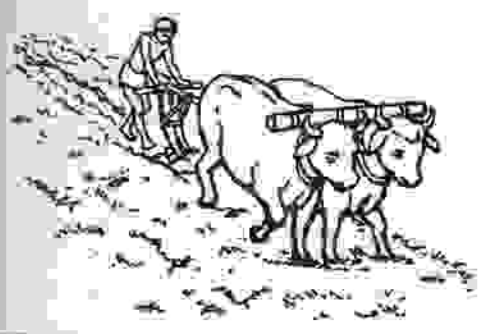
Cutting big trees

Ploughing/tilling land

Ploughing is the opening up of land by digging.

- It is done to make soil loose and soft for easy growth of roots.
- It allows water and air to enter the soil.
- It controls weed growth.
- It removes roots left by previous plants.

It is done using a hoe, tractor, oxen-plough and forked hoe





Activities done during ploughing

1. Digging

Reasons why digging is done in crop growing

- To loosen soil.
- To improve soil aeration.
- To remove weeds for better planting conditions.

2. Harrowing. This is the breaking of big lumps of soil which are left after ploughing.

-It opens up land for easy detrashing.

-It controls weeds that would grow after ploughing.

It can be done using; a hoes, tractor, oxen-plough, forked hoe and garden fork





3. Detrashing. This is the removal of unwanted materials from the soil. These materials include stone particles, tree stumps, plastic materials, polythene bags, and metallic scraps. It is done using a rake.

-It helps to level the soil.

-It improves water infiltration.

-It improves aeration in the soil.

Controlled bush burning

A farmer should make fire breaks to prevent fire from spreading to other areas.

Fire breaks are made by; slashing or digging around the part or field one wants to clear.

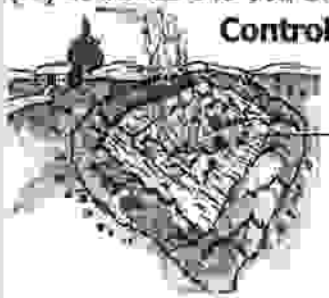
Burning as a method of clearing land is not good because;

(i) It destroys humus in the soil.

(ii) It destroys soil texture.

(iii) It kills useful soil organisms.

(iv) It bakes the soil and makes it hard to cultivate.



Controlled bush burning

Fire breaks



slashing

Seed selection.

Seed selection is the removal of damaged seeds from the healthy ones.
It is also called sorting seeds.

Reasons why farmers sort seeds before planting them.



- To increase crop yields.
- To reduce chances of gap filling.
- To choose the right variety.
- To get viable seeds.
- To prevent wastage of space in the garden.
- To prevent wastage of time and labour.
- To promote proper utilisation of space in the garden.

Qualities/ characteristics of viable seeds for planting.

- They are free from physical damage.
- They should be mature enough.
- They should of reasonable size.
- They should not be broken.
- They should be obtained from health parent plants.

Furrowing

This is the making of holes and furrows in the garden for planting.
It can be done using a hoes, garden fork or dibber.

Planting

Planting is the practice of burying seeds or stem cuttings in the soil to grow.
Planting is done during rainy season.

Reasons.

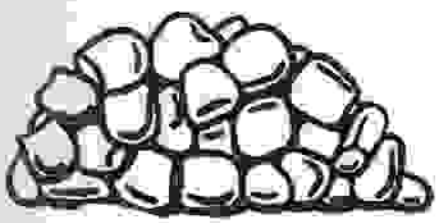
There is plenty of rainfall to support the growth of crops.
The soil is soft for easy growth of roots.

Advantages of early planting.

- Crops make full use of rainfall.
- It controls crop pests and diseases. Crops grow and mature before pests multiply.
- Crops mature earlier and get good market.
- It controls the outbreak of famine at home.

Examples of planting materials are;

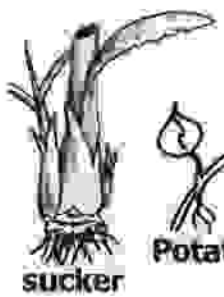
seeds, stem cuttings, potato vines suckers, and stem tubers



Seeds



stem cutting



sucker

Potato vine

Methods of planting

Row planting.

This is the planting of seeds or stem cuttings in lines.

It can be done using a garden line/string or tape measure.



Uses of a calendar to crop farmers.

- It helps farmers to know the right time for planting.
- It ensures timely harvesting.
- It helps farmers to monitor weather changes for better land preparation.
- It helps farmers to know the right period for high demand of harvest on market.
- It helps farmers to prepare for wet or dry season.

Advantages of planting crops in rows.

- ❖ It makes weeding easy.
- ❖ It enables crops to get enough sunlight.
- ❖ It controls wastage of seeds.
- ❖ It makes harvesting of crops easy.
- ❖ It makes application of fertilisers easy.
- ❖ It promotes proper spacing of crops.
- ❖ It makes the movement of the farmer in the garden easy.

Disadvantages of planting crops in rows.

- ↕ It takes a lot of time to plant seeds.
- ↕ It needs a lot of labour.
- ↕ It requires a large piece of land.
- ↕ It is tiring.
- ↕ It needs some skills of proper crop.

Broadcasting

This is a way of planting seeds by randomly scattering them.

It is suitable for crops like millet, simsim, ground nuts, sorghum, peas and rice





Advantages of broadcasting method

- It is time saving
- It needs less labour.
- It is easy and quick.
- It does not waste nutrients in the soil.

Disadvantages of broadcasting

- It makes weeding difficult.
- It makes harvesting difficult.
- Pests and diseases easily spread.
- It wastes seeds.
- It burdens the crop farmer to thin.
- It encourages competition for nutrients and sunlight.

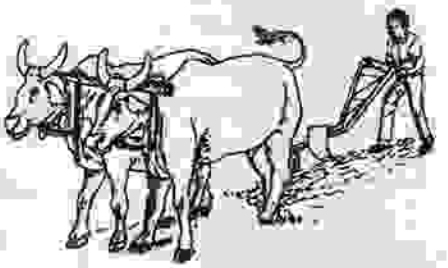
Evaluation activity 2.4

1. Write down any **two** examples of crop growing practices.
2. Give any **one** reason why crop farmers clear their land during dry season.
3. (a) Apart from planting crops in rows, give another method of planting crops.
   (b) Give any **two** advantages of planting maize in rows.
   (c) Name any **one** example of a planting material used by farmers.

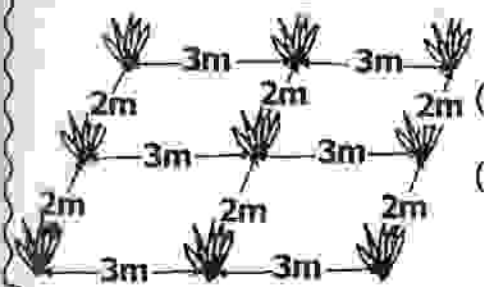
The diagram below shows a method of planting crops. Use it to answer questions **4** and **5**.



4. Identify the method of planting shown above.
5. State any **one** advantage of the method of planting crops shown in the diagram above.



(a) Identify the crop growing practice shown in the diagram above.
(b) Give any **two** advantages of carrying out the crop growing activity shown above.
(c) Name the group of animals used to carry out the crop growing activity shown in the diagram above.



(a) Name the method of planting crops shown in the diagram above.
(b) State any **two** advantages of spacing crops as shown above.
(c) Name any **one** garden tool that can be used for spacing crops as shown above.

Key words:
• hardening • nursery • trowel • thinning • seedling
• transplant

Caring for crops

Nursery bed

A nursery bed is a small garden prepared for raising seedlings before transplanting.

A seedbed is a piece of land prepared to plant seedlings for further growth.

Crops whose seeds are first planted in a nursery bed

Tomatoes, eggplants, passion fruits, carrots, coffee, cabbage, onions and kale plants

Reasons why farmers plant some seeds in a nursery bed

- It gives a farmer enough time to prepare the garden.
- It allows a farmer to select healthy seedlings.
- It protects seedlings from harsh weather conditions.
- It prevents wastage of seeds.
- It allows proper care for the seedlings.

Preparing a nursery bed

Activities done when preparing a nursery bed

Site selection, clearing land, digging, harrowing, adding fertilisers, constructing the shade

Materials needed: dry grass, hoe, seeds, slasher, poles, water and manure.

Steps taken:

(i) Clear the area using the slasher.
(ii) Dig and loosen the soil, mixing it with manure to improve fertility.
(iii) Construct a raised bed using poles and dry grass to create a shade.
(iv) Sow the seeds evenly in the prepared soil and water as needed.



- Dry grass provides the shade and protects the seedlings from sunshine and heavy raindrops.
- The grass is spread slightly to allow water pass and reach the seedlings during rain or watering.
- The shade should be raised to protect the seedlings from direct sunshine.
- The soil should be slightly raised to improve water drainage and prevent waterlogging.

Ways of caring for the seedlings in a nursery bed.

- Watering seedlings
- Spraying seedlings.
- Regular weeding
- Thinning
- Fencing the nursery bed

Factors to consider when choosing a site for a nursery bed

Factor	Reason for its consideration
Fertility of the soil	Fertile soil helps seedlings grow strong, well and healthy.
Nearness to water source.	For easy access to water needed to keep the seedlings alive.
Nearness to the home.	It makes it easy to care for and inspect the seedlings.
Drainage of the soil.	Water should not stay in the soil for too long to avoid rotting of the roots.

Hardening off.

Hardening off is the gradual exposure of seedlings to different weather conditions before transplanting.

These conditions include; strong sunshine, strong wind and heavy rainfall.

Hardening off is done to increase resistance of seedlings against harsh weather.

Hardening off enables seedlings to adopt external garden conditions.

Activities done during hardening off

Gradual removal of the shade and reduced watering.



Transplanting

Transplanting is the transfer of seedlings from the nursery bed to the main garden.

Transplanting should be done in the evening

Reason. To prevent wilting of seedlings. There is low rate of transpiration.

The correct garden tool for transplanting is a **garden trowel.**

A garden trowel is ridged to easily scoop seedlings from the soil without damaging the Roots.

Steps taken when transplanting seedlings

(i) Water the nursery bed before scooping seedlings from the soil.
Reason. To loosen the soil and reduce damage on the seedlings.

(ii) Scoop the seedlings carefully using a garden trowel to avoid root damage.
Reason. To preserve roots for proper growth after transplanting.

(iii) Dig the holes in the main garden at an appropriate spacing.
Reason. To ensure proper spacing and avoid overcrowding.

(iv) Plant seedlings, cover the roots with soil and water immediately.
Reason. To secure the seedlings in the soil and promote quick growth.

Ways of caring for the seedlings in a nursery bed.

- Watering seedlings
- Spraying seedlings.
- Regular weeding
- Thinning
- Fencing the nursery bed

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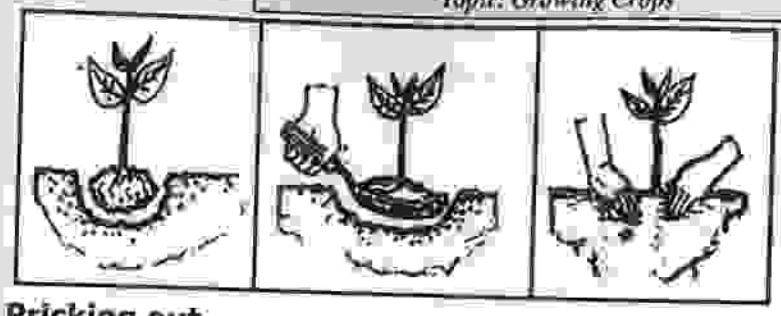
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Reason. To secure the seedlings in the soil and promote quick growth.





Garden trowel

Pricking out.

Pricking out is the practice of removing overcrowded seedlings from a nursery bed to be planted elsewhere.

It is done soon after the sown seeds have germinated.

Advantages of pricking out

- It reduces competition for sunlight, space and nutrients.
- It prevents easy spread of diseases among seedlings.

Gap filling.

Gap filling is the replanting of fresh seeds in the garden where they did not germinate after planting.

Factors that may make planted seeds fail to germinate

-Damaged seeds by pests.

-Poorly dried seeds.

-Overstayed seeds.

-Immature seeds.

Ways of keeping seeds viable for a long time.

-Proper drying of seeds before storage.
-Dusting seeds.
-Proper storage.
-Seed dressing.

Advantages of gap filling

- It prevents wastage of space in the garden.
- It maintains crop population.
- It increases crop yields.

Weeding

Weeding is the removal unwanted plants growing in the garden.



The unwanted plants that grow in the garden are called **weeds.**

Examples of weeds are;
spear grass, blackjack,
elephant grass,
pigweeds, couch grass,
star grass and wandering Jew.


Wandering jew


Blackjack

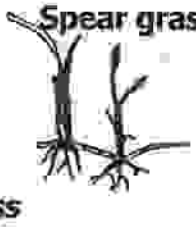

Thorn apple


Yellow wood sorrel


Couch grass


Nut grass


Love grass


Spear grass


Tickberry


Elephant grass

Ways in which weeds can lower the yields of crops in the garden
(i) They compete with crops for sunlight, space and nutrients.
(ii) They are hiding places for crop pests.
(iii) They spread diseases to crops.

Methods of weeding/ways of controlling weeds
- Slashing using a slasher/mower.
- Spraying herbicides.
- Cover cropping
- Use of bioherbicides.
- Uprooting weeds.
- Digging weeds.
- Crop rotation
- Mulching

Biological methods of controlling weeds.
- Grazing livestock on weeds.
- Planting cover crops.
- Use of bioherbicides.
- Use of beetles to feed on water hyacinth.


Uprooting weeds


Slashing using a mower


Spraying with herbicides

Advantages of timely weeding the garden

- It eases harvesting.
- It eases application of manure.
- It controls crop pests and diseases.
- It destroys hiding/breeding places for pests which spread diseases.
- It breaks the life cycle of pests.
- It helps in easy management of diseases by spraying.
- It enables crops to have more strength to fight diseases.
- It reduces competition for plant nutrients, space and sunlight.
- It eases spraying.

Thinning

Thinning is the removal of excess seedlings from the garden.





Before thinning



After thinning

Advantages of thinning

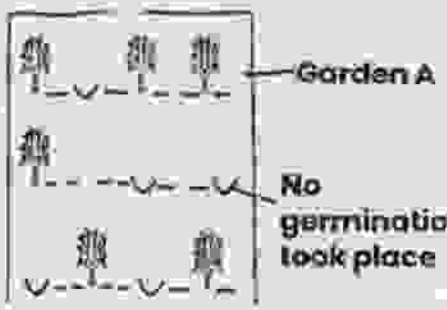
- It makes weeding, spraying and harvesting easy.
- It controls easy spread of crop pests and diseases.
- It improves crop yields and the quality of harvests.

Evaluation activity 2.5

1.(a) Give the meaning of the following terms:
(i) Nursery bed (ii) Seedbed
(b) State any **two** activities carried out when preparing a tomato nursery bed.
2.(a) Write down any **two** factors to consider when choosing a site for a nursery bed.
(b) Give any **two** reasons why farmers first plant some seeds in a nursery bed.
3. (a) State any **two** ways in which weeds can lower the yields of crops.
(b) State any **one** way in which the growth of weeds can be controlled in the garden.
(c) Apart from controlling weeds, give any **one** other way in which farmers can care for their crops.
4.(a) Name the correct garden tool for transplanting seedlings.
(b) State the reason why evening time is the best for transplanting seedlings.
(c) Mention any **two** advantages of thinning seedlings in the nursery bed.

Evaluation activity 2.5 Continued

The diagram below shows a garden with seedlings growing. Use it to answer question **5**.



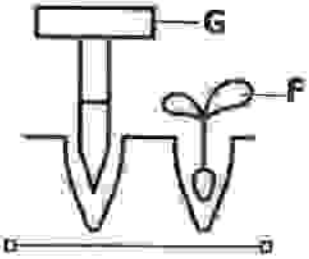
5. Identify the crop growing activity that should be done in the garden above.
6. What is the importance of gap filling in crop growing?
7. Write down any **one** activity done to the seedlings during hardening off.
8. Give any **one** way in which timely weeding of crops helps to control diseases.
9. Give any **one** way in which crop farmers can keep seeds viable for a long time.

The diagram below shows a crop growing practice. Use it to answer questions **10** and **11**.



10. Name the garden to tool marked **Q**.
11. Identify the crop growing practice shown in the diagram above.

The diagram below shows an activity done in the garden by farmers. Use it o answer questions **11** and **12**.



12. Identify the garden tool marked **G**.
13. Name any **one** crop that can planted as the one marked **F**.

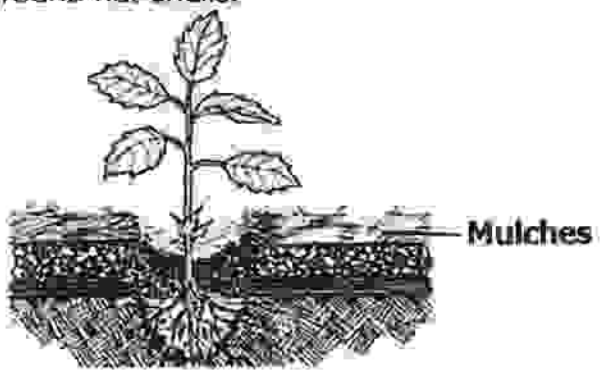
Mulching

Mulching is the covering of topsoil using dry plant materials.

The dry plant materials are called mulches.

Examples of mulches

dry grass, dry banana leaves, dry coffee husks, dry cereal stalks and dry ground nut shells.





Advantages of mulching

❖ Mulching controls soil erosion.
Mulches reduce the speed of flowing water.
Mulches trap soil from flowing with water.

❖ Mulching improves soil fertility.
Mulches decompose and form humus in the soil.

❖ Mulching conserves moisture in the soil.
Mulches minimise the rate of evaporation of water in the soil.

❖ Mulching controls the growth of weeds.
Mulches cut off the supply of sunlight for photosynthesis to the weeds.

Disadvantages of mulching.

✓ Mulching is tiring.
✓ Mulches are fire hazard.
✓ Mulches hide some crop pests e.g. termites
✓ Some mulches can grow into weeds.

Watering/irrigation

It is the artificial application of water on crops especially seedlings.

It is mainly done during dry season.

It should be done in the morning and in the evening.

Methods of irrigation.

Using a watering can, sprinkler irrigation, horse pipe irrigation and canal irrigation.

Watering crops is done using a watering can.

Importance of watering crops

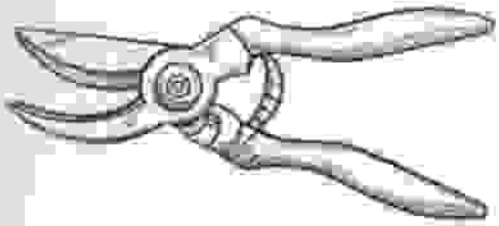
- Water enables crops to make food.
- Water enables crops to grow well.
- Water makes soil soft and loose for roots to penetrate.





Pruning

Pruning is the removal of excess branches or leaves from the plant.
The correct garden tool for pruning is secateurs



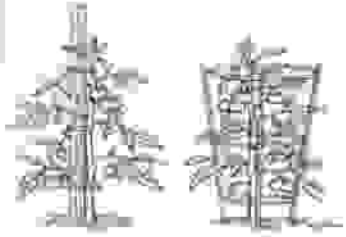


Advantages of pruning crops

- It enables crops to grow well.
- It enables crops to get enough sunlight.
- It reduces the weight of crops.
- It improves the quality of yields.
- It creates space for other management practices like spraying.
- It reduces hiding places for crop pests.

Staking

Staking is the providing of support to crops with weak stems.
Crops which are staked are; passion fruits, watermelon, irish potatoes and bananas.



Staking tomato plants



Trellising passion fruits



propping a banana plant

Advantages of staking crops

(i) It makes weeding and harvesting easy.
(ii) It enables crops to get enough sunlight.
(iii) It prevents ground pests from damaging fruits.
(iv) It prevents crops with heavy bunches from breaking.



Earthing up potato plants

Earthing up

Earthing up is the piling/heaping of soil around the base of a crop.
It is done using a hoe.

Crops which are earthed up

Sweet potatoes, cassava, onions, irish potatoes and maize

Importance/advantages of earthing up

✓ It provides extra support to the plant.
✓ It prevents the growth of weeds.
✓ It retains moisture in the soil.
✓ It improves on crop yields.

Harvesting crops

Harvesting is the collection of mature crops from the garden.

Harvesting of crops is determined by;

• Maturity of crops • Garden tools used. • Season of harvesting.

Signs of when different crop types are ready for harvesting

Crop	Signs
Vegetables	• Leaves are big enough • Colour changes • Size is full • Stems start drying
Legumes	• Pods turn brown • Pods become dry • Seeds rattle in pod • Leaves fall off
Fruit Crops	• Fruit changes colour • Fruit becomes soft • Fruit has full size • Smells nice
Cereals	• Leaves turn yellow • Grains harden • Heads bend down • Plant dries

Crop	Signs
Tuber Crops	• Leaves turn yellow • Stems dry • Skin hardens • Tuber is big

Early harvesting of crops may give poor seeds or fruits.
Late harvesting may lead to wastage of crop yields.
It can cause rotting of crops in the garden
It can lead to germination of seeds in the garden.
The season of harvesting crops depends on the type of crops to be harvested.
Cereals and legumes are best harvested during dry season.
Reason. There is plenty of sunshine to allow drying of harvested crops.
Tuber crops can also be harvested during wet season.
Reason. The soil is soft for easy uprooting or digging with less damage on the tubers.
Vegetables and fruits are also harvested during wet/rainy season to keep them fresh.

Methods of harvesting crops

Method	Crops harvested
Uprooting	Beans, peas, ground nuts, carrots, cassava and turnips
Digging out	Cassava, sweet potatoes, irish potatoes, turnips and carrots
Hand picking	Coffee, tea, cotton, tomatoes, oranges, guavas, avocados, maize
Cutting	Rice, banana, millet, simsim, sorghum and wheat


Uprooting

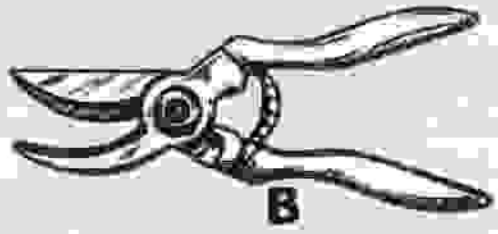

Digging out


Hand picking

Evaluation activity 2.6

1. Mention any **two** advantages of mulching a tomato garden.
2. State **one** way in which mulching helps to control the growth of weeds in the garden.
3. State any **two** reasons why pruning of crops is important in crop growing.
4. The diagrams below show garden tools used to carry out crop growing practices. Study and use them to answer the questions that follow.





(a) Name the crop growing activity that requires the use of garden tool marked **A** and **B.**
(b) Give any **one** way in which a crop farmer can care for the garden tool marked **A.**
(c) State **one** way in which rusting affects the garden tool marked **B.**
5. Identify the best season for harvesting beans.
6. State any **one** factor a farmer should consider before harvesting crops.
7. (a) Apart from digging out, name any **one** other method of harvesting crops in the garden.
(b) Write down any **two** examples of crops harvested by digging out.
(c) State any **one** disadvantage of harvesting crops by digging out.

Preserving crops

Preserving crops is the way of keeping crops safe for a long time without going bad.

❖ Sun-drying.

Heat from the sun dehydrates crops.

Crops that are sun-dried include; beans, maize, ground nuts, cassava, peas

Importance of sun-drying harvested crops

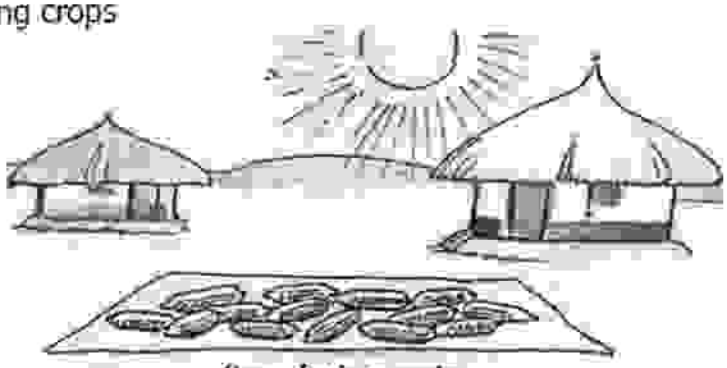
- It prevents crops from rotting when stored.
- It prevents crops from germinating from the store.
- It helps to maintain seed viability.
- It enables good processing, milling and grinding.
- It prevents crops from being attacked by germs.

❖ Burying in the soil under the shade.

It keeps the tubers fresh for some time.

❖ Pickling vegetables.

- Sprinkling water on vegetables.
- Refrigerating crops



Sun-drying maize

Importance of preserving crops.

- ✓ It ensures food security.
- ✓ It adds value to crops.
- ✓ It increases on the life span of crops.
- ✓ It makes it easy to transport crops.
- ✓ It prevents crops from developing new shoots when stored.

Threshing/shelling

This is the act of separating seeds or grains from their husks or pods.
Methods of threshing are; hand threshing and machine threshing.
Crops that can be threshed are; maize, beans, millet, soya beans, rice, peas.





Winnowing.

This is separating of seeds or grains from chaff or husks.
It is done to crops like millet, rice, maize, beans, sorghum and wheat.



Storing Crops

- They can be stored in granaries. The granary should have rat guards.

Reason. To prevents rats from entering the granary to spoil stored crops.

Crops stored in the granary are; beans, maize, millet, ground nuts and soya beans.

- Flour should be packed in dry sacks.
- The sacks should be placed on dry pieces of wood that are slightly raised

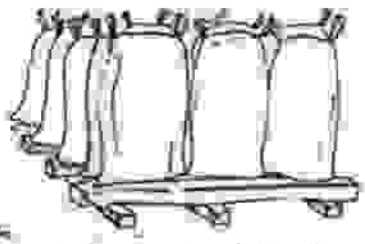
Reason. To prevent the sacks from absorbing moisture from the floor.

Note well. If the sacks are placed on the floor directly, the flour absorbs moisture from the floor and gets spoilt.

Using silos.

Using baskets and buckets.





Sacks on raised stands



Buckets of stored crops

Qualities of a good store for storing dried crops.

❖ It should have rat guards.

Reason. To prevent rats from climbing into the granary to destroy stored crops.

Rat guards are smooth, slippery and slanted to prevent rats from climbing into the granary.

❖ It should be leakproof.

Reason. To prevent rain from falling in the granary which leads to rotting of crops.

❖ It should have ample space.

Reason. To accommodate enough crops.

❖ It should be well ventilated.

Reason. To promote free circulation of air in the granary.

❖ It should be raised above the ground.

Reason. To prevent dampness from the ground.

Reasons why farmers store crops

- To reserve seeds for planting in the next season.
- To be sold when market prices are high.
- To keep food for eating in the dry season.

Activities carried out to make crops fit for market.

Processing, sorting, grading, packaging, standardizing and labelling

Record keeping

A crop record refers to information about activities carried out in crop production.

Examples of records that can be kept on a crop farm

Labour records. Track the number of workers, tasks done and salaries paid.
Expenditure records. Document money spent on farm activities and inputs.
Production records. Show details of crop yields and harvested quantities.
Health records. Monitor the conditions of crops, pests and diseases treated.
Inventory records. Record tools, seeds and other farm inputs in stock.

A sample of crop record (production record)

Season	Crop Type	Planting Date	Harvesting Date	Field size	Quantity Harvested	Remarks
Season 1	Maize	01.01.24	30.03.24	3 acres	3,500 kg	Good yields
Season 2	Beans	01.04.24	30.06.24	2.5 acres	1,200 kg	Pests controlled
Season 3	Rice	01.07.24	30.09.24	4 acres	4,800 kg	Adequate Rainfall
Season 4	Ground nuts	01.10.24	15.12.24	2 acres	950 kg	Low yields due to drought

Instructions

Planting Date and Harvesting Date. To note the respective dates for each season.
Field Size. To specify the areas used for planting.
Quantity Harvested. To record the total yield for the season.

Importance of record keeping to a crop farmer

- It helps in proper timing for planting and harvesting.
- It helps a farmer to calculate profits and losses made.
- It helps a farmer to know the expenditure.
- It helps a farmer to plan for his crop farms.
- It helps a farmer to practice crop rotation.
- It helps a farmer to get loans from the bank.

Evaluation activity 2.7

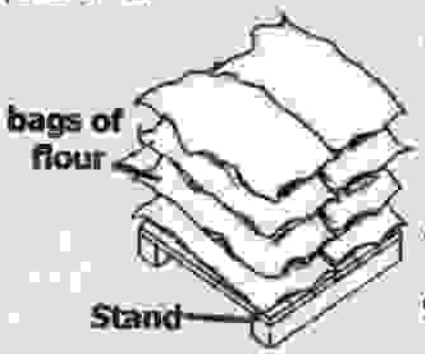
1. State any **one** reason why farmers are encouraged to preserve food stuffs like sweet potatoes.
2. State any **one** way of keeping vegetables fresh for some time.
3. Identify the activity done to remove chaff from millets grains at home after threshing.
4. (a) Apart from using sacks, state any **two** other ways in which farmers store their crops after harvesting.
   (b) Mention **one** way in which farmers can prevent sacks containing stored crops from absorbing moisture from the ground.

Evaluation activity 2.7 Continued

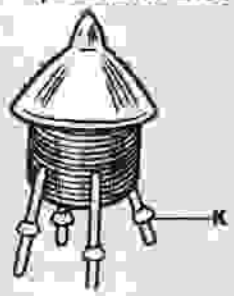
(c) Give any **one** reason why farmers store their crops after harvesting.

5. State any **two** qualities of a good store for keeping harvested crops after sun-drying.

The diagram below shows a bag of maize flour in the store. Use it to answer question **6**.



6. Why are bags of maize flour put on such a raised stand?
7. Write down any **two** activities done by farmers to make their harvested crops fit for marketing.
8. Give any **two** ways in which record keeping is important to a crop farmer.
9. Give any **two** reasons why farmers sun-dry their harvested crops properly before storing them.
10. State **one** reason why rat guards are able to prevent rats from entering the granary.
11. Apart from sun-drying, mention any **one** other way in which sweet potatoes can be preserved at home.
12. The diagram below shows a food storage facility. Use it to answer the questions that follow.



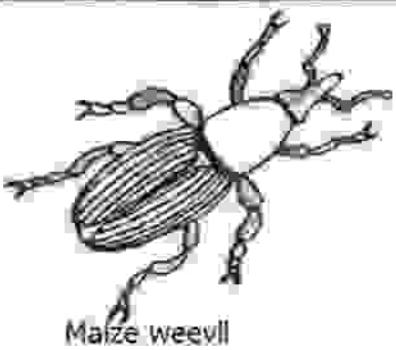
(a) Name the farm storage facility shown above.
(b) Give the importance of the structure marked **K** on the granary.
(c) Mention any **two** reasons why most farmers in rural areas use the food storage facility above for storing their harvested crops.

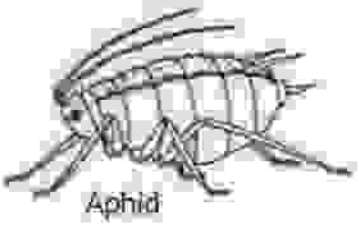
Crop Pests and Diseases

Pests are organisms that destroy crops.

Crop Pests	Crops affected
Aphids	Vegetables, beans, sorghum, carrots, peas
Cutworms	Maize, beans, tomatoes, wheat, rice and cabbage
Stalk borer	Maize, sugarcanes and sorghum
Mealy bug	coffee
Weevils	Maize, sweet potatoes, beans, wheat, peas
armyworm	Rice, cassava and carrots
White fly	Cassava (leaves)
Mole rats	Potatoes, cassava, carrots


Rhino beetle


Maize weevil


Aphid


Potato weevil


Armyworm


Caterpillar


Termite


Rat


Whitefly

Parts of the crops affected by pests.
Stems, leaves, roots and fruits

Types of crop pests
Field pests. These affect crops when still in the garden.
They include; weaver birds, monkeys, moles, caterpillars and aphids
Storage pests. These affect crops in the store.
They include; rats, weevils, termites and mice.
Signs of crops affected by pests
- Rotten plant parts.
- Deformed plant parts.
- Cut off buds
- Holes in leaves, fruits and seeds
- Leaf yellowing
- Leaf curling
- Spots on leaves, fruits and tubers

Effects of pests and diseases on crops.
- They reduce on the crop yields.
- They cause rotting of crops.
- They damage the crops.
- They cause holes on crops.

- They cause retarded growth to crops.



Caterpillars on the plant



Leaves affected by pests

Ways in which pests destroy crops

- ❖ Biting the plants.
  They include; caterpillars and termites
- ❖ Sucking the plant's juice or food.
- ❖ Pecking the seeds on cobs
  They include; weaver birds and monkey.
- ❖ Eating the roots and leaves
  They include armyworms. They eat carrots and cassava tubers
- ❖ Cutting the plants.
  They include termites

Methods of controlling pests in crops

Mechanical methods	• Physical guarding by the farmer. • Fencing the garden. • Setting mouse-traps. • Hanging scarecrows in the garden. • Uprooting the infected crops.
Biological methods	• Taming a cat to kill and eat rats. • Keeping a dog to kill squirrels and moles. • Use of ladybird to feed on aphids. • Use of wasps to control caterpillars.
Cultural methods	- Practising crop rotation. - Early planting and harvesting. - Timely weeding. - Planting resistant crop species. - Planting disease-free cuttings or vines. - Proper spacing of crops.
Chemical methods	• Spraying pesticides. • Dusting the affected crops. • Using rat poison to kill rats.



Table showing biopesticides

Predator	Target pest
Lady bird	Aphids
Wasp	Mealy bugs
Praying mantis	Grasshoppers
Red ants	White scales
Chicken	Cotton stainers
Cats	Moles, rats and mice



Use of scare crow



Spraying

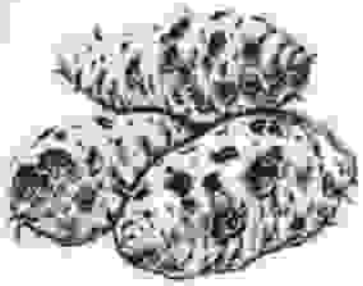


Use of mouse-trap

Disease	Crops affected
Maize streak Maize smut Leaf spot Rust	Maize
Cassava mosaic (virus) Leaf spot Brown streak	Cassava Note: cassava mosaic is spread by a white fly.
Rosette	Ground nuts
Potato blight Tomato blight	Sweet potatoes Tomatoes
Leaf rust Coffee berry disease	Coffee
Smut	Maize and sorghum
Bacterial wilt	Sweet potatoes, carrots and tomatoes
Ratoon stunting disease Smut, Rust	Sugarcanes
Panama disease Cigar end rot	bananas

Signs of crops affected by diseases

- Poor growth
- Yellowing of leaves
- Deformed leaves.
- Wilting of crops.
- Brown or black spots on leaves.
- Rotten crop parts
- Stunted growth of crops



Sweet potato affected by diseases

Ways of controlling diseases in crops

Spraying pesticides to kill pests.

Pesticides kill pests that spread diseases to crops.

Uprooting affected crops and burning them.

It prevents the disease affecting the uprooted crops from spreading to other crops.

Early planting

It helps crops to grow before pests multiply and spread diseases.

Timely weeding

It reduces the hiding places for pests that spread diseases to crops.
It breaks down the life cycle of crop pests. It starves pests to death.

Planting clean materials.

Planting disease-free seeds, stem cuttings or suckers controls the spread of diseases in the garden.

Evaluation activity 2.8

1. Give the meaning of the term crop pest.
2. Mention any **two** examples of insect crop pests.
3. State any **two** cultural methods commonly used by farmers to control crop pests.
4. Identify any **one** effect of pests and diseases on crops.
5. Give any **two** crops that are affected by bacterial blight disease.
6. Name the germ that causes mosaic disease in cassava crops.
7. Which part of cassava is attacked by cassava mosaic disease?

The diagram below shows a method of controlling crop pests. Use it to answer questions **8** and **9**.



8. Identify the method of controlling crop pests shown in the diagram above.

Evaluation activity 2.8 Continued

9. State any **one** advantage of using the method above over poisoning.

The diagram below shows a common tuber crop pest that affects cassava plant. Use it to answer questions **10** and **11**.



10. Identify the crop pest shown above.
11. Name the disease of cassava transmitted by the pest shown above.
12. Give any **two** ways in which crop pests destroy crops in the garden.
13. Mention any **two** signs of crops damaged by pests and diseases.
14. Identify **one** perennial crop affected by the mealy bug.
15. State any **one** way in which each of the following helps to control crop diseases.

(a) Early planting

(b) Early weeding

16. (a) Apart from lack of necessary conditions for germination, give any **two** other factors that can make a seed fail to germinate.

(b) Write down any **two** ways in which farmers can control pests in the garden without using chemicals.

MEANING OF KEY TERMS IN GROWING CROPS

Annual crops: Crops that complete their life cycle within a year.
Mulching: The covering of topsoil with dry plant materials.
Weeding: The removal of unwanted plants from the garden.
Crop pest: An organism that spoils crops.
Pruning: The removal of excess branches or leaves from the plant.
Transplanting: The transfer of seedlings from the nursery bed to the main garden.
Weeds: Unwanted plants that grow in the garden.
Harvesting: The removal of mature crops from the garden.
Mulch: Dry plant materials used to cover the soil surface in the garden.
Thinning: The removal of excess seedlings from the garden.
Ploughing: Turning the soil over to make it easy for planting crops.
Irrigation: The artificial method of watering plants.
Gap filling: The planting of seeds in holes where they did not germinate.
Pricking out: Removing seedlings from the nursery bed to be replanted elsewhere forfurther growth.
Uprooting: Pulling a plant out of the ground.
Row planting: Planting seeds in lines.
Broadcasting: Scattering seeds in the garden during planting.

3 THEME: OUR ENVIRONMENT
TOPIC 3: WEATHER CHANGES AROUND US

Key words:

- atmosphere
- weather
- rainy
- average
- surround
- cirrus

Changes in Weather

Weather is a condition of the atmosphere at a given time and place.
Atmosphere is the layer of gases surrounding the earth.
Sky is the area of the atmosphere above the ground.
Climate is the average weather condition of a place recorded for a long period of time.
During the day, the atmosphere is warmed by the sun. At night, the atmosphere cools.

Elements of weather.

These are factors that lead to changes in weather.
They are measured at a place called weather station.

The elements and types of weather

Elements of weather	Types of weather
sunshine	sunny weather
cloud cover	cloudy weather
rainfall	rainy weather
wind	windy weather
temperature	hot or cold weather
humidity	humid weather
air pressure	


Sunny weather


Rainy weather


Windy weather


Cloudy weather

Sunshine

Sunshine is the light and heat from the sun reaching the earth's surface.

Effects of sunshine in the environment.

- It makes the day bright.
- It causes evaporation.
- It increases temperature.
- It dries materials.

- It supports solar energy.
- It causes melting of ice.
- It influences wind and rainfall.

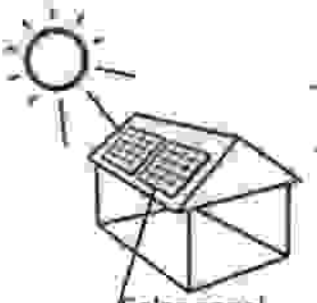


Importance of sunshine

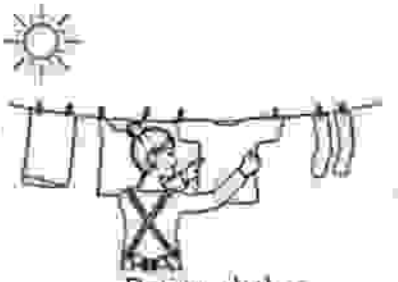
- It provides heat for drying clothes.
- Sunlight enables plants to make food.
- It provides heat for drying harvested crops.
- It enables the skin to make vitamin D.
- It provides light used to generate solar electricity.



Drying harvested crops



Solar panel



Drying clothes

Activities people do during sunny weather.

Harvesting crops, sun-drying crops, drying wet clothes, clearing land, sun-bathing, mulching, skate boating and watering crops/irrigation.

Dangers of too much sunshine.

- It causes crops to wither and dry up.
- It dries up pasture for animals.
- It causes water sources to dry up.
- It causes drought.



Dried up river



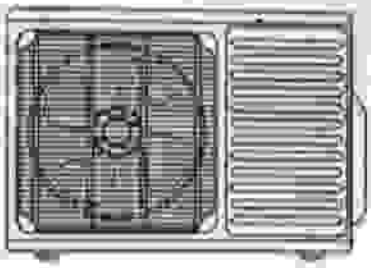
Dried up pasture



A wilting plant

Ways of managing sunny weather.

- Wear hats and sunglasses.
- Wear light clothes.
- Drink plenty of boiled water.
- Stay in shaded areas.
- Use fans and air conditioning to cool indoor spaces.



air conditioning

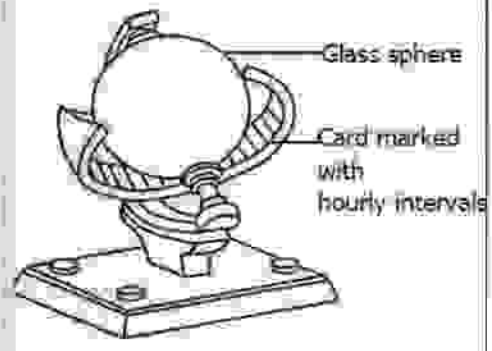


sunglasses

Sunshine recorder.

It is used to show the duration of sunshine received in an area.

It should be placed in an open place in order to get the accurate duration of sunshine. The duration of sunshine is recorded in **hours**.



The glass sphere focuses sunrays.

Cloud cover

Clouds are very small droplets of water or ice floating in the sky.

The height of clouds in the sky is measured using a **ceilometer**.

Clouds are grouped according to their shape, height and colour.

Types of clouds

➢ cirrus clouds ➢ cumulus clouds ➢ nimbus clouds ➢ stratus clouds.

Characteristics of different types of clouds

Types of clouds	Characteristics
Cirrus clouds	- They are the highest clouds in the sky. - They appear thin and spread out. - They show fine/fair and clear weather/nice warm day. - They look like white feathers in the sky.
Cumulus clouds	• They appear flat at the bottom and round at the top. • They look like cotton wool/piles. • They are large and clear on sunny days. • They show fine weather. • They can develop into thunder clouds and indicate rain. *Note: Cumulus clouds are usually seen in the afternoons.*
Nimbus clouds	• They are big and dark grey in colour. • They are the heaviest clouds in the sky. • They do not have any special shape. • They bring rain.

Types of clouds	Characteristics
Stratus clouds	• They appear at the lowest level. (Nearest to the earth.) • They often bring light mists and rain. • They are grey in colour. • They look like a large blanket spread all over the sky. • They are arranged in horizontal layers.


Cirrus clouds


Cumulus clouds


Nimbus clouds


Stratus cloud

Effects of cloud cover in the environment

- It cools temperature.
- It blocks sunlight.
- It reduces evaporation.
- It reduces heat and light intensity.
- It increases humidity.
- It lowers transpiration rate.

Dangers of clouds

- They cause lightning which kills people.
- They disrupt space navigation causing plane crushes.
- They can cause heavy storms which spoil property and crops.

Evaluation activity 3.0

1. The diagrams below show shapes of clouds in the sky. Study them and answer the questions that follow.

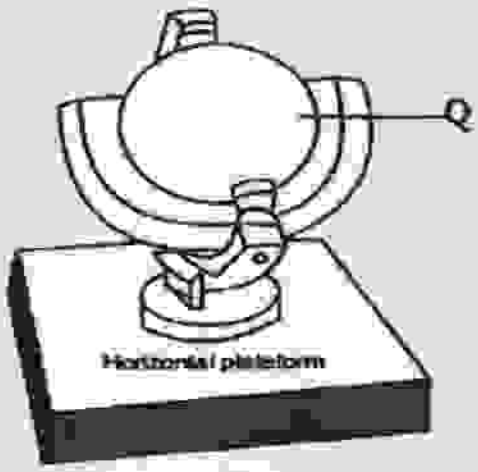

H


K

(a) Name the types of clouds with shapes marked **H** and **K**.
(b) State any **one** characteristic of the type of clouds with the shape marked **H**.
(c) Give any **one** way in which crop farmers benefit from the clouds with the shape marked **K**.
2. Name the type of cloud which is the highest in the sky.
3. Give any **two** effects of cloud cover in an area.
4. State the difference between the sky and the atmosphere.

Evaluation activity 3.0 Continued

5. (a) State the meaning of the term weather.
   (b) Write down any **two** factors that cause weather changes.
   (c) Which type of weather favours the harvesting of beans?
6. The diagram below shows a weather instrument. Use it to answer the questions that follow.



(a) Name the weather instrument shown above.
(b) Identify the element of weather that requires the use of the above instrument.
(c) State the role of part marked **Q** on the weather instrument above.
(d) Give **one** reason why the weather instrument above should be placed in an open place.

Key words:

- received
- gauge
- cylinder
- winnow
- millimetre
- rainwater

Rainfall

Rainfall is the amount of rainwater in the ground.
It is the total amount of rainwater received in a specific area over time.
Rain is the water droplets that fall from clouds to the ground.
Rain is the main natural source of water on earth.

Reasons.

All water used on earth comes directly or indirectly from rain.
Rain supplies water to other sources of water.

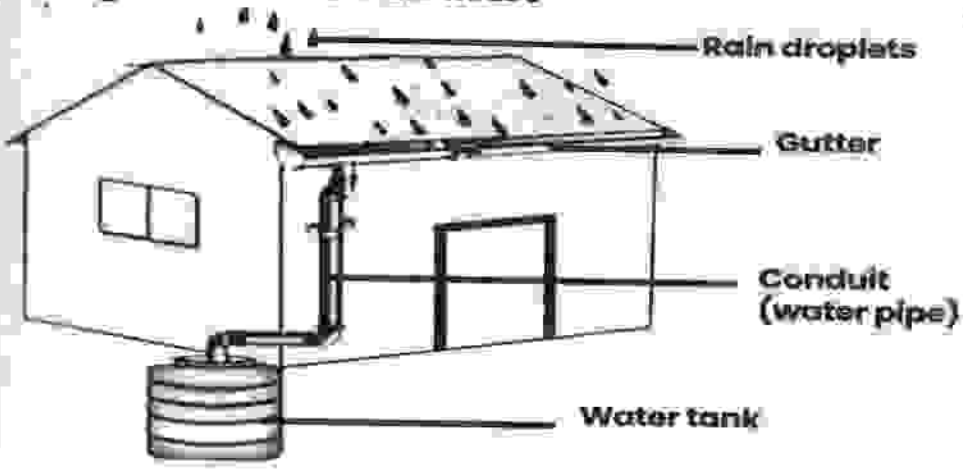
Importance of rainfall in the environment.

- Rain supplies water to water sources.
- Rainwater is used for irrigating crops.
- Rain removes dust from the atmosphere.
- Rainwater is used for domestic uses like washing
- Rainfall enables plants to make food.
- Rainfall supports plant growth.

Ways in which rainfall supports plant growth.

- It helps plants to make food.
- It dissolves mineral salts for roots to absorb.
- It provides water for germination.
- It softens soil for the roots to grow.
- It cools plants.

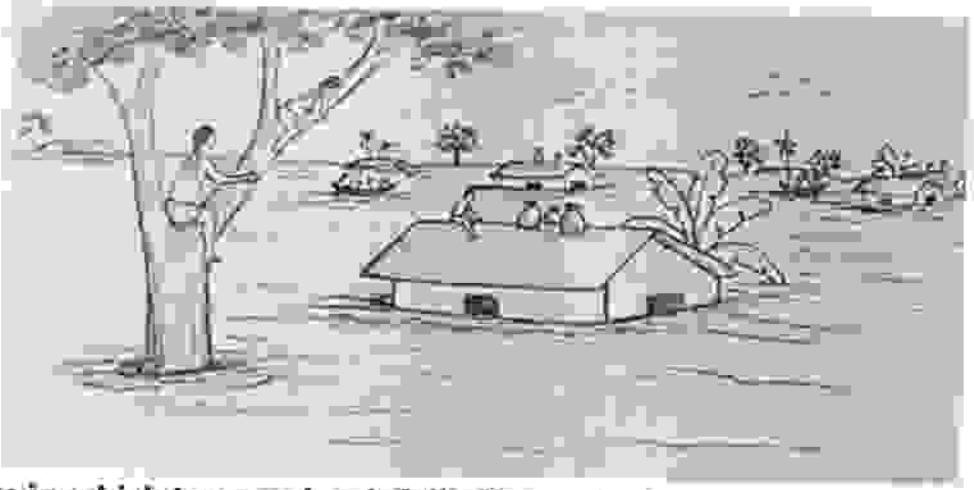
Activities done during rainy weather.
planting, harvesting rainwater and weeding
Harvesting rainwater from the house



The **gutter** traps and directs rainwater to the water pipe.
The **water pipe** directs rainwater to the water tank.
The **water tank** stores water for future use.

Dangers/disadvantages/negative effects of too much rainfall

(i) It causes floods.
(ii) It causes damage to crops e.g., rotting.
(iii) It destroys infrastructure.
(iv) It causes landslides in mountainous areas.
(v) It causes soil erosion.



Ways in which too much rainfall affects plants

(i) It causes floods which destroy plants.
(ii) It causes soil erosion.
(iii) It causes leaching of plant nutrients.
(iv) It can cause stunted growth of plants.

Ways of managing rainy weather

- Use of umbrellas and raincoats.
- Avoid flooded areas.
- Stay indoor during heavy rain.
- Drink warm liquids like tea.
- Build proper drainage systems.
- Wear gumboots.
- Build houses with slanting roofs.
- Put lightning conductors on tall buildings.

Materials used to manage rainy weather

Gumboots. It protects feet from mud and water.
Raincoat. It keeps the body dry from rain.
It keeps the body warm during rainy weather.
Umbrella. It shields the body from raindrops.
Sweater. It keeps the body warm during rainy weather.


gumboots


raincoat

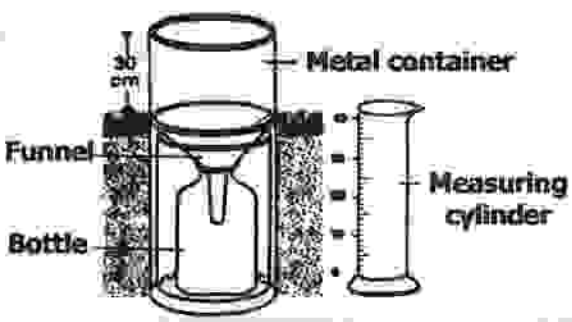

umbrella






sweater

Measuring rainfall.
A rain gauge is used to measure the amount of rainfall received in an area.
Rainfall is measured in units called **millimetres(mm)**.
Reason. To find how deep rainwater has entered the soil.
Components of a rain gauge.
Funnel. It directs rainwater into the measuring cylinder.
Measuring cylinder. It collects and measures the volume of rainwater for accurate readings.
Bottle. It collects the rainwater for measurement of its volume.
Metal container. It holds the measuring cylinder.



Points to note about the rain gauge
It should be put in an open ground when using it.
Reasons.
To get the actual measurements of rainfall.
To avoid obstruction of raindrops.

It should be raised 30cm above the ground when using it

Reasons.

To prevent rain splash from entering the rain gauge.

To get accurate measurements of rainfall.

Steps taken when measuring rainfall using a rain gauge.

(i) Place the rain gauge in an open area away from obstacles.

(ii) Ensure the rain gauge is levelled and securely fixed to the ground.

(iii) After rainfall, check the water level in the gauge using the scale marked.

(iv) Record the measurement in millimetres to track the rainfall amount.

How to make a simple rain gauge.

Materials needed: Tin, glass, funnel and ruler

Steps taken.

(i) Place the funnel into the glass in the tin.

(ii) Put them on the ground in an open place away from buildings and trees.

(iii) Using a ruler to measure, place the funnel 30cm above the ground. This prevents splashing of rainwater from the ground from entering the rain gauge.

(iv) The amount of water in the glass is measured using a measuring cylinder.

Set up:



tin



glass



funnel



ruler



Measuring cylinder

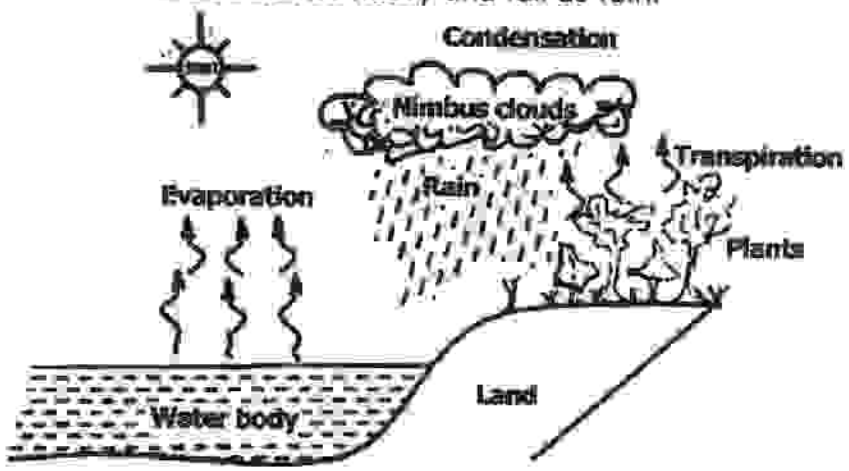
The Water Cycle.

Water cycle is the constant circulation of water between the atmosphere, the land and the oceans.

Water cycle is the natural process by which rain is formed in the environment. It can also be called **hydrological cycle**.

Describing how rain is formed.

(i) The sun rays heat water bodies and plants.
(ii) Evaporation and transpiration occur forming water vapour.
(iii) The water vapour rises and condenses to form nimbus clouds.
(iv) The nimbus clouds become heavy and fall as rain.



Roles of some components in rain formation.

The sun. It causes evaporation and transpiration.
Plants. They add water vapour to the atmosphere through transpiration.
Water body. It adds water vapour to the atmosphere through evaporation.
Nimbus clouds. They condense and form rain.

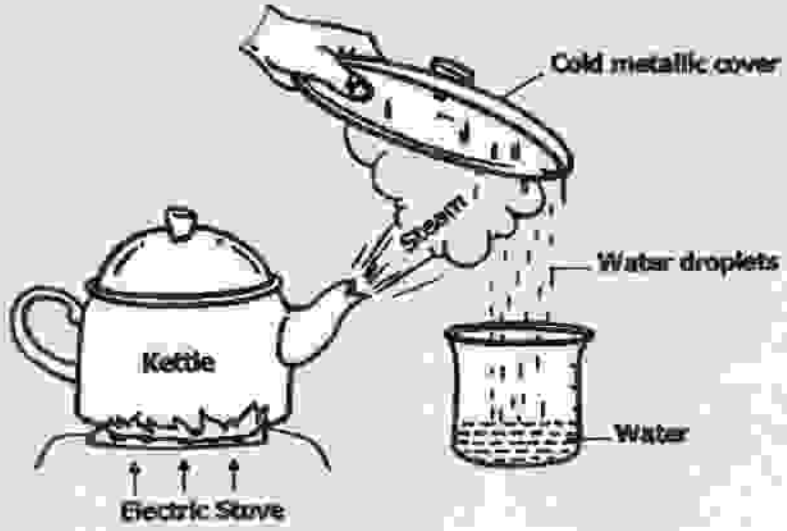
An experiment to show water cycle.

Materials needed: electric stove/Charcoal stove, kettle, metallic cover and a mug

Steps taken.

(i) Boil water in a kettle.
(ii) Cover the top part until water evaporates.
(iii) Hold a metallic cover at the opposite end where steam passes.
(iv) Steam passes out to reach the cold metallic cover and condenses back to liquid in the mug.

Set up:



Points to note about the experiment.

- *The electric stove represents the sun. it causes evaporation of water in the kettle.*
- *The cold metallic cover represents the nimbus clouds. It condenses steam to water.*
- *Steam represents water vapour. Steam condenses to form water.*
- *The droplets of water falling in the mug represent rain.*
- *The water collected in the mug represents rainfall/rainwater.*

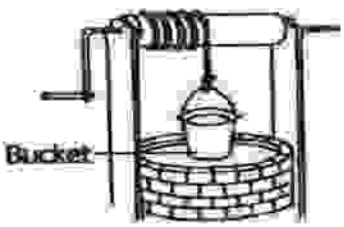
Other sources of water in the environment

Natural sources are; rivers, lakes, springs, seas, streams, swamps and oceans,

Artificial sources of water are; water pumps, ponds, dams and wells



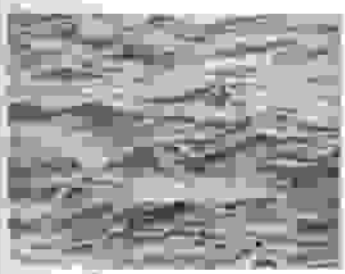
A water pump



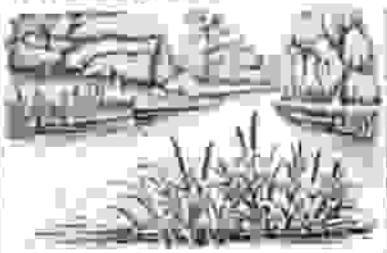
An underground well



A river



An ocean



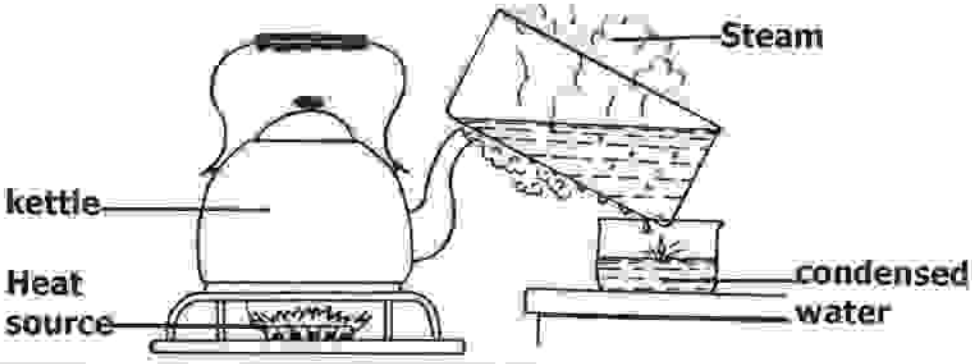
A swamp



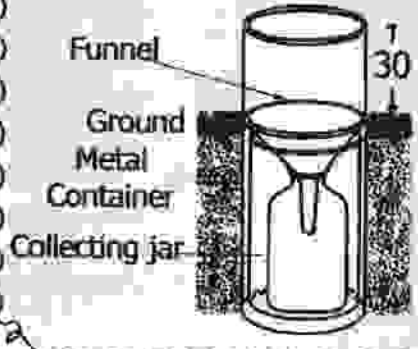
A lake

Evaluation activity 3.1

1. In which way is the sun useful during rain formation?
2. Name the weather instrument used to measure the amount of rainfall received in an area.
3. Give any **one** reason why rain is considered as the main nautral source of water on earth.
4. (a) Apart from rainfall, give **one** other element of weather necessary for plant growth.
   (b) In which **two** ways is rainfall important for plant growth?
   (c) State any **one** disadvantage of too much rainfall to plants.
5. State the reason why rainfall is measured in millimetres.
6. The diagram below shows an experiment carried out to demonstrate a certain process in the environment. Study and use it to answer the questions that follow.



   (a) Identify the process in the environment demonstrated by the experiment above.
   (b) What does each of the following demonstrate in the experiment above?
   (i) Kettle
   (ii) Steam
   (c) State **one** material that can be source of heat in the above experiment.
7. In **four** sentences, describe how rain is formed in the environment.
8. State the difference between rain and rainfall.
9. The diagram below shows a weather instrument. Study and use it to answer the questions that follow.



   (a) Name the weather instrument shown in the diagram above.
   (b) What is the use of the weather instrument above at a weather instrument?
   (c) Give a reason why people are advised to;
   (i) Put the instrument in an open ground when using it.
   (ii) Raise the instrument 30cm above the ground when using it

Evaluation activity 3.1 Continued

10. The diagrams below show materials used to manage weather conditions. Study and use them to answer the questions that follow.



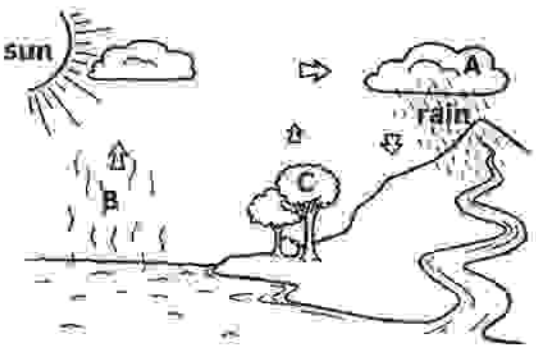
Q



R

(a) Identify the weather condition in which the material marked **Q** is commonly used.
(b) Give any **one** way in which the material marked R is useful to people during weather conditions.
(c) State any **two** ways of caring for the materials marked Q and R.

11. The diagram below shows water cycle. Study and use it to answer the questions that follow.



(a) Name the type of clouds represented by letter **A**.
(b) State the process taking place at **B** and **C**.
(c) What is the importance of the sun in the above diagram?

12. (a) Give **one** way in which people use the following materials to manage rainy weather.
(i) Gumboots. (ii) Umbrella
(b) Apart from gumboots and umbrella, name any **two** other materials used to manage rainy weather.

Key Words:

• motion • pressure • winnowing • louvres • measure • capsize

Wind

Wind is air in motion.

Ways in which wind is useful in the environment.

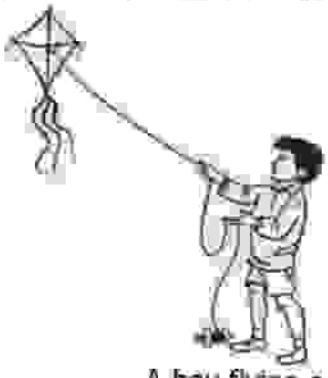
- ✓ Wind helps in winnowing of seeds and grains.
- ✓ Wind helps to sail boats on water bodies.
- ✓ Wind helps in the formation of rain.
- ✓ Wind helps to dry harvested cops.
- ✓ Wind helps to dry wet clothes.
- ✓ Wind helps in pollination.
- ✓ Wind helps in seed dispersal.
- ✓ Wind helps in pumping water using a windmill.

Activities done during windy weather

Winnowing, flying kites, pumping water using windmill and windsurfing



Winnowing



A boy flying a kite



A boat sailing on water

How to make a kite to show that wind flies a kite in the sky

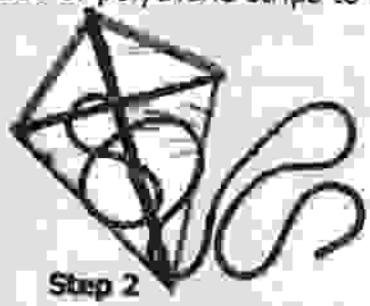
Materials needed: polythene paper, sticks, string

Steps taken.

(i) Cut the polythene paper into a big diamond shape.
(ii) Tie two sticks together to make a cross and attach them to the paper.
(iii) Fix a long string to the centre of the cross flying the kite.
(iv) Add a long tail made of polythene strips to help the kite balance.



Step 1



Step 2



Step 3

Roles of the materials in the experiment

Polythene paper. Makes the body of the kite.
It is light to enable the kite to fly easily.
Sticks. Provide support and shape to the kite.
They should be light to enable the kite fly easily when released.
String. Helps the user to control and fly the kite.

Points to note about the kite:

When a kite has many holes on it, it cannot fly. This is because air passes through, reducing lift. A kite works on the property of air known as; Air exerts pressure.

Dangers of strong wind in the environment

- It causes accidents.

Wind capsizes boats.

Wind breaks trees which fall on people.

Wind breaks houses which kills people.

- It causes food contamination.
- It leads to soil erosion.
- It spreads airborne diseases.
- It causes shame when it lifts off people's clothes.
- It causes air pollution.
- It raises dust in the atmosphere.
- It destroys crops and property.
- It breaks trees.



Dust raised by wind



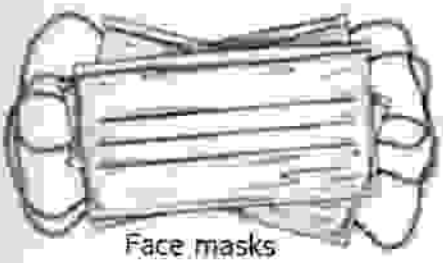
A boat capsizing in water



Trees broken by Wind

Ways of managing windy weather

- Plant trees to act as windbreaks.
- Build houses with strong roofs and walls.
- Wear helmets and face masks.
- Stay indoors during storms.
- Wear sunglasses.
- Avoid staying near tall trees.
- Build houses with flat roofs.



Face masks



Helmet

Types of wind

Light wind, strong wind, gale wind and stormy wind

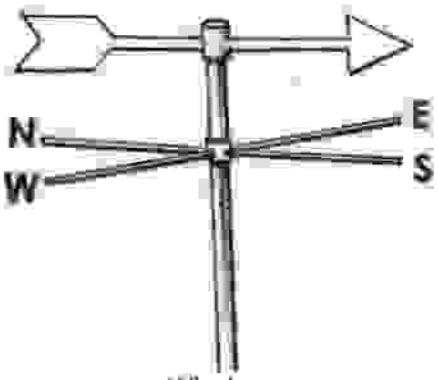
Aspects used in the measurement of wind.

Speed of wind, direction of wind and strength of wind

Weather Instruments used in the Measurement of Wind

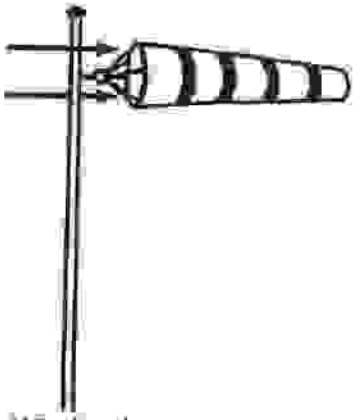
Weather Instruments	Use of the instrument
Wind vane	It is used to show the direction of wind. The arrow shows where wind is coming from. Wind direction is where the wind is blowing from.
Anemometer	It is used to measure the speed of wind.
Windsock	It is used to show the strength of wind.
Weathercock	It is used to show the direction of wind.

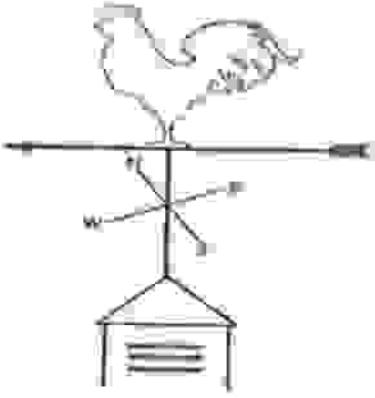
Note. The above instruments should be placed in an open place in order to prevent obstruction of wind from buildings and trees.



Wind vane


Anemometer


Windsock


Weathercock

How to make a simple/local windsock

Materials needed: stick, string and polythene bag

Steps taken.

(i) Cut one side of the polythene bag to make it open at both ends.

(ii) Tie the open end of the polythene bag to one end of the stick using a string.

(iii) Hold the stick and place it where the wind can blow through the polythene bag.

(iv) The polythene bag will be inflated and become straight when wind passes through.

How to make a simple/local windsock

Set up diagram:



Roles of the materials in the experiment

Polythene bag. Catches the wind and shows its direction.

Stick. Holds the polythene bag upright and steady.

String. Ties the bag to the stick securely.

How to make a simple/local anemometer

Materials needed: 4 disposable plastic cups, sticks, bottle tops and nail

Steps taken.

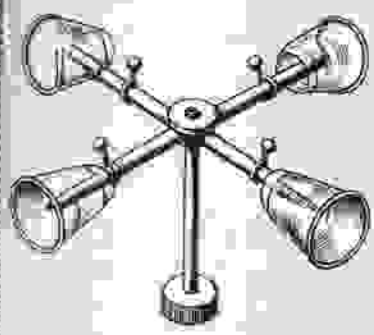
(i)Take four disposable plastic cups and fix them on the ends of four sticks.

(ii) Join the sticks together in the middle with a nail so they can spin.

(iii) Fix a bottle top under the nail to hold the sticks in place.

(iv) Blow wind on the cups and watch them spin to show wind speed.

Set up diagram:



Roles of the materials in the experiment

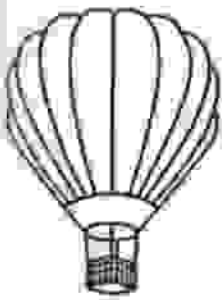
Plastic cups. They catch the wind.

Sticks. They hold the plastic cups in place.

Nail. It allows the sticks to spin.

Bottle top. It keeps the sticks steady.

Devices that use wind to work



Air balloon



Windmill

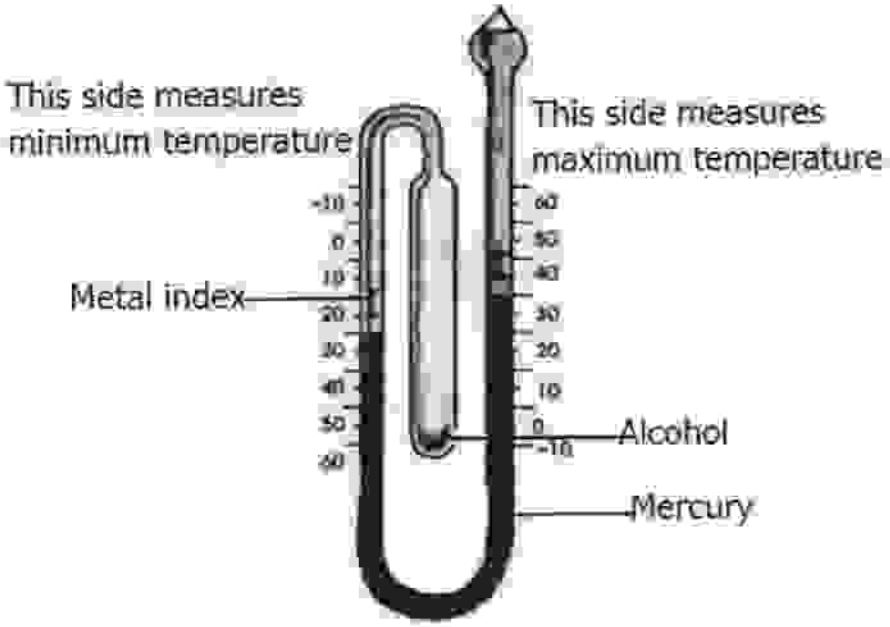


Boat

Other Weather Instruments

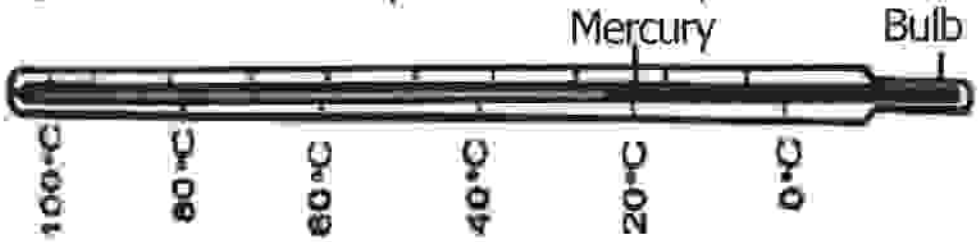
Six's thermometer (maximum and minimum thermometer)

It is to measure the highest and lowest temperature of the day.
It has the right arm and the left arm.
The right arm measures the highest temperature of the day.
The left arm measures the lowest temperature of the day.
The Six's thermometer uses both mercury and alcohol.
It was invented by James Six.
Temperature is the degree of hotness or coldness of an object or place.
It is measured in units called degrees Celsius or degrees Fahrenheit.



The weather thermometer

It is used to measure the temperature of the atmosphere.



Hygrometer.

It is used to measure humidity.
Humidity is the amount of water vapour in the atmosphere.
It is also used in egg incubators and saunas.



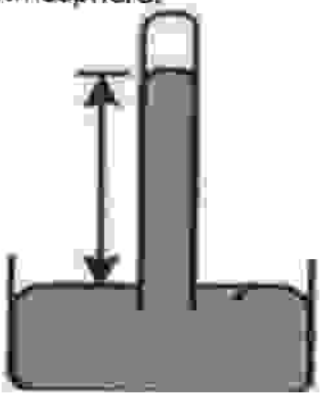


Barometer.

It is used to measure air pressure/atmospheric pressure.

Air pressure is the pressure exerted by air in the atmosphere.





Storage of the delicate weather instruments.

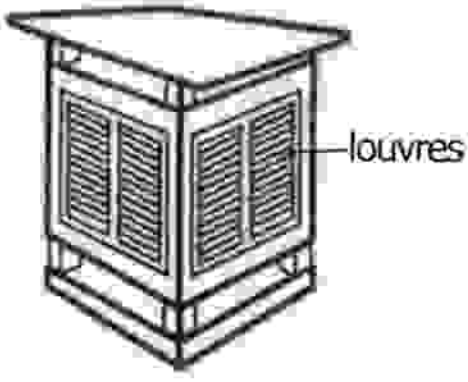
A Stevenson screen is used to keep delicate weather instruments.

These delicate weather instruments are; hygrometer, barometer and Six's thermometer.

The measuring cylinder of the rain gauge and a record book and pen are also kept in the Stevenson screen.

The Stevenson screen is painted white in order to reflect heat from the sun so as to remain cool inside.

It has louvres to allow free circulation of air.



Summary of the elements of weather, instruments and their uses

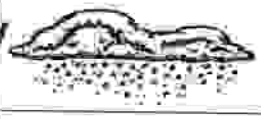
Element	Instrument used	Its use
Rainfall	Rain gauge	Measures the amount of rainfall received in an area.
Wind	Wind vane Windsock Anemometer	Shows the direction of wind. Shows the strength of wind. Measures the speed of wind.
Temperature	Thermometer	Measures the hotness or coldness of the atmosphere.
Humidity	Hygrometer	Measures the amount of water vapour in the atmosphere.
Sunshine	Sunshine recorder	Shows the duration of sunshine received in an area.
Air pressure	Barometer	It measures the air pressure.
Clouds	Ceilometer	It measures the height of clouds.

Adaptations of weather instruments

Weather Instrument	Adaptation/Feature
Sunshine recorder	It has a glass ball to focus sunlight and burn a card.
Rain gauge	It has a funnel to direct rainwater into a measuring cylinder.
Hygrometer	It has a special material that changes with air moisture.
Six's thermometer	It has two liquids to record hot and cold temperatures.
Barometer	It has a sealed metal box that expands or contracts with air pressure.
Windsock	It is light and open, so wind can easily lift and move it.
Wind vane	It has an arrow that turns to show wind direction.
Anemometer	It has cups that spin faster when the wind is strong.

A weather chart.

A weather chart records weather changes at different times in the day.
It helps one to know the weather changes.

Day	Morning	Afternoon	Evening
Monday			
Tuesday			
Wednesday			
Thursday			
Friday			

Key


Rainy weather


Sunny weather


Windy weather

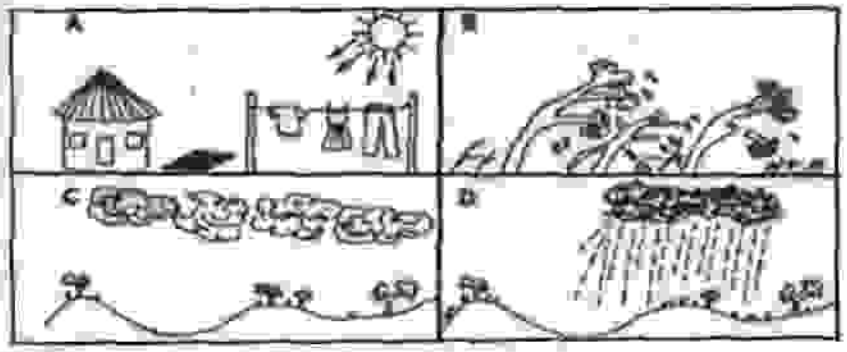

Cloudy weather

Guiding Questions on the Weather Chart.

1. Name any **two** activities that can be done on Monday;
   (i) Morning
   (ii) Evening
2. Give **one** reason why a crop farmer would carry out winnowing on Thursday afternoon.
3. State any **two** problems caused by the type of weather experienced on Wednesday morning and afternoon.
4. Give any **one** way in which people can manage the type of weather experienced on Friday afternoon.
5. Identify the day that is suitable for harvesting maize from the garden.
6. Mention any **two** types of weather experienced on Thursday.
7. State any **one** danger of the type of weather experienced on Tuesday in the environment.
8. Identify any **one** day that experienced three different types of weather.
9. State the reason why a crop farmer would plant his bean seeds on Monday evening.
10. Name any **one** activity that can be done on Tuesday.
11. Give any **one** reason why you would wear a sweater on Friday evening.
12. Give any **one** material that people can use to manage the type of weather experienced on Tuesday.

Evaluation activity 3.2

1. The diagram below shows types of weather. Study and use it to answer the questions that follow.



(a) Name the type of weather marked **C.**
(b) State **one** way in which the type of weather in A helps in formation of the type of weather in **D.**
(c) Give **one** way in which people protect themselves from the effects of the type of weather in **B.**
(d) State any **one** way in which too much of the type of weather in **A** is dangerous to the environment.

2.(a) Name the aspect of wind measured by;
   (i) Windsock (ii) Anemometer

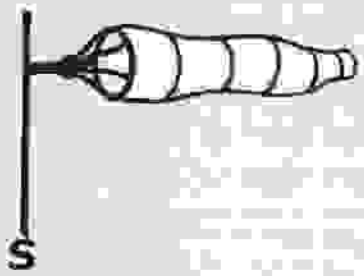
Evaluation activity 3.2 Continued

(b) State any **two** dangers of strong wind to people.

3. The table below shows weather elements, instruments and their uses. Study and complete it correctly.

Element	Instrument	Use
Rainfall	.....	Measures the amount of rainfall.
.....	Hygrometer	Measures the amount of water vapour in the atmosphere.
Wind	.....	Measures the speed of wind.
Temperature	Thermometer	.....

4. The diagrams below show weather instruments. Study and use them to answer the questions that follow.





(a) Name the weather instruments marked **S** and **T**.

(b) State the use of the weather instrument marked **S** to a pilot.

(c) Give **one** reason why the weather instrument marked **T** is kept in a Stevenson screen.

MEANING OF KEY TERMS IN WEATHER CHANGES AROUND US

Sky: The area of atmosphere above the ground.

Fog: Thick cloudy of tiny water droplets near the ground. It reduces visibility.

Mist: Thin cloud of water droplets near the ground. It has better visibility than fog.

Rain: The amount of water droplets in the atmosphere that falls from the sky.

Wind: Air in motion.

Clouds: Condensed masses of water droplets floating in air

Thermometer: The instrument used to measure temperature.

Atmosphere: Air around the earth.

Weather: The condition of the atmosphere at a given time and place.

Environment: The surrounding of an organism.

Temperature: The degree of hotness or coldness of an object or place.

Air pressure: The pressure exerted by air in the atmosphere.

Humidity: The amount of water vapour in the atmosphere.

Barometer: The weather instrument used to measure air pressure.
Rainfall: The amount of rainwater that falls in a certain area at a certain time.
Climate: The average weather condition of a place recorded for a long period of time.

4 THEME: HUMAN HEALTH
TOPIC 4: PERSONAL HYGIENE

Key Words:
• hygiene • bathe • scrub • pumice • scabies • regularly

Personal hygiene is the way of keeping the body, clothing and bedding clean. Personal hygiene is the element of Primary Health Care that promotes the cleanliness of the human body.

Parts of the body that should be kept clean
Hands, skin, ears, teeth, eyes, fingernails and hair.

Things that should be kept clean
Bedding, clothing, toothbrush, sandals and shoes

Reasons why school children should keep their bodies clean.
- To remove dirt from the body.
- To remove germs from the body.
- To kill parasites and vectors like lice.
- To prevent diseases like scabies.
- To remove bad smell from the body.
- To keep healthy.

Activities that school children can do to keep their bodies clean.

❖ **Regular bathing**
It helps to remove dirt and germs from the body.
It prevents skin diseases like ringworm infection and scabies.
It prevents bad body smell.





❖ **Regular washing of hands with clean water and soap.**
This can be done from the tippy-tap, wash basin (sink) or a water tap.
It removes dirt and kills germs. It prevents the spread of diarrhoeal diseases.

Conditions that can lead to washing of hands
(i) After visiting a latrine or toilet.
(ii) Before and after eating food.
(iii) After picking rubbish
(iv) Before serving food.





❖ **Regular brushing of teeth using toothpaste.**

It prevents bad breath from the mouth.

It prevents diseases of teeth.

Importance of using toothpaste

- Tooth paste prevents bad breath
- Toothpaste prevents tooth decay.
- Toothpaste makes teeth strong.
- Toothpaste kills germs in teeth.







Steps taken when brushing teeth.

(i) Wet the toothbrush and apply a small amount of toothpaste.

(ii) Brush your teeth upwards and downwards.

(iii) Brush your tongue to remove food remains and freshen your breath.

(iv) Rinse your mouth and toothbrush thoroughly with water.

1	2	3	4
			
Things used for brushing	Brushing upwards	Brushing downwards	Brushing the tongue

The tongue should be brushed to remove food remains from the tongue.

To prevent bad breath. To kill germs on the tongue.

The keeping of one's mouth clean is called **oral health/oral hygiene.**

❖ **Regular washing of clothes.**

It removes parasites like lice from clothes.

It removes bad smell from clothes.

It kills germs that cause skin diseases like ringworm.

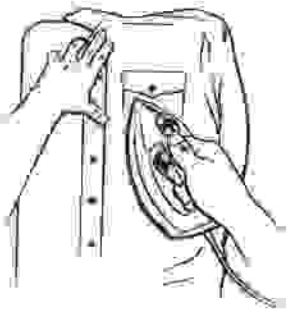
It keeps clothes clean.
Soap is used to dissolve dirt or stains from clothes.





❖ **Ironing clothing and bedding.**
It kills parasites like lice.
It kills germs from clothes and bedding.





❖ **Grooming fingernails and toenails.**
It helps to remove hiding places for germs.
It helps to prevent cuts on the skin.





Note: Long fingernails act as breeding and hiding places for germs.

❖ **Combing hair regularly.**

It helps to remove hiding places for parasites like head lice.

It prevents bad smell from hair.





❖ **Shaving/trimming off overgrown hair.**

It helps to remove hiding places for parasites like head lice.





❖ **Regular washing of the face.**

It helps to remove rheum from eyes.

It prevents eye infections like conjunctivitis.

It promotes the health of the skin.

It refreshes the eyes and face.







Evaluation activity 4.1

1. Name the element of Primary Health Care that promotes human body cleanliness.
2. (a) Give the meaning of the term personal hygiene.
   (b) Write down any **two** activities that school children can do to promote personal hygiene.
   (c) State any **one** way in which it is important to promote personal hygiene among school children.
3. (a) Apart from washing hands, mention any **one** other way of keeping our hands clean.
   (b) Give **one** reason why we should regularly wash our hands with clean water and soap.
   (c) Identify any **two** occasions that require you to wash your hands.
4. The diagrams below show activities done to promote personal hygiene. Study and use them to answer the questions that follow.



R



S



T

   (a) Identify the personal hygiene activities done by the persons at **R** and **S**.
   (b) State any **two** reasons why the person at **T** is carrying out that activity.
   (c) Name any **one** disease controlled by carrying out the activity marked at **S**.
5. State **one** reason why the use of soap makes washing clothes easy.
6. Give any **two** ways in which toothpaste promotes oral health.
7. State any **one** way of controlling the breeding of germs in hands.
8. State **one** danger of having long fingernails on the hands.
9. Mention **one** reason why you are encouraged to brush the tongue as well when cleaning the teeth.

Things Used for Keeping our Bodies Clean.

Loofah/bathing sponge. For scrubbing dirt from the body.

Toothbrush. For brushing teeth.

Razorblade. For grooming fingernails and toenails. For shaving hair.

Soap.

It is used for washing hands and clothes.

It is used for bathing.

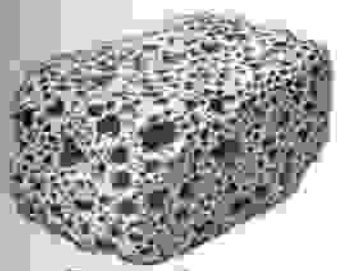
Comb. It is used for combing hair.

Pumice stone. It is used for scrubbing feet.

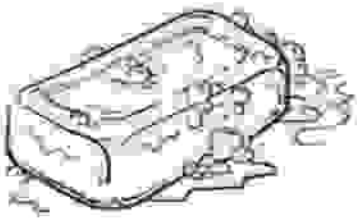

Loofah


Toothbrush


Razorblade

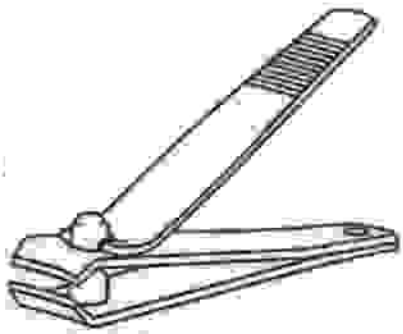

Pumice stone


Comb


Soap

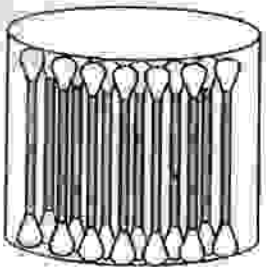
Water. It is used for washing hands, clothes and bedding.
Towel. It is used for drying the body after bathing.
Nail clipper/cutter. It is used to cut fingernails and toenails short.


Water


Nail clipper


Towel

Cotton swabs. For removing excess wax from the human ear.
Brush. For scrubbing dirt on sandals, shoes and heavy clothes.
Handkerchief. For removing mucus from the nose. For drying hands after washing them.


Cotton swabs


Brush


Handkerchief

Learn, unlearn and relearn: earbuds are small earphones inserted in the ears to listen to music without disrupting other people. Ear pads are spongy materials that cover the earphones. Therefore, they are not useil to clean ears.

Toilet paper. For cleaning oneself after defecation.
Slippers. For wearing after bathing to keep the feet clean.
Shoe brush. For brushing shoe polish on shoes.
Shoes polish. For polishing shoes to keep them shiny.


Toilet paper


Slippers

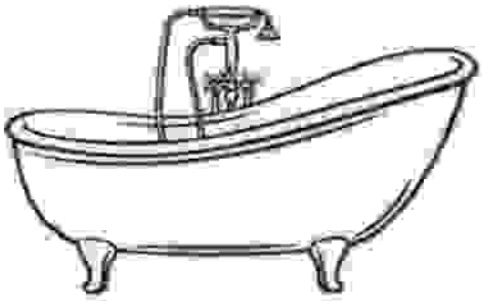

Shoe brush

Good practices of using the materials used to keep the body clean.

- Washing them.
- Scrubbing them.
- Ironing the towels and handkerchiefs.
- Replacing them when they are old.
- Keeping them in clean places.
- Use them for their right purposes.
- Disinfecting them to kill germs.
- Proper drying after use.

Places and materials used in promoting personal hygiene

Bathroom, bathtub, wash basin, tippy-tap and washing machine.


Bathtub


Wash basin/sink


Washing machine

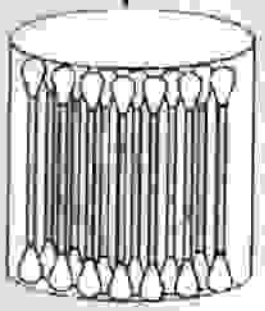

Bathroom


Tippy-tap

Evaluation activity 4.2

1. (a) Give the importance of each of the following materials in the promotion of personal hygiene.
   (i) Towel (ii) Pumice stone
   (b) State any **two** good practices of using the towel when promoting personal hygiene.
2. Mention any **two** ways in which water is useful in the promotion of personal hygiene.
3. State any **one** danger of sharing a nail cutter when promoting personal hygiene among school children.
4. State any **one** use of a razorblade in promoting personal hygiene.
5. Give **one** reason why school cleaners usually put toilet papers in latrine or toilets.
6. The diagrams below show items used to keep the body clean. Study and use them to answer the questions that follow.

Y



Z



(a) Name the items marked **Y** and **Z.**
(b) Identify the part of the body cleaned using the item marked **Y.**
(c) Give any **one** proper practice of using the item marked **Z.**

7. The table below shows parts of the body, items used to clean them and the use of each item. Study and complete it correctly.

Body part	**Item**	**Use of the item.**
Skin	.....	For scrubbing dirt
....	Toothpaste	For removing stains.
Ears	Cotton swabs	.....
Fingers	.....	For cutting fingernails short.

Diagrams showing clean children in school uniforms





Dangers of failing to maintain personal hygiene.

(i) It can lead to the spread of skin diseases.
(ii) It can lead to breeding of germs and parasites on the body.
(iii) It leads to bad smell from the body.

Diseases that can break out due to poor personal hygiene.

Scabies, athlete's foot, ringworm infection, tooth decay, candidiasis, trachoma and conjunctivitis

Signs of poor personal hygiene in school children

- Dirty school uniform
- Long and uncombed hair.
- Long fingernails.
- Dirt accumulated on the skin.
- Too much creases on the school uniform.
- Presence of lice in uniform or on the body.
- Bad smell from the body.
- Bad breath from the mouth.
- Presence of rheum in the eyes.
- Dirty shoes and socks.



Dirty uniform



Uncombed hair

Keeping bedding and clothing clean.

Bedding includes;

bedsheets,
pillowcases,
duvet covers,
blankets and mattresses


Bedsheets


Pillowcases


Blankets

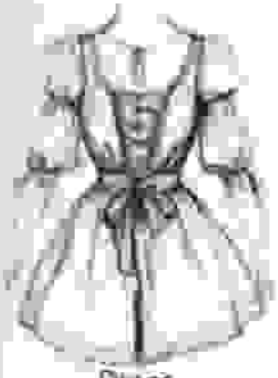
Clothing includes; shirts, caps, blouses, underwears, socks, dresses, shorts, petticoats, scarves, sweaters and brassieres.


T-shirt


Socks


Shirt


School uniform


Dress


Scarfe


Sweater


School shoes


Shoes

Ways of caring for bedding and clothing

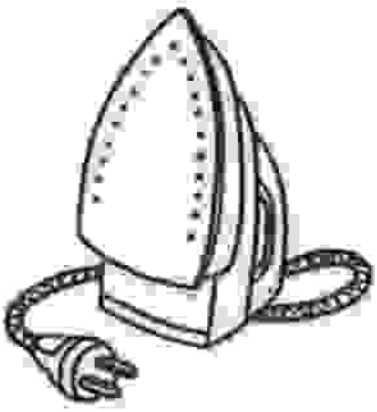
- Washing them regularly.
- Ironing them.
- Spraying them with disinfectants.
- Laying the bed every morning.
- Spreading them in sunshine to kill parasites.

Things used to manage bedding and clothing.

soap/detergent, washing brushes, water, charcoal iron/flat iron, suit cases



Suit case



Electric Iron



Charcoal iron

vectors that breed in bedding and clothing.

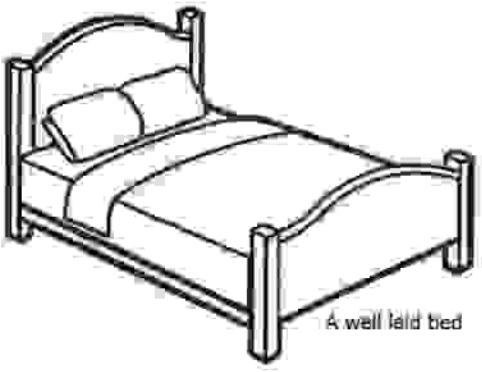
Lice, fleas, mites, bedbugs and cockroaches.

Ways of keeping the bedroom clean.

- ❖ Sweeping the floor
- ❖ Organising bedroom property.
- ❖ Mopping the bedroom.
- ❖ Scrubbing the floor of the bedroom.
- ❖ Removing cobwebs from the bedroom.



A clean bedroom

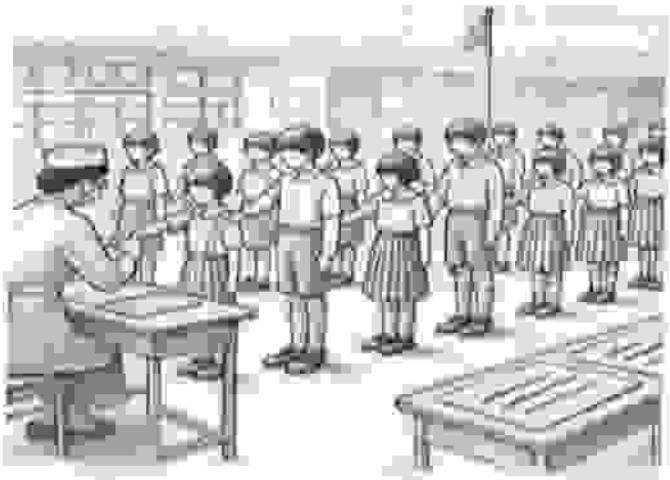


A well laid bed

Roles of the School Health Club in promoting personal hygiene.

- Organising school health parades.
- Displaying posters with personal hygiene messages.
- Providing handwashing facilities.
- Organising talk shows on personal hygiene.

- Teaching pupils proper handwashing techniques.
- Distributing hygiene materials like soap or sanitisers.



Health parade

Messages that can be displayed at school to promote personal hygiene of pupils.

- Wash your hands with clean water and soap after using the toilet.
- Keep your fingernails short and clean.
- Brush your teeth twice a day.
- Cover your mouth when coughing or sneezing.
- Bathe daily to stay healthy.
- Always wear clean uniforms.

Health programmes that promote personal hygiene at school.

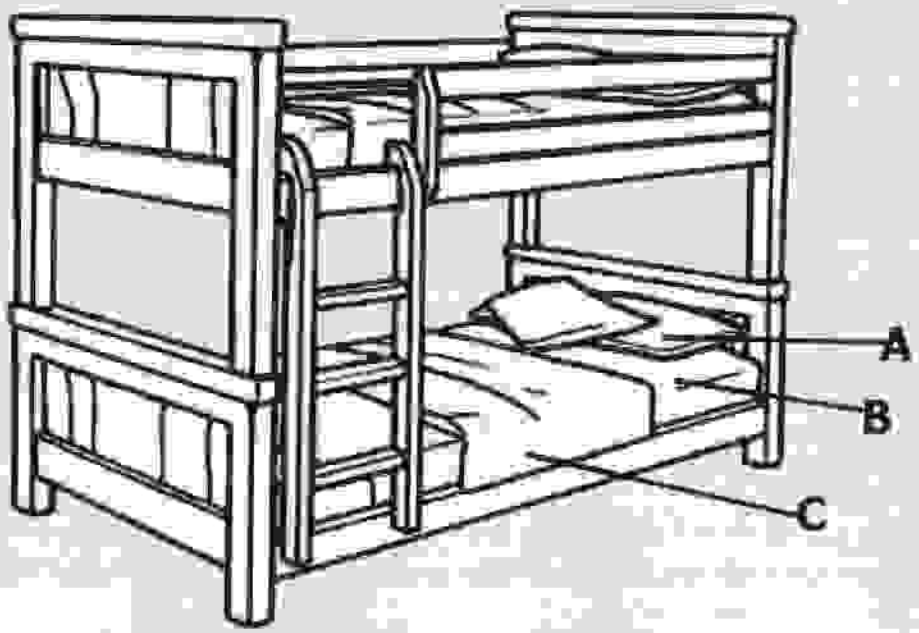
Health parades, child to child programme and health education

Evaluation activity 4.3

1. (a) Mention any **two** activities a School Health Club can do to improve the personal hygiene of children at school.
   (b) Write down any **two** messages that can be displayed in the school compound to promote personal hygiene among school children.
2. (a) List down any **one** example of bedding used by school children.
   (b) State any **two** ways in which school children keep their bedding clean.
   (c) Name any **one** vector that can breed in bedding when not kept clean.
3. The diagram below shows a school decker. Study and use it to answer the questions that follow.

Evaluation activity 4.3

3. The diagram below shows a school decker. Study and use it to answer the questions that follow.



(a) Name the materials marked **A** and **B** on the decker.
(b) Give any **one** way of keeping the material marked **C** clean.
(c) Identify any **one** danger of sharing the material marked **B** with a friend.

5 THEME: HUMAN HEALTH
TOPIC 5: OUR FOOD

Key words
• nourish • proteins • nutritional • substance • carbohydrates
• values

FOOD
Food is anything taken into the body to nourish it.
Food also means a substance that adds nutritional value in the body when taken in.
Nutrition is the process by which food is taken in and used by the body.

Learn, unlearn and relearn: *It is wrong to say that food is anything/something/substance good to eat or drink and has values in the body because there is no element of nutritional value/ nourish in the definition. Additionally, even essential drugs add values in the body yet they are not food and some are drunk e.g., syrup. Thirdly, not every kind of food people eat is good! Some food may be half cooked, have dirt, some people like Chinese eat snake, Congolese eat caterpillars which many of the Ugandans cannot eat.*

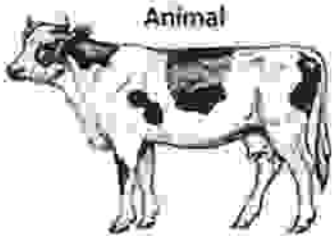
Ways in which food is taken into the body.
• Through eating. • Through drinking. • Through feeding tubes.

Reasons why people eat food.
✓ To have a healthy body.
✓ To enable the body get energy.
✓ To make the body grow strong.
✓ To promote proper body growth

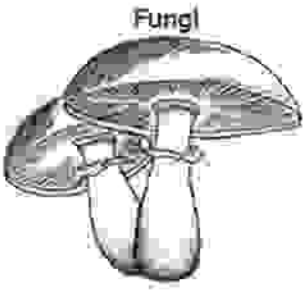
Learn, unlearn and relearn: *It is wrong to say that we eat food to be happy, it is a habit and for hospitality because they are not connected to the THEME "HUMAN HEALTH" instead they are just cultural perspectives.*
Ignore the 5Hs in teaching this topic since it is not reflected anywhere in the curriculum.

Ways in which food is important in the body.
❖ Food provides energy to the body.
❖ Food keeps the body healthy.
❖ Food provides heat in the body.
❖ Food builds the body.
❖ Food promotes body growth.
❖ Food repairs damaged tissues.

Sources of food in the environment
Plants, animals and fungi







CLASSES OF FOOD.

Food is classified according to the chemical composition and nutrient value in the human body. Classes of food include; carbohydrates, proteins, vitamins, mineral salts and fats and oil

mnemonic used: **CPV FM**

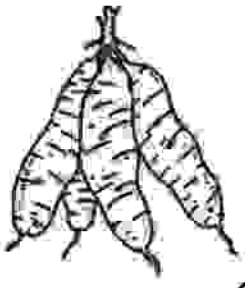
Carbohydrates

These are foodstuffs that are broken down into the body to produce energy.

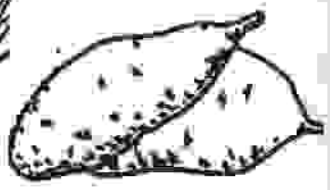
Sources of carbohydrates

- cassava
- sugar cane
- bread
- honey
- potatoes
- posho
- maize
- rice
- yams
- millet
- sorghum
- chapatti
- bans
- cakes


Bread


cassava


maize


Sweet potatoes

Proteins

These are foodstuffs that help to build the body.

Proteins are grouped as animal proteins and plant proteins.

Sources of animal proteins

- eggs
- meat
- fish
- grasshoppers
- crickets

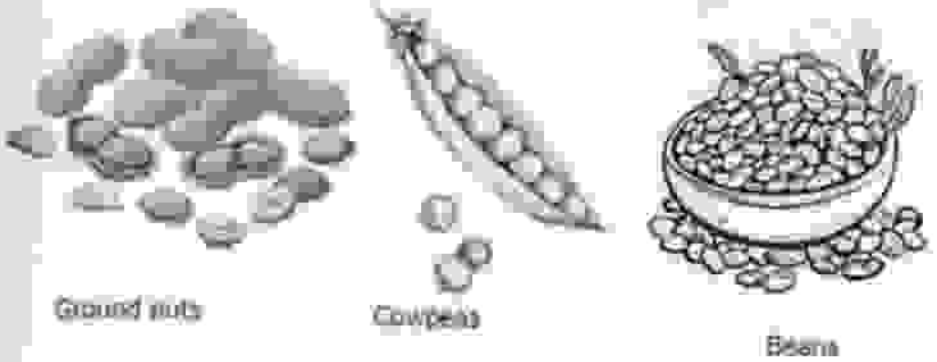

meat


eggs


fish

Sources of plant proteins

Beans, ground nuts, soya beans and cowpeas



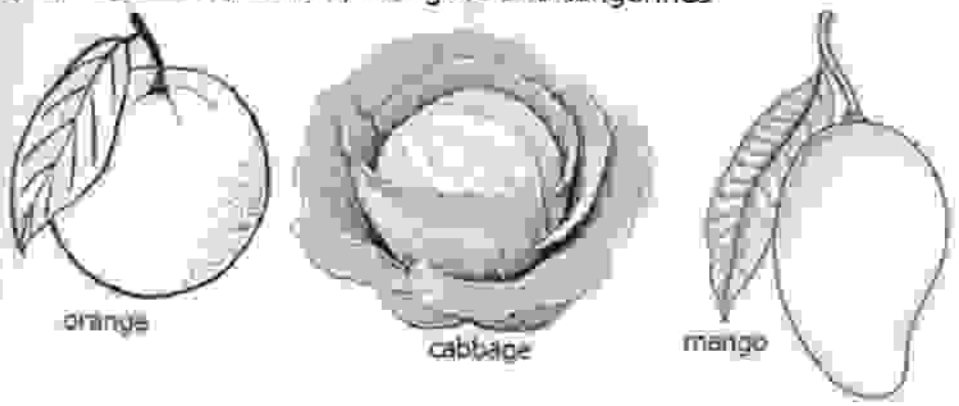
Mushrooms are sources of fungi proteins.

Vitamins

These are foodstuffs that give health and protection to the body against diseases.

Sources of vitamins.

Oranges, vegetables, carrots, mangoes and tangerines



Note well. Most fruits are eaten in raw form to maintain their vitamin value. Most fruits are not cooked because cooking destroys their vitamin value.

Mineral salts

Mineral salts are foodstuffs that help to make bones strong, form tissues and blood in the body.

Sources of mineral salts.

Beans, meat, eggshells, tomatoes, liver, milk and green vegetables are rich in iron.

Iron is used in the formation of blood.

Smoked fish, meat, eggs, green vegetables, eggshells, cheese are rich in calcium.

Calcium makes bones and teeth strong.

Iodised salt, seawater, cray fish, sea fish, shrimps and lobsters are rich in iodine.

It helps in the proper working of the thyroid glands.



Iodised salt



eggshell

Fats and oils (lipids)

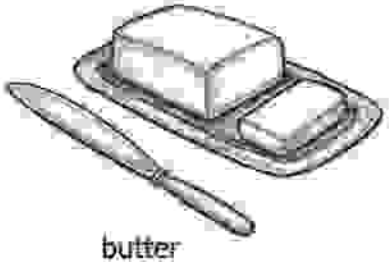
These are foodstuffs that provide heat in the body.

Fats are solids at room temperature while oils are liquids at room temperature.

Lipids is the collective name for fats and oils.

Sources of fats and oils.

Lard, olive oil, simsim, palm oil, ghee, margarine, egg yolk, cheese and butter



butter



simsim



cheese

Learn, unlearn and relearn: Go food, Grow food and Glow food are not classes of food.

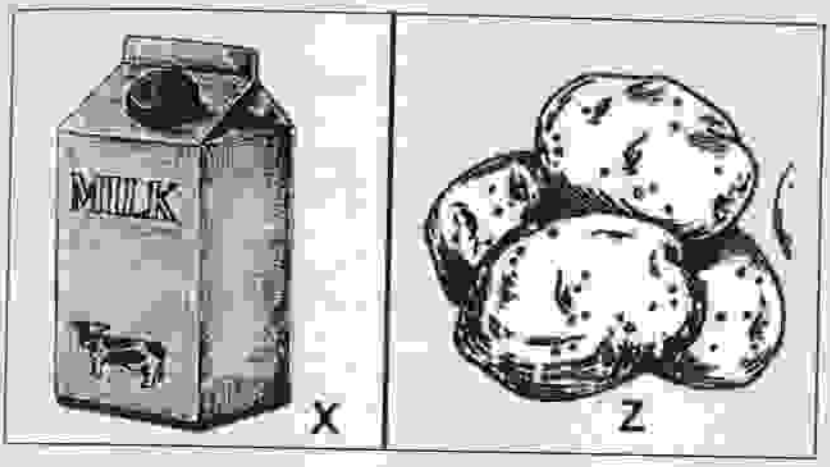
Food is classified according to chemical composition and nutrient value in the body. Not the resultant positive effect, so grow, go and glow are just descriptions of what the nutrients do in the human body.

Evaluation activity 5.1

1. (a) Give the meaning of the following terms:
   (i) Food
   (ii) Nutrition
   (b) State any **two** reasons why people eat food.
2. Give any **two** ways in which food is important in the human body.
3. Apart from proteins, name any **two** other classes of food commonly eaten by people.
4. What food value do you get when you eat posho at school regularly?
5. Name the class of food that helps in the hardening of human teeth and bones.
6. Give any **two** sources of animal proteins.

Evaluation activity 5.1 Comtinued

7. The diagrams below show foodstuffs. Study and use them to answer the questions that follow.



(a) Name the class of food obtained from the foodstuffs marked **X** and **Z**.
(b) State the main source from which the foodstuff marked **X** is obtained.
(c) Give any **one** way in which foodstuff marked **Z** is useful in the human body.

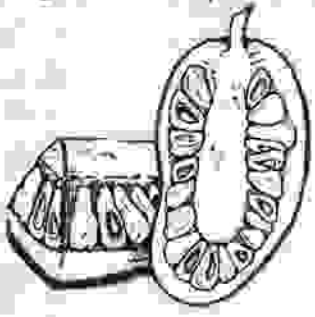
8. In the table below, part **A** has foodstuffs and their uses in the human body in part **B** of the table.

Part A: Foodstuffs	**Part B: Their uses in the body.**
Carrots	It helps in the formation of blood.
Honey	It helps to build the body.
Yoghurt	It helps to provide energy to the body.
Chicken liver	It helps to protect the body against diseases.

Match the foodstuffs to their uses in the body in part B of the table in the spaces provided below correctly.

(i) Carrots .....
(ii) Honey .....
(iii) Yoghurt .....
(iv) Chicken liver .....

9. Identify the class of food got when a person eats the foodstuff below.



Key words.
• diet • dietary • nutrients • roughage • nutritional
• repair

FOOD VALUES

Food values are nutritional and dietary components of food.
Nutrients are substances which are essential for healthy growth.

Examples of food values and their uses in the human body.

Food values	Uses in the human body
proteins	• They build the body. • They help in body growth. • They repair worn out body tissues. • They build body cells. • They replace worn out body cells. • They repair body cells.
carbohydrates	❖ They provide energy to the body.
Vitamins	• They help the body to remain healthy.
Mineral salts	• Calcium makes the bones and teeth strong. • Iron helps in the formation of blood in the body.
Fats and oils	• They generate heat in the body. • They provide energy in the body.

Learn, unlearn and relearn: Proteins do not repair "worn out" body cells but repair body cells.

Importance of proteins to different vulnerable people.

Group of people.	Uses of proteins
Babies	• They promote proper body growth. • They build the body of the baby.
Elderly people	• They repair worn out body tissues. • They build body cells. • They replace worn out body cells
Pregnant mothers	• They repair worn out body tissues. • They build body cells. • They replace worn out body cells



Points to note: carbohydrates, fats and oils and proteins provide energy in the body.

Did you know? Fats and oils provide more energy to the human body than carbohydrates.

Learn, unlearn and relearn: Roughages and water are not food values and classes of food. This is because they do not provide any nutritional values(nutrients) in the body.

BALANCED DIET

A balanced diet is a diet that contains all food nutrients in their right amounts. A diet refers to the overall pattern of food people consume over a period of time.

A meal is a single sitting of food eaten at a specific time.

Types/examples of meals

Breakfast, lunch, dinner and supper

Importance of a balanced diet

✓ It protects the body against malnutrition.
✓ It promotes proper body growth.

Components of a balanced diet

- carbohydrates
- proteins
- vitamins
- mineral salts
- fats and oils
- roughages
- water



ROUGHAGES

Roughages are plant-fibre materials in food.

They are also called dietary fibres.

They help to prevent constipation by making the stool soft.

Importance of roughages in the hu-man body.

They allow easy digestion of food.

They add bulk to the diet.

Sources of roughages

- ☐ sweet potatoes
- ☐ bread
- ☐ wheat bran
- ☐ green leafy vegetables
- ☐ unpolished cereals
- ☐ fresh fruits

WATER

Importance of water in the human body.

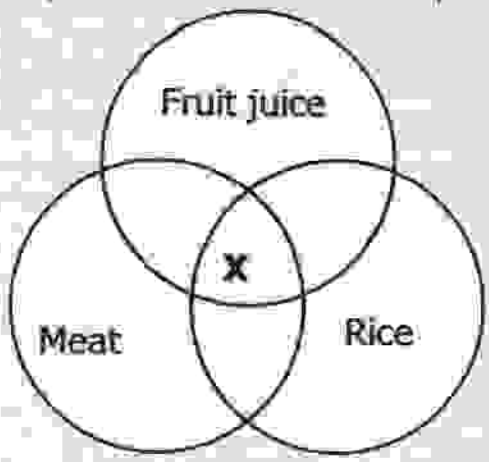
✓ It aids digestion of food.
✓ It regulates body temperature.
✓ It lubricates joints.
✓ It removes wastes like sweat and urine.
✓ It transports nutrients and oxygen.
✓ It increases the volume of blood.

Sources of water in the human body

- fruit juice
- milk
- fresh fruits
- soup
- coffee tea
- soda

Evaluation activity 5.2

1. (a) What is meant by the term balanced diet?
   (b) Write down any **two** examples of food values needed by the human body to remain healthy.
   (c) State any **one** reason why it is important for a person to feed on a balanced diet.
2. Give any **one** use of proteins in the human body.
3. Name the class of food that helps in the repair of body cells.
4. Apart from carbohydrates, name any **one** other class of food that gives the body energy.
5. Give **one** reason why babies should be fed on more food rich in proteins than the elderly people.
6. Give **one** reason why fruits like mangoes are eaten in raw form.
7. The diagram below shows foodstuffs eaten by family members in a certain home. Study and use it to answer the questions that follow.



(a) What does the part marked **X** represent in the diagram above?
(b) How does the father from the family who is a builder benefit from eating rice in the diagram above?
(c) Which foodstuff from the diagram above would you recommend the mother to give her children to prevent kwashiorkor?
(d) Name the food nutrient that the family members get from drinking fruit juice shown on the foodstuff above.

Key words

• deficiency • marasmus • stunted • nutrients • kwashiorkor • starvation

DEFICIENCY DISEASES

Deficiency diseases are diseases caused by lack of food nutrients in the human body.

Kwashiorkor



Additional information

Sometimes children do not get enough proteins because;

- Protein rich foods are expensive.
- Ignorance on how to feed children properly.
- Giving children more carbohydrates than proteins.
- Shortage of food at home.
- Kwashiorkor is usually got by children who are no longer breastfeeding because they easily get carbohydrates but not proteins.

Children should be given more food rich in proteins thanother food to help children get enough proteins. for proper growth.
Kwashiorkor is common in children due to rapid use of proteins for body growth.

Disease	Signs/symptoms	Prevention of the disease.
Kwashiorkor It is caused by deficiency of proteins in the human body.	▪ Swollen moon-like face. ▪ Stunted growth. ▪ Swollen belly. ▪ Swollen hands and feet. ▪ The hair takes on reddish colour ▪ The skin cracks and becomes scaly.	Feed a child on food rich in proteins.

Note: When you press the skin of a child with kwashiorkor, it takes long to come back to its position.

Reason. The whole body is swollen.

Disease	Signs/symptoms	Prevention of the disease.
Marasmus It is caused by Deficiency of all nutrients in in the human body.	• A child has a bony face. • A child is always hungry. • Swollen potbelly. • The child grows very thin. • Old man's face. • Wasted muscles. • Loss of body weight.	Feed a child on a balanced diet.

Note: A child who is affected by marasmus mostly do not have enough food to eat or the food eaten does not have the required nutrients needed by the body.



A child suffering from marasmus

VITAMIN DEFICIENCY DISEASES

Night blindness

This is a health condition in which a person is unable to see properly when there is little light. They may not see properly in the evening when it is getting dark or in a room which is not well lit. Night blindness is caused by lack of vitamin A in the human body.

Signs of night blindness.

- Poor night vision.
- Dry eyes.
- Unhealthy skin.
- Reduced night vision.
- Sore eyes

Prevention of night blindness

Feed on food rich in vitamin A.
Such food includes carrots, milk, liver, fish, spinach, kale, mangoes, red pepper and pumpkins

Scurvy

- Scurvy is a skin deficiency disease.
- It is caused by deficiency of vitamin C in the human body.
- A child who does not eat enough vegetables and fruits may get scurvy.
- This is because the child lacks enough vitamin C in the body.

Signs and symptoms of scurvy

- Bleeding gums.
- Pain in the joints.
- Swelling of gums.
- Weak teeth.
- Dry/scaly skin.
- Wounds on the skin.
- Poor healing of wounds.

Prevention of scurvy

Feeding on food rich in vitamin C.
Such food includes fresh fruits like mangoes, oranges, tangerines and leafy vegetables.

Beriberi

It is caused by deficiency of vitamin B1 in the human body.
It affects the brain.

Signs and symptoms

- Loss of appetite
- Heart failure.
- Decreased muscle formation.
- Loss of feelings in the limbs.
- Difficulty in speaking.
- Mental confusion.
- Swelling of the arms and legs.
- Tiredness.

Prevention of beriberi.

Feeding on food rich in vitamin B1.
Such food include; yeast and beans.

Pellagra

It is caused by deficiency of vitamin B3 in the human body.

Signs and symptoms of pellagra.

- Skin sores at the mouth and eyes.
- General body weakness.
- Rough scaly skin

Prevention of pellagra.

Feeding on food rich in vitamin B3.

Such food includes; chicken, salmon, tuna, brown rice, whole grains and legumes.

Learn, unlearn and relearn: Lack of vitamin B2 in a diet does not cause pellagra but it causes ariboflavinosis which is not part of primary concept.

Rickets

- Rickets is caused by deficiency of vitamin D in the human body.
- It is also caused by lack of calcium and phosphorus in a diet.
- Rickets affects bones in children.
- Vitamin D can be got from fatty food like fish, butter, milk and eggs.
- Sunlight enables the skin to make vitamin D.

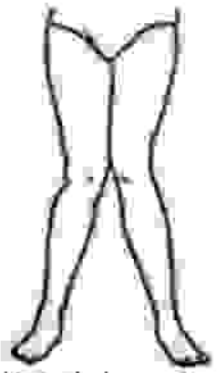
Signs of rickets in children.

- Bow-shaped legs, Knock knee bones, Brittle bones and teeth, Poor teeth formation.





Bow-shaped legs



Knock-knee legs

Prevention of rickets in children.

Sunbathing in sunshine for a short time in the morning.

Feed the child on food rich in vitamin D.

Such food include; margarine, fatty fish, cod liver oil, egg yolks, beef liver, cheese and mushrooms.

Evaluation activity 5.3

1. (a) What is meant by the term deficiency diseases?
   (b) Mention any **two** examples of vitamin deficiency diseases.
   (c) State the best way of preventing deficiency diseases.
2. Which deficiency disease is caused by lack of vitamin B1 in a diet?
3. (a) Name the disease which results from the deficiency of vitamin C in the human body.
   (b) State any **two** signs of vitamin C deficiency in the human body.
   (c) Give any **one** example of food which gives us vitamin C.
4. In the table below, part **A** shows of deficiency diseases and their signs in part **B** of the table.

Part A: deficiency diseases	Part B: Signs
Scurvy	Reduced night vision.
Rickets	Heart failure.
Beriberi	Bow-shaped legs.
Night blindness	Bleeding gums.

Use the signs in part B of the table to match the deficiency diseases below.

(a) Scurvy .....
(b) Rickets .....
(c) Beriberi .....
(d) Night blindness .....

Key words

- goitre
- anaemia
- malnutrition
- nutrients
- vital
- pale

MINERAL DEFICIENCY DISEASES

Goitre

Goitre is a health condition in which one's glands in the throat swell.
It is caused by deficiency of iodine in the human body.

Signs and symptoms of goitre	Prevention of goitre
• Swollen neck.	Feeding on food rich in iodine. Such food includes; iodised salt, shrimps, sea fish ground water, egg yolk and cheese

Anaemia

Anaemia is a health condition in which a person lacks blood in the body.
Anaemia is caused by deficiency of iron in the human body.

Other causes of anaemia

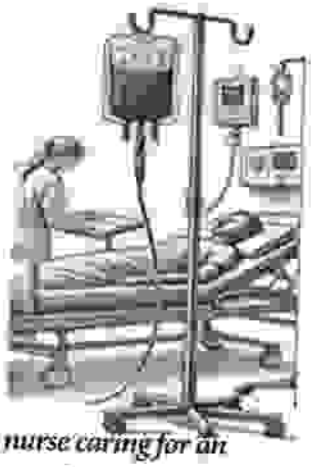
• Malaria infection. Malaria parasites destroy the red blood cells.

- Hookworm infestation. Hookworms feed on blood.
- Jigger infestation. Jiggers feed on blood.
- Excessive bleeding. This causes to loss of too much blood in the body.
- Bilharzia. Blood flukes feed on blood.

Signs and symptoms of anaemia	Prevention of goitre
• Pale skin • White fingernails. • Feeling tired and weak. • Pale gums. • Pale eyelids and the tongue.	❖ Feeding on food rich in iron. Such food include; Meat, eggs, liver, fish, spinach, ground nuts beans, peas and tomatoes. ❖ Swallowing iron supplement tablets Such tablets include; feosol, fergon, niferex.



A person suffering from goitre



A nurse caring for an anaemic person

MALNUTRITION

Malnutrition is a condition of poor health caused when the body does not get the right or required amount of nutrients.

Malnutrition comes from two words; **Mal** which means abnormal and **nutrition** meaning feeding.

Causes of malnutrition in children.

Cause/factor	How it causes malnutrition
Tapeworm infestation	Tapeworms feed on digested food.
Improper breastfeeding	Inadequate breastfeeding deprives infants of vital nutrients.
Food insecurity	It causes shortage of access to nutritious food.
Floods and drought	Disrupt food production and supply. Destroy food in the garden.
Poverty	It restricts access to nutritious food.

Cause/factor	How it causes malnutrition
Food taboos	Limit some vital nutrient intake.
Diseases like diarrhoea	Impair nutrient absorption and appetite for food.
Wars	They limit access to nutritious food. They disrupt food production.
Improper weaning	Introducing solid food too early or too late to the baby can result into nutrient deficiency.

Signs of malnutrition in children.

- Retarded growth.
- Body weakness
- Chronic tiredness
- Low body immunity
- Loss of body weight.



A malnourished child

Effects of malnutrition to a child.

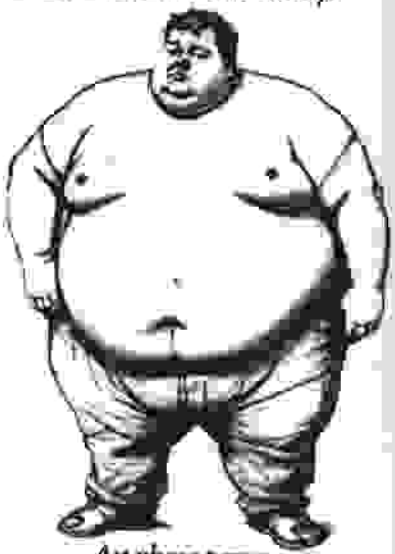
- ✓ It causes stunted growth in children.
- ✓ It lowers the immunity of the child.
- ✓ It prevents proper brain development.
- ✓ It causes chronic fatigue.
- ✓ It leads to loss of interest in work.

OBESITY

Obesity is a chronic condition caused by having excess fats in the body.

Factors that can lead to obesity

- ➢ Over eating.
- ➢ Lack of physical exercises.
- ➢ Eating a lot of fatty food.
- ➢ Medication like antidepressants.



An obese person

Effects of obesity to human health.

- ▪ It can lead to high blood pressure.
- ▪ It can lead to diabetes.
- ▪ It leads to reduced life expectancy.
- ▪ It causes breathing complications.
- ▪ It may lead to death.
- ▪ It causes oedema.

Signs of good health in a child.

- ❖ Mental alertness.
- ❖ Good eyesight.
- ❖ Strong bones and teeth.
- ❖ Having energy.

Evaluation activity 5.4

1. Give any **one** example of mineral deficiency disease in humans.
2. Which food nutrient helps to prevent anaemia in children?
3. What is meant by the term malnutrition?
4. Give any **two** factors that can lead to malnutrition in children.
5. How does tapeworm infestation lead to malnutrition in humans?

The diagram below shows health condition in humans. Use it to answer questions **6** and **7**



6. Name the health condition in humans above.
7. State any **one** cause of the health condition above in humans.
8. State any **two** effects of malnutrition in children.
9. (a) Name the disease which results from the deficiency of iodine in the human body.
   (b) State any **two** signs of iodine deficiency disease in humans.
   (c) Give any **one** example of food which gives us iodine.

Key words:

- contaminate
- substance
- dysentery
- foodborne
- poisoning
- expire

FOOD CONTAMINATION

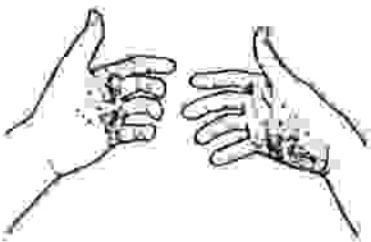
Food contamination is the presence of harmful substances in food that can make it unsafe to eat.

Substances that contaminate food.

- germs
- dust
- poison
- glass particles
- metal shards
- vectors

Ways in which food can get contaminated.

- Serving food with dirty hands.
- Leaving food uncovered.
- Serving food in dirty containers.
- Serving food from dirty places.
- Cooking food in dirty places and saucepans.
- Covering food with dirty containers.



Dirty hands



Uncovered food



Dirty plates

Dangers of eating contaminated food to human health

- It causes food poisoning.
- It can lead to death of a person.
- It makes one to suffer from foodborne diseases.
- It can lead to vomiting.
- It causes stomach cramps.

Food poisoning

Food poisoning is an illness caused to human health by eating contaminated food.

Causes of food poisoning

- ➢ Presence of germs in food.
- ➢ Putting toxins in food.
- ➢ Cross-contamination.
- ➢ Eating spoilt or expired food.
- ➢ Eating raw shellfish.
- ➢ Presence of vectors on food.
- ➢ Poor food hygiene.
- ➢ Undercooking food.
- ➢ Improper storage of food.
- ➢ Drinking unboiled milk or water

Ways of preventing food poisoning in human health.

- ✓ Eating steamed food.
- ✓ Cook food thoroughly.
- ✓ Store food at safe temperatures.
- ✓ Practice good food hygiene.
- ✓ Eat well washed fruits and vegetables.
- ✓ Refrigerate leftover food promptly.
- ✓ Check for expiry date on packed food.
- ✓ Clean kitchen surfaces daily.
- ✓ Ensure proper food packaging.
- ✓ Keep raw and cooked food separate.

Foodborne diseases

These are infections caused when a person eats contaminated food.

Examples of foodborne diseases

Diarrhoea, dysentery, typhoid, cholera, typhoid and tuberculosis

Ways in which food contamination can be prevented.

- ↕ Covering food properly.
- ↕ Serving food from clean places.
- ↕ Preparing food in clean containers.
- ↕ Proper sanitation
- ↕ Proper disposal of rubbish.
- ↕ Proper sensitisation of people about hygiene.
- ↕ Serving food in clean plates.
- ↕ Washing hands before touching food.
- ↕ Covering food with clean lids.
- ↕ Raising stalls above the ground.

Note well:

We wash hands before eating food to remove germs and dirt from hands.
We wash hands after eating food to remove food remains from hands.
Tuberculosis is a foodborne disease because it can spread through drinking unboiled milk from an infected cow.

FOOD HYGIENE

Food hygiene is the general cleanliness of food and things used when preparing food.

Ways of promoting food hygiene.

- Serving food in clean containers
- Serving food from clean places.
- Washing hands before touching food.
- Keeping food in clean places and plates.



Washing utensils



Clean kitchen



Washing food

Roles played by the school health club to ensure food hygiene at school.

(i) Educating pupils on proper food handling.
(ii) Displaying posters on food hygiene practices.
(iii) Providing sanitation facilities near eating areas.
(iv) Encouraging proper waste disposal near food areas.
(v) Inspecting the school kitchen and dining areas regularly.
(vi) Monitoring the hygiene of food vendors at school.

Importance of food hygiene to human health.

-It prevents food poisoning. -It prevents foodborne diseases.

Evaluation activity 5.5

1. (a) What is meant by the term food contamination?
(b) Mention any **two** ways in which food may get contaminated after it has been prepared.
(c) State any **one** way in which you can prevent food from getting contaminated after serving it.
2. What name is given to an illness caused by eating contaminated food?
3. Write down any **three** substances that contaminate food.
4. State any **two** causes of food poisoning at home.
5. State any **two** ways in which contaminated food affects the health of humans.
6. Why should we wash our hands before eating food?
7. Give any **two** examples of foodborne diseases.
8. State any **one** way of controlling food poisoning at home.
9. Give any **two** roles played by school Health Club to ensure food hygiene at school.

Key words:

- refrigerate
- mould
- irradiate
- pasteurise
- preservative
- pickle

FOOD PRESERVATION

Food preservation is a way of keeping food safe for a long time without going bad.

Organisms like bacteria and fungi make food to go bad.

Changes in food that show that it has gone bad.

- Develops bad smell
- Grows moulds
- Unpleasant taste
- Becomes sticky
- Change of colour
- Bubbles form in food

Traditional/local methods of preserving food.

Sun-drying

This involves putting/spreading food in sunshine to dry.
Heat from the sun dehydrates food. /it drains moisture from food.
Sun-drying prevents food from rotting by dehydrating it.
Foodstuffs that are sun-dried include; fish, cassava, maize, millet, beans, sorghum.

Note. Wind also helps to preserve food by increasing the rate of evaporation in food.

Smoking

This involves using slow burning heat to dry food.
Heat from smoke dehydrates food.
Foodstuffs preserved by smoking are fish and meat.

Salting

This involves soaking food into salt solution or sprinkling salt on food.
Salt dehydrates food.
Examples of foodstuffs preserved by salting are meat and fish.

Note: These traditional methods of food preservation are recalled using 3Ss.






Foodstuffs preserved by refrigeration include; meat, fish, fruits, vegetables, milk, onions

Pickling

This involves soaking food in acids like vinegar (acetic acid)

It reduces the growth of bacteria in food.

Examples of foodstuffs preserved by pickling are cucumber, carrots, onions, eggs, peppers, lemons, watermelons, fish, meat, garlic, ginger and olives.



Refrigeration



Pickling

Irradiation

This is a process of exposing food to strong heat.

Radiant heat kills bacteria in food.

Examples of foodstuffs preserved by irradiation are; chicken, shellfish, vegetables

Sterilisation

Food like milk is heated at high temperature, cooled and sealed in sterile containers.

Bacteria are killed by maximum boiling.

Pasteurisation

Food like milk is heated at controlled temperature and allowed to cool.

Pasteurisation kills bacteria in milk.

Freezing

This involves keeping food at a very low temperature in order to preserve it.

The cold temperature prevents the breeding of bacteria in food.

How freezing is applied when preserving food.

- Boiling, cooling and then freezing vegetables.
- Freezing berries quickly individually before packaging.
- Using liquid nitrogen to freeze food like meat.
- Using household freezers to store meat, prepared meat and leftovers.

Using food preservatives.

They prevent bacteria and mould from breeding in food.

Examples of food preservatives

salt, sugar, local salt and vinegar, sodium benzoate.

Ice-glazing

This involves covering food with a thin layer of ice, which prevents dehydration and freezer burn.

Examples of foodstuffs ice-glazed are fish, shrimps and fruits.

Ice cubes are put in fresh fish.

Advantages of modern methods of preserving food.

- ❖ Longer shelf life.
- ❖ Reduce food waste.
- ❖ They are convenient to use.
- ❖ They improve food safety.
- ❖ They are faster in killing bacteria.
- ❖ They maintain the nutritional value of food.

Disadvantages of modern methods of preserving food.

- ❖ Some items used are not easily accessible e.g. vinegar.
- ❖ Some modern methods require electricity to work.
- ❖ They are expensive to use e.g. refrigeration
- ❖ The flavour in food can easily get lost.

Reasons why food should be preserved.

- ✓ To ensure food security.
- ✓ To increase on the lifespan of food.
- ✓ To add nutritional value to food.
- ✓ For easy transportation.
- ✓ To keep food for future use.
- ✓ To maintain the taste of food.

FOOD SECURITY

Food security is a situation in which there is enough food available all time for consumption.

Ways of promoting food security

- ➢ Proper storage of food.
- ➢ Storing excess food.
- ➢ Preserving food.
- ➢ Growing more food crops.
- ➢ Growing drought resistant crops.
- ➢ Early planting

Importance of food security.

-It prevents famine.

-It prevents malnutrition.

Problems faced by people when getting food.

(i) Contamination of food.
(ii) High prices of food.
(iii) Accidents along the way.
(iv) Wrong labelling of food.
(v) Low quality food.
(vi) Poor transport.
(vii) Failure to get fresh food.
(viii) Long distances to the market.
(ix) Overstayed food.

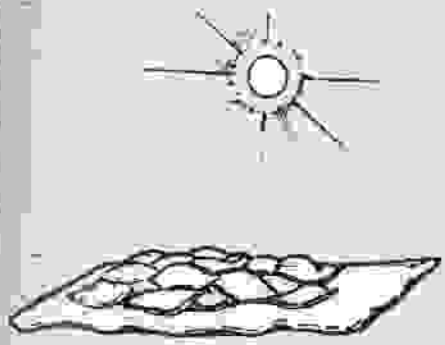
Evaluation activity 5.6

1. (a) Give the meaning of the following terms:
   (i) Food preservation
   (ii) Food security

(b) State any **two** ways in which parents can ensure the availability of enough food at home.

Evaluation activity S.6

2. The diagram below shows methods of preserving food. Study and use them to answer the questions that follow.



W



X

(a) Name the methods of preserving food marked **W** and **X**.
(b) Give any **one** vegetable that can be preserved using the method marked **W**.
(c) State any **one** advantage of using method marked X over method marked **W** when preserving food at home.
3. How is sun-drying helpful in the preservation of food crops?
4. How does smoking control the effect of bacteria in food preservation?
5. (a) Name any **one** organism that makes fresh meat go bad.
(b) State any **three** ways of preventing fresh meat from going bad
6. State any **two** changes in food which show that food has gone bad.
7. Mention any **two** methods of preserving food that;
(a) Do not require heat: .....
(b) Keep food in cold conditions: .....

Key words:

• matoke • mingle • roast • dish • stew • consumption

PREPARATION OF SIMPLE DISHES USING LOCAL FOODS

Food preparation is the way of making food ready for consumption
A dish is a combination of staple foods and sauce.

Examples of dishes that can be prepared at home

- mashed matoke
- posho
- chips
- fried rice (pilau)
- boiled potatoes
- stew
- millet bread
- roasted cassava
- salads

Methods that can be used to prepare simple dishes at home.

Method	Examples of food prepared
Boiling	Porridge, rice, potatoes, cassava, meat, milk tea
mingling	Millet bread, posho, cassava bread, matoke
frying	Meat, fish, rice, matoke

Method	Examples of food prepared
roasting	Meat, ground nuts, cassava, maize, fish
baking	Cakes, bread, bans, chapattis
steaming	Matoke, rice, meat, fish, vegetables
stewing	Matoke, meat



Boiling



Mingling



Roasting

Steps taken to prepare a meal at home.

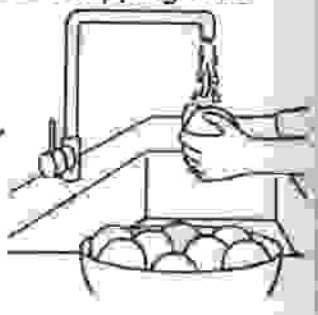
Sorting food, washing food, peeling food, slicing food and chopping food



Slicing food



Peeling food



Washing food

How to Prepare some Dishes at Home

Preparing matoke mixed with groundnuts.

Materials needed:

- ☑ 16 fingers of matoke
- ☑ ¼ kg of pounded groundnuts
- ☑ salt
- ☑ water
- ☑ saucepan
- ☑ onions, tomatoes (one if big)
- ☑ knife

Steps taken:

(i) Peel the 16 fingers of matoke and cut them into pieces.

(ii) Place the matoke in a saucepan, add water, and a pinch of salt, then place it on the charcoal stove to cook.

(iii) While matoke is cooking, grind the ¼ kg of groundnuts, chop the onions and tomatoes.

(iv) Once the matoke is soft, mix in the groundnuts, onions, and tomatoes, and let it cooks until the mixture is thick and well combined.

Preparing posho at home

Materials needed:

- ½ kg of maize flour
- water
- mingling stick
- saucepan

Steps taken:

(i) Boil water in a saucepan and gradually add maize flour to the boiling water while stirring continuously to avoid lumps.
(ii) Reduce the heat and keep stirring until the mixture thickens.
(iii) Cover the saucepan and let it simmer on low heat for about 10-15 minutes, stirring occasionally.
(iv) Remove the posho from heat and let it sit covered for a few minutes before serving.

Demonstration:



Preparing sweet potatoes

Materials needed:

- Sweet potatoes (at least 5 big tubers)
- water
- banana leaves
- knife
- saucepan
- banana stalks

Steps taken:

(i) Wash and peel the tubers, then cut them into halves.
(ii) Line the saucepan with banana stalks and banana leaves.
(iii) Place the sweet potatoes on the banana leaves, add some water, and cover them with more banana leaves.
(iv) Place the saucepan on the charcoal stove and cook until sweet potatoes are soft and ready to eat

Demonstration:



Preparing boiled rice at home

Materials needed:

- ½ kg of rice
- 1 litre of water
- 1 litre of cooking oil
- mingling stick
- saucepan
- charcoal stove

Steps taken:

(i) Measure ½ kg of rice and rinse it with clean water.
(ii) Pour 1 litre of water into the saucepan and place it on the charcoal stove to boil.
(iii) Once the water is boiling, add the rinsed rice into the saucepan.
(iv) Cover the saucepan and let the rice cook on low heat till all the water is absorbed and the rice is soft.

Demonstration:



Preparing millet/cassava bread at home

Materials needed:

- ¼ kg of millet flour
- water
- mingling stick
- saucepan
- ½ kg of cassava flour

Steps taken:

(i) Pour half of a litre of water into a saucepan and place it on a charcoal stove to boil
(ii) Add the ¼ kg of millet flour to the boiling water and stir it using the mingling stick.
(iii) Slowly add the half kg of cassava flour, stirring continuously to avoid lumps.
(iv) Keep stirring till the mixture thickens, then remove the saucepan from the stove, let it cool, and serve.

Demonstration:



Reasons for cooking food.

- To make it soft.
- To make it easy to digest
- To kill germs in food.
- To make it tasty.

Eating habits practised by people.

Good eating habits	Bad eating habits
▪ Chewing food with mouth closed. ▪ Having meals at regular times. ▪ Putting small lumps of food in the mouth at a time. ▪ Sitting upright when eating. ▪ Eating food while quiet. ▪ Avoid eating in between meals.	▪ Eating food with unwashed hands. ▪ Talking with food in the mouth. ▪ Chewing food with the mouth open. ▪ Eating while walking. ▪ Bending while eating food. ▪ Swallowing big lumps of food at a time.

Dangers of bad eating habits.

- It causes choking.
- It causes indigestion.
- It causes stomach pain.

Evaluation activity 5.7

1. (a) Write down any **two** methods that can be used to prepare simple dishes at home.
   (b) State any **two** steps that can be used when preparing local dishes at home.
2. (a) Name any **two** local dishes that people commonly prepare at home using local food.
   (b) Give any **two** reasons why people cook food at home.
3. The sentences below show steps used when preparing posho at home in wrong order.

(i) Cover the saucepan and let it simmer on low heat for about 10-15 minutes stirring occasionally.
(ii) Remove the posho from heat and let it sit covered for a few minutes before serving.
(iii) Boil water in a saucepan and gradually add maize flour to the boiling water while stirring continuously to avoid lumps.
(iv) Reduce the heat and keep stirring until the mixture thickens.

Write the sentences above in the spaces below by arranging them correctly to show the steps used when preparing posho.

(i) .....

(ii) .....

(iii) .....

(iv) .....

MEANING OF KEY TERMS IN OUR FOOD:

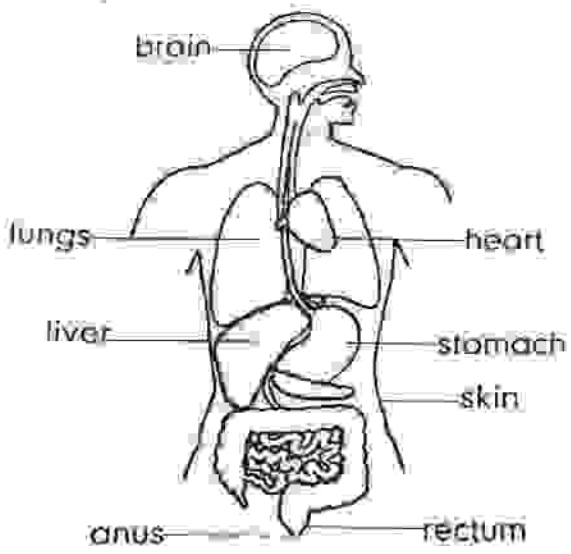
Food: A substance that adds nutritional value when taken into the body.
Carbohydrates: Energy giving food.
Balanced diet: A meal containing all food values in their right amounts.
Food preservation: A way of keeping food safe for a long time without going bad.
Food hygiene: General cleanliness of food and things used to prepare it.
Malnutrition: A condition when the body does not receive essential nutrients.
Deficiency diseases: Diseases that spread due to poor feeding.
Proteins: Body building foods.
Vitamins: Health giving food.
Food values: Nutritional and dietary components of food.
Pickling: It involves soaking food in vinegar to preserve it.
Smoking: It is the exposure of food to slow burning heat to preserve it.
Diet: Refers to the overall pattern of food a person consumes over a period of time.
Meal: A single sitting of food eaten at specific time.
Food security: A situation in which there is enough food available all time for consumption.
Foodborne diseases: Infections caused when a person eats contaminated.
Food contamination: The presence of harmful substances in food that can make it unsafe to eat.
Nutrients: Substances which are essential for healthy growth.
Nutrition: The process by which food is taken in and used by the body.
Goitre: The swelling of the thyroid glands due to lack of iodine.
Anaemia: A condition in which the body lacks blood in it.

6 THEME: HUMAN BODY
TOPIC 6: HUMAN BODY ORGANS

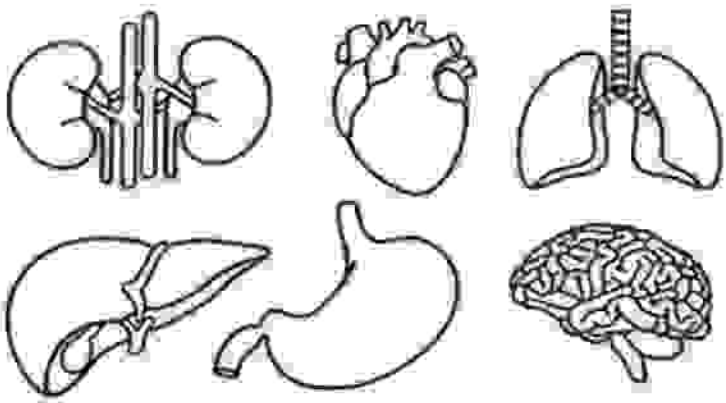
Key words:
• nasal • cilia • trachoma • orbit • conjunctivitis
• tongue

Major Organs of the Human Body

An organ is a group of different tissues which work together to do a particular function in the body.



Human body organs

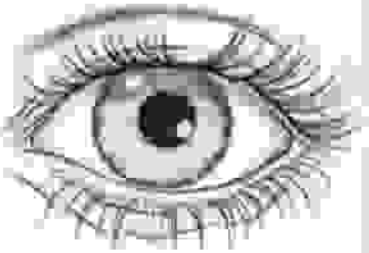


The major human body organs include;

- the eyes
- the brain
- the ears
- the stomach
- the heart
- the urinary bladder
- the lungs
- the liver
- the tongue
- the nose
- the kidneys.

The External Human Body Organs

The eyes



- The eye is the sense organ for seeing/sight.
- The eyes help us to identify the sizes, height and colours of objects in the environment.
- The eyes use light energy for them to function.
- The human eye is protected by the orbit (eye socket)

Diseases that attack the human eye

Night blindness, river blindness, trachoma, conjunctivitis and stye.
Night blindness is caused by lack of vitamin A in a diet. It is the deficiency disease that attacks the eyes.
River blindness is spread by a black fly.
Trachoma is spread by a housefly and caused by bacteria.
Conjunctivitis is caused by bacteria or virus.
Stye is an inflammation on the human eye due to bacterial invasion.

The disorders of the eyes

Short sight, long sight, blindness, old age sight and squint

Ways of caring for the eyes

- Regular washing of the eyes with clean water and soap. It helps to remove rheum from the eyes.
- Do not touch the eyes with dirty hands.
- Avoid looking at the sun directly.
- Avoid reading under too dim or too bright light.
- Having regular optical check-ups.

The nose.



The nose is the sense organ for smelling.
The nose has holes called nostrils.
The nose has cilia and mucus which filter air and trap dust.

Diseases of the nose

Sinusitis, nasal allergies and influenza

Disorders of the nose.

Nose bleeding, nasal congestion and foreign bodies in the nose

Care for the nose

- Avoid pushing objects into the nose.
- Removing mucus regularly.
- Wearing a face mask.
- Regular washing of the nose.
- Avoid staying in dusty places.
- Avoid rough games like slide kicks.
- Clean the nose using a clean handkerchief.



The ear

The ear is the body organ for hearing.
The ear uses sound energy for it to function.
The ear is also used for body balance.
The human ear has wax which traps dust in the human ear.



Diseases of the ears.

Earache, otitis, ear cancer and otalgia

Disorders of the ears

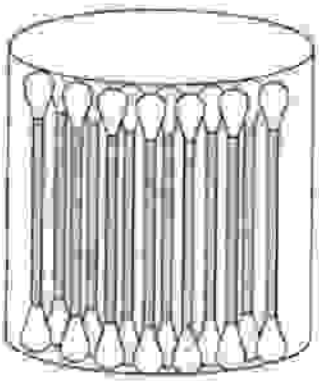
Ear boils, ringing in the ear, deafness, itching inside the ear.

Care for the ears

- Going for regular medical check-ups.
- Clean the ear using ear cotton swabs.
- Avoid playing rough games like slide kicks.
- Listen to music at a low volume.
- Insert earplugs in the ear to protect the ear from loud noise.

Learn, unlearn and relearn: Ear buds are small earphones inserted in the ear for listening music thus are not used to clean the ear. Additionally, ear pads are cushioned parts of headphones or earphones or headsets that rest against or around your ears.

Ear cotton swabs



Evaluation activity 6.1

1. Name human body organ that requires light energy for it to function.
2. Apart from hearing, give **one** other function of the ear to a person.
3. Write down any **one** vector that affects the human eyes.
4. State the difference in function between the human nose and the eyes.
5. Give any **one** reason why it is dangerous to clean the ears using sharp objects.

The diagram below shows a material used to clean the human ears. Use it to answer questions **6** and **7**.



6. Identify the material used to clean the human ears shown above.
7. Why is it advisable to use the above item once when cleaning the human ears?
8. Apart from the nose, mention any **one** other external organ of the human body.
9. State the importance of the orbit on the human body.
10. State the importance of the hair in the nose of a person.

Key words:

▪ brain ▪ taste ▪ bile ▪ regulate ▪ cirrhosis ▪ churn

The Internal Human Body Organs

The brain.

The brain is protected by the skull.

The brain organises different organs to work together as a whole body.

The brain controls all other body organs.



Functions of the brain

- It stores information.
- It controls all body activities.
- It is used for body balance.
- It is used for remembering/ recalling/memory.
- It is used for thinking.
- It is used for reasoning.
- It is used for learning new things.
- It is used to interpret information.
- It is used for decision making.

Diseases that attack the brain

Cerebral malaria, meningitis, epilepsy, brain cancer

Note: Amnesia is a disorder of the brain that leads to memory loss.

Care for the brain

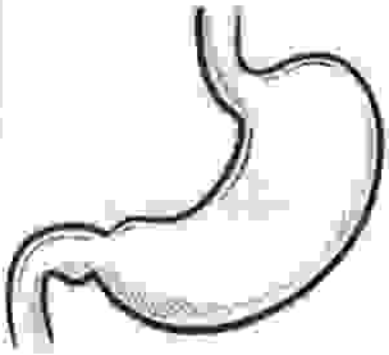
- Have enough rest and sleep. It helps to refresh the brain.
- Feed on a balanced diet.
- Do regular physical exercises.
- Immunise children against meningitis.
- Avoid tobacco smoking and alcoholism.
- Early treatment of malaria.

The stomach

The stomach is part of the digestive system.

Chemical digestion of proteins begins from the stomach.

Digestion is the breakdown of food into simple soluble particles that the body can absorb.



Functions of the stomach

- It stores food for some time.
- It churns food.
- It absorbs alcohol.
- It produces hydrochloric acid that kills germs in food.
- It produces gastric juice.

Diseases that affect the stomach.

Peptic ulcers, gastritis and gastric cancer

Care for the stomach

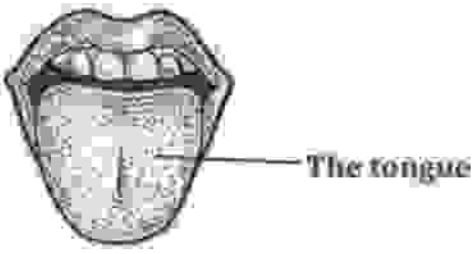
- Limit alcohol intake.
- Do regular physical exercises
- Eat food in time.
- Limit intake of fatty food.

The tongue

It is found in the mouth.

Functions of the tongue

- It used for tasting.
- It is used in talking.
- It rolls food into bolus.
- It mixes food with saliva.



Diseases of the tongue

Oral herpes, oral thrush and oral cancer

Disorders of the tongue

Burns, cuts, blisters, taste loss

Care for the tongue

- Do not eat too hot food. *It destroys taste buds.*
- Brush the tongue daily. *It removes food remains from the tongue.*
- Do not put sharp objects in the mouth.
- Rinse the mouth to remove leftover food particles.
- Avoid smoking tobacco.
- Use a tongue scraper to remove dead cells and food remains.

The liver

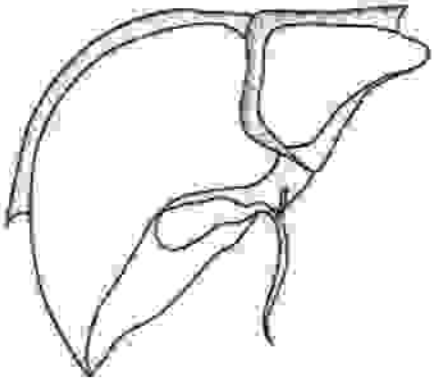
The liver is a reddish-brown organ in the body.
It is the largest internal organ in the human body.

Functions of the human liver.

- ✓ It regulates blood sugar.
- ✓ It produces bile juice.
- ✓ It stores food and blood.
- ✓ It removes poisonous substances from food and blood.

Diseases that affect the human liver.

Hepatitis B, yellow fever, liver cancer, liver cirrhosis, liver abscess and hepatitis A.
Hepatitis B, and yellow fever are immunisable diseases.
Hepatitis A is a waterborne immunisable disease
Liver cirrhosis is caused due to alcohol consumption.



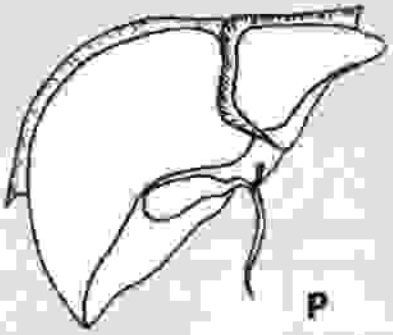
Care for the liver

- Feeding on a balanced diet.
- Doing regular physical exercises.
- Going for regular medical check-ups
- Avoid alcohol consumption.

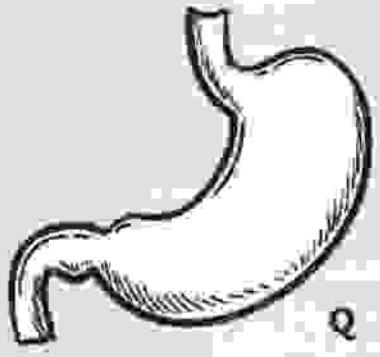
Evaluation activity 6.2

1. Name the human body organ which helps a person to know whether there is enough sugar in tea or not.
2. State **one** way in which the function of the brain is similar to that of the ear.
3. Give any **two** ways of caring for the stomach.
4. Which immunisable disease attacks the human liver?
5. Name the body organ that requires one to have enough rest and sleep.

The diagrams below show human body organs. Use them to answer the questions that follow.



P



Q

(a) Name the body organs marked **P** and **Q**.
(b) Give any **one** function of the organ marked **P** in the human body.
(c) Which human body system does the organ marked **Q** form?

6. State **one** way in which it is important to have enough rest and sleep.
7. How is the skull useful to the brain?
8. Name **one** disease that affects the stomach because of alcoholism.

Key words:

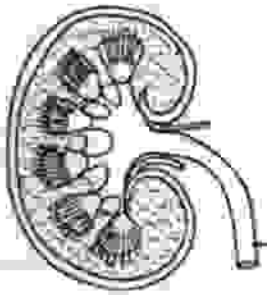
• Bladder • Vapour • Urine • Excretory • Respiratory
• Influenza

The kidney

The kidney is bean-shaped and reddish brown in colour.
The kidney works hand in hand with the urinary bladder.
The urinary bladder stores urine for some time.
The kidney is part of the excretory system.



Urinary bladder



The kidney

The ureter takes urine to the urinary bladder.

Functions of the human kidney.

- It filters blood.
- It balances the amount of water in the body.
- It regulates salt levels in the body.

Diseases that affect the kidneys

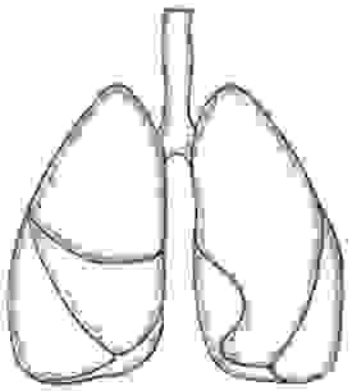
Kidney stones, kidney cancer, kidney failure, bilharzia and gonorrhoea

Care for the kidneys.

- Feed on a balanced diet.
- Do regular physical exercises.
- Drink enough boiled water.
- Empty the urinary bladder when full.
- Limit salt intake.
- Going for regular medical check-ups.

The lungs

The lungs are used for gaseous exchange.
Lungs are used for breathing.
The lungs are protected by the rib cage.

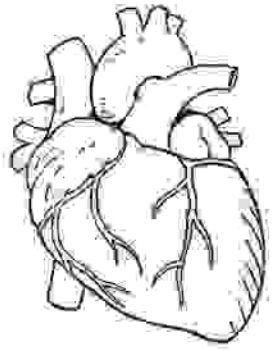


Diseases of the lungs

Lung cancer, tuberculosis, influenza, diphtheria, whooping cough, asthma and pneumonia.
Lung cancer is caused by tobacco smoking.

The heart.

The heart pumps blood in the body.
The heart is protected by the rib cage.



The human Heart

How the Human Body Works

How the body works	System used
Takes in food and oxygen. The processes involved are digestion, blood circulation and breathing.	• Digestive system • Respiratory system • Circulatory system
Carries food and oxygen to different parts of the body.	• Respiratory system • Circulatory system
Uses food and oxygen to get energy. This process is called respiration.	• Respiratory system • Circulatory system
Removes wastes. The process is called excretion.	• Excretory system

Examples of wastes removed from the body

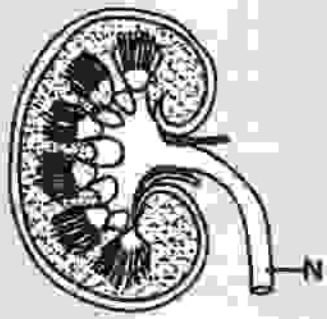
Wastes	Organ that removes it	Process
Urine	Kidney	Urination
Sweat	Skin	Sweating
Carbon dioxide	Lungs	Respiration
Bile pigments	Liver	Excretion
Water vapour	Lungs	Respiration

Note: sweating cools the body. Respiration enables the body to get energy.

Evaluation activity 6.3

1. Name the human body organ that balances the amount of water needed in the human body.
2. State the main function of the heart in the human body.
3. Name the excretory organ that removes water and carbon dioxide from the human body.
4. Name any **one** human body organ protected by the rib cage.
5. By what process does the body produce energy from food and oxygen?

The diagram below shows the human body organ. Use it to answer questions **6** and **7**.



6. Name the organ shown above.
7. Give the function of the part labelled **N**.
8. The sentences below show how the human body works in wrong order.
   (a) Carries food and oxygen to different parts of the body.
   (b) Removes wastes.
   (c) Takes in food and oxygen.
   (d) Uses food and oxygen to get energy.

Write the sentences above in the right order.

Examples of wastes removed from the body

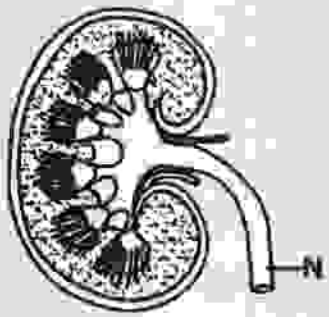
Wastes	Organ that removes it	Process
Urine	Kidney	Urination
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Carbon dioxide	Lungs	Respiration
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Note: sweating cools the body. Respiration enables the body to get energy.

Evaluation activity 6.3

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The diagram below shows the human body organ. Use it to answer questions **6** and **7**.



6. Name the organ shown above.
7. Give the function of the part labelled **N**.
8. The sentences below show how the human body works in wrong order.
   (a) Carries food and oxygen to different parts of the body.
   (b) Removes wastes.
   (c) Takes in food and oxygen.
   (d) Uses food and oxygen to get energy.

Write the sentences above correctly in the spaces provided below to show how the human body works.

.....

.....

.....

.....

Evaluation activity 6.3 Continued

9. The table below shows human body organs, the system to which they belong and the organ function. Study and complete it correctly.

ORGAN	SYSTEM	ORGAN FUNCTION
Heart	Circulatory	.....
....	Excretory	Formation of urine.
Pancreas	.....	Produces pancreatic juice.
Lungs	Respiratory	.....

10. Write down any **two** examples of wastes that are removed from the human body.

MEANING OF KEY WORDS IN HUMAN BODY ORGANS

Organ: A group of tissues performing the same function.
Brain: The human body organ for storing and interpreting information.
Skin: The human body organ for feeling.
Lungs: The human body organs for gaseous exchange.
Heart: The human body organ that pumps blood.
Kidney: The human body organ that filters blood.
Liver: The largest internal human body organ.
Stomach: The bag-like structure in which food from the stomach collects.
Body wastes: The metabolic materials removed from the body.
Digestion: The process by which food is broken down into smaller particles.
Excretion: The removal of metabolic waste materials from the human body.
Sweating: The process of removing sweat from the body.
Urination: The removal of urine from the human body.
Respiration: The breakdown of digested food to produce energy.
Breathing: The intake or outtake of air by an organism.

7 THEME: HUMAN BODY
TOPIC 7: THE TEETH

Key words:

• canine • biting • incisor • chisel • tear • permanent

Teeth are hard bone-like structures in the mouth of an animal.
Teeth help to break down food into small particles for easy swallowing.

Sets of teeth

Milk teeth and permanent teeth.

Milk teeth

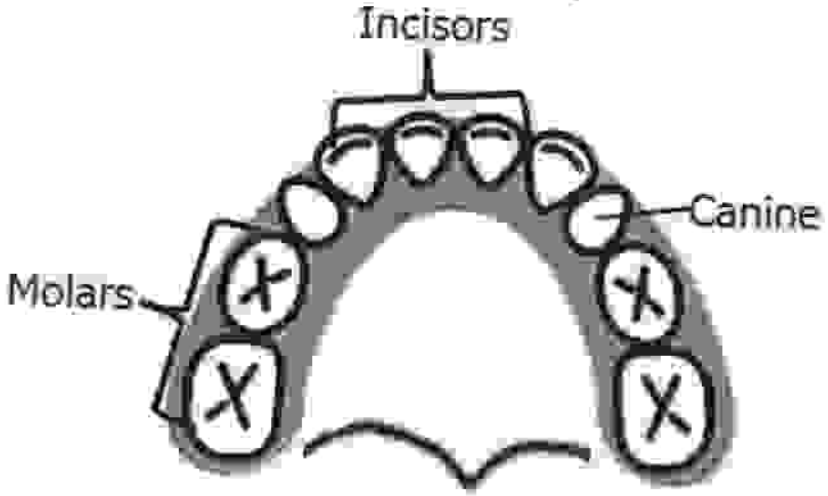
These are the first set of teeth to grow in the mouth.
In a child, they are 20 teeth.
At around 5 years of age, these teeth begin to fall out.
This is called teeth shedding.
Milk teeth are also called deciduous teeth or primary teeth.

Dental formula for milk teeth

Type of teeth	Upper jaw	Lower jaw	Total teeth
Incisors	4	4	8
Canines	2	2	4
Premolars	0	0	0
Molars	4	4	8

Total in one jaw = 10 teeth
Total in the mouth = 20 teeth
Missing teeth: premolars

Arrangement of milk teeth in the upper jaw



Points to note:
Premolars are missing in milk set because they are not needed for grinding food at an early age, as a baby primarily consumes soft or liquid diet.
On the other hand, molars are present in milk teeth to help grind and chew soft solid foods as the baby transitions to a varied diet, unlike premolar, which are more specialised and develop later to support the stronger chewing needs of permanent set.

Permanent teeth

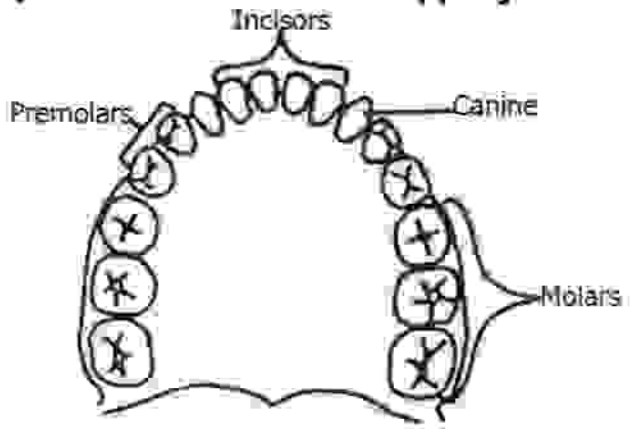
These are the second and the final/last set of teeth to grow in the mouth of a person.
They are also called secondary teeth.
An adult person has 32 permanent teeth in the mouth.
Permanent teeth begin developing at the age of 13 years.

Dental Formular in Permanent Teeth

Type of teeth	Upper jaw	Lower jaw	Total teeth
Incisors	4	4	8
Canines	2	2	4
Premolars	4	4	8
Molars	6	6	12

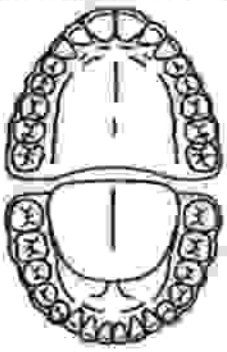
8+4+8+12 = 32 teeth.

Arrangement of permanent teeth in the upper jaw.



Arrangement of milk teeth and permanent teeth in the mouth


Milk teeth


Permanent teeth

A SIMPLE METHOD TO ENABLE YOU REMEMBER THE TYPES OF TEETH IN EACH SET

A teacher asks learners these questions and then they give you answers.

1. When Ian is a baby, can he manage to pick mangoes from a tree?

Then you reply, No, Ian can't manage (ICM), So, no picking, no premolars.

2. When Ian is old, can he pick mangoes from a tree?

Then you reply, Yes, Ian can pick mangoes. (ICPM)

In summary, the method is;

Milk set: Ian can't manage (ICM)

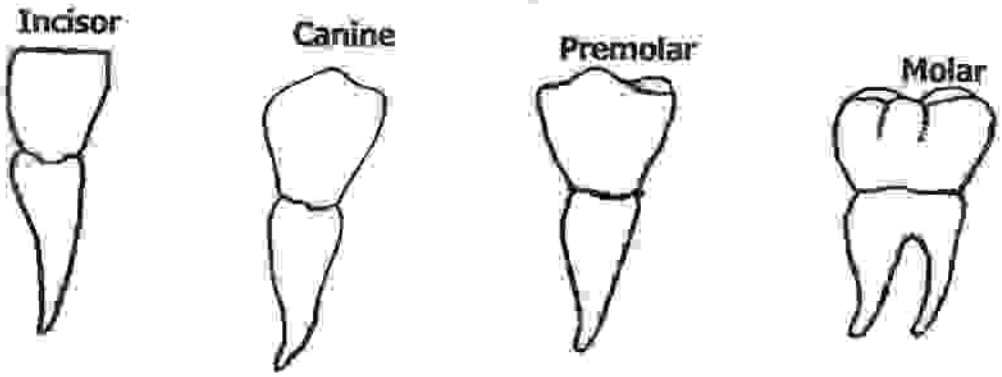
Permanent set: Ian can pick mangoes. (ICPM)

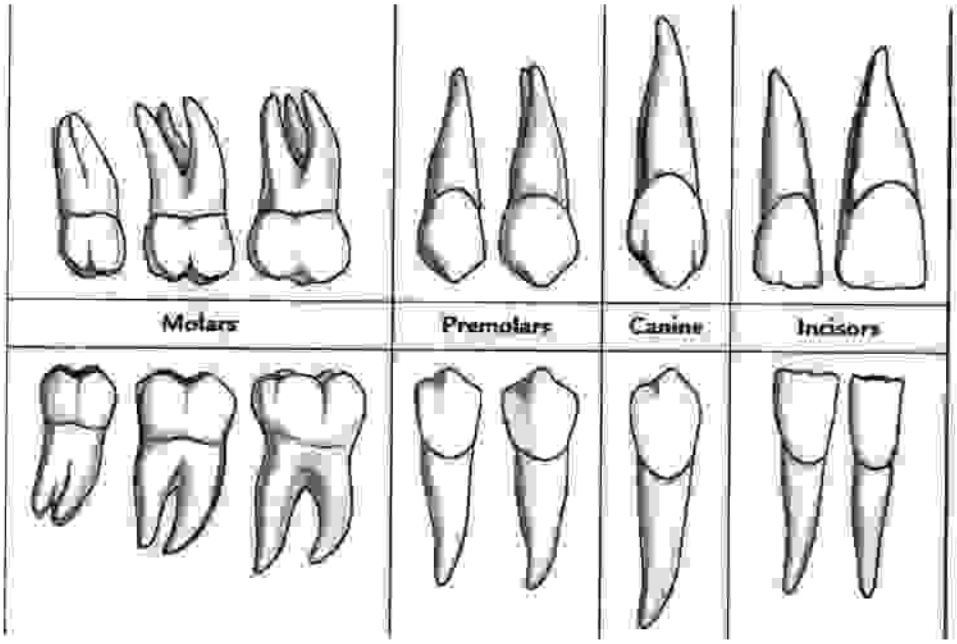
In brief, the milk set lacks premolars.

Types of Teeth and their Functions

Type of teeth	Characteristics	Function and adaptation
Incisor teeth	- First type of teeth to grow in human beings and first to fall out. -They are chisel shaped. -They have straight edges. -They have one root.	For cutting or biting food. **Adaptation.** They are chisel-shaped to cut food.
Canine teeth	- They are sharp and pointed on top. - They have one root. - They are cone-shaped. *Note: They are found at the corner of the mouth.*	For tearing flesh. **Adaptation.** They are sharp and pointed to tear flesh
Premolar teeth	-They have blunt broad ridges. - They are smaller than molars. - They have one root. - They have cusps.	For crushing, grinding and chewing food. Adaptation -They have blunt broad ridges to grind and chew food. -They are broad to increase the surface area for chewing food. -They have cusps for chewing food.

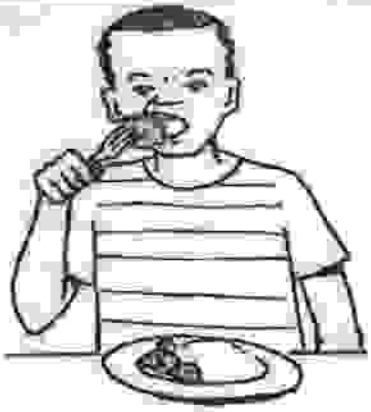
Type of teeth	Characteristics	Function and adaptation
Molar teeth	- They are the last type of teeth to grow in human beings. -They have blunt broad ridges. -They are bigger than premolars. -They have two or three roots. Molars in lower jaw have two roots while those in upper jaw have one root -They have rough blunt tops. -They have many cusps and ridges. -They are the largest teeth.	For crushing, grinding and chewing food. **Adaptation** They have blunt broad ridges to grind and chew food. -They are broad to increase the surface area for chewing food. -They have cusps for chewing food.







A girl biting a yellow banana



A boy tearing meat.

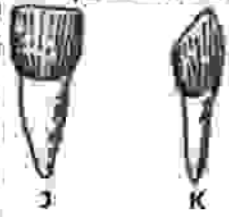
Points to note:

Incisors are well developed in rats, squirrels, moles, mice and rabbits.

Canines are well developed in dogs, cats, lions, leopards and foxes.

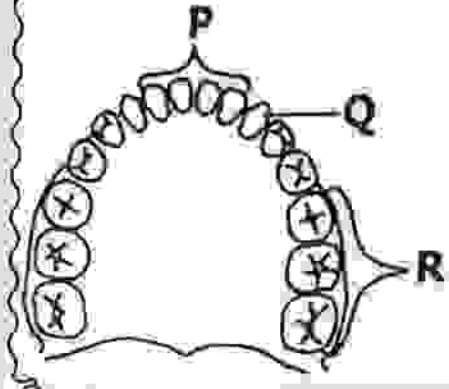
Evaluation activity 7.1

1. Name the type of teeth which develops last in human beings.
2. State any **one** characteristic of incisor teeth which makes them suitable for their function.
3. Name the type of teeth that has;
   (i) Sharp pointed crowns.....(ii) Chisel shaped crowns.....
4. The diagrams below show types of teeth. Use them to answer the questions that follow.



   (a) Name the types of teeth labelled **J** and **K**.
   (b) Give the function of the type of tooth labelled **J**.
   (c) State **one** way in which the type of tooth marked **K** is able to carry out its function during feeding.

5. The diagram below shows the arrangement of permanent teeth in the lower jaw of a human being. Study the diagram and use it to answer the questions that follow.

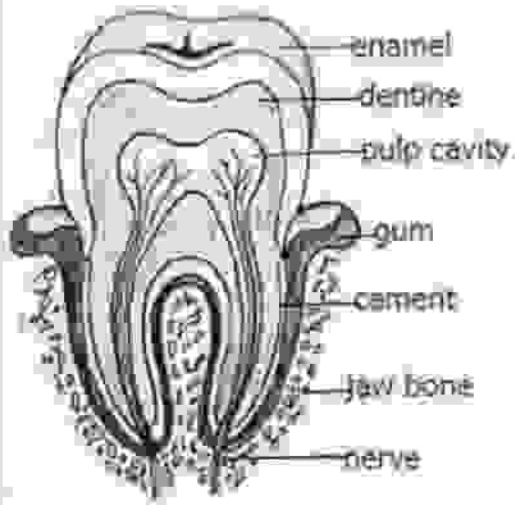


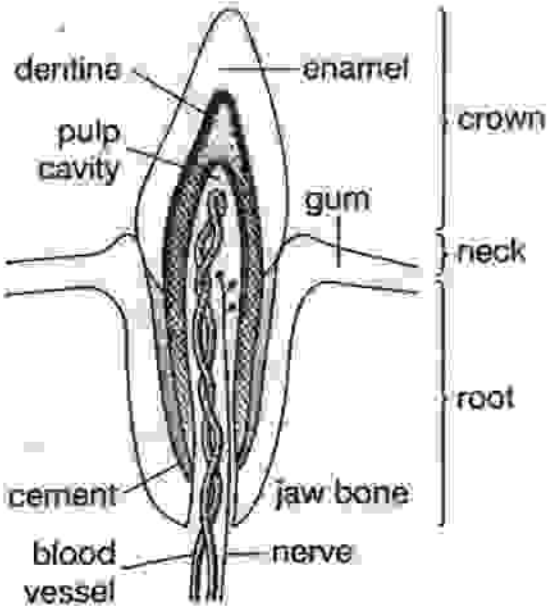
   (a) Name the type of teeth marked **Q**.
   (b) State the function of the types of teeth labelled **P** and **Q**.
   (c) Identify any **one** difference between the structure of the types of teeth labelled **Q** and **R**.

Key words:

• enamel • capillaries • phosphorus • dentine • calcium • pulp

Teeth structure





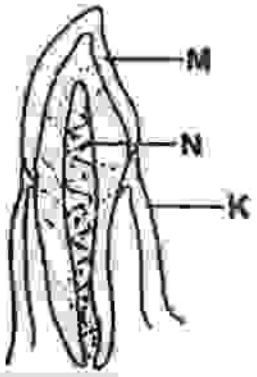
Functions of each part of the tooth.

Part	Description and function
Enamel	• Enamel is the hardest part of the tooth/substance in the body. • It covers the crown of the tooth. • It is the outer most part of the crown. • It is made up of calcium and phosphorus. • It is the white part of the tooth. • It protects the inner parts of the tooth. • It prevents wear and tear of the tooth. **Adaptations to its function.** - *It is hard which makes it resistant to wear and tear.* - *It has smooth surfaces which prevent bacteria from easily sticking onto it.* - *It is thick to withstand chewing and biting forces.*
Dentine	• It is the largest part of the tooth and contains living cells. • It is covered by both the enamel and cement. • It protects the pulp cavity. • It reinforces the enamel and supports the structure of the tooth. • It helps in the formation of the enamel.

Part	Description and function
Pulp cavity	• It contains blood capillaries and nerve endings. • Blood capillaries supply food and oxygen to the tooth • It is the most sensitive part of the tooth. • It contains living cells. It is the living part of the tooth.
Cement	• It holds the tooth firmly into the jawbone. • It covers the root of the tooth.
Jawbone	• It holds the tooth in position. • It gives extra support to the tooth in the jawbone.
Gum	• It contains fibres for holding the tooth in the jawbone. • It covers the jawbone.
Nerves	• Nerves are sensitive to pain, heat or cold. • They carry electrical messages about heat and pain to the brain.

Evaluation activity 7.2

1. Name the part of the human tooth that contains blood capillaries and nerve endings.
2. State **one** function of the enamel in the human tooth.
3. Name the part of the tooth that covers the dentine in the following regions.
   (i) Crown (ii) Root
4. Identify the hardest part of the human tooth.
5. State the difference in function between the pulp cavity and the cemen tum in the tooth.
6. Give the reason why the pulp cavity is regarded as the living part of the tooth.
7. The diagram below shows a section through a human tooth. Study and use it to answer the questions that follow



(a) Name the parts marked **M** and **N**.
(b) Which mineral salt helps to form part **M**?
(c) Give the function of part labelled **N**.

Evaluation activity 7.2 Continued

8. The table below shows parts of the human tooth in part **A** and their functions in part **B**.

Part A: Parts	Part B: Function
Jawbone	Sends impulses of pain to the brain.
Cementum	Holds the tooth in position.
Dentine	Replaces the enamel when it wears off.
Nerves	Fixes the tooth firmly in the jawbone.

Write the correct function to the part of the tooth in the spaces provided below.

(i) Jawbone .....

(ii) Cementum .....

(iii) Dentine .....

(iv) Nerves .....

Signs/indicators of dirty teeth

- Bad breath.
- Bleeding gums.
- Tartar build-up.
- Dental plaque.
- Yellow stains.
- Discolouration.

Care for Our Teeth

❖ **Brushing teeth regularly.**

It helps to remove food remains that attract bacteria in teeth.

Brushing teeth is recommended in the morning and every after a meal.

It is done using a toothbrush, toothpaste, charcoal paste and chewed stick.

Importance toothpaste

- Tooth paste prevents bad breath
- Toothpaste prevents tooth decay.
- Toothpaste removes plaque or stains from teeth.
- Toothpaste makes teeth strong.
- Toothpaste kills germs in teeth.



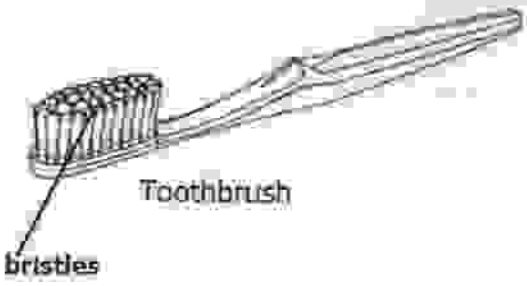
Toothpaste



Charcoal paste



Chewed stick




Brushing teeth

Steps taken when brushing teeth

i. Wet the toothbrush and apply a small amount of toothpaste.
ii. Brush your teeth using gentle circular motions, covering all surfaces-front, back and chewing surfaces.
iii. Brush your tongue to remove food remains.
iv Rinse your mouth and toothbrush thoroughly with water.

Reasons for brushing teeth.

- To remove food remains that attract bacteria in teeth.
- To prevent bad oral breath.

❖ **Dental flossing**

Dental flossing it the removal of plaque that forms between teeth using a thread like material.
It helps to prevent gum disease.



Dental floss


Dental flossing

❖ Using toothpicks to remove food remains from teeth.
❖ Rinsing the mouth with water and salt. Salt kills bacteria.


Toothpicks

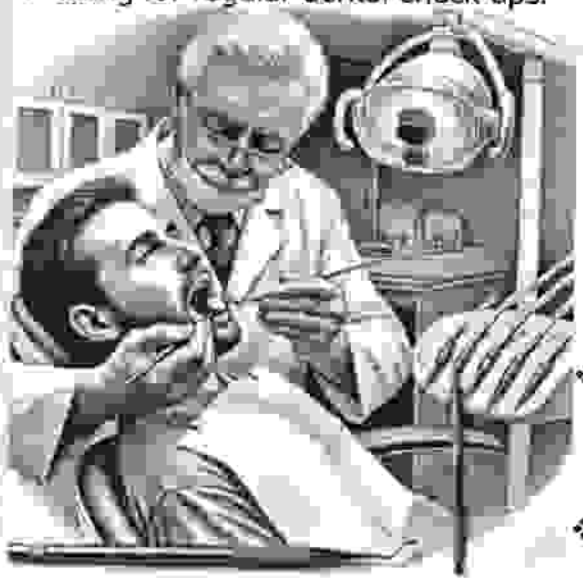

Rinsing the mouth

❖ Eating correct food.
❖ Avoid eating very hot and cold food. This weakens the enamel.
❖ Avoid eating too much sweets and biscuits.
They attract bacteria which cause tooth decay.





❖ Going for regular dental check-ups.



Advantages of regular dental check-ups.
-It helps to detect oral health issues early.
-It helps to maintain fresh breath and clean teeth.
-It helps to reduce the risks of gum disease.

❖ Avoid opening hard objects with teeth.
It causes breaking or cracking of the enamel.
❖ Feeding on plenty of fruits and vegetables to clean the mouth.

❖ Avoid tobacco smoking.
Tobacco smoking causes; Bad oral breath, Tooth discolouration, Build-up of plaque and Gum disease and oral cancer.

Evaluation activity 7.3

1. Name any **one** material that can be used to brush teeth.
2. State any **two** ways in which toothpaste is important when brushing teeth.
3. Give **one** reason why children are discouraged from eating too much sweets.
4. Write down any **two** practices that can destroy teeth in one's mouth.
5. Mention any **two** ways in which tobacco smoking affects teeth in the mouth.
6. State any **one** reason why children are encouraged to brush their teeth regularly.

Evaluation activity 7.3 Continued

The diagram below shows materials used to clean the teeth. Use it to answer questions **7** and **8**.



6. Name the materials shown above.
7. Give any **one** way in which the above materials help to clean teeth.

Teeth Diseases and Disorders.

Tooth decay (dental caries)

It is caused by bacteria.
It is characterised by dental cavity in the teeth.
Foods like chocolates, cakes and sweets attract bacteria that cause tooth decay.



A tooth affected with dental caries

Gum disease (gingivitis)

It is caused by bacteria. It results from poor dental hygiene.

Signs of gum disease

- Swollen gums.
- Bleeding gums when brushing teeth.
- Bright red gums.
- Bad breath from the mouth.

Root abscess

This is a pus-filled infection at the root of a tooth.
It is caused by bacteria.

Signs and symptoms of root abscess

- Severe toothache.
- Fever
- Swelling.
- Sensitivity to hot or cold food.

Periodontitis

This is a severe gum infection that damages the soft tissue and bone-supporting teeth.
It is caused by bacteria.

Signs and symptoms of periodontitis

- Swollen gums.
- Bad breath.
- Bleeding while brushing.
- Loose teeth.

Use mnemonic: GACP *to recall teeth diseases*

Ways in which diseases that attack teeth can be controlled/ prevented.

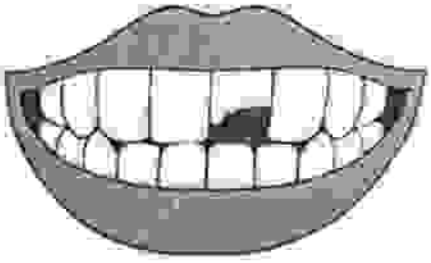
- Regular and proper brushing of teeth.
- Limit the intake of sugary food.
- Regular dental check-ups.
- Dental flossing.
- Avoid smoking.
- Use antibacterial mouthwash.

Dental Disorders

Broken or cracked teeth

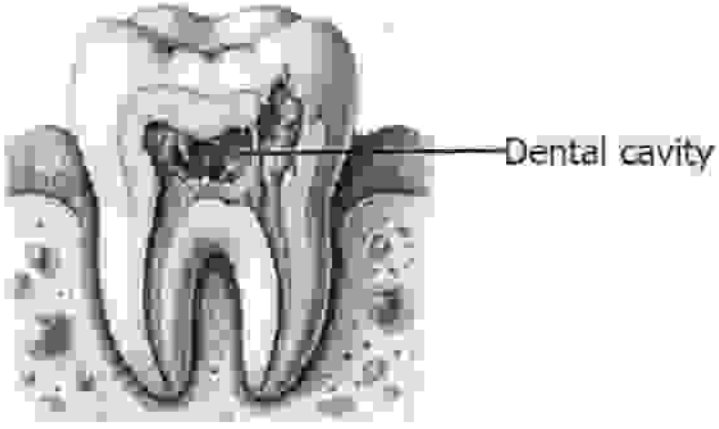
Causes of broken teeth are;

- Falls
- Little calcium
- Opening bottles using teeth
- Eating hard food stuffs like sugarcane.



Dental cavities.

These are holes in the tooth which are caused by bacterial infection.
The holes can be dental filled using dental amalgam.

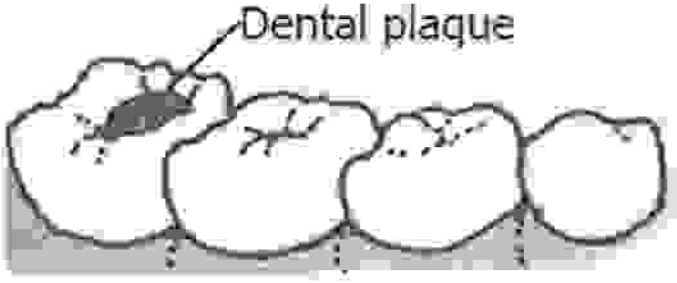


Dental plaque

This is a sticky film of food debris, saliva and bacteria which form on teeth after meals.
Dental plaque forms **tartar** when not removed by brushing.
Tartar is a hard yellowish or brownish substance on teeth caused by plaque.

Dangers of tartar on teeth.

It leads to gum disease and tooth decay.
It gives teeth a very bad appearance.

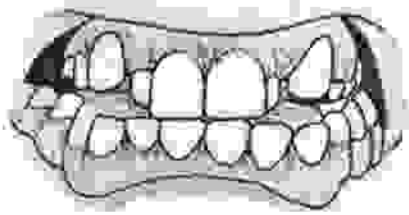


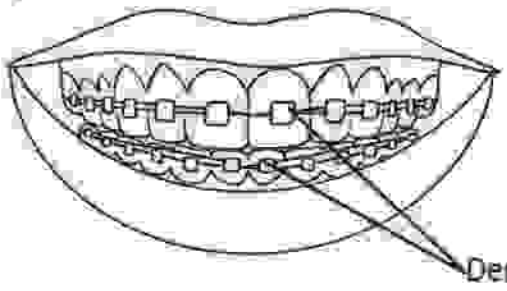
Improper growth of teeth

This is a dental condition in which teeth grow in a wrong way.

Causes of improper growth of teeth.

- Dental accidents.
- Mouth breathing.
- Fingernail biting.
- Lip biting.
- Thumb sucking.
- Early loss of milk.





A dental brace is used to correct misaligned teeth

Signs of improper growth of teeth

- Too much space between teeth.
- Missing teeth.
- Misaligned teeth.
- Some teeth grow on top of others.

Dangers of improper growth of teeth

- Difficulty in chewing.
- Dental diseases can easily develop.
- Poor facial appearance.
- Speech problems.

Other dental disorders

- False teeth.
- Discolouration of teeth.
- Impacted teeth.
- Bruxism (Grinding teeth)
- Malocclusion (Misalignment of teeth in the jaws)

Evaluation activity 7.4

1. Name the germ that causes dental caries.
2. State any **one** characteristic of incisor teeth that makes them suitable for their function.
3. State any **one** way in which toothpaste helps to promote oral health.
4. State any **two** signs of gum disease in humans.
5. Write down any **two** diseases that attack human teeth.

The diagram below shows ill health on teeth. Use it to answer questions **6** and **7**



6. Name the substance marked **D**.
7. How are the above teeth able to grind food?
8. State the importance of sharp pointed crowns to the canine teeth.
9. Apart from brushing, give any **one** other way of preventing dental caries.

MEANING OF KEY WORDS IN TEETH

Enamel. The hardest part of the tooth.
Dental. Relating to teeth.
Tartar. Hard deposit formed on teeth and not removed by brushing.
Dentist. A doctor who treats teeth.
Dentistry. The care and treatment of teeth, gums and oral health.
Canines. Sharp cone-shaped teeth used to tear food.
Incisors. Sharp chisel-shaped teeth at the front of the mouth used to cut food.
Molars. Blunt broad ridged teeth with larger surface area used to grind and chew food.
Dental formula. The number and types of teeth in one's mouth.
Dentition. The arrangement of teeth in the jaws.
Oral health. The general cleanliness of the mouth.
Malocclusion. The misalignment of teeth in the jaws.
Dental plaque. A substance containing bacteria that forms on teeth surface.
Dental cavity. Holes in the teeth.

8 THEME: HUMAN HEALTH
TOPIC 8: SANITATION

Sanitation is the general cleanliness of the environment.
It is the way of keeping an area clean.

Activities that school children can do to promote sanitation

- ✓ Mopping classrooms.
- ✓ Sweeping the compound.
- ✓ Draining stagnant water.
- ✓ Smoking latrines.
- ✓ Scrubbing toilets.
- ✓ Proper disposal of human wastes.
- ✓ Dusting tables and chairs.
- ✓ Clearing bushes around the school.
- ✓ Scrubbing verandas and floors.
- ✓ Constructing urinals.
- ✓ Flushing toilets after use.
- ✓ Proper disposal of rubbish









Materials used to promote sanitation.

Material	Use in promoting sanitation
water	For mopping houses For scrubbing toilets/latrines/paved compounds.
Broom	For sweeping rubbish in the house or compound
Rag/mop	For mopping the floor to remove dirt.

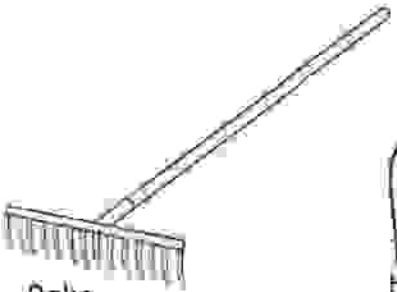
Material	Use in promoting sanitation
Slasher	For cutting tall grass in the compound.
Liquid soap	For mopping and scrubbing the floor and toilets. For disinfecting toilets/latrines
Squeegee	For drying water on floor after mopping.
Floor brush	For scrubbing dirt on the floor.
Detergents	For mopping, scrubbing and disinfecting the floor.
Rake	For collecting rubbish in the compound.
Hoe	For trimming the compound.
Dustpan	For collecting rubbish in the house after sweeping.

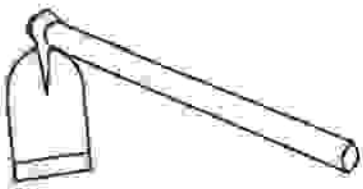

Mop and water


Squeegee


Broom


Floor brush


Rake


Hoe


Slasher


dustpan

Learn, unlearn and relearn: *A squeezer is not used for drying the floor. It is used for making passion fruit juice. Additionally, the steel wire is not used for scrubbing saucepans and other metallic utensils. Steel wool is used for making window frames, door frames. The fibre like material used for scrubbing saucepans is called steel wool.*

Element	Importance/use
Clean latrines/ toilets	For proper disposal of human wastes. Note well. A latrine is a structure built on a pit for disposing human wastes. A toilet is a waterborne facility for disposing human wastes.

Element	Importance/use
Clean bathrooms	For bathing
Rubbish pit/dust-bin	For proper disposal of rubbish.
Plate stand/rack	For proper drying of utensils and cutlery.
Tippy-tap/hand washing facility	For washing hands.
Animal houses.	For sheltering farm animals.

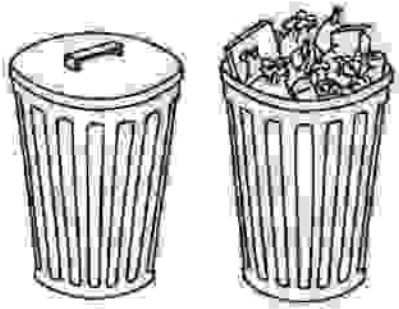
Hand washing facility



Plate stand



Dustbins

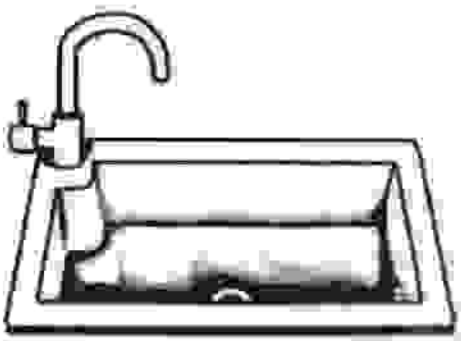




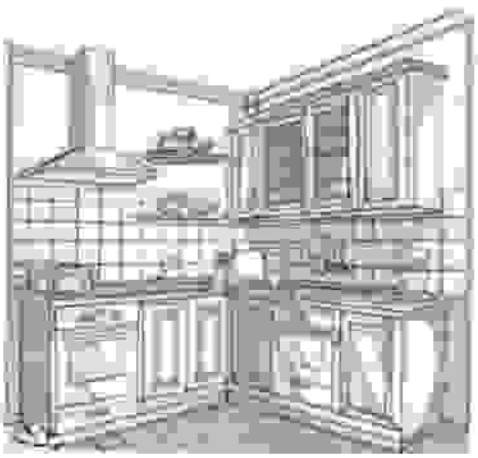
Flush toilet



Pit latrine



Wash basin/sink



Kitchen

Points to note.

The stand has to be raised at least 1 metre above the ground

Reasons

-To prevent animals and young children from dirtying washed utensils and cutlery.

-To prevent water splashes from dirtying utensils in case it is raining.

-To prevent dust from dirtying washed utensils.

A sink is used for washing utensils and cutlery.

A dustbin should have a lid

Reasons.

To prevent rubbish from coming out.

To prevent bad smell from the dustbin from coming out.

To prevent easy spread of germs by houseflies.

Places at home and school that need to be kept clean regularly

- Kitchen
- Latrines/toilets
- Houses
- Compound
- Classrooms
- Bathrooms
- dormitories

Indicators of poor sanitation

- ✓ Bushy compound.
- ✓ littered compound.
- ✓ Poor disposal of faeces, rubbish, domestic wastes.
- ✓ High number of houseflies in the compound.
- ✓ Dusty classrooms.
- ✓ Pools of water in the compound.

Dangers/disadvantages of sharing the house with domestic animals.

- It leads to the spread of ectoparasites.
- It leads to competition for oxygen.
- It leads to bad smell in the house.
- It leads to spread of some diseases.
- Animals can cause noise pollution in the house
- Animals destroy household

Dangers of poor sanitation in an area.

- ↕ It leads to easy spread of diarrhoeal diseases.
- ↕ It leads to water and air pollution.
- ↕ It leads to contamination of food by houseflies.
- ↕ It encourages the breeding of vectors and germs.

Importance of promoting proper sanitation

- ❖ It prevents easy spread of diarrhoeal diseases.
- ❖ It prevents air and water pollution.
- ❖ It reduces the breeding of disease vectors and germs.
- ❖ It controls bad smell in the environment.

Messages that can be displayed in the school compound to promote sanitation at the school.

- Do not litter the compound.
- Keep the compound clean daily.
- Flush the toilets after use.
- Dump rubbish in the dustbins/rubbish pits.
- Empty dustbins when full.
- Report leaking hand washing facilities.
- Use the toilets/latrines properly.

Activities a School Health Club can do to promote sanitation at school.

- Providing dustbins at school.
- Providing handwashing facilities.
- Organizing smoking of pit latrines.
- Conducting talks on sanitation practices
- Digging a rubbish pit at school.
- Creating posters and signs having sanitation messages.

Evaluation activity 8.1

1. (a) Give the meaning of the term sanitation.
(b) Identify any **two** activities that school children can do to promote sanitation at school.
(c) Mention any **one** place at school that needs to be kept clean.
2. The diagram below shows a sanitary item. Study and use it to answer the questions that follow.



(a) Name the sanitary item shown above.
(b) How is the above item useful in the promotion of sanitation at home?
(c) State any **two** ways of caring for the sanitary item above.

3.(a) State any **two** indicators of poor sanitation in a school.
(b) Give any **two** diseases that can break out due to poor sanitation in a school.

4.(a) Give **two** reasons for placing the utensils on a rack.
(b) Why should a rack be constructed with stands at least 1 metre above the ground?
(c) Give any **one** danger of leaving utensils outside for a long time on the rack.

5. The diagram below shows a hand washing facility. Study and use it to answer the questions that follow.



(a) Name the hand washing facility shown above.
(b) How does the above hand washing facility help to prevent the spread of diarrhoeal diseases?
(c) What is the importance of the stick on the above hand washing facility?
(d) State any **one** way of keeping the hand washing facility above clean.

Key words

• inhale • microscope • protozoa • hairy • influenza • microbe

GERMS AND DISEASES

Germs are organisms that cause diseases.

Germs are also called **pathogens**.

Most germs are microbes/microscopic because they cannot be seen using our naked eyes.

A microscope is used to see germs.

It is commonly used by health workers to see germs in blood, stool and saliva. This enables them to give us the correct treatment.

Characteristics of germs.

- They cause diseases.
- Most of them are microbes.
- They reproduce faster.



A microscope

Examples/types of germs

Germs	Diseases they cause
bacteria	tuberculosis typhoid scabies cholera
fungi	ringworm infection athlete's foot dhobi's itch jock itch
viruses	AIDS influenza polio measles
protozoa	malaria sleeping sickness
Worms	bilharzia, river blindness and elephantiasis

Places where germs can be found.

• soil • air • latrines/toilets • rubbish pits
• rotting matter • unboiled water

Ways in which germs spread.

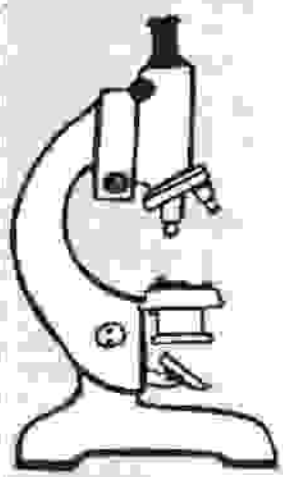
- Through drinking unboiled water.
  Clean water is not safe water. Clean water still contains germs.
- Through body contact.
  Body parts through which germs can spread are skin, hands, mouth and shoulders.

- Through bites of insects or dogs.
  Such insects include; mosquitoes, tsetse flies, ticks and lice.
- Through inhaling contaminated air.
  Bacteria and viruses can spread through air.
  Diseases like Influenza, COVID 19, tuberculosis, whooping cough and mumps can spread through inhaling contaminated air.
- Through sharing skin cutting objects.
  Some germs are found in blood. Objects like razorblades, needles, syringes and safety pins can spread germs in the body.
  Medical cutting or piercing equipment are sterilised to kill germs. Alcohol is mostly used.
- Through sharing clothes, combs, towels and bedding.
  Fungi and bacteria can easily spread through sharing such materials.
  Ringworm and scabies can spread fast when such materials are shared.
- Through the hairy body of some insects.
  Houseflies and cockroaches have a hairy body on which germs attach.
  Hairs are sticky for germs to easily attach onto them.

Evaluation activity 8.2

1. Name the group of organisms that cause diseases in human beings.
2. (a) What is meant by the term germs?
   (b) Apart from bacteria and viruses, name any **two** other examples of germs.
   (c) State any **two** ways in which human beings contract germs.
3. (a) List down any **two** places where germs can be found.
   (b) State any **two** characteristics of germs.
4. How is a housefly able to spread germs?
5. Write down any **two** diseases that can spread through air.

The diagram below shows a medical equipment. Study and use it to answer questions **6** and **7**.



6. Name the medical equipment shown above.
7. How is the medical equipment shown above useful to a doctor?
8. State any **one** danger of sharing stockings with your fellow pupils at school.
9. Why are germs called microscopic organisms?
10. Mention any **one** body fluid where germs can be found.
11. Why should health workers sterilise medical cutting equipment before using them to other patients?
12. Name any **two** insects that spread germs through bites.
13. How do cockroaches spread germs to people?

Key words

• fingers • faeces • hairy • diarrhoeal • disposal • rubbish

THE 4Fs GERM PATH

4Fs germ path refers to the routes through which houseflies spread germs that cause diarrhoeal diseases to humans.

A housefly has a hairy body to carry germs.

The 4Fs in germ path stands for Faeces, Flies, Food and Fingers.

How to explain the 4Fs germ path.

(i) Germs breed in faeces.

(ii) Flies pick germs from faeces using their hairy bodies onto the food.

(iii) The food is then contaminated by the germs put by the houseflies.

(iv) The person uses fingers to eat the contaminated food causing diarrhoeal diseases.



Diseases that are spread through the 4Fs germ path.

These diseases are called diarrhoeal diseases.

Examples of diarrhoeal diseases are; diarrhoea, dysentery, typhoid and cholera.

Ways in which the spread of germs can be controlled in an area.

❖ Covering leftover food properly.
It prevents houseflies and cockroaches from landing on food.
It prevents dust from getting on food.

❖ Collecting and burning rubbish.
It destroys the breeding of houseflies and cockroaches.
Houseflies lay eggs in decaying matter.

Reasons.

- To enable their larvae get food after being hatched.
- Presence of warmth for eggs to hatch.
- Presence of moisture for the eggs to hatch.

❖ Washing hands with clean water and soap.
To remove and kill germs from hands.

❖ Serving, cooking and keeping food in clean places and containers.

❖ Drinking boiled water.
Boiled water is free from germs.
Boiling water kills germs.

Ways in which rubbish can be disposed in an area.

✓ Burning rubbish. ✓ Burying rubbish.

✓ Wearing face masks. ✓ Dumping rubbish in dustbins or rubbish pits.

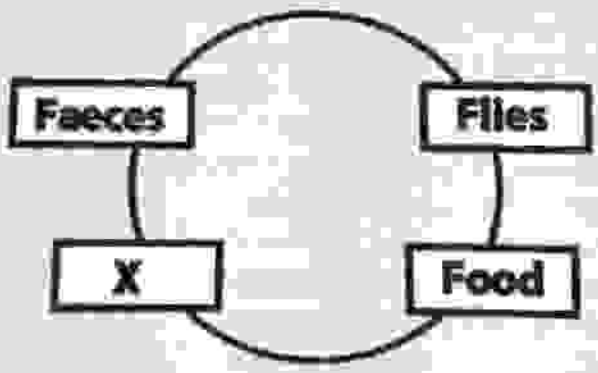
✓ Using decayable rubbish in production of biogas.
✓ Using decayable rubbish in making manure.

Learn, unlearn and relearn: ***It is wrong to say that we wash our hands with clean water to kill germs because water has no chemicals which kill germs on hands. It is only soap which kill germs on hands. Therefore, we wash hands with clean water, to remove germs from hands.***

Additionally, we wash our hands with clean water and soap to remove and kill germs from the hands.

Evaluation activity 8.3

1. Write down any **two** of the 4Fs in the transmission of diseases.
2. What type of diseases are spread through the 4Fs germ path?
3. Which vector is active in the 4Fs germ path?
4. State any **two** ways in which germs can spread through the 4Fs germ path.
5. How is a housefly able to spread germs to human beings?
6. The diagram below shows how germs can spread through the 4Fs. Study and use it to answer the questions that follow.



(a) Identify the word which should be in the box **X**.
(b) State any **two** activities that can be done to prevent the germ path shown in the diagram above.
(c) What structure on the body of flies in the germ path above enables it to carry germs onto food?
7. What health habit should you do to after visiting a latrine/toilet?
8. Why do houseflies lay eggs in decaying matter?
9. Write down any **two** diseases that can spread through the 4Fs germ path.
10. How does proper covering of food control germs spread by houseflies?

Germs cause rotting

Rotting is the breakdown of dead organic matter by bacteria or fungi.
It is also called decomposition or decaying.
The fungi that cause rotting are called **moulds.**
It requires warmth and moisture.

Organic materials that can rot are; food, animal dung, paper, faeces, plant leaves, dead bodies

Importance of rotting.

✓ It produces humus from dead organic matter.
✓ It destroys garbage heaps.
✓ It destroys faeces in latrines and sewage systems.

Note. It is not advisable to pour acid in latrines and septic tanks.
Reason. Acid kills bacteria that cause rotting of faeces.

Dangers of rotting.

- It causes wounds to become septic.
- It makes food to go bad.
- It is a source of germs.
- It attracts vectors.

Signs of rotting on food.

- Growth of moulds on food.
- Bad smell.
- Change in colour.
- Presence of maggots

Ways in which rotting can be prevented in an area.

- Heating food.
- Putting food in refrigerators.
- Smoking food.
- Proper sun drying of food.
- Eating food immediately after cooking.

Ways of preventing diseases without using drugs.

- Feeding on a balanced diet.
- Doing regular physical exercises.
- Having enough rest and sleep.
- Avoid smoking and alcoholism

Evaluation activity 8.4

1. What is meant by the term **rotting**?
2. State any **one** condition favourable for rotting to take place.
3. Name any **one** organism that causes rotting.
4. Give any **two** ways in which rotting is useful in the environment.
5. State any **two** dangers of rotting in the environment.
6. Why is it not advisable to pour acid in latrines or septic tanks?
7. Write down any **two** substances that can rot in the environment.
8. State any **two** changes on food that show that it is rotting.
9. Give any **two** ways of preventing diseases without using essential drugs.
10. Identify the process that reduces the volume of faeces in latrines and septic tanks.
11. State any **two** ways of preventing rotting of food at home.
12. Name the fungus that normally grows on leftover food.
13. In **four** sentences, describe an experiment to show rotting of a ripe banana fruit.

Evaluation activity 8.4 Continued

13. The diagrams below show equipment used to promote sanitation at home. Study and use them to answer the questions that follow.



R



S

(a) Name the equipment marked **R** and **S**.
(b) State the importance of the equipment marked S in the promotion of sanitation at home.
(c) Give any **one** reason why most homes do not use the equipment marked **R** to promote sanitation.

MEANING OF KEY WORDS IN SANITATION

Sanitation. The general cleanliness of the environment.
Germs. Organisms that cause diseases.
Mower. A device used to cut grass in fields or lawns.
Rag. A piece of old cloth used for mopping.
Microscope. An instrument used to see germs.
Latrine. A place for urination and defecation.
Mop. An implement that consists of a bundle of thick loose strings or a sponge attached to a handle, used for wiping floors.
Excreta. Human wastes especially urine and faeces.
Rotting. The breakdown of organic matter by bacteria or fungi.
Squeegee. A tool for washing floors, that has a long handle with two thick pieces of soft materials at the end used to remove water or other liquids from the floor.
Environment. The surrounding of an organism.

9 THEME: HUMAN HEALTH
TOPIC 9: COMMUNICABLE INTESTINAL DISEASES AND WORM INFESTATION

Key words:
- intestinal
- communicable
- diarrhoeal
- dysentery
- amoeba
- microscope

Communicable intestinal diseases are infections that attack the intestines of humans.
Communicable diseases are diseases that spread from one person to another.
They are also called infectious Intestinal diseases.
They are caused by germs like; bacteria, viruses and protozoa.

DIARRHOEAL DISEASES.
Diarrhoeal diseases are infections that make a person to pass out watery stool frequently

Ways in which communicable intestinal diseases spread.
- By houseflies or cockroaches.
- Eating contaminated food.
- Drinking contaminated water.
- Using contaminated utensils.
- Through the 4Fs germ path.

Factors that can lead to the outbreak of diarrhoeal diseases in an area.
- Poor disposal of human wastes.
- Poor disposal of human wastes.

The above vectors increase the breeding of houseflies which spread germs that cause diarrhoeal diseases.
Note: Heavy population in an area leads to poor disposal of human wastes and rubbish which attracts house flies which spread germs.

Examples of diarrhoeal diseases
- dysentery
- cholera
- typhoid

Diarrhoea
Diarrhoea is the frequent passing out of watery stool.
Diarrhoea can be caused by bacteria or viruses

Signs/symptoms of diarrhoea
- dehydration
- abdominal pain
- headache.
- general body weakness
- watery stool



Effects of diarrhoea to human health.
- ✓ It causes dehydration.
- ✓ It causes loss of potassium and sodium.
- ✓ It causes emaciation.
- ✓ It can lead to death through dehydration.

3Ds i.e. diarrhoea, dehydration and death.

Dysentery

Dysentery is the frequent passing out of blood-stained stool.

Dysentery is caused by bacteria or protozoa

Forms of dysentery in humans

- bacillary dysentery
- amoebic dysentery

Signs and symptoms of dysentery

- Blood-stained stool
- Painful egestion
- Vomiting
- Diarrhoea
- Dehydration
- Abdominal pain
- Loss of appetite
- Fever

Cholera

Cholera is a very dangerous disease which kills many people within a short time. Cholera is caused by bacteria.

Signs and symptoms of cholera.

- Severe vomiting
- Severe diarrhoea
- Dehydration
- Abdominal pain
- General body weakness.

Typhoid

Typhoid is caused by bacteria.

Signs and symptoms of typhoid.

- Vomiting
- headache
- fever
- stomach aches
- diarrhoea
- persistent fever
- loss of appetite

Evaluation activity 9.1

1. Name the intestinal disease that causes frequent emptying of bowels.
2. State any **one** way in which the spread of diarrhoea can be controlled.
3. Name any **one** mineral salt lost in the human body during diarrhoea.
4. Write **DDD** in full as used in communicable intestinal diseases.

The diagram below shows a child suffering from an intestinal disease. Use it to answer questions **5** and **6**



5. Name any one intestinal disease that the child could be suffering from.
6. State any **one** effect of the intestinal disease above to a child.
7. (a) Name the germ that causes typhoid.
   (b) Write down any **two** ways in which you can identify a patient suffering from typhoid.
   (c) State any **one** way of controlling the spread of typhoid in human beings.
8. Name the communicable intestinal disease characterized by passing out blood-stained stool.
9. (a) Name the germ that causes cholera.
   (b) Explain why cholera is common in heavily populated communities (2 marks)
   (c) Apart from proper use of latrines, mention any **one** other way of controlling the spread of cholera in the community.

Key words

▪ germ ▪ microscope ▪ hairy ▪ faeces ▪ intestines ▪ fingers

How some common intestinal communicable diseases are spread through 4Fs.

THE 4Fs GERM PATH

4Fs germ path refers to the routes through which houseflies spread germs that cause diarrhoeal diseases to humans.

A housefly has a hairy body to carry germs.

The 4Fs in germ path stands for Faeces, Flies, Food and Fingers.

How to explain the 4Fs germ path.

(i) Germs breed in faeces.

(ii) Flies pick germs from faeces using their hairy bodies onto the food.

(iii) The food is then contaminated by the germs put by the houseflies.

(iv) The person uses fingers to eat the contaminated food causing diarrhoeal diseases.



Diseases that are spread through the 4Fs germ path.

• diarrhoea • dysentery • typhoid • cholera

Ways in which the spread of communicable intestinal diseases can be controlled.

- Proper disposal of rubbish and wastes.
- Scrubbing toilet/latrines
- Drinking boiled water.
- Serving food in clean container.
- Spraying insecticides to kill houseflies.
- Washing hands after visiting the latrine/toilet.



Evaluation activity 9.2

1. Name the intestinal disease that causes frequent emptying of bowels.
2. (a) Write down any **two** of the 4Fs in germ path.
   (b) Which vector that transmits germs in the 4Fs germ path.
   (c) State any **one** way in which the spread of diarrhoea can be controlled.
3. The sentences below show the ways in which houseflies carries germs through the 4Fs germ path in wrong order.

(a) The food is then contaminated by germs carried by the houseflies.
(b) The fingers touch contaminated food.
(c) Germs breed in faeces.
(d) Houseflies pick germs from faeces with their hairy bodies.

Write the sentences above in the spaces provided below to show the correct ways in which houseflies carry germs through 4Fs germ path.

(a) .....
(b) .....
(c) .....
(d) .....

4. The diagram below shows a good health habit practiced by a child. Study and use it to answer the questions that follow.



(a) Name any **one** health condition controlled by the above health habit.
(b) State any **two** conditions that can make a child to practice the health habit shown above.
(c) Give any **one** reason why the child is practising the above health habit.

Key words

- intravenous - sodium - potassium - severe - diarrhoea - fluid

DEHYDRATION

Dehydration is a health condition in which the person does not have enough water in the body.

During dehydration, water, mineral salts and energy are things lost in the person's body.

Causes of dehydration in humans

Severe diarrhoea

The body loses too much water through frequent watery stool.

Severe vomiting

The body loses water through vomited fluids.

Severe burns or scalds

The skin loses water when it is badly burnt.

Signs and symptoms of a dehydrated patient.

- Dry lips.
- Sunken eyes.
- Little or no urine.
- Sunken fontanelle. (soft spot on the head)
- Little or no tears.

Experiment to demonstrate dehydration in a baby

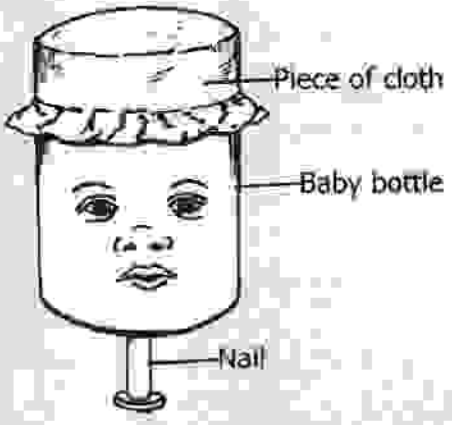
Making a baby bottle

Materials needed: a plastic bottle, a nail, a piece of cloth, water, a rubber band

Steps taken.

(i) Cut the top of a plastic bottle and make it flat on top.
(ii) Make a hole at the bottom of the bottle using a nail.
(iii) Fill the bottle with water while the hole is closed.
(iv) Place a piece of cloth on the top of the bottle and open the hole by removing the nail.

Set up :



Observation.

When the nail is removed slightly from the bottle, the water pours out and the piece of cloth becomes sunken. This proves that the soft spot on the baby's head/fontanelle becomes sunken when the baby becomes dehydrated.
The piece of cloth acts as the fontanelle/soft spot on the head.
The baby bottle acts as the baby's head/baby's body.

Effects of dehydration to human health

- It leads to death of a person.
- It leads to loss of sodium and potassium.
- It makes one's body weak.

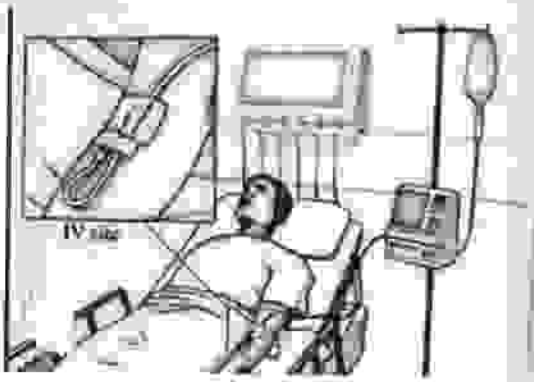
Prevention and Treatment of dehydration in a patient.

-Giving the patient oral rehydration solution.
-Giving the patient a lot of fluids like fruit juice.

- Giving the patient a lot of fruits to eat e.g. watermelon.
- Using drips (intravenous injection)



Giving oral rehydration solution to baby



Giving intravenous injection to a dehydrated patient.

Rehydration is the replacement of the lost water in the body of a dehydrated patient.

Substances that can be used during rehydration include; water, fruit juice and fruits

How to prepare salt sugar solution for treating dehydration at home.

Materials needed: Salt, sugar, teaspoon, boiled water and clean container.

Steps taken.

(i) Wash hands with clean water and soap.
(ii) Measure one litre of boiled water into a container.
(iii) Measure one teaspoonful of salt and eight teaspoonfuls of sugar.
(iv) Stir the mixture to form a solution.

Set up :



Demonstration Set up.



Points to note:

- Sugar and salt are solutes used to make salt sugar solution and water is the solvent.
- Salt is used to replace the lost mineral salts in the body. These mineral salts are sodium and potassium.
- Sugar is used to replace the lost energy in the body.
- Water replaces the lost fluids in the body.
- Washing hands helps to remove germs and dirt from hands
- Using a clean container helps to prevent contamination of the solution.
- A baby should take ¼ cup of the solution every after a stool while an adult child should take ½ cup of the solution every after a stool.
- The solution should be served within 24 hours. After this time, the solution goes bad.
- Dehydration in humans is similar to wilting in plants. Both cause water loss.

How to prepare oral rehydration solution (ORS) from factory-made packets for treating dehydration at home.

Materials needed. Sachet of oral rehydration salts, 1 litre of boiled water and clean container.

Steps taken:

(i) Wash hands with clean water and soap.
(ii) Measure one litre of boiled water into a clean container.
(iii) Open the sachet of oral rehydration salts and empty it into the water.
(iv) Stir the mixture well and give the solution to the patient.

Set up:

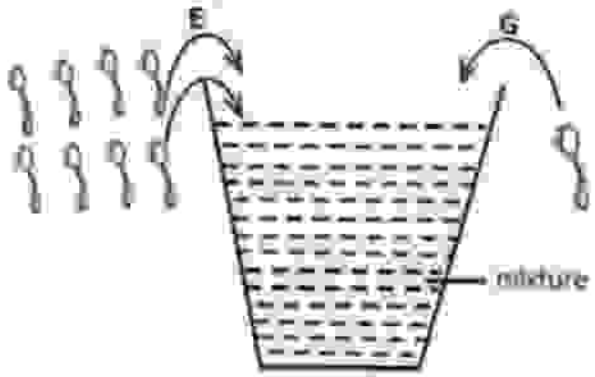


Evaluation activity 9.3

1. (a) Give the importance of each of the following components of oral rehydration solution (ORS) to a patient.
   (i) Sugar .....
   (ii) Water .....

(b) Apart from using ORS, give any **two** other ways of replacing fluids in a patient.

2. The diagram below shows substances used in preparation of a certain medical mixture. Study and use it to answer the questions that follow.



(a) Name the substances marked **E** and **G**.
(b) State the importance of the substance **E** when preparing the medical mixture above.
(c) Which human health condition makes a patient to drink the medical mixture above?

3. You are provided with the following materials:
Salt, sugar, teaspoon, boiled water and clean containers. After washing your hands, write four other steps you will take to prepare salt sugar solution for treating dehydration at home.
(i) .....
(ii) .....
(iii) .....
(iv) .....

Key words:

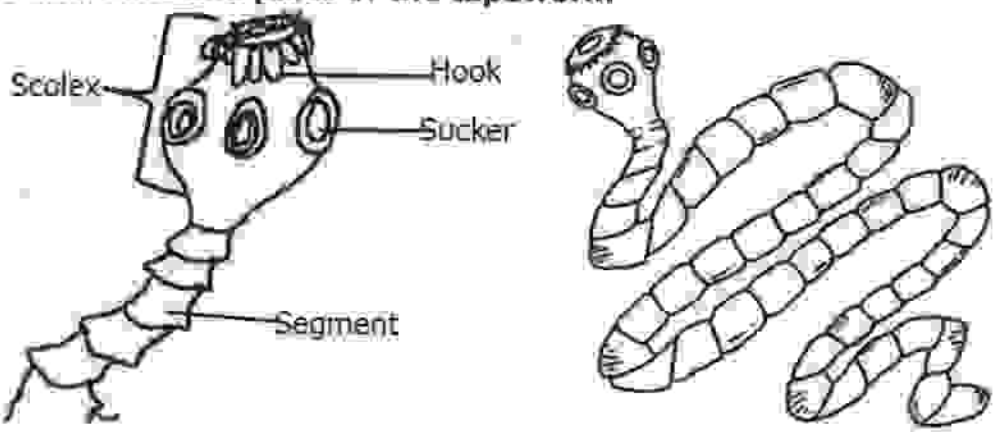
- parasite
- segment
- scolex
- intestinal
- tapeworm
- infestation

WORM INFESTATION

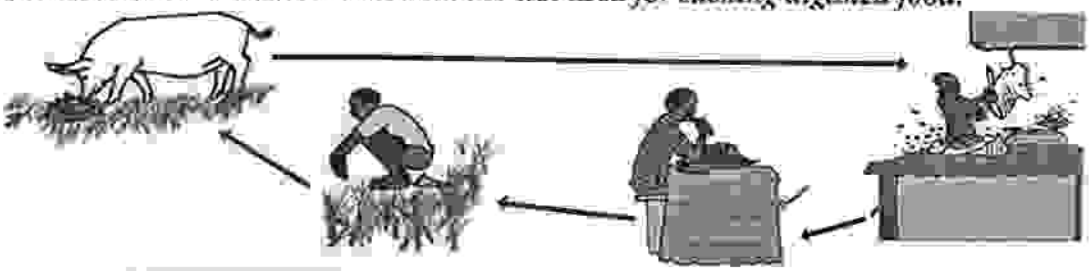
Worm infestation is the presence of worms in the intestines which affect the proper working of the human body.
Intestinal worms are endoparistes/internal parasites because they live in the intestines of the host.
A host is an organism which harbours endoparasites like tapeworms.
Other endoparasites are liver flukes and blood flukes.

Endoparasites live in the liver, kidney and intestine of the host.
A parasite is an organism that depends on another organism for food and shelter.
Intestinal worms obtain digested food, blood and shelter from the host
Examples of common intestinal worms that affect human health.
- tapeworms
- roundworms
- pinworms

Tapeworm infestation
This is a health condition characterised by the presence of tapeworms in the small intestines of the human body.
Tapeworms enter the body through eating undercooked meat. (pork, beef, chevon or mutton)
Tapeworms feed on digested food. This is because they lack the digestive system.
Tapeworms cause malnutrition. They feed on digested food in the body.
The structure and parts of the tapeworm



Functions of parts of the tapeworm.
Hooks: For attachment onto the intestinal walls.
Suckers: For attachment onto the intestinal walls.
Segments:
For storing eggs/for reproduction.
For absorbing digested food from the host.
Segments are porous to absorb digested food from the ileum.
Scolex: It holds the hooks and suckers together. A scolex is the head of a tapeworm.
Learn, unlearn and relearn: suckers are not used for sucking digested food.



Ways in which tapeworms are able to live in the small intestines.

- ✓ They have hooks and suckers for attachment.
- ✓ They have segments to absorb digested food.
- ✓ They have flat and segmented bodies to fit in the ileum.
- ✓ They can grow long and reproduce when inside the host.

Signs and symptoms of tapeworm infestation.

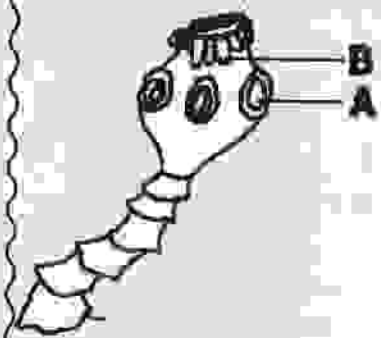
- ❖ Stomach pain.
- ❖ Loss of body weight.
- ❖ General body weakness.
- ❖ Feeling hungry all the time.
- ❖ Pieces of worms in faeces.

Ways in which tapeworm infestation can be controlled/prevented.

- ↕ Eating well cooked meat.
- ↕ Regular deworming.
- ↕ Eating properly roasted meat.
- ↕ Proper disposal of faeces.

Evaluation activity 9.4

1. Name the health condition caused by the presence of tapeworms in the intestines.
2. Apart from digested food, give **one** other way in which tapeworms benefit from the small intestines.
3. State any **one** way in which tapeworm infestation affects human health.
4. Write down any **one** foodstuff that can lead to tapeworm infestation when not properly cooked.
5. How does tapeworm infestation lead to malnutrition in humans?
6. The diagram below shows the head of a tapeworm. Study and use it to answer the questions that follow.



(a) Name the parts marked **A** and **B**.
(b) How is part marked **B** useful to a tapeworm?
(c) What kind of food does the worm above feed on?

7. (a) Mention any **one** sign of tapeworm infestation in humans.
(b) Give any **two** ways of controlling tapeworm infestation in humans.

(c) State any **one** way in which tapeworms are able to live in the small intestines.

2. In which way is the function of the suckers similar to that of the hooks on a tapeworm?
3. Why are tapeworms called endoparasites?
4. Name the group of organisms that depend on the other organisms for food an shelter.
5. A butcher slaughtered a cow and found tapeworms in the intestines of the cow.
   Identify the following from the statement above.
   (i) Parasites:.....(ii) Host:.....

Key words:

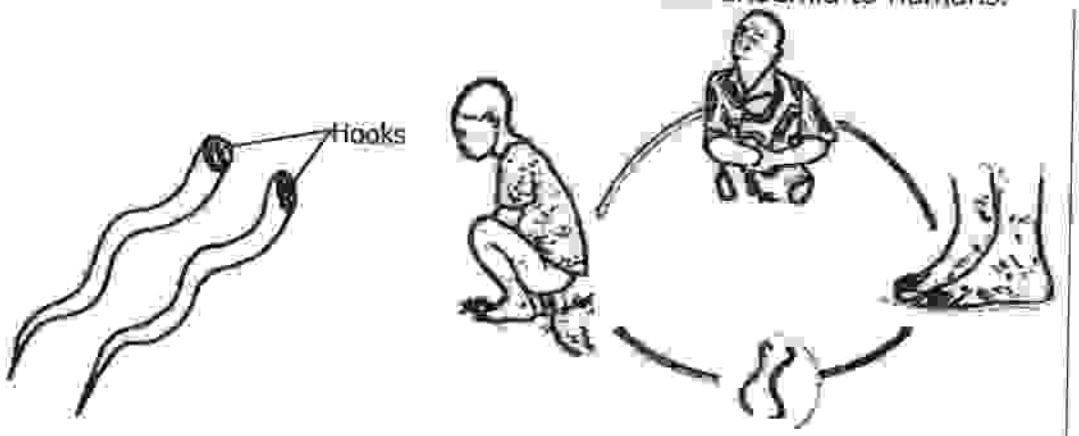
- hook
- ascari
- abdominal
- itching
- pale
- presence

Roundworm infestation

This is a health condition characterised by the presence of hookworms and ascaris in the intestines of the human body.

Hookworms enter the body through penetrating the bare skin. They have hooks for attachment onto the intestinal walls.

Hookworms feed on blood. This causes hookworm anaemia to humans.



Learn, unlearn and relearn: hookworms do not enter the body through walking bare footed because we do not walk in the residential house while wearing shoes or sandals.

Pinworms (threadworms)

Characteristics of pinworms.

- They live in large intestines.
- They are white in colour.
- They lay eggs outside the anus.
- They cause itching around the anus.
- They look like threads.
- They feed on blood.



Signs and symptoms of pinworms.

- Itching around the anus.
- Stomach ache
- Cough
- Restlessness
- Loss of appetite.
- Weight loss

Evaluation activity 9.5

1. Name the intestinal worm that penetrates through bare skin.
2. Mention any **two** signs of roundworm infestation in humans.
3. (a) Write down any **two** things that intestinal worms obtain from the human body.

(b) State any **two** ways in which intestinal worms affect the health of a child.

4. State the difference between the feeding of tapeworm and that of a hookworm.
5. How does hookworm infestation lead to anaemia in humans?

Evaluation activity 9.5 Continued

The diagram below shows intestinal worms. Study and use it to answer questions **6** and **7.**



6. Name the intestinal worms shown in the diagram above.
7. How do intestinal worms above enter the intestines of humans?
8. Name the intestinal worms that cause severe itching at the anus of a child.
9. State any **two** characteristics of threadworms.
10. Write down any **two** intestinal worms that feed on blood in the human body.

Meaning of key terms in communicable intestinal diseases and worm infestation

Host: An organism that harbours a parasite.
Scolex: The head of a tapeworm.
Parasite: An organism that depends on another organism for food and shelter.
Anaemia: A condition when the body lacks blood.
Diarrhoea: The frequent emptying of bowels.
Dysentery: The passing out of blood-stained stool.
Dehydration: A condition in which the body lacks water.
Rehydration: The replacement of the lost water in the body.
Deworming: Introduction of drugs in the body to kill endoparasites.
Communicable diseases: Diseases that spread from one person to another.
Diarrhoeal diseases: Diseases that makes a person to frequently pass out watery stool.
Worm infestation: The presence of parasitic worms in the body.

10 THEME: HUMAN HEALTH
TOPIC 10: VECTORS AND DISEASES

Key words:

▪ vector ▪ spread ▪ hairy ▪ maggot ▪ dysentery ▪ bedbugs

Common Vectors.

Vectors are organisms that spread germs to humans.
Germs are organisms that cause diseases.

Examples of common vectors

- blackflies
- houseflies
- tsetse flies
- cockroaches
- water snails
- ticks
- lice
- fleas
- mosquitoes



mosquito



housefly



snail



tick



tsetse fly



flea



cockroach



louse

Characteristics of disease vectors.

- They have hairy bodies.
- They have specialised feeding habits.
- They have complete or incomplete life cycles.
- They breed in water, dirty places or dark corners.
- Germs are able to survive in their bodies.
- Some vectors bite.

Ways in which vectors spread diseases to humans.

✓ Through bites.
✓ Through body contact.
✓ Through contaminating food.

BLACKFLIES AND COCKROACHES

BLACKFLY

It is also called simulium fly, simulid fly, Jinja fly

Characteristics of a blackfly.

- It has broad and colourless wings.

- A blackfly lays eggs in fast flowing water.

Reason.

Its eggs and larva need highly oxygenated water to develop.

- A black fly undergoes complete metamorphosis.
- It spreads river blindness to people.

River blindness is caused by worms.

- They can be carried in blood to the eyes causing blindness.



Signs and symptoms of river blindness

- Itching of the eyes.
- The eyes turn red.
- Blindness
- Swelling of the skin.
- Inflammation of the iris

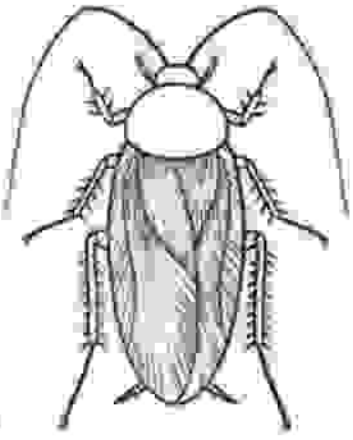
Prevention and control of river blindness.

- Spray insecticides to kill blackflies.
- Treat the infected people.
- Clear vegetation on the banks of rivers.
- Stay away from river banks.

COCKROACHES

Characteristics of cockroaches.

- They have long flattened bodies to fit in small spaces.
- They are mostly brown in colour.
- They have mandibles for chewing food materials.
- They are active at night.
- They live in dark places because they fear light.



Places where cockroaches live.

Drawers, cupboards, bookshelves, latrines rubbish and cracked walls in corners

Substances eaten by cockroaches.

Papers, food, soft wood, faeces

Ways in which cockroaches spread diseases.

- They carry germs on their hairy bodies.
- Landing on food.
- They leave their droppings on food.
- They spit or vomit on food.

Diseases transmitted by cockroaches.

- typhoid
- dysentery
- diarrhoea
- cholera
- polio
- leprosy

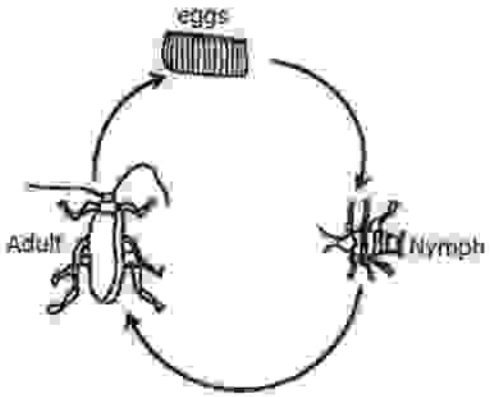
Note well: The cockroaches spread bacteria that cause food poisoning.

The above diseases are caused by bacteria except polio which is caused by viruses.

The lifecycle of a cockroach.

- It undergoes incomplete metamorphosis. This is because the nymph resembles the adult in structure and function.
- The stages are eggs, nymph and adult.
- The eggs are laid in hard brown cases in dark places which stick on doors, cupboards and drawers to keep eggs from being eaten by predators.

- The eggs hatch into the nymph. The nymph resembles the adult stage but it lacks wings.
- Nymphs grow by moulting. After last moulting, the nymph becomes an adult and wings grow.



Note well. Incomplete metamorphosis is the type of lifecycle in insects where the nymph resembles the adult in structure and function.

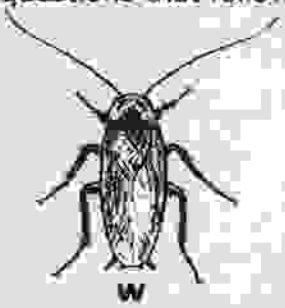
All insects that have mandibles (chewing insects) undergo incomplete metamorphosis.

Prevention and control of cockroaches.

- Covering all the food properly.
- Keeping the house clean.
- Smoking the latrine regularly.
- Spraying insecticides to kill cockroaches.
- Putting mothballs in clothes to chase cockroaches.

Evaluation activity 10.0

1. Name the vector that transmits polio disease to people.
2. How do cockroaches spread diseases to people?
3. Apart from cockroaches, list down any **two** other common vectors to people.
4. The diagrams below show disease vectors. Study and use it to answer the questions that follow.


W


X

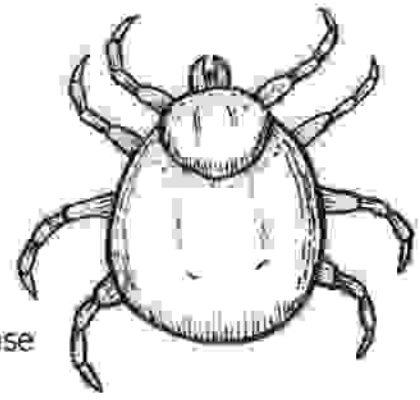
(a) Name the disease vectors marked **W** and **X** above.
(b) State any **one** way in which the vector marked W transmits diseases to humans.
(c) Identify the disease transmitted by the vector marked **X** to people.
5. Which vector causes food poisoning to people?
6. State any **two** ways in which the breeding of bedbugs can be controlled at home.
7. (a) State any **two** ways in which vectors spread diseases to humans.
(b) Give any **two** ways of preventing vectors from spreading diseases to humans.
8. Name the vector that spreads river blindness to people.
9. Give the reason why blackflies lay eggs in fast flowing water.

Key words:
▪ relapsing ▪ scabies ▪ bubonic ▪ typhus ▪ acaricide ▪ plague

TICKS, FLEAS AND HOUSEFLIES

TICKS

Characteristics of ticks



- They have four pairs of jointed legs.
- They feed on blood.
- They have two main body divisions.
- They have no wings.
- They spread germs through bites.
- They live on dogs, cattle, goats, sheep.

Diseases spread by ticks to humans.

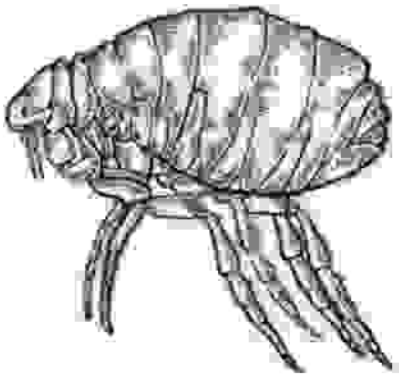
➢ relapsing fever ➢ lyme disease

Prevention and Control of ticks.

- Dipping farm animals.
- Spraying animals with acaricides.
- Hand picking of ticks from livestock.
- Burning old pasture to kill ticks.
- Avoid lying in grass where animals graze.

FLEAS

Characteristics of fleas



- Fleas are very small insects without wings.
- They have proboscis for sucking blood.
- They spread diseases through bites.
- They cause itching of the skin.
- They jump from one place to another.
- They undergo complete metamorphosis.

Fleas have long powerful hind legs to enable them jump from one place to another

Places where fleas live.

clothes, mattresses, cracks of walls, dust, rats, mice and hair of dogs

Diseases transmitted by fleas

- typhus fever
- bubonic plague

Bubonic plague is spread by rat fleas and caused by bacteria.

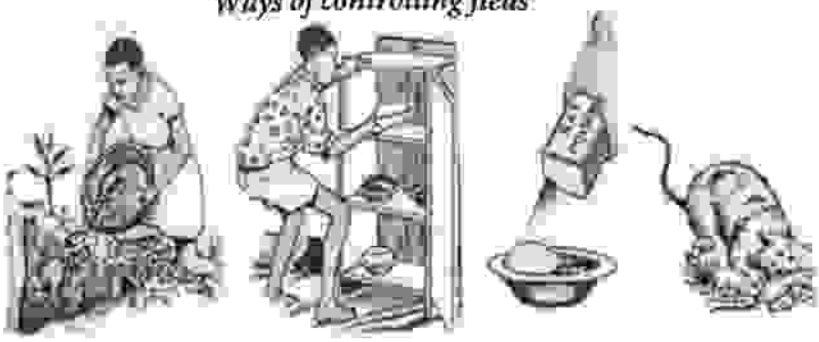
Signs and symptoms of bubonic plague

- Chills
- Swollen glands
- Chest pain
- Muscle pain
- Headache
- High fever
- Swelling in the neck and armpits.

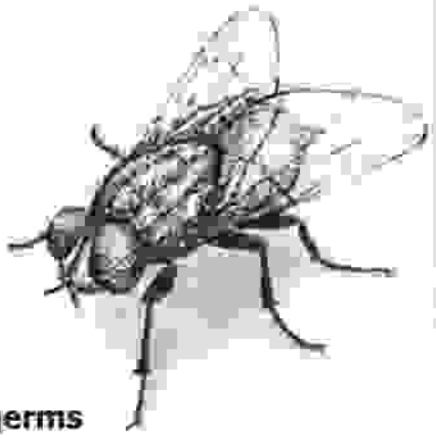
Prevention and control of fleas

❖ Spraying fleas with insecticides.
❖ Keeping the floor clean daily.
❖ Smearing walls with cow dung.
❖ Cementing the floor to reduce dust.
❖ Killing rats.

Ways of controlling fleas



HOUSEFLIES



Characteristics of houseflies.
- They are insect vectors.
- They have a hairy body.
- They have a pair of wings.
- They have proboscis for sucking fluids.
- They feed on rotting or sugary liquids.
  The liquid material is softened by its saliva.
- They spread their wings at rest.

Habitats for houseflies

Dirty places. Such places are rubbish, latrines

Ways how houseflies are able to spread germs

✓ They have a hairy body to carry germs.
✓ They leave their dropping on food.
✓ They spit or vomit on food. ✓ Through faeces and landing on food.

Ways in which houseflies contaminate food.

- Spitting on food.
- Defecating on food.
- Vomiting on food.
- Laying eggs on food.
- Landing on food.

Diseases spread by houseflies

- cholera
- typhoid
- diarrhoea
- dysentery
- trachoma. This affects the eyes.

These diseases are caused by bacteria. Dysentery is also caused by protozoa.

The lifecycle of a housefly.

- A housefly undergoes complete metamorphosis.

Reason. The larva is completely different from the adult in structure and function.

- It shows all the four stages of development i.e., eggs, larva, pupa and adult.
- The eggs and pupa are dormant stages because they do not feed and move.
- The larva and the adult are the active stages because they feed and move.
- The larva stage of a housefly is called **maggot**.
- Houseflies lay their eggs in decaying matter/moist decomposing places.

Reasons. - To enable their larvae get food after being hatched.

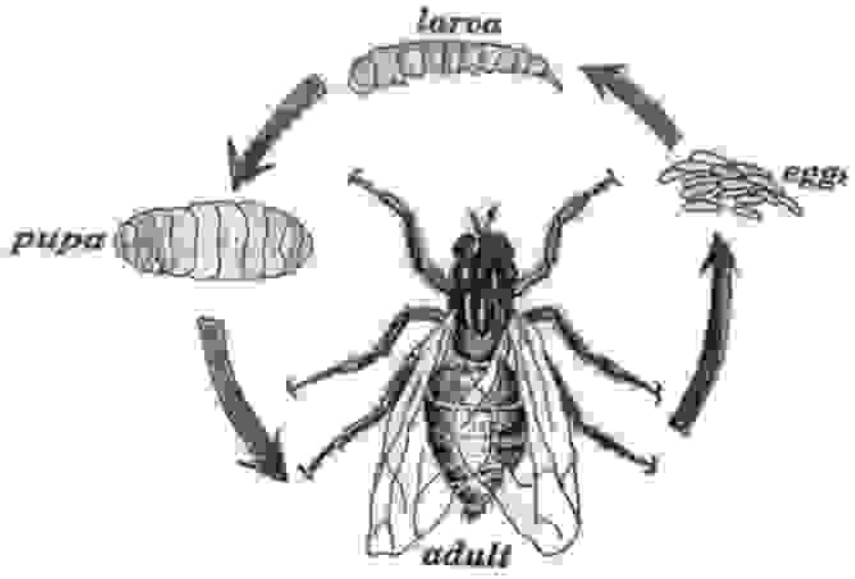
- Presence of warmth and moisture for eggs to hatch.

▪ The larvae (maggots) are the useful stage of the housefly.

▪ Maggots help to reduce the volume of faeces in latrines.

Note. It is not advisable to pour chemicals like acids in pit latrines
Reason. Chemicals kill maggots and bacteria that reduce the volume of faeces in latrines.

▪ The adult is the imago. It is the harmful stage of the housefly.



Note well. Complete metamorphosis is the type of lifecycle where the larva is completely different from the adult in structure and function.

Ways in which the breeding of houseflies can be controlled.

- Proper disposal of rubbish.
- Proper disposal of human wastes.
- Cleaning latrines/toilets.
- Spraying insecticides to kill adult houseflies.
- Smoking ordinary pit latrines. Smoking keeps out smell which attracts houseflies.
- Use of a fly swatter. This uses electricity.



A fly swatter

Activities carried out in the control of houseflies



Spraying houseflies.



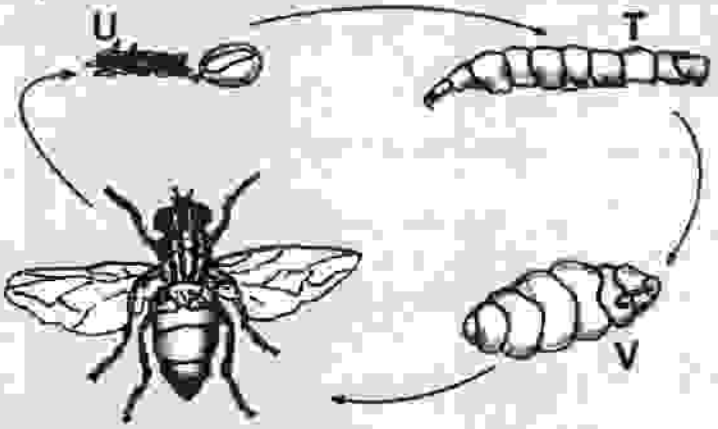
Smoking pit latrine.



General cleanliness.

Evaluation activity 10.1

1. Which vector spreads relapsing fever to people?
2. How can the breeding of houseflies be controlled in our compound?
3. State the difference between the feeding of a tick and that of a housefly.
4. Why do houseflies lay their eggs in decaying matter?
5. Name the dormant stage of a housefly after hatching.
6. The diagram below shows the lifecycle of a housefly. Study and use it to answer the questions that follow.



(a) Name the growth stages marked **U** and **V**.
(b) How is the stage marked **T** useful in the environment?
(c) Why is the above lifecycle regarded as the complete lifecycle?

7.(a) Write down any **one** disease that houseflies spread to humans.
(b) State any **two** ways in which houseflies contaminate food.
(c) Give **one** characteristic of a housefly that makes it to spread germs to humans.

8. (a) Identify the stage in the lifecycle of a housefly that reduces the volume of faeces in the latrine.
(b) Why is it not advisable to pour acids in the pit latrines?

Evaluation activity 10.2 Continued

8 (c) Apart from the stage identified in (a) above, name **one** other group of organisms that reduce the volume of faeces in pit latrines.

9. The list below shows different disease vectors. Study and use it to answer the questions that follow.

tick, housefly, flea, cockroach

(a) Identify **two** vectors from the list above which suck blood from the human body.

(b) What type of lifecycle does the cockroach from the list above undergo?

(c) Apart from spreading diarrhoeal diseases, state **one** other way in which houseflies are harmful to human health.

Key words:

- rabid
- relapsing
- nits
- louse
- scabies
- itch mite

MTES AND LICE

MITES

Characteristics of mites

- They are very small parasites.
- They have no wings.
- They have eight jointed legs.
- They have two main body divisions
- They feed on blood.
- They spread scabies to people.
- They spread scabies through bites.

Scabies

Scabies is an infectious disease that affects the human skin.

It is caused by **itch mites.** They are parasites. Itch mites affect the sides of feet, elbows, behind knees,waists, writs and between fingers and armpits.



How scabies spreads among people.

- Sharing clothing.
- Sharing basins or buckets with an infected person.
- Through body contact with an infected person.

Signs and symptoms of scabies.

- Skin rash
- Blisters on the skin.
- Itchy skin.
- Thick crusts on the skin.
- Swelling of the skin.



Itch mite

Ways of controlling scabies.

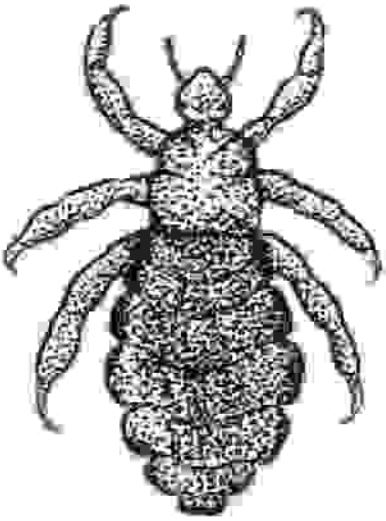
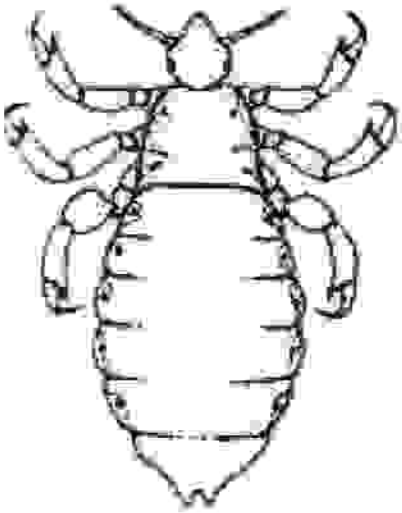
- Regular bathing.
- Regular washing of clothes.
- Ironing clothing and bedding.
- Early treatment of scabies.
- Avoid sharing clothes, towels, bedding or combs.
- Using chemicals to kill mites.

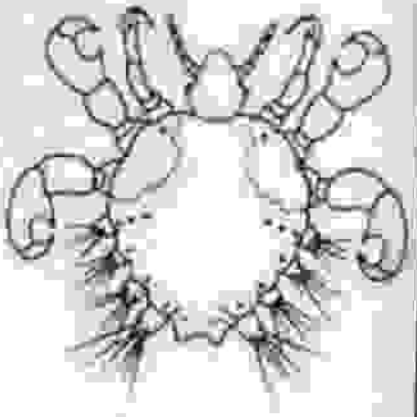
LICE

Characteristics of lice

- They are wingless insect vectors.
- They suck blood causing itching and irritation.
- They spread germs through body contact.
- They undergo incomplete metamorphosis.
- They use proboscis to suck blood.
- Their eggs are called nits.
- They spread germs through bites.
- They spread relapsing fever and typhus fever to people. They are caused by bacteria

Types of lice in humans

Type	Features	Diagram
Body lice	- They are cream-white in colour. - They live in dirty clothes. - The female lice lay eggs in clothes. - They are common in clothes which are dirty and not ironed.	
Head lice	- They are the most common lice. - They are black in colour. - They live in dirty hair and heads. - They spread by sharing combs.	

Type	Features	Diagram
Crab lice	- They live in hair around the private parts. - They spread through body contact during sexual intercourse.	

Prevention and control of lice.

- Regular bathing.
- Regular washing of clothes.
- Shaving hair.
- Regular ironing of clothes.



Bathing



Washing clothes



Shaving hair

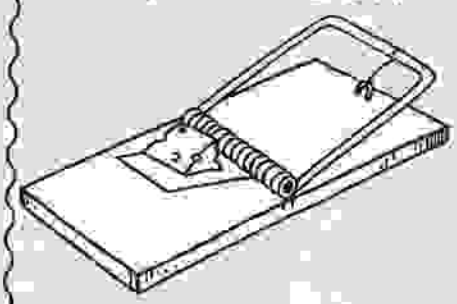


Ironing clothes

Evaluation activity 10.3

1. Which disease is spread to humans by rat fleas?
2. State any **one** way of controlling the breeding of lice at home.
3. How is the feeding of lice similar to that of mites?
4. Apart from lice, name **one** other vector that spreads relapsing fever to people.
5. Give any **two** dangers of rats to people.

The diagram below shows a way of controlling vectors at home. Use it to answer questions **6** and **7**.



6. Name the structure shown in the diagram above.
7. Which vector is controlled using the method above?

Evaluation activity 10.3 Continued

8. Name the type of lice prevented by regular combing of hair.
9. What name is given to the eggs laid by nits?
10. In the table below, part A shows vectors and part B has diseases they spread to humans.

Part A: Vectors	Part B: diseases
Mite	Dysentery
Louse	Bubonic plague
Rat flea	Scabies
Housefly	Relapsing fever

Use the diseases in part B of the table to match the vectors in the spaces provided below.

(a) Mite .....
(b) Louse .....
(c) Rat flea .....
(d) Housefly .....

Key words:

- rabid
- bark
- agitation
- paralysis
- rabies
- infected

RABID DOG

A rabid dog spread rabies to humans.
Rabies is caused by virus and affects the brain.
Rabies is also spread by infected cats or bats.

Signs and symptoms of rabies in humans.

- Barking like a dog.
- Excessive salivation.
- Difficulty in swallowing.
- Fever and headache
- Muscle weakness
- Agitation and anxiety.
- Fear of water.
- Paralysis in severe cases.



Prevention of rabies in humans.

- Vaccinate dogs and cats.
- Avoid body contact with infected dog or cats.
- Seek immediate medical care if bitten by a dog.
- Kill the infected dogs or cats.

Evaluation activity 10.4

1. (a) Name any **one** domestic animal that can spread rabies to people.
(b) Write down any **two** ways in which a patient of rabies can be identified.
(c) State any **one** way of controlling the spread of rabies among people.
2. The table below shows diseases, the vectors that spread them and disease germs. Study and complete it correctly.

Disease	Vector	Germs
Cholera	houseflies	.....
Typhus fever	.....	bacteria
.....	Infected cats	virus
River blindness	Blackflies	.....

Key words:

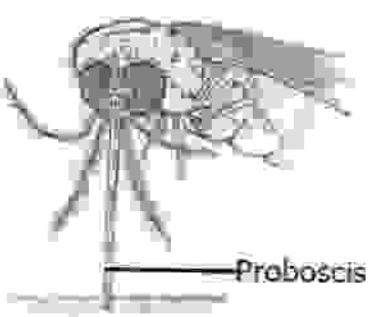
- plasmodium
- aedes
- elephantiasis
- filarial worm
- trypanosome
- tsetse fly

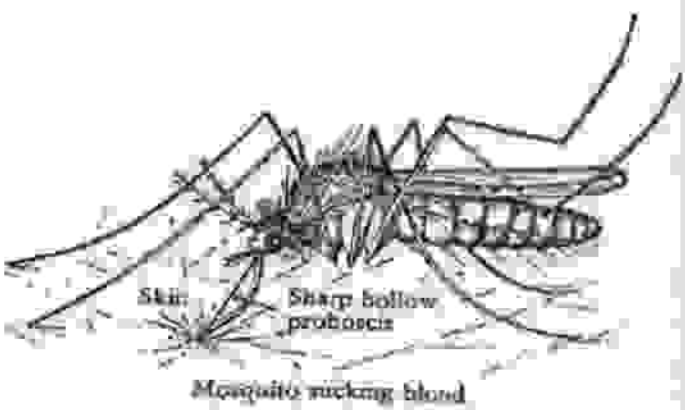
MOSQUITOES, WATER SNAILS AND TSETSE FLIES

MOSQUITOES

Characteristics that make mosquitoes suitable vectors.

- They have sharp proboscis for sucking blood.
- They feed on human blood.
- Germs survive in their bodies.

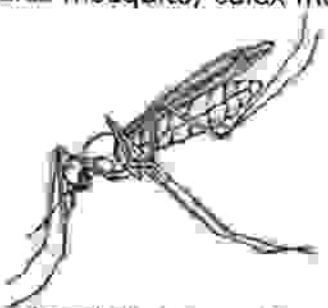




Mosquito sucking blood

Types of mosquitoes

Anopheles mosquito, culex mosquito and aedes/tiger mosquito


Anopheles mosquito


Culex mosquito

Note well. Both aedes mosquito and culex mosquito lie parallel while at rest.

Places where mosquitoes live.

Bushes, stagnant water, swamps and houses

Feeding habits in mosquitoes.

• Female mosquitoes feed on blood.

Reason. To develop eggs that hatch properly. Proteins help to develop eggs. Iron helps the eggs to hatch.

The female mosquitoes follow the direction of the sense of smell to reach people.

They follow the warm air produced in the mouth or nose.

• Male mosquitoes feed on plant juice/nectar.

Mosquitoes have proboscis for sucking blood and plant juice.

Lifecycle of mosquitoes

Mosquitoes undergo complete metamorphosis.

Mosquitoes lay their eggs in stagnant/still water.

Reasons.

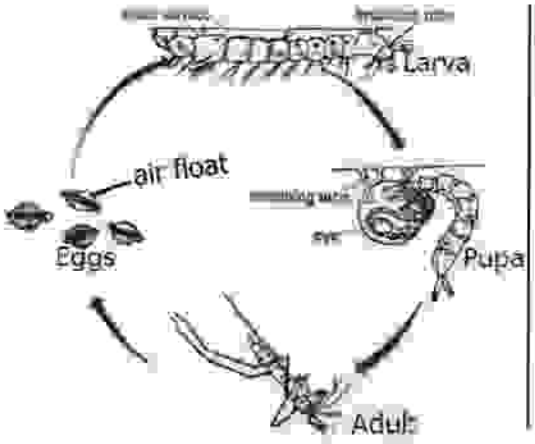
- Stagnant water keeps their eggs in one position.
- Stagnant water enables their larvae to get food.
- Stagnant water enables their larvae to attach to the surface of water.

↕ The larvae of mosquitoes are the wrigglers.

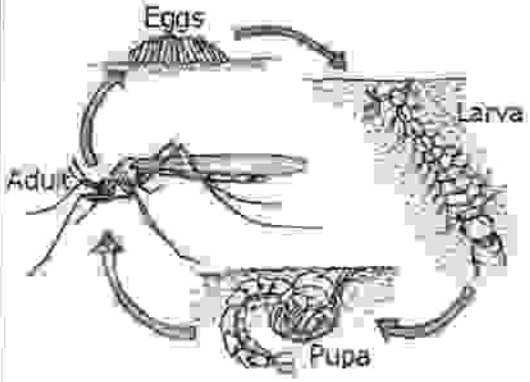
↕ The larvae move by swimming while the pupae move by wriggling.

↕ The larvae breathe through the breathing trumpets and pupae breathe through the siphon.

Life cycle of anopheles mosquito



Life cycle of culex mosquito



Characteristics of lifecycles of mosquitoes.

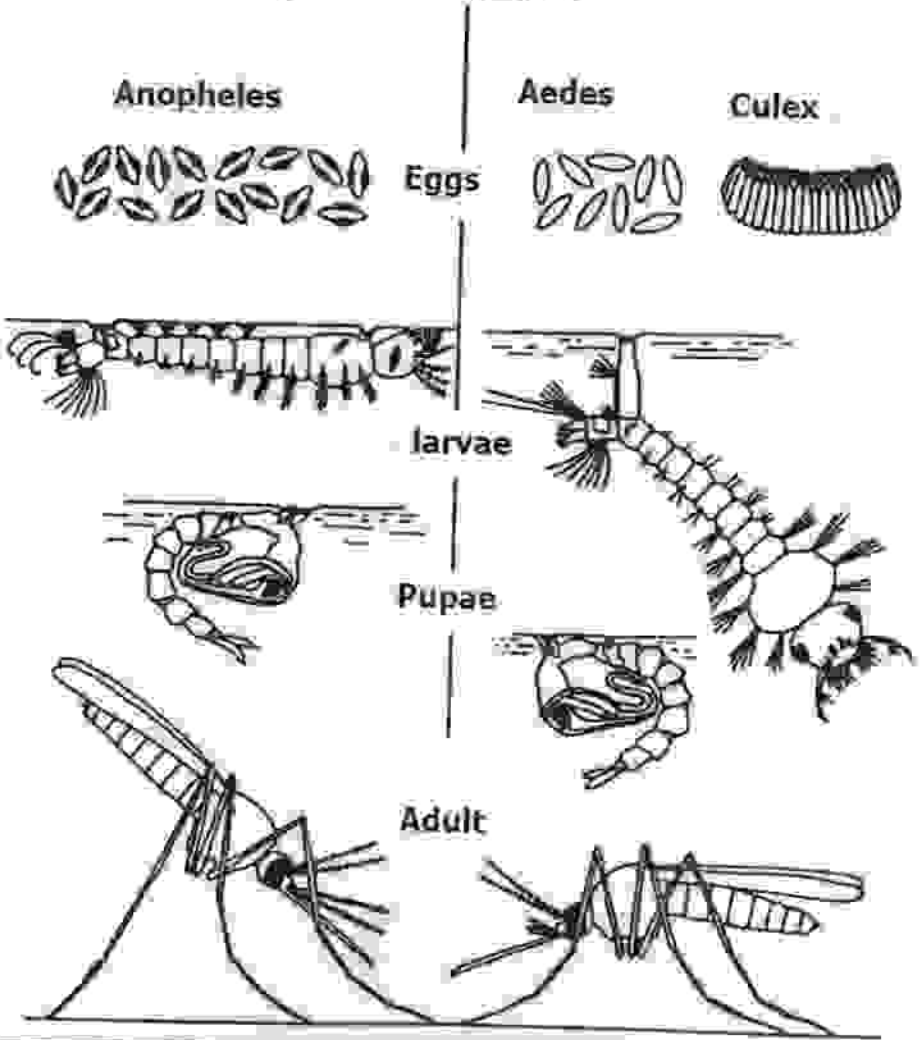
Female anopheles mosquitoes

- They lay eggs singly with air floats on either side.
- Air floats enable eggs to float on stagnant water.
- Adult lies to the ground at an angle of 450/sloppy position while at rest.
- The larva stage lies parallel to the water surface when breathing.
- They are active at night.

Culex mosquitoes
- They lay eggs in a raft/batch.
- The adult lies horizontally or flat on the surface while at rest.
- The larva attaches itself onto stagnant water with the head pointing downwards.
- The adult culex mosquito lies parallel to the surface when at rest.
- They are active during day and night.

Aedes/tiger mosquitoes
- They lay eggs singly.
- The adult lies parallel or flat to the ground while at rest.
- The adult has white and black spots.
- The larva lies at an angle to the surface source when breathing.
- They are more active during day time.

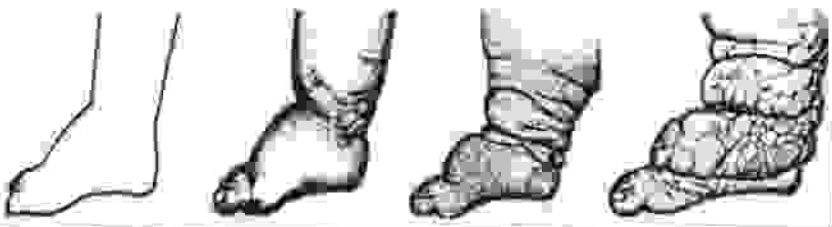
Summary of the life cycle of mosquitoes



Diseases spread by mosquitoes

Mosquitoes spread diseases through bites.

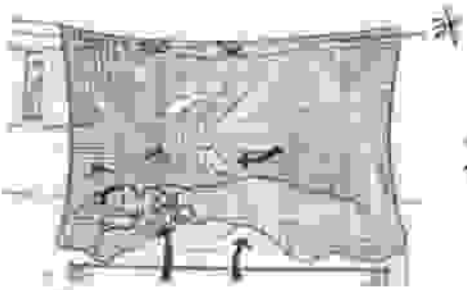
Mosquito	Diseases	Germs	Body organs /part affected
female anopheles	malaria	protozoa	brain
culex mosquito	elephantiasis	worms	legs, genitals and breasts
aedes mosquito	yellow fever dengue fever	viruses	liver

Disease	Signs and symptoms
malaria	• Headache • Abdominal pain. • Vomiting • Pain in joints • Fever • Body weakness ***Note: untreated malaria leads to anaemia.*** ***How?*** ***Malaria parasites destroy the red blood cells.*** ***Children suffering from malaria can shiver and chatter their teeth.*** ***Reason. Due to fever as the body tries to raise its temperature to fight the infection.***
elephantiasis	• Severe swelling in limbs or genitals. • Thickened and hardened skin. • Pain in affected areas. • Limited movement *Stage 1* *Stage 2* *Stage 3* *Stage 4* 
yellow fever	• Fever and chills • Muscle pain • Jaundice/yellowing of eyes. • Headache and nausea

Ways in which mosquitoes can be controlled at home.

Control measure	How it controls mosquitoes/malaria
❖ Draining stagnant water.	- It destroys the breeding grounds for mosquitoes
❖ Pouring oil on stagnant water	- Oil cuts off oxygen supply to the mosquito larvae hence killing them. - It blocks breathing tubes which suffocates and kills mosquito larvae.

Control measure	How it controls mosquitoes/malaria
❖ Sleeping under treated mosquito nets. This controls the spread of malaria not mosquitoes.	- Mosquito nets prevent mosquitoes from biting and transmitting malaria to people.
❖ Slashing tall grass around the compound.	- It destroys the hiding places for mosquitoes.
❖ Spraying insecticides.	- It kills adult mosquitoes that would breed.
❖ Keeping fish in nearby ponds.	- Fish feed on mosquito larvae.
❖ Using mosquito repellents. They include coils which are burnt to produce repelling smell. Other are applied on the skin e.g., citronella and lemon grass.	-The repelling smell chases away mosquitoes.
❖ Using protective nets or screens in windows.	-They prevent mosquitoes from entering the house.
❖ Early closing of windows and doors.	-It prevents mosquitoes from entering the house.



Sleeping under a mosquito net



Spraying mosquito



mosquito coil

Biological methods of controlling malaria/mosquitoes.

❖ Keeping fish in nearby ponds.
Fish feed on mosquito larvae.

❖ Allowing amphibians to stay in ponds.
Amphibians like frogs, toads, salamanders and newts feed on mosquitoes and their larvae.

Examples of insects that control mosquitoes are dragon flies and damselflies.

❖ Planting mosquito repellent plants
They produce strong unpleasant smell that chases away mosquitoes.
Examples of mosquito repellent plants are; citronella, lavender, marigold, lemon balm rosemary, catnip and basil.
❖ Allowing geckoes to stay in the house.
Geckoes feed on mosquitoes which spread malaria.

Ways in which the breeding of mosquitoes can be controlled.

- Pouring oil on stagnant water.
- Draining stagnant water.
- Allowing amphibians to stay in ponds.
- Keeping fish in ponds.

Roles a school Health Club can play in the control of malaria.

✓ Staging talk shows about malaria control.
✓ Composing songs about malaria control.
✓ Acting plays about malaria control.
✓ Planting mosquito repellent plants.
✓ Displaying posters about malaria control.
✓ Educating the community about malaria control.

Examples of animals that help to control mosquitoes

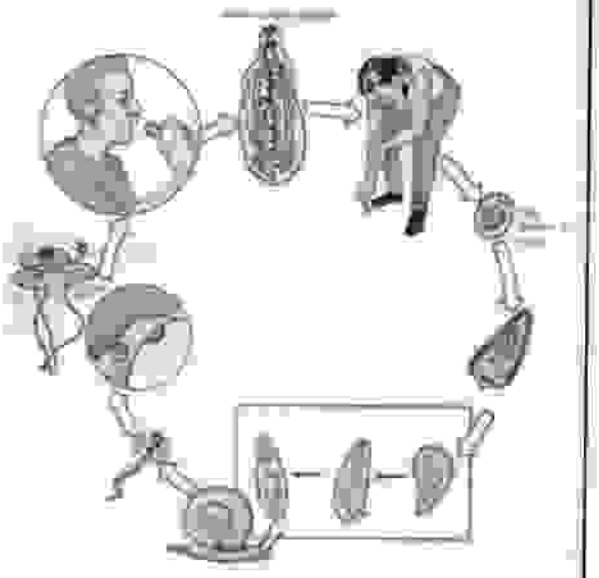
Frogs, toads, fish, geckos

Examples of drugs used to treat malaria.

Artemether, Quinine, Chloroquine and Mefloquine

WATER SNAILS

- They live in water.
- They spread bilharzia.
- It affects the kidney.
- Bilharzia is caused by blood flukes. Blood flukes feed on blood causing anaema.



Signs of bilharzia in humans.

- Blood in urine or stool.
- Difficulty urination.
- Abdominal pain.
- Diarrhoea
- Skin rash.
- Muscle aches.
- Coughing
- Weight loss.
- Fever

Prevention and control of bilharzia

- Avoid swimming in fresh water bodies in endemic areas.
- Use safe water for bathing and drinking.
- Boil water before drinking.
- Wear protective clothing when working in contaminated water.
- Control snail populations in freshwater bodies.

TSETSE FLIES

Characteristics of a tsetse fly as a vector.

- A tsetse fly feeds on blood.
- A tsetse fly has a sharp proboscis to suck blood. It has forward pointed proboscis.
- The germs do not die in its body.



Note: Its wings fold over the body like a pair of closed scissors.

Points to note:

-A tsetse fly spreads sleeping sickness through bites. It bites only during day time.

-Sleeping sickness is caused by protozoa.

-A tsetse fly also spreads nagana to farm animals.

-It crosses its wings when at rest.

-Tsetse flies live in bushy areas, forests, near water bodies, swamps, thickets and jungles.

The lifecycle of a tsetse fly

A tsetse fly does not lay eggs.

Reason. It gives birth to live larva.

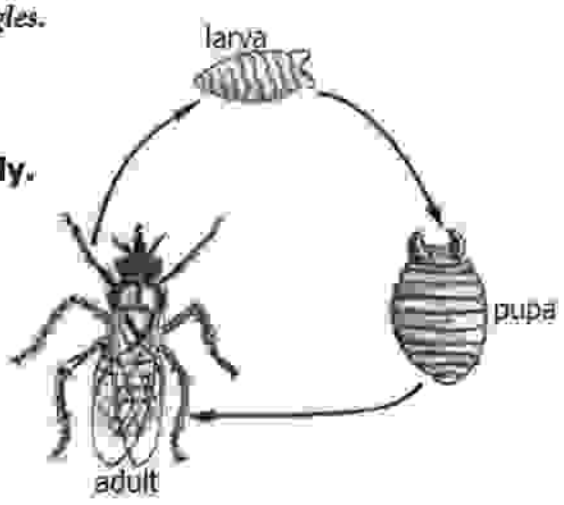
Process of life cycle of a tsetse fly.

(i) The female tsetse fly produces a fertilised egg.

(ii) The egg develops into a larva in the mother's uterus. The larva feeds on a milky substance secreted by the mother.

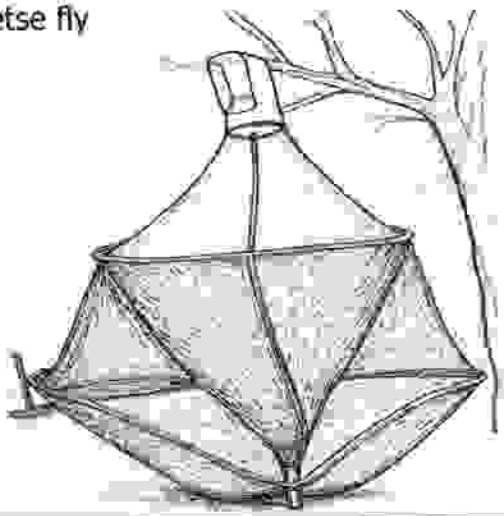
(iii) The larva burrows into the ground and develops a hard outer shell. This shell is called **puparial case.**

(iv) The pupa hatches and the adult tsetse fly emerges.



Signs and symptoms of sleeping sickness

- Consistent fever
- General body weakness
- Feeling sleepy all the time.
- Loss of appetite.
- Loss of body weight.



Ways of controlling tsetse flies at home.

- Using tsetse fly traps.
  The traps have colours which attract tsetse flies.
- Spraying insecticides to kill adult tsetse flies.
- Clearing bushes near homes.
- Keeping away from forested areas.

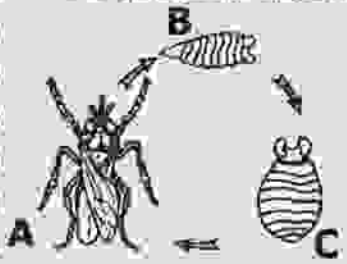
Vector	Disease (s)	Cause
cockroach	cholera, diarrhoea, typhoid, dysentery	bacteria protozoa
tick	relapsing fever	bacteria
rat flea	bubonic plague	bacteria
housefly	cholera, diarrhoea, typhoid, dysentery, trachoma	bacteria, protozoa
mite	scabies	Itch mites
louse	typhus fever, relapsing fever	bacteria
female anopheles mosquito	malaria	protozoa
blackfly	river blindness	worms
rabid dog	rabies	virus
culex mosquito	elephantiasis/filariasis	worms
infected cat	rabies	virus
water snail	bilharzia	blood flukes
tsetse fly	sleeping sickness	protozoa
Aedes mosquito	Yellow fever and dengue fever	virus

Evaluation activity 10.5

1. Name the vector that transmits elephantiasis to people.
2. Which organ in the human body is affected by yellow fever?
3. Give the reason why a person suffering from bilharzia develops anaemia.
4. (a) Identify any **two** characteristics of the female anopheles mosquito that makes it a good vector for malaria.
   (b) How do the following methods help in controlling the spread of malaria?
   (i) Pouring oil on stagnant water.
   (ii) Sleeping under a mosquito net.
5. (a) Which vector spreads sleeping sickness to humans?
   (b) Write down any **two** signs of sleeping sickness in humans.
   (c) State any **one** way of controlling sleeping sickness in humans.

Evaluation activity 10.5 Continued

6. The diagram below shows the life cycle of a vector. Study it to answer the questions that follow.



(a) Name the vector whose life cycle is shown above.
(b) Name the stage marked **B**.
(c) What does stage **A** feed on?
(d) How does the vector whose life cycle is shown above spread disease?

7. The diagram below shows the eggs laid by different mosquitoes. Study and use it to answer the questions that follow.



U



V

(a) Name the type of mosquito that lays eggs marked **U** and **V**.
(b) State the importance of air floats on the eggs marked **U**.
(c) How does the mosquito that lays eggs marked **V** affect human health?

The diagram below shows a patient suffering from a disease. Use it to answer questions **8** and **9**.



8. Name the disease that the patient above is suffering from.
9. Which vector transmits the disease the above patient is suffering from?

MEANING OF KEY TERMS IN DISEASES AND VECTORS

Nits: Eggs laid by a louse.

Vector: An organism that spreads germs.

Germ: A microbe that causes diseases.

MEANING OF KEY TERMS IN DISEASES AND VECTORS

Nymph: The second stage in the life cycle of a cockroach.

Imago: An adult insect.

Plasmodia: The germ that causes malaria.

Proboscis: The part on the mouth of some insects used for sucking fluids.

Mandible: The part on the mouth of some insects used for chewing food.

Metamorphosis: Body changes in the growth of an insect.

Biological control: Use of an organism to eliminate another organism.

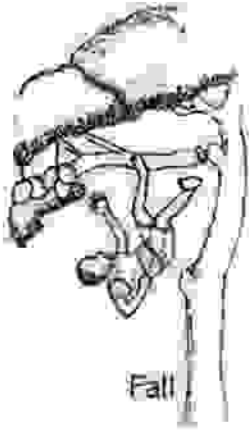
11 THEME: HUMAN HEALTH
TOPIC 11: ACCIDENTS, POISONING AND FIRST AID

Key words:
- sudden
- precaution
- accident
- injury
- casualty
- victim

ACCIDENTS
An accident is a sudden happening that can cause injury or death to a victim.
An accident victim is called a **casualty**.
First aid is the first help given to a casualty before getting further treatment.

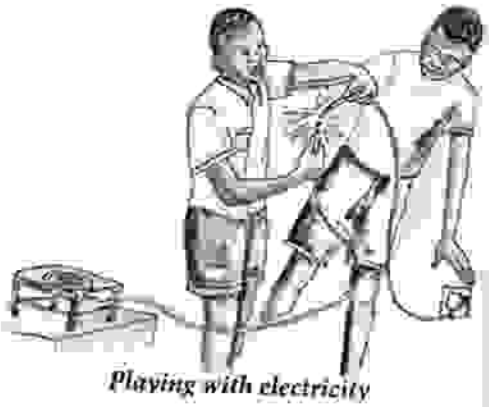
Accidents at home, on the way and at school.
They include; traffic accidents, cuts, falls, scalds, lightning, snake bites, poisoning, Burns and nose bleeding.
Note well: All injuries are accidents but not all accidents are injuries.

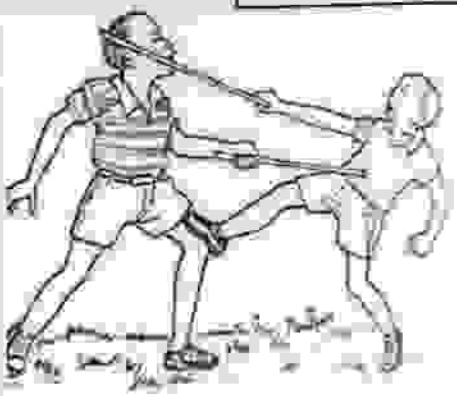

Cut


Fall


Traffic accident

Common causes of accidents.
- ✓ Failure to follow safety precautions.
- ✓ Handling sharp objects carelessly.
- ✓ Playing bad games.
- ✓ Ignorance of dangerous objects.
- ✓ Poor conditions
- ✓ Playing with electricity.
- ✓ Throwing stones
- ✓ Stumbling over the stone.
- ✓ Unawareness


Playing with electricity



Rough games



Disturbing a dog

Common forms/types of accidents

Form of accident	First aid.
Insect stings Bees and wasps sting when disturbed. Their stings have venom. The infected part itch, swell and pain severely.	- Scrap the stung area with a blunt object to remove the stinger. - Apply Vaseline on the stung area. - Apply lemon juice to neutralise the venom.
Dog bite This happens when a dog is disturbed. It can lead to the spread of rabies in case it is a rabid dog.	- Wash the wound using soap and clean water.
Snake bite. An injury caused when a snake bites a person. Some snakes like cobra, mamba and viper have venom. They use fangs to bite and inject venom. Snakes bite is identified by; - Two or four fang marks in the skin. - Bleeding from the wound. - Numbness - Swelling at the bitten site.	- Tie a bandage or tourniquet slightly above the bitten part. This prevents venom from flowing to the heart to weaken cardiac muscles. -Apply a black stone in case it is available. ***Note well:*** ***Never make a casualty to run. This makes the venom to reach the heart very fast.***

Form of accident	First aid.
Nose bleeding. This is a condition in which blood flows from the nostrils. **Causes of nose bleeding** -Overblowing of the nose -Fractured skull -Strong sneezing -Punch on the nose -Nose picking.	-Lean the casualty forward. **Reason.** To prevent blood from flowing into the throat and airways. When blood flows through the throat or airways can cause irritation or choking. -Pinch the soft part of the nose. **Reason.** To promote blood clotting. Apply cold compress. **Reason.** To cool temperature in the nose. Encourage the casualty to breathe through the mouth. **Reason.** To reduce the pressure of blood flowing through the nose.
Electric shock. This is a sudden discharge of electricity through the body. **Causes of electric shock** -Body contact with live electric wires. -Body conductors connected to the source of electricity.	-Switch off electricity as soon as possible. -Use the dry wood to remove the casualty.



Snake bite



Stumbling over the stone



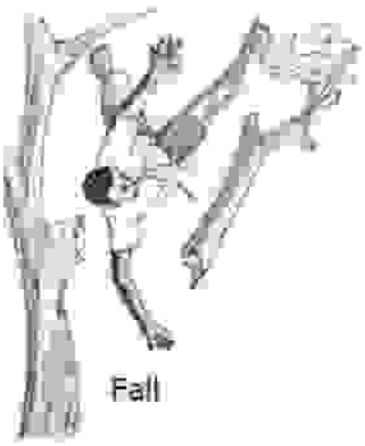
Nose bleeding



Electric shock



Bee stings



Fall

Evaluation activity 11.1

1. (a) Give the meaning of the following terms:
   (i) Accident
   (ii) First aid

(b) State any **two** conditions that can lead to outbreak of accidents at school.

2. Write down any **two** causes of falls at home.

The diagram below shows a form of accident. Use it to answer questions **3** and **4.**



3. Name the form of accident shown in the diagram above.
4. Give any **one** cause of the form of accident above to people.
5. (a) Write down any **two** causes of nose bleeding.

(b) Why is a person whose nose is bleeding advised to;
(i) Lean forward his/her nose?
(ii) Pinch the soft part of his/her nose?

6. State any **one** danger of playing with electricity at home.

Evaluation activity 11.1

7. Why is it not advisable to make a casualty of a snake bite to run?
8. In the table below, part **A** shows forms of accident and their first aid in part **B** of the table.

Part A: Form of accident	Part B: First aid
Electric shock	Apply Vaseline on the affected part.
Noses bleeding	Tie the bandage slightly above the affected part.
Snake bite	Use a dry wooden stick to remove the casualty.
Bee sting	Lean the casualty forward.

Match correctly the forms of accidents with their importance in the spaces provided below.

(i) Electric shock .....
(ii) Nose bleeding .....
(iii) Snake bite .....
(iv) Bee sting .....

Key word:

⊡ inflammable ⊡ bleeding ⊡ injury ⊡ bruise ⊡ wound ⊡ tissue

Causes of accidents at home.

-Throwing hard or sharp objects anyhow.
-Wet surfaces
-Failure to switch of the gas after use.
-Poor lighting of fire.
-Leaving candles near flammable substances.
-Over drinking of alcohol.
-Keeping flammable liquids like petrol in the house.

Common accidents at home

Falls, dog bites, scalds, falls, insect stings, snake bites, burns, cuts

Common accidents on the way

Road traffic accidents, drowning and near drowning.

Common accidents at school.

Falls, insect bites, cuts and scalds.

Effects/dangers of accidents.

✓ They can cause lameness.
✓ They can cause death.
✓ They cause a lot of pain to the victim.
✓ They increase costs of treating the victim.

TYPES OF INJURIES

An injury is a harm or damage done to a person's body.

Examples/types of injuries.

- fracture
- bruises
- cuts
- wounds
- sprains
- strains
- dislocation
- burns
- scalds

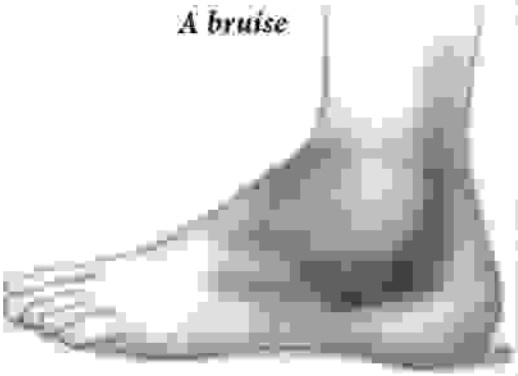
Bruise

A bruise is a body swelling caused by internal bleeding

First aid

Apply ice pack wrapped in a towel around the bruise.

A bruise



Cut

A cut is an opening on the skin caused by a sharp object.

Such objects include; nails, knife, pin broken glass and razorblade.





Cuts

Types of cuts

Minor/simple cuts. They do not go deep in the skin.

Deep cuts. They go deeper in the skin

Effects of cuts

-They cause wounds.

-They cause bleeding.

First aid for deep cuts

- Use the index finger to press hard on the bleeding spot.
- Tie the bandage at the cut to reduce bleeding.
- Raise the injured part higher than the heart. This reduces the flow of blood to the injured part.

Note: a tourniquet is used to control excessive bleeding at the cut or wound.

Minor cuts.

- ❖ Hold the cut hard with your fingers to stop bleeding
- ❖ Put surgical spirit to prevent infection.
- ❖ Apply a plaster/bandage to reduce bleeding pressure.

Ways in which cuts can be prevented.

- Throwing broken glasses into deep pits.
- Keeping sharp tools in safe places.
- Use chopping boards when cutting meat.
- Use hard hand gloves to pick broken sharp objects.



Bandaging the cut



Wounds

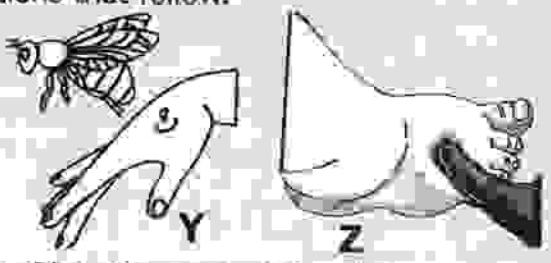
A wound is a tear of the body tissues.

Type of wound	Description
Incised wounds	Wounds caused by sharp objects that cause open bleeding. Such objects are razorblades and knives.
Lacerated wounds	Wounds caused by objects with irregular edges. Such objects are; barbed wires, animal teeth and animal claws.
Contused wounds	Wounds caused by direct blows by some objects. Such objects are hard walls.
Punctured wounds	Wounds which have a small opening but very deep. Objects that caused such wounds are; needles, nails, arrows and spears.

Type of wound	Description
First aid for wounds 	Put surgical spirit around the wound. Apply plaster/bandage to reduce bleeding ***Points to note:*** ***Abrasion is a wound on the skin caused by rubbing on a hard surface*** ***Avulsion is an injury in which the body structure is torn off.***

Evaluation activity 11.2

1. (a) Write down any **two** examples of common accidents at home.
   (b) State any **two** causes of common accidents at home.
2. The diagram below shows types of injuries. Study and use them to answer the questions that follow.



(a) Name the types of injuries marked **Y** and **Z**.
(b) State any **one** first aid that can be given to the type of injury marked **Y**.
(c) Give any **one** cause of the type of injury marked **Z**.

3. In the table below, part **A** shows types of wounds and part **B** has their meanings.

Part A: Type of wound	Part B: Meaning
Incised wound	Wounds which have a small opening but very deep.
Lacerated wound	Wounds caused by objects with irregular edges.
Contused wound	Wounds caused by sharp objects that cause open bleeding.
Punctured wound	Wounds caused by direct blows by some objects.

Use the meanings in part B of the table above to match the types of wounds in the spaces provided below.
(a) Incised wound .....
(b) Lacerated wound .....
(c) Contused wound .....
(d) Punctured wound .....

Fracture

A fracture is a broken or cracked bone in the human body.

Causes of fracture

- Falls
- Playing football
- Knocks
- Kick boxing
- Rough games.

Types of fracture

- Simple fracture
- Compound fracture

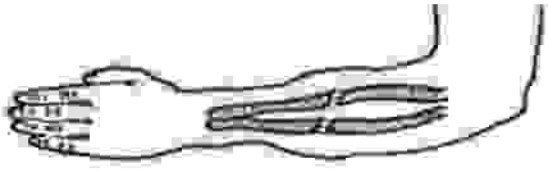
Simple fracture

This is where the broken bone remains inside the skin.

Signs and symptoms of simple fracture

- ❖ Swelling of the injured limb.
- ❖ Pain at the joint.
- ❖ Internal bleeding
- ❖ Failure to move the injured limb.
- ❖ The broken bone remains inside the skin.





Point to note

Green stick fracture is a form of simple fracture in babies. Babies have soft bones It makes the bones of the baby to bend. The bone does not break completely.

Compound fracture

This is where the broken bone tears the flesh.

Signs and symptoms of compound fracture.

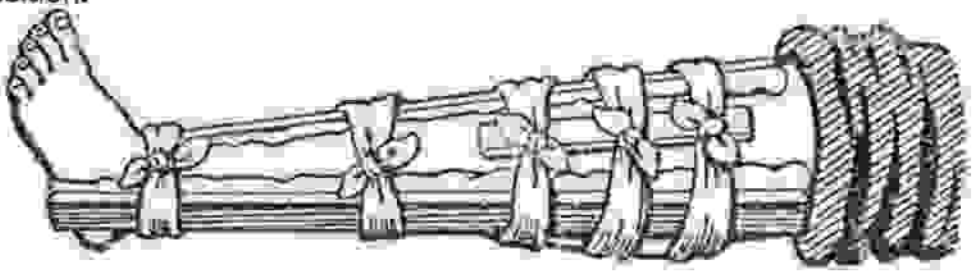
- The broken bone comes of the skin.
- External bleeding.
- Damaged tissue at the injured part.
- Swelling of the fractured part.
- Severe pain around the injured part.
- Failure to move the injured part.





First aid for fractures

(i) Tie splints around the injured part. Splints keep the fractured bone in one position.



(ii) Tie the armsling around the injured arm.
Armsling keeps the broken bone in one position.



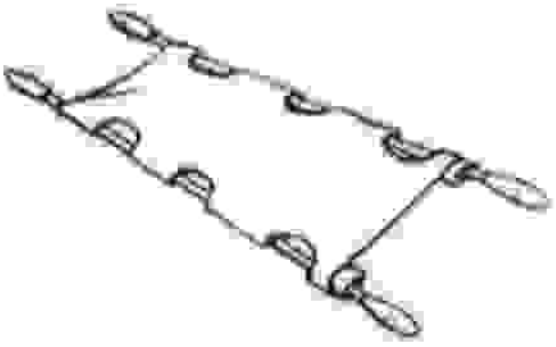
(iii) Give the casualty crutches to support him/her when walking.

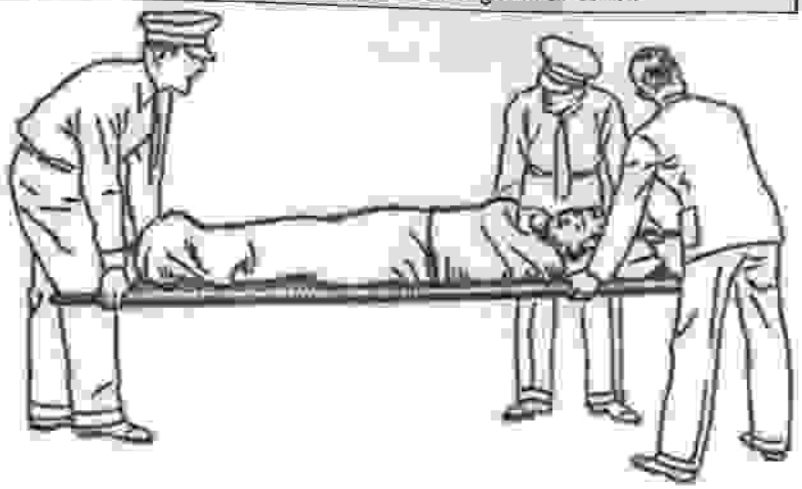


Crutches

(iv) Use the stretcher to carry the casualty to the health centre.
It is used to carry a casualty whose injured limb cannot move.







Dislocation, sprains and strains

A sprain is a torn/twisted ligament

A sprain is torn/twisted muscle.

A dislocation is an injury in which the bone moves away from its normal position.

Activities that cause sprains, strains and dislocations

- Playing football
- Athletics
- Playing netball
- Kick boxing
- Wrestling
- Walking down stairs

Signs and symptoms of sprains, strains and dislocation

- Pain around the injured part.
- Swelling of the injured part.
- Redness around the injured part.
- Difficulty in moving the injured part.

First aid for sprains, strains and dislocation

Apply cold compress around the injured part.

This is called **RICE**

First Aid	Importance
Rest the injured part.	It prevents further injuries.
Apply a block of ice at the injured part.	It numbs pain and reduces swelling around the limb.
Compress the injured part.	It reduces swelling and provides support. Support is provided by stabilising the injured area which reduces movement.
Elevate the injured part.	It reduces swelling by improving blood flow. It allows fluids to drain away from the injured area.

Materials used when giving first aid for sprains, strains and dislocations.

- splints
- stretcher
- ice
- armsling
- crutches
- plaster of Paris

Body parts affected by sprains, strains, dislocations and fractures

- ↕ bones
- ↕ backbone
- ↕ arms
- ↕ muscles
- ↕ hands
- ↕ shoulders
- ↕ joints
- ↕ legs
- ↕ head
- ↕ fingers

Note. Sprains are more common in the ankles of footballers than the one of the gardeners.

Reason. A footballer does more vigorous exercises than a gardener.

Reason. A footballer does more vigorous exercises than a gardener.

Burns and scalds

A burn is a skin injury caused by dry heat.



Sources of dry heat

hot electric iron, hot charcoal stove, hot gas cooker, acid, hot saucepan and burning charcoal

Note well. An injury caused by acid is a burn because acid chemically reacts with and damages the skin tissues.

A scald is a skin injury caused by wet heat.

Sources of wet heat

hot water, hot soup, hot porridge, hot sauce, hot milk and steam

Causes of burns	Caused of scalds
- skin contact with hot metals. - Skin contact with acid. - Skin contact with burning charcoal.	- Skin contact with steam. - Skin contact with hot liquids.

First aid for burns and scalds.

Put the burnt part in clean cold water.

Reason. To cool temperature around the injured part of the skin.

Ways in which burns and scalds can be prevented at home.

- ✓ Use insulators to carry hot objects.
- ✓ Cook from raised places.
- ✓ Keep acids away from people's reach.
- ✓ Put out fire after use.
- ✓ Switch off burning candles in house after use.

Evaluation activity 11.3

1. Give the difference between a sprain and a strain.
2. State any **one** sign of compound fracture.
3. In which way are the crutches helpful to a fractured casualty during walking?
4. State the importance of splints when giving first aid to a casualty whose thigh bone is broken.
5. Write down any **two** activities that can lead to sprains.

Evaluation activity 11.3 Continued

The diagram below shows the type of fracture. Study and use it to answer questions **6** and **7.**



6. Name the type of fracture shown in the diagram above.
7. State any **one** first aid for the type of fracture above.
8. Which type of accident requires the use of splints when giving first aid?
9. Why should the burnt part with burning charcoal be dipped in cold water?
10. A child suddenly removed the cover of a saucepan full of boiling water. One of the child's arms was burnt.

(a) Name the substance that burnt the child.
(b) What kind of accident did the child get?
(c) State the first aid you would give to the child.
(d) Give any **one** way in which the child would avoid the accident named above.

Key words

- traffic
- pedestrian
- lollipop
- cyclist
- repair
- vehicle

ROAD TRAFFIC ACCIDENTS

These are accidents that occur among the road users.
Traffic refers to the movement of vehicles and people in a particular place.

Factor	How it causes road traffic accidents
Speeding vehicles	It is very difficult to stop vehicles where there is a problem.
Driving under alcohol influence	Alcohol makes drivers lose their normal senses to control the movement of the vehicle. It causes one to make mistakes easily while driving or riding.
Unskilled drivers or cyclists	It causes lack of proper control and decision-making leading to errors while driving or riding.
Driving overloaded vehicles	The driver can easily lose control of the vehicle and causes accident.
Bad roads	Potholes or poor surfaces make it difficult to maintain control, increasing accident risks.

Factor	How it causes road traffic accidents
Careless driving	Reckless actions increase the likelihood of collisions.
Careless crossing of the roads	Pedestrians may enter the road without warning causing accident.
Failure to follow road signs.	Ignoring traffic rules leads to confusion and crashes.
Driving vehicles in dangerous mechanical conditions.	Faulty brakes or tyres lead to loss of control.
Traffic jam	It increases frustration, leading to sudden or unsafe manoeuvres and rear-end collisions.
Playing on the road.	It distracts drivers and increases the risk of hitting pedestrians.
Floods across the road	Flooding can cause vehicles to hydroplane reducing control and visibility. Floods cause collisions or cars being swept off the road.
Broken bridges	Damaged or collapsed bridges disrupt traffic flow.
Mist, heavy smoke and rain	They prevent drivers from seeing the roads clearly while driving.
Overtaking poorly.	Some drivers overtake in wrong places like corners where they cannot see in front.

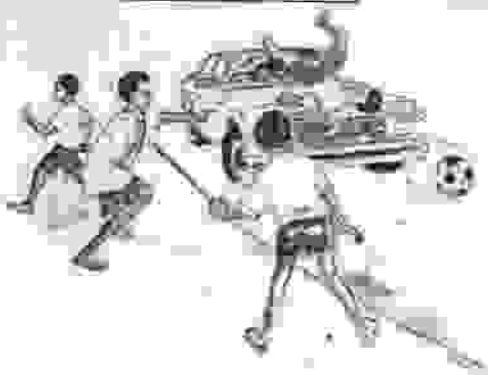
Some situations of car road traffic accidents.



Illustrations of some causes of road traffic accidents.







Groups of people that normally get involved in road traffic accidents.
Pedestrians, passengers, cyclists, motorists (drivers and aviation staff)

Prevention of road traffic accidents.

- ❖ Following road signs.
  Road signs guide road users.
- ❖ Putting humps on roads.
- ❖ Avoid driving overloaded vehicles.
- ❖ Driving vehicles in good mechanical conditions.
- ❖ Crossing the road carefully.
- ❖ Repairing damaged roads.
- ❖ Arresting drivers who speed cars.

Effects of accidents to an individual

- They can lead to death.
- They may lead to lameness.
- They can lead to deformities.
- They lead to injuries.
- They can lead to loss of some body parts.
- They can lead to loss of blood from the body.

Safety rules on the road.

- Always cross the road at the zebra crossing point.
- Elders should help children to cross roads.
- People should wait the vehicles to stop before getting in or out.
  Entering or getting out of a moving car can cause one to knock oneself against the metals around the seats inside the car.
  It can cause one to fall as one gets out of the car.
- People should not drink and drive.

Ways in which chances of road accidents can be reduced when crossing roads.

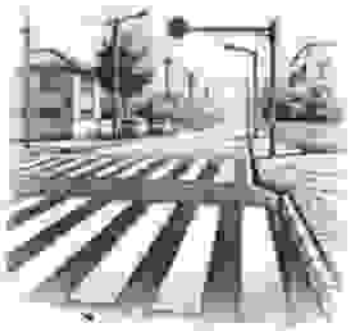
- ↕ Crossing the road at a zebra crossing.
- ↕ Crossing the road at traffic islands.
- ↕ Crossing the road where there are lollipops.
- ↕ Crossing the road at the humps.
- ↕ Crossing the road at traffic lights.
- ↕ Using flyovers to cross the road

How to cross the busy road

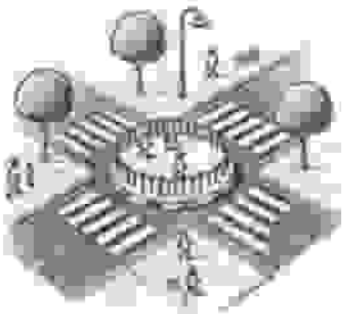
(i) First stop alongside the road.
(ii) Look right, look left.
(iii) Look right again.
(iv) If the road is clear, then cross quickly but do not run.

Places where children can cross the busy road from.

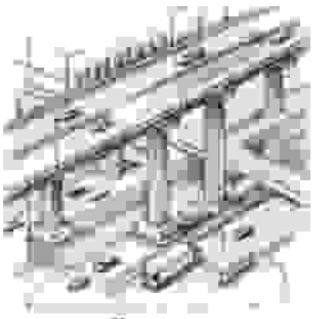
zebra crossing, flyovers, traffic lights and road islands



Zebra crossing



Road islands



flyovers

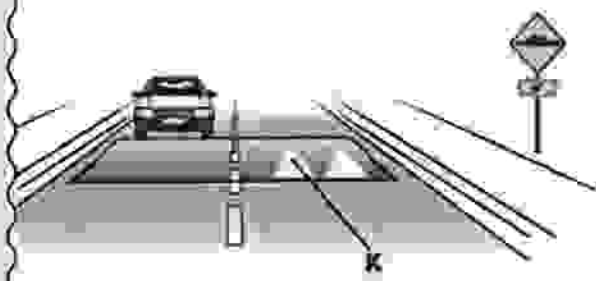
Ways of preventing road traffic accidents.

- ❖ Following road signs.
- ❖ Avoid playing on the road.
- ❖ Repairing damaged roads.
- ❖ Driving a vehicle at the moderate speed.
- ❖ Indicate signs when changing direction.
- ❖ Cross busy roads at the zebra crossing.
- ❖ Driving cars in good mechanical conditions.
- ❖ Driving vehicles under sobber state

Evaluation activity 11.4

1. State any **two** causes of accidents on roads.
2. State any **two** effects of accidents to an individual.
3. Why is the ankle of a footballer more likely to sprained than that of a gardener?
4. How does each of the following lead to road traffic accident?
   (i) Playing on roads (ii) Floods on roads.
5. Give any **two** ways in which you can reduce chances of getting accidents when crossing a road.

The diagram below shows a tarmac road. Study and use it to answer questions **6** and **7**.



6. Name the structure marked **K** on the diagram above.
7. In which way is the structure marked **K** help to reduce road traffic accidents?
8. Give the reason why many drivers cause accidents when they are drunk.
9. Give any **one** reason why is dangerous to enter or get out of the moving car.
10. Apart from zebra crossing, name any **one** other place where pedestrians can use to cross the busy road.
11. Explain how driving under the influence of alcohol causes road traffic accidents. *(2marks)*

Key words:

• poison • toxic • expire • dilute • overdose • reach

POISONING

Poisoning is the act of taking toxic substances into the body.

Poison is any substance that is harmful to the body once taken.

Poisoning is very common among young children.

Reason. Some adults or older children keep drugs in the reach of children.

Examples of poison are;

• agrochemical • rat poison • methanol • paraffin • drugs • expired food

Poison	First aid
Insects and rat poison	-Make the casualty to vomit. Vomiting helps to expel the toxin from the body before it is absorbed. This is done by; (i) Making the casualty drink water with a lot of soap. (ii) Pushing the finger into the casualty's throat. -Take the casualty to the nearest health center after vomiting.
Paraffin, petrol, jik or bleach	Make the victim to drink a lot of milk, water or juice. **Reason.** To dilute the poison content in the body. ***Warning! For poisoning like paraffin, petrol, jik and bleach,*** ***Never make the victim to vomit.*** **Reason.** It can lead to more damage of the stomach and gullet.
Snake bites	-Tie the tourniquet between the bitten part and the heart. **Reason.** To prevent venom from reaching the heart very fast. Tying a tourniquet reduces the speed of movement of the venom in blood. ***Warning! Do not make the casualty to walk. It increases the speed for the flow of blood with venom to the heart.***

Poison	First aid
Expired food products	- Drink plenty of fluids like water to prevent dehydration. - Encourage the person to rest and allow their body to recover.
Poisonous chemicals inhaled	- Move the victim to fresh air at once. - If breathing has stopped, perform artificial respiration. - Keep the affected person warm and at rest.






Signs and symptoms of a poisoned victim;

- Rapid breathing
- Loss of body balance
- Feeling a lot of thirst.
- Excessive vomiting
- Fever and sweating
- Mental confusion

Causes of poisoning at home.

- Eating expired food.
- Taking expired drugs.
- Taking overdose.
- Ignorance.
- Keeping unlabeled drugs.
- Keeping drugs in reach of children.
- Keeping chemicals in containers which are commonly used.

The major cause of child poisoning is keeping chemicals near the reach of children.



First Aid for poisoning

Give a casualty a lot of fluids like fruit juice and milk to dilute the concentration of poison.

In case of paraffin, diesel, petrol, jik and bleach, do not make the victim to vomit.

Reason. Vomiting may cause damage to the gullet, throat and stomach.

Prevention of poisoning

- Label dangerous chemicals.
- Keep chemicals out of reach of children.
- Read and follow instructions on chemicals.
- Provide proper storage of chemicals

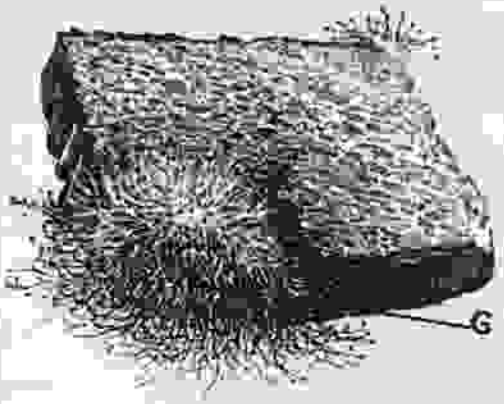
- Clear bushes around homes to prevent snakes.
- Avoid eating expired food.
- Wear the face mask while spraying agrochemicals.
  A mask prevents the person from inhaling contaminated air.



Evaluation activity 11.5

1. Name the condition caused by taking toxic substances into the body.
2. Why should chemicals like rat poison be kept away from the reach of children?
3. Why is it not advisable to make a victim who has taken paraffin to vomit?
4. Write down any **two** examples of poisonous substances.
5. State **one** way in which a mask is important in the prevention of poisoning at home.

The diagram below shows a bread. Study and use it to answer questions **6** and **7**.



6. Name the substance marked **K** on the bread above.
7. State the danger of eating the bread above to human health.
8. Give any **two** causes of poisoning at home.
9. Apart from controlling excessive bleeding at the cut, state **one** other way in which a tourniquet is useful when giving first aid.
10. Give any **two** ways in which poisoning can be prevented at home.

Key words:

• initial • tourniquet • scissor • excessive • stitch • surgical

FIRST AID AND FIRST AID KIT

First aid is the initial help given to a casualty before getting further treatment. A casualty is an accident victim.

Reasons why casualties should be given first aid

✓ To save life of the casualty. ✓ To reduce pain.
✓ To reduce excessive bleeding. ✓ To promote quick recovery.
✓ To prevent further injuries.

Factors to consider before giving first aid to a casualty.

➢ Nature of the accident ➢ Severity of the accident.
➢ Hygiene of the area and first aider. ➢ Size of the casualty
➢ Cause of the accident. ➢ Status of the casualty e.g., bleeding
➢ Items to use when giving first aid.
➢ Purpose of giving first aid for the accident.

ABC in giving first aid

Airway: The trachea has to be kept open.
Breathing: Check for chest movement during breathing.
Circulation: The person's pulse rate should be easily felt as a sign of blood circulation.

Ways in which giving first aid is important to human health.

-It saves life of the casualty. -It promotes quick recovery
-It prevents further injuries. -It reduces excessive bleeding.

First Aider

A first aider is a person who gives initial help to an accident victim.

Responsibilities of the first aider.

- To provide appropriate first aid treatment.
- To ensure safety of the casualty.
- To assess the situation quickly and calmly.
- To call for medical help if needed.
- To monitor the condition of the casualty.

Qualities that a first aider should possess

Quality	Reason for its possession
Should be clean	To prevent infection and ensure hygiene when treating wounds or injuries.
Quick in decision-making	To act promptly in emergencies which can save lives.
Should be observant	To notice all symptoms or changes in the casualty for proper care.
Should be skilful	To apply first aid techniques correctly and effectively.

Quality	Reason for its possession
Should be knowledgeable	To know the right first aid procedures for different situations.
Should be courageous	To remain calm and take action, even in difficult situations.
Should be kind	To comfort the casualty and keep him/her calm during stressful situation.

First aid box

First aid box is a container where first aid items are kept.
It can be found in schools, clinics, aeroplanes, vehicles, factories, petrol stations and hospitals.
It is painted with bright colours e.g., red for easy identification.
It has a cross to represent medical assistance and emergency care.
A first aid kit is a set of materials used to give initial help to an accident victim.



First aid box



First aid kit

Reasons why a first aid kit is necessary in different places.

- ✓ It makes it easy to give first aid to the casualties.
- ✓ It reduces the risk of infection of an injury.
- ✓ It ensures the safety of people in case of an accident.
- ✓ It keeps the first aid items safe, clean and dry.

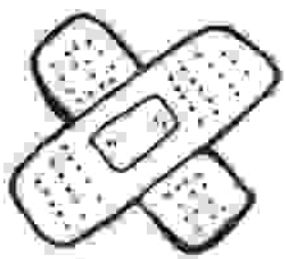
Components of the first aid kit and their uses when giving first aid.

Component	Its use/importance
Cotton wool	For cleaning the wound/cut of the casualty.
Razorblade	For cutting short fingernails to expose the wound. For shaving hair to expose the wound/cut. For cutting plasters, gauze or bandage.
Scissors	For shaving hair around the injured part of the head. For cutting plasters, gauze or bandage.
Arm sling	For holding fractured bones of the arm in one position.

Component	Its use/importance
Bandage	For covering wounds. It is used for holding the injured part. For fastening the splints. It is used as tourniquet or armsling.
Safety pins	For pinning/stitching/holding/fastening the bandage. For removing foreign bodies from the skin.
Adhesive plaster	For covering small cuts or wounds. This keeps away dirt and germs.
Iodine solution	For cleaning wounds to remove and kill germs.
Surgical spirit	For killing germs on the wound or cut. It prevents contamination of the wound by the first aider.
Surgical gloves	For protecting hands against germs while dressing the wounds.
Tweezers	For picking up tiny objects like hair from the wound. For removing bee stinger from the skin of the casualty. For holding sterile gauze and cotton wool.
Cotton gauze	For absorbing pus and other fluids from the wound. For protecting the wound from infections. For reducing bleeding. For dressing wounds.
Pain killers	For relieving pain to the casualty. They include asprin and Panadol.
Sterile eye band	For covering the eye in case of an eye injury.
Clinical thermometer	For measuring the body temperature of the casualty.
Antiseptics	For washing hands to kill germs when giving first aid. For washing some first aid equipment after use e.g., scissors
Syringe	For injecting drugs into the body.



Surgical gloves



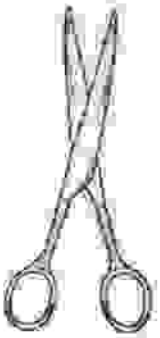
Adhesive plaster



Bandage



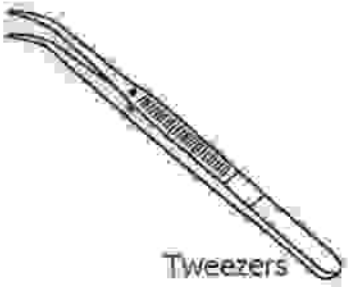
Razorblade



Scissors



Safety pin



Tweezers



Surgical spirit

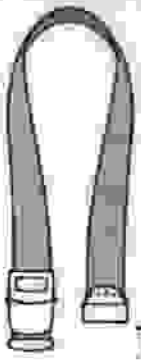


Pain killers

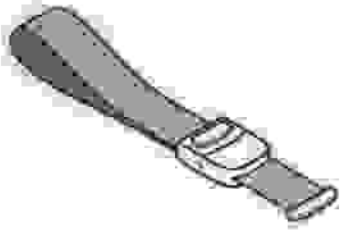


Cotton wool

Material	Accident	Use of the material
Plaster of Paris	fracture	It immobilises the broken bone as it heals. For dressing wounds.
Splints	Fracture	They keep the broken bone in one position.
Crutch	Fracture Dislocation sprain Strain	It provides support to the casualty who cannot walk alone.
Tourniquet	Wound, cut, fracture, snake bite	It is used to control excessive bleeding at the wound or cut.
Stretcher	Fracture Dislocation sprain Strain	It is used to carry a casualty who cannot walk.
First aid triangle		For making arm sling bandages.



Tourniquet





Splints

Points to note. The Plaster of Paris is first dipped in water.
Reasons. To make it wet for easy application on the fractured part. To make it hard.











An ambulance. It provides quick and safe way of transporting casualties to the hospital.



Making an emergency stretcher

Materials needed:

-2 straight pieces of wood
-2 sacks
-Strings

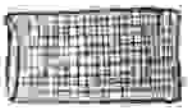
Steps taken:

(i) Get the two straight pieces of wood.
(ii) Get a sack and pass the two pieces of wood through the inverted edges.
(iii) Use the string to tie the sacks firmly onto the pieces of wood.
(iv) Make the size of the stretcher depending on the size of the casualty

Set up:


string


pieces of wood

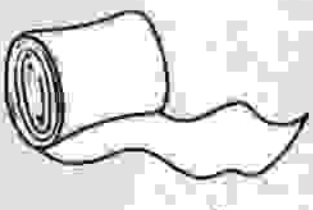

sack


stretcher

Evaluation activity 11.6

1. The diagrams below show first aid materials. Study and use them to answer the questions that follow.


W


X

(a) Name the first aid materials marked **W** and **X**.
(b) Identify any **one** kind of accident requires the material marked W.
(c) State the function of the material marked **X** when giving first aid.

2. Name the material used to control excessive bleeding at the cut.
3. What kind of accident requires the use of splints when giving first aid?

Evaluation activity 11.6 Continued

4.(a) What is meant by the term first aid?

(b) Give the importance of the following items when giving first aid.

(i) bandage.....(ii) scissors .....

(c) State any **one** quality of a person who gives first aid to a victim.

5. Kwashiorkor fell off a bicycle and broke his thigh bone.

(a) Name the injury which Kwashiorkor got.

(b) State any **two** ways in which Kwashiorkor can be given first aid.

(c) Give the importance of giving first aid to Kwashiorkor.

6. (a) Give any **two** reasons why a casualty should be given first aid.

(b) State any **two** factors considered before giving first aid to a casualty.

7. How is a safety pin useful in a first aid kit?

8. Give a reason why a person giving first aid to a casualty should be;

(i) Clean .....(ii) Quick .....

9. The table below shows first aid materials, accident and use of the material. Study it carefully and complete it correctly.

Material	Accident	Use of the material
Surgical spirit	Cut	.....
.....	Fracture	Used to keep the broken bone in place
Stretcher	.....	Used to carry a casualty who can't walk
Tourniquet	Wound	.....

MEANING OF KEY TERMS IN ACCIDENTS, POISONING AND FIRST AID.

Cut. An opening on the skin caused by a sharp object.
Sprain. A torn ligament.
Strain. A torn muscle.
Bruise. A body swelling caused by internal bleeding.
Burn. A skin injury caused by dry hot substance.
Scald. A skin injury caused by wet hot substance.
Wound. A tear of the body tissues.
Casualty. An accident victim.
Accident. A sudden happening that cause harm or death.
First aid. The first help given to a casualty before getting further treatment.
First aid kit. A set of first aid equipment.
First aid box. A container in which first aid equipment are kept.
Poisoning. Taking toxic substances into the body.
Dislocation. An injury in which a bone moves away from its normal position at a joint.
Fracture. A broken or cracked bone in the body.

12 THEME: SCIENCE IN HUMAN ACTIVITIES AND OCCUPATIONS

TOPIC 12: ANIMAL LIFE

Key Words:

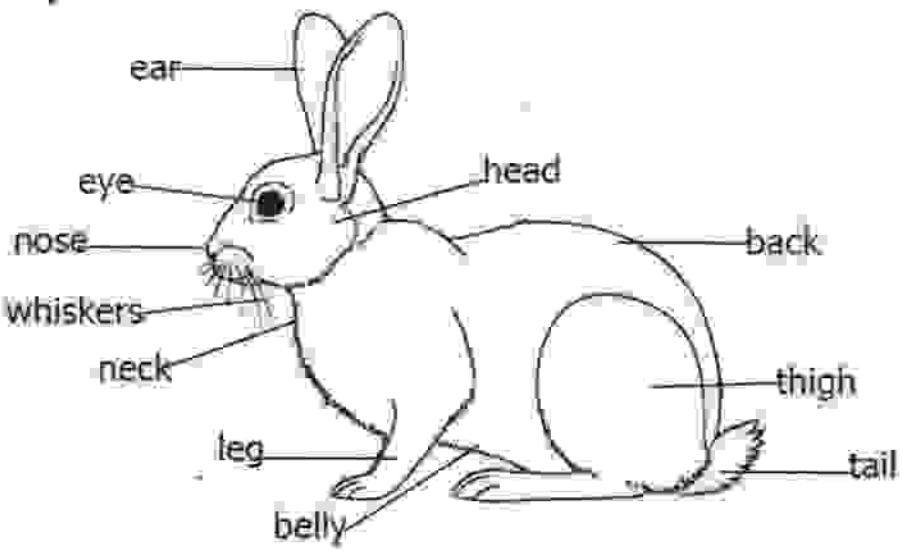
▪ Fur ▪ Exotic ▪ Californian ▪ Rabbitry ▪ Cuniculture ▪ Chinchilla

Keeping Rabbits

Cuniculture is the practice of rearing domestic rabbits.

Rabbitry is a farm of domestic rabbits.

External parts of a rabbit



Functions of some parts to the rabbit

Whiskers. They help the rabbit to sense its surroundings in the dark.

Tail. It is used for body balance during hopping.

Eyes. For sight

Nose. For smelling.

Legs. For hopping.

Ears. For hearing.

Reasons why people keep rabbits

✓ To get meat.
✓ To get fur/cony hair.
✓ To sell and get money.
✓ For practical studies in schools.
✓ Rabbits are kept as pet animals

Advantages of keeping rabbits over goats.

(i) Rabbits grow faster than goats.
(ii) Rabbits reproduce faster than goats.
(iii) Rabbits need less space than goats.
(iv) Rabbits are cheaper to buy than goats.

Types of rabbits

Local rabbits and exotic rabbits

Characteristics of the types of rabbits

Characteristics of local rabbits	Characteristics of exotic rabbits
• They are small in size • They grow slowly. • They produce little meat. • They are disease resistant. • They are mostly wild rabbits. • They run very fast. • They live in burrows.	❖ They produce a lot of meat. ❖ They grow big in size. ❖ They grow very fast. ❖ They have a lot of fur on their bodies.





Examples of exotic breeds of rabbits.

Chinchilla rabbits, Angora rabbits, Rex rabbits, New Zealand white rabbits, Californian rabbits, Dutch rabbits and Lop rabbits

Mnemonic used: **DR. CLANC**

Note. Angora rabbits are mainly kept for fur/cony hair.



Angora rabbit



Californian rabbit



New Zealand white rabbit

Differences between local rabbits and exotic rabbits.

-Local rabbits have different colours while exotic rabbits have the same colour.

-Local rabbits grow slowly while exotic rabbits grow faster.

-Local rabbits are small in size while exotic rabbits are big in size.

Ways in which rabbits are useful/important to people.

✓ They are source of meat.

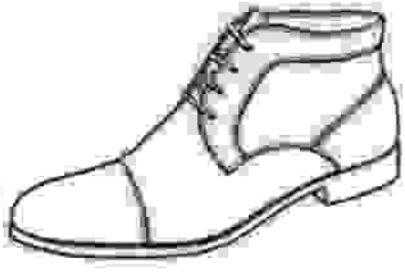
Rabbit meat is highest in proteins and lowest in fats. Meat from rabbits is called **rabbit.**

Learn, unlearn and relearn. Meat from rabbits is NOT called rabbit pie but simply called rabbit.

- ✓ They are sold for money.
- ✓ They are source of fur used in textile industries.
- ✓ Their skin is used to make leather products.
  These are; bags, shoes and belts.



Leather bag



Leather shoe



Leather belt

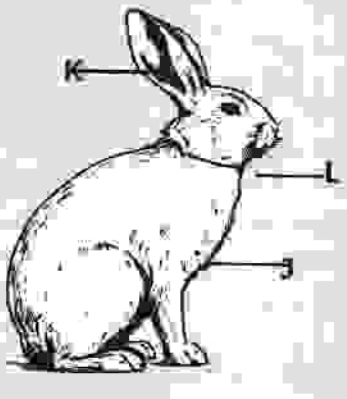
- ✓ Their droppings are used as farmyard manure in gardens.
- ✓ Rabbits are used for practical studies at school.

Products got from rabbits

Meat, skin and fur

Evaluation activity 12.1

1. Name the domestic animals kept by practising cuniculture.
2. State any **two** reasons why some farmers would prefer rearing rabbits to goats.
3. Give any **two** reasons why people keep rabbits in their homes.
4. Name the product from rabbits that is used in;
(i) Leather industry
(ii) Textile industry
5.(a) Write down any two products got from domestic rabbits.
(b) Give any **two** exotic breeds of rabbits commonly kept by people.
6. The diagram below shows external parts of the rabbit. Study and use it to answer the questions that follow.



(a) Name the parts marked **J** and **K**.
(b) State the function of part marked **L** to a rabbit.
(c) On the diagram above, label the part that has the same function as the nostril of a fish.

Key words:

▪ ventilate ▪ circulation ▪ raised ▪ dampness ▪ hutch ▪ morant

Construction of a Rabbit Hutch (Housing)



Materials used when building a hutch

Wire mesh. To allow free air circulation in the hutch
To let light into the hutch

Grass/iron sheets. For roofing the hutch to protect rabbits from harsh weather.

This harsh weather includes rain, sunshine and wind.

Other materials are; timber, nails, mud and poles.

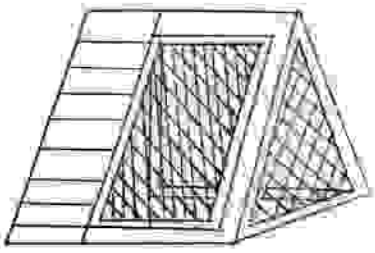
Qualities of a good hutch for keeping rabbits

Quality	Reason (s)
Raised above the ground.	To protect the floor from dampness. To protect rabbits against wild animals. To prevent rabbits from stepping in their droppings.
Well roofed/leak-proof	To protect rabbits from harsh weather.
Built with wire mesh on the sides.	To allow free air circulation in the hutch. To let light into the hutch.
Slightly sloppy floor.	To allow easy flow of urine to the ground.
Well ventilated	To allow free air circulation in the hutch.
Strong enough	To protect rabbits from wild animals and rain.
Cemented floor.	To prevent rabbits from burrowing on the floor.

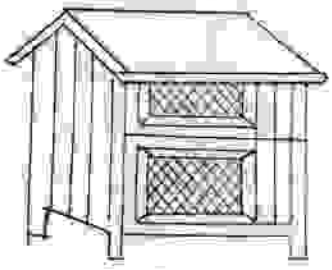
Ways of improving air circulation in a hutch.

- Using a wire mesh when building the hutch.
- Position the hutch in the direction of wind.
- Ensuring proper ventilation.

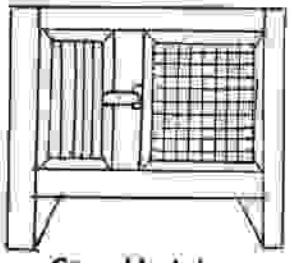
Types of hutches used in keeping rabbits.



Morant hutch



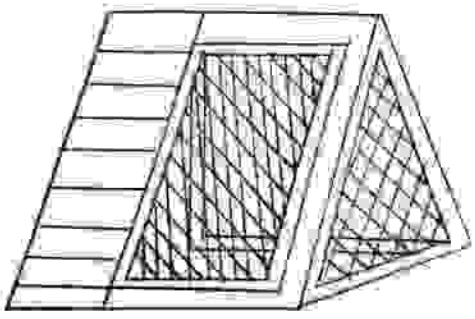
Traditional hutch



Caged hutch

Morant hutch.

It involves moving rabbits from one place to another.
It is used for feeding rabbits on grass.



Morant hutch

Disadvantages of using a morant hutch

- Limited space for rabbit hopping.
- Exposes rabbits to wild animals e.g., snakes
- It is less durable.
- Easily affected by dampness from flowing water.
- Can be cramped if multiple rabbits are housed.
- Difficult to clean.

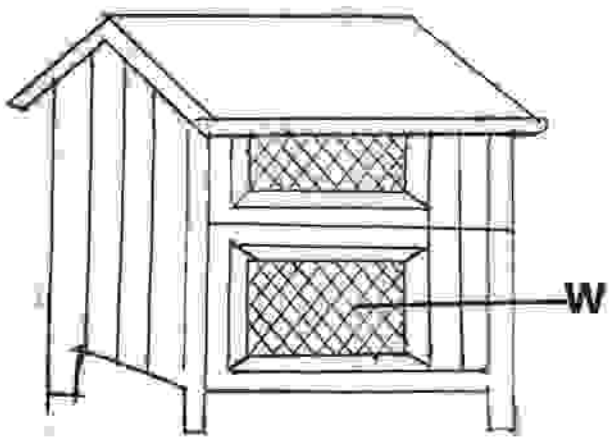
Ways of caring and cleaning the hutch.

- Regular sweeping of the hutch.
- Repairing damaged parts of the hutch.
- Ensure the hutch is dry all the time.
- Remove all leftover feeds from the hutch.
- Disinfecting the hutch.

Evaluation activity 12.2

1. Name the habitat used by farmers when rearing domestic rabbits.
2. State any **one** reason why the rabbit should be built with a sloppy floor.
3. Give any **two** ways in which a farmer can prevent dampness in the hutch during construction.
4. Name the type of hutch that has the same features as the fold system of keeping poultry.
5. Give any **two** reasons why the rabbit hutch should be raised above the ground.
6. Apart from raising above the ground, state any **two** other features of the good rabbit hutch.
7. State any **one** way in which the farmer can ensure proper circulation of air in the rabbit hutch.

The diagram below shows the rabbit hutch. Study and use it to answer questions **8** and **9**.



8. Identify the type of rabbit hutch shown in the diagram above.
9. Give any **one** way in which the structure marked W is useful to the hutch above.
10. State any **two** ways of ensuring proper care for the rabbit hutch.

Key words:

• Pellet • Kindling • Restless • Lick • Buck • doe

FEEDING RABBITS

Materials used to feed rabbits at home.

- green vegetables
- carrots
- sweet potato leaves
- plant peelings
- cereals
- milk grass
- pellets

Pellets are animal feeds produced in factories.

Salt. Salt makes their bones strong and healthy.


Pigweed

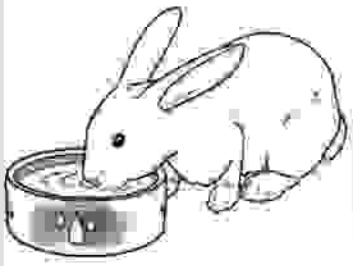

Blackjack


Sweet Potato leaves


Carrot


Banana peeling


Pellets


A rabbit drinking water.


A rabbit eating pellets.


A rabbit licking salt.

Rabbits should be given hard things to bite.
This stops their teeth from growing very long.

BREEDING OF RABBITS

Breeding is the process through which animals multiply through giving birth to young ones.

Factors to consider when selecting a good rabbit to breed.

- Healthy rabbits
- Well-formed bodies
- Age of 4-5 months old.
- Good body size
- High fertility rate.

A doe is a female rabbit.
A buck is a male rabbit.

Signs of a doe on heat

-It is restless.
-It moves up and down a lot.
-It lies on one side of its body.
-It rubs its body against the wall of the hutch.
- It is excited when it sees other rabbits.



A rabbit lying on one side

REPRODUCTION IN RABBITS

- Kindling is the act of giving birth in rabbits.
- When the doe is pregnant, it should be kept in its own hutch.

Reason. To protect her and the litter from potential harm after kindling.

- A nest should be made inside its hutch.
- A nest is a place where the doe will keep its litter.
- Young rabbits are called kits.
- A group of kits born at the same time is called litter.
- A doe takes 30 days (1 month) to kindle.
- When the doe is about to produce, it prepares a soft bed for its young ones.
- It uses hair from its body to make the bed.
- The buck should not be kept with the litter after producing.

Reason. The buck can kill the kits.

- The doe should be fed well.
- It helps the doe to produce milk for its kits.
- It keeps the health of the doe in good condition.



Point to note. When carrying an older rabbit, it Should be carried by holding it by ears firmly.

Reason. To prevent injuries and reduce pain.

BRAIN CHECK

1. (a) Give the meaning of the following terms:
(i) Doe
(ii) Buck
(b) State any **two** factors to consider when selecting a breed of rabbits to keep.
2. Write down any **two** materials that can be used to feed rabbits.
3. Why should rabbits be given a block of salt to lick?
4. Give the reason why a doe removes fur from its body when it is about to kindle.
5. State the difference between a kit and litter in rabbit keeping.
6. What is the gestation period for a doe?
7. Why is it not advisable to keep the buck in the same hutch with the kits?
8. What name is given to the manufactured feeds for rabbits?

Key words:

- snuffle
- pneumonia
- difficulty
- coccidiosis
- shiny
- runny

HYGIENE IN RABBITS

Ways of maintaining the hygiene of rabbits

- Sweeping the hutch regularly.
- Removing dirty bedding.
- Cleaning all nest boxes before using them.
- Keeping the hutch dry all the time.
- Protecting the hutch from rain.
- Washing feeding containers.

Signs of a healthy rabbit

✓ It has clear and bright eyes.
✓ It has normal appetite and drinking.
✓ It has a dry and clean nose.
✓ It has smooth shiny fur.
✓ Its droppings are in form of pellets.
✓ It is alert and has quick movements.

Signs of a sick rabbit.

❖ It has dull or cloudy eyes.
❖ It loses appetite for food.
❖ Its fur starts falling off.
❖ It sleeps all the time.
❖ It has rough and patchy fur.

Common Diseases that attack rabbits

➢ pneumonia ➢ ear canker ➢ coccidiosis ➢ colds ➢ snuffles

Remembered as: **PECCS**

Disease	Cause (germ)	Signs and symptoms.
Pneumonia	Bacteria or virus It affects the lungs	- Difficulty in breathing - Shivering - High temperature - Loss of appetite - Loss of weight - Nasal discharge
Ear canker	Bacteria Pest mites	↕ Wounds in their ears. ↕ Itching of their ears. ↕ Ears bend downwards. ↕ The rabbit shakes its head and scratches its ears.
Coccidiosis	Protozoa	❖ Rough fur. ❖ Swollen abdomen. ❖ Sudden death. ❖ Diarrhoea with blood. ❖ Loss of weight.
Colds	virus	▪ Sneezing a lot. ▪ Runny nose.

Disease	Cause (germ)	Signs and symptoms.
Snuffles	Bacteria	• Runny nose. • Mucus from nostrils. • Sneezing • Difficulty in breathing. • Coughing • Wheezing • Watery eyes blocked ears. • Sudden death

Points to note:

Coccidiosis and pneumonia attack both rabbits and poultry.

Pneumonia attack rabbits, human beings, poultry, goats, sheep and cattle.

Conditions that cause diseases in rabbits.

- ➢ Poor hygiene in the hutch.
- ➢ Overcrowding of rabbits.
- ➢ Germ invasion.
- ➢ Contaminated food.
- ➢ Poor feeding of rabbits.
- ➢ Parasites like fleas and mites.
- ➢ Inadequate ventilation.
- ➢ Lack of proper vaccinations

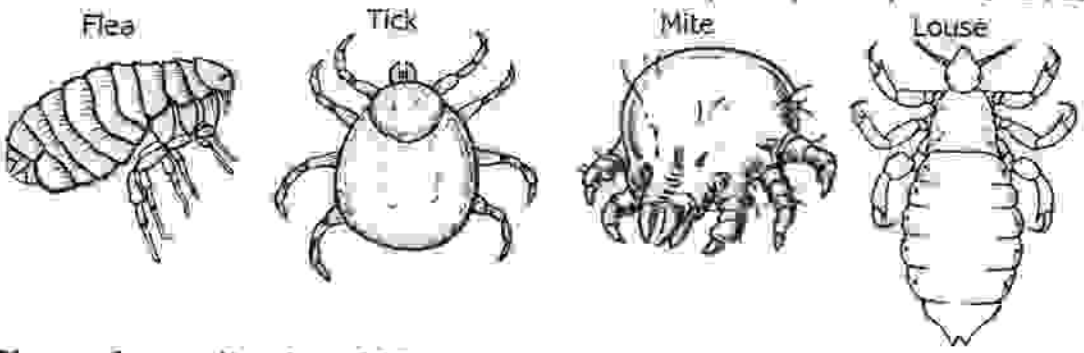
Ways in which diseases that attack rabbits can be controlled/ prevented.

- ✓ Proper ventilation of the hutch.
- ✓ Proper stocking of rabbits.
- ✓ Keeping rabbits in a warm place.
- ✓ Isolating sick rabbits from healthy ones.
- ✓ Providing clean feeds to rabbits and in clean containers.
- ✓ Keeping the hutches clean and dry regularly.
- ✓ Proper feeding of rabbits.
- ✓ Use of canker powder to the ears.

PARASITES THAT AFFECT RABBITS

A parasite is an organism that depends on another organism for food and shelter.

Examples of parasites that affect rabbits are; ticks, mites, fleas and lice



Signs of parasites in rabbits.

- Rubbing the body on the walls.
- Losing some of the fur.
- Losing weight.

Effects of parasites to rabbits

- ❖ They spread diseases to rabbits.
- ❖ They suck blood causing anaemia.

- ❖ They lower the quality of rabbit products.
- ❖ They damage the skin of rabbits.
- ❖ They cause irritation to the rabbits.

Ways of controlling parasites in rabbits.

- Spraying acaricides.
- Dusting rabbits.
- Regular deworming.
- Hand picking.
- Washing rabbits with water containing chemicals.



Washing rabbits with chemical water.



Spraying chemicals on a rabbit.

KEEPING RABBIT RECORDS

Farm records are written information about various activities done on the farm.

Date of mating	Doe number	Buck used	Date of expected birth	No. of kits	
				BORN	DIED
18/06/2024	Doe no. 12	No. 9	22/07/2024	10	4

Examples of farm records.

Production records.

They provide information about rabbit yields.

Feeding records.

They show the amount of feeds given to animals in a specific period of time.

Health records.

They show cases of diseases, their diagnosis and treatment.

Breeding records

They show the breeds of animals kept and varieties of rabbits kept.

Sales records

They show the income, daily expenditure and total sales.

Ways in which farm records are important to a farmer.

- They help a farmer to plan for the rabbits.
- They help a farmer to know income and expenditure.
- They help a farmer to know profits and losses made.
- They help a farmer to get loans from the bank.
- They help a farmer to keep the history of the farm.

STARTING A RABBIT KEEPING PROJECT.

Cuniculture is the keeping of domestic rabbits.

Rabbitry is a farm of domestic rabbits.

Factors to consider when starting a rabbit project.

Factor	Importance of the factor
land	It is used for setting up the rabbitry. It is used for growing pasture for feeding rabbits.
capital	It is used to buy the farm tools and equipment. It is used for paying workers. It is used to buy rabbits. It is used to buy feeds.
labour	It is needed to look after animals and carry out farm activities like vaccination
management	It is needed to carry out the day today activities on the farm.
market	For selling livestock products like meat and milk. For buying farm requirements like tools, drugs and feeds.

Mnemonic: LCM can used to recall these factors

ROUND UP

1. Which rabbit disease causes wounds in their ears?
2. State any **one** sign of snuffles in rabbits.
3. Name the disease of rabbits that attacks;
(i) Both rabbits and humans.
(ii) Both rabbits and cattle.
(iii) Both rabbits and poultry.
4. (a) Write down any **two** signs of a healthy rabbit.
(b) Name the disease in rabbits that attacks their sense of hearing.
(c) State any **one** way in which a farmer can ensure the hygiene of rabbits in the hutch.
5. (a) Name **two** diseases that cause sudden death in rabbits.
(b) State any **two** ways in which the farmer can control diseases in rabbits.
6. The diagram below shows a farm record on the rabbit farm. Study and use it to answer the questions that follow.

Date	Health problem	Animal (s) affected	Treatment/ Control method	Cost of treatment
04/06/2024	Pneumonia	Rabbits No. 2 and No. 6	Antibiotics	Shs. 15,000
14/09/2024	Worm infestation	Kits (03)	Dewormers	Shs.32,000
05/10/2024	Coccidiosis	All rabbits	Vaccination	Shs. 40,000

ROUND UP Cont...

(a) Name the farm record above.
(b) Apart from coccidiosis, mention any **one** other disease which is controlled by the method used on 05/09/2024.
(c) Give any **two** reasons why keeping such a record is important to farmers.
7. (a) Write down any **one** example of parasites that attack rabbits.
(b) Give any **two** ways of controlling parasites in rabbits.
(c) State any **one** factor to consider when starting a rabbit project.

MEANING OF KEY WORDS IN ANIMAL LIFE:

Kit: A young one of a rabbit. Also called kitten.
Doe: A female rabbit.
Buck: A male rabbit.
Litter: A group of kits born at the same time by the same doe.
Hutch: A house of domestic rabbits.
Burrow: A house of wild rabbits.
Pellets: Factory-made feeds for domestic rabbits.
Kindling: The act of giving birth in rabbits.
Rabbitry: A farm of domestic rabbits.
Rabbit: Meat got from rabbits.
Cuniculture: The rearing of domestic rabbits.

Congratulations, You Have Completed Primary Four Syllabus!